

*An Introduction to
Metric Spaces and
Fixed Point Theory*

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*An Introduction to
Metric Spaces and
Fixed Point Theory*

**MOHAMED A. KHAMSI
WILLIAM A. KIRK**



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Preface

This text is primarily an introduction to metric spaces and fixed point theory. It is intended to be especially useful to those who might not have ready access to other sources, or to groups of people with diverse mathematical backgrounds. Because of this the text is self-contained. Introductory properties of metric spaces and Banach spaces are included, and an appendix contains a summary of the concepts of set theory (Zorn's Lemma, Tychonoff's Theorem, transfinite induction, etc.) that might be encountered elsewhere in the text. Most of the text should be accessible to reasonably mature students who have had very little training in mathematics beyond calculus. In particular a very elementary treatment of Brouwer's Theorem is given in Chapter 7. At the same time later chapters of the book contain a large amount of material that might be of interest to more advanced students and even to serious scholars.

Readers with a good background in elementary real analysis should skip Chapters 1 and 2, and those who have had a course in functional analysis should also skip Chapter 6. Those who have had a course in set theory will have little use for the Appendix. Most readers will find something new in the remaining chapters and they might find the inclusion of this other material helpful as well.

Although a number of exercises are included, only rarely are important details of the major developments left to the reader. However in order to focus on the main development some peripheral material is included without proof, especially in later chapters.

Despite the fact that the text is largely self-contained, extensive bibliographic references are included.

In terms of content this text overlaps in places with three recent books on fixed point theory: *Nonstandard Methods in Fixed Point Theory* by A. Aksoy and M. A. Khamsi (Springer-Verlag, New York, Berlin, 1990, 139 pp.), *Topics in Metric Fixed Point Theory* by K. Goebel and W. A. Kirk (Cambridge Univ. Press, Cambridge, 1990, 244 pp.), and *Nonlinear Functional Analysis and its Applications I: Fixed-Point Theorems* by E. Zeidler (Springer-Verlag, New York, Berlin, Heidelberg, Tokyo, 1986, 897 pp.). However, in addition to the inclusion of exercises, the level of presentation and the comprehensive development of what is known in a purely metric context (especially in hyperconvex spaces) is unique to this treatment. Among other things, it has been our hope especially to illustrate the richness and depth of the abstract metric theory. Also a number of Banach space results are included here which appear in none of the above books,

some by choice and some because they describe more recent developments in the subject.

Material on the general theory of Banach space geometry is drawn from many sources but one is worth special mention: *Introduction to Banach Spaces and their Geometry, Second revised edition*, by B. Beauzamy (North-Holland, Amsterdam, New York, Oxford, 1985).

This book could easily serve as a text for an introductory course in metric and Banach spaces. In this case material should be drawn selectively from Chapters 1 through 4 along with Chapters 6 and 7, and the Appendix as needed. A number of exercises have been included at the end of each of these chapters.

The second author lectured on portions of the material covered in the text to students of the I. C. T. P.- Trieste Diploma Program in Mathematics during May 1998. He wishes to thank them for providing an attentive and critical audience. Both authors express their deep gratitude to Rafael Espínola for calling attention to a number of oversights in the penultimate draft of this text.

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by Mohamed A. Khamsi and William A. Kirk
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Part I

Metric Spaces

Chapter 1

Introduction

1.1 The real numbers $\mathbb{R}$

It would be an overstatement to say the real number system $\mathbb{R}$ is thoroughly understood despite its seeming simplicity. It is well known that $\mathbb{R}$ consists of both rational and irrational numbers, but beyond that questions arise almost immediately. A number is said to be *algebraic* if it is the solution of a polynomial equation with rational coefficients. All other numbers are said to be *transcendental*. The irrational number $\sqrt{2}$ is algebraic since it is a solution of the polynomial equation $x^2 - 2 = 0$. It is known (and these are deep facts) that π and e are transcendental. However, to this day, these are the only typical numbers that are known to be transcendental; indeed it is not even known whether or not such basic constants as $\pi + e$, $\frac{\pi}{e}$, or $\ln \pi$ are even irrational. (But surely no one really believes they are rational!) The best that can be said with certainty, at least now, is that they cannot satisfy any polynomial equation of degree eight or less with integer coefficients of average size less than 10^9 . And apart from its listing to a few million places virtually nothing is known about the decimal expansion of π . It is possible, but not likely, that all but finitely many of the terms in its decimal expansion are in fact 0's or 1's. (For other bizarre facts about π , see the recent article by Borwein, Borwein, and Bailey [15].)

Fortunately details such as the above, while curious, are not relevant for an understanding of what follows. Elementary mathematical analysis as it is usually treated rests on basic properties of $\mathbb{R}$ that can be easily understood. One such property is the so-called *least upper bound (lub) property*—which deserves special attention because of the crucial role it plays in the development of analysis. A nonempty set S of real numbers is said to be *bounded above* if there exists a number m such that for each number $x \in S$ it is the case that $x \leq m$. Such a number m is said to be an *upper bound* for S . A number u is said to be the *least upper bound* or *supremum (sup)* of a set S if (i) u is an upper bound for S and (ii) $u < m$ for any upper bound of S for which $m \neq u$. Notice by this

definition a set S can have at most one least upper bound (supremum). The sup axiom states:

(sup) Each nonempty set S of real numbers which is bounded above has a supremum.

The dual notions of *bounded below* and *greatest lower bound* or *infimum* (inf) are defined in the obvious manner, by replacing 'above' with 'below' and reversing the inequalities. The fact that any nonempty set which is bounded below has an infimum follows upon applying the sup axiom to the set $-S = \{-x : x \in S\}$. Thus for a set S which is bounded below, $\inf S = -\sup(-S)$.

Important Note: The terms 'least upper bound (lub)' and 'supremum (sup)' are usually used interchangeably, as well as the terms 'greatest lower bound (glb)' and 'infimum (inf)'.

The sup axiom is precisely the axiom that distinguishes the real numbers from the rational numbers. Consequences of the sup axiom are manifold. It assures, for instance, that any increasing [respectively, decreasing] sequence of real numbers which is bounded above [respectively, below] must have a limit. The following is an immediate consequences of this fact.

Proposition 1.1 *If $I_1 \supseteq I_2 \supseteq I_3 \supseteq \cdots$ is a descending sequence of nonempty closed intervals in $\mathbb{R}$, then $\bigcap_{n=1}^{\infty} I_n \neq \emptyset$.*

Proposition 1.1 leads directly to what is called the Bolzano-Weierstrass Theorem. The relevant definition is this: If S is a subset of $\mathbb{R}$, then a number p in $\mathbb{R}$ is said to be an *accumulation point* of S if every open interval which contains p also contains a point of S distinct from p .

Proposition 1.2 *If $[a, b]$ is a closed interval in $\mathbb{R}$ and if $S \subseteq [a, b]$ contains an infinite number of points, then some point of $[a, b]$ is an accumulation point of S .*

Proof. Since S has an infinite number of points, an infinite number of points of S must lie in at least one of the half-intervals $\left[a, \frac{a+b}{2}\right]$ or in $\left[\frac{a+b}{2}, b\right]$. Select any one of the two which contains an infinite number of points of S and call it I_1 . Now divide I_1 in half and let I_2 be one of the half-intervals of I_1 which necessarily contains an infinite number of points of S . Continue by induction. Given I_n let I_{n+1} be one of the half-intervals of I_n which contains an infinite number of points of S . In this way obtain a descending sequence of nonempty closed intervals $[a, b] \supseteq I_1 \supseteq I_2 \supseteq I_3 \supseteq \cdots$. By Proposition 1.1 $\bigcap_{n=1}^{\infty} I_n \neq \emptyset$. In fact, since the length of I_n is $\frac{b-a}{2^n}$, it must be the case that $\bigcap_{n=1}^{\infty} I_n$ consists

of a single point p . Any open interval containing p must contain I_n for some n and hence must contain an infinite number of points of S ; hence a point of S distinct from p . ■

A slightly more subtle fact which we take up in Chapter 4 is also true.

Proposition 1.3 *Any family of closed bounded subintervals of $\mathbb{R}$, each two of which intersect, must have a point in common.*

Another interesting consequence of Proposition 1.1 involves the ‘cardinality’ of $\mathbb{R}$.

Proposition 1.4 *There does not exist a function defined on $\mathbb{N}$ whose range is all of $\mathbb{R}$.*

Proof. Suppose $f : \mathbb{N} \rightarrow \mathbb{R}$ and suppose $f(\mathbb{N}) = \mathbb{R}$. Then in particular f maps a subset of $\mathbb{N}$ onto $[0, 1]$, and it is easy to replace f with a new function $\bar{f}$, which has the property $\bar{f}(\mathbb{N}) = [0, 1]$. (Set $\bar{f}(n) = 1$ if $f(n) \notin [0, 1]$ and $\bar{f}(n) = f(n)$ otherwise.) Consequently $[0, 1] = \{x_1, x_2, \dots\}$, where $\bar{f}(n) = x_n$, $n = 1, 2, \dots$. Obviously it is possible to choose a nonempty closed subinterval I_1 of $[0, 1]$ for which $x_1 \notin I_1$. Having chosen I_1 , it is now possible to choose a nonempty closed subinterval I_2 of I_1 such that $x_2 \notin I_2$. (If $x_2 \notin I_1$, simply choose $I_2 = I_1$.) Continuing in this way it is possible to choose a descending sequence $I_1 \supseteq I_2 \supseteq I_3 \supseteq \dots$ of nonempty closed intervals in $[0, 1]$ such that $x_n \notin I_n$, $n = 1, 2, \dots$. By Proposition 1.1 there exists at least one number $x \in \bigcap_{n=1}^{\infty} I_n$. Since $x \neq x_n$ for each n , x cannot be in the range of f . ■

In light of Proposition 1.4 $\mathbb{R}$ is said to be *uncountable*.

1.2 Continuous mappings in $\mathbb{R}$

The concept of ‘continuity’ arises almost immediately in the study of calculus.

Definition 1.1 *If S is a subset of $\mathbb{R}$ and if $f : S \rightarrow \mathbb{R}$, then f is said to be continuous on S if for each $a \in S$,*

$$\lim_{x \rightarrow a} f(x) = f(a).$$

In precise terms, this means the following: For each $a \in S$ and $\varepsilon > 0$ there exists a number $\delta > 0$ (depending on both a and ε) such that if $x \in S$ and if $|x - a| < \delta$, then $|f(x) - f(a)| < \varepsilon$.

Early in the study of calculus one also encounters two topological theorems on which much of the theory depends. The first is a consequence of a result published in 1917 by the Czech mathematician and philosopher Bernard Bolzano¹ (1781–1848).

¹It appears that historically the mathematical community has been slow to fully recognize Bolzano’s many contributions. In addition to the Intermediate Value Theorem and the

Theorem 1.1 (*Intermediate Value Theorem*) Let $[a, b]$ be a closed interval in $\mathbb{R}$ and let $f : [a, b] \rightarrow \mathbb{R}$ be continuous. Then for each number ξ which is between $f(a)$ and $f(b)$ there is a number c between a and b such that $f(c) = \xi$.

Theorem 1.2 (*Maximum Value Theorem*) Let $[a, b]$ be a closed interval in $\mathbb{R}$ and let $f : [a, b] \rightarrow \mathbb{R}$ be continuous. Then there is a number $c \in [a, b]$ for which $f(x) \leq f(c)$ for each $x \in [a, b]$.

The proofs of each of the above theorems rest on the important sup axiom of $\mathbb{R}$. To prove the Intermediate Value Theorem, suppose ξ is any number between $f(a)$ and $f(b)$; in particular assume $f(a) \leq \xi \leq f(b)$. If $f(a) = \xi$ there is nothing to prove. Otherwise the set

$$S = \{x \in [a, b] : f(x) < \xi\}$$

is nonempty and bounded above by b , so $\sup S$ exists. If $c = \sup S$ then it is easy to see that for each $\varepsilon > 0$, $(c - \varepsilon, c] \cap S \neq \emptyset$ and this implies

$$\lim_{x \rightarrow c^-} f(x) = f(c) \leq \xi.$$

But if $f(c) < \xi$, the fact that

$$\lim_{x \rightarrow c^+} f(x) = f(c)$$

shows that there must exist numbers $x > c$ for which $f(x) < \xi$, and this contradicts the definition of c . Therefore it must be the case that $f(c) = \xi$.

The proof of the Maximum Value Theorem is a little longer. The first step is to establish the following:

Step 1. If $f : [a, b] \rightarrow [a, b]$ is continuous, then there exists a number M such that $f(x) \leq M$ for each $x \in [a, b]$.

Assume, for the moment, that Step 1 has been established (Exercise 1.1). Then the set

$$W = \{y \in \mathbb{R} : y = f(x) \text{ for some } x \in [a, b]\}$$

is bounded above so $k = \sup W$ is well defined. We wish to show that there is a number $c \in [a, b]$ such that $f(c) = k$. The proof is by contradiction. Suppose no such c exists; that is, suppose $f(x) < k$ for each $x \in [a, b]$. Then the function

$$g(x) = \frac{1}{k - f(x)}$$

Bolzano-Weierstrass Theorem, it seems that Bolzano discovered the modern definitions of convergent sequences and even the notion of a Cauchy sequence. See [140]; also [35], pp. 48–49.

is defined and continuous on $[a, b]$. By Step 1 there is a number $M > 0$ such that $g(x) < M$ for each $x \in [a, b]$. But this is equivalent to the assertion that

$$f(x) < k - \frac{1}{M}$$

for every $x \in [a, b]$, contradicting the assertion $k = \sup W$.

Already we are in a position to state our *first fixed point theorem*.

Theorem 1.3 *Let $[a, b]$ be a closed interval in $\mathbb{R}$ and let $f : [a, b] \rightarrow [a, b]$ be continuous. Then there exists a number ξ in $[a, b]$ for which $f(\xi) = \xi$.*

Proof. Introduce the mapping $T : [a, b] \rightarrow \mathbb{R}$ by defining $T(x) = x - f(x)$ for each $x \in [a, b]$. Then T is also a continuous mapping and since $f(a) \geq a$ it must be the case that $T(a) \leq 0$. Similarly, $f(b) \leq b$ so it must be the case that $T(b) \geq 0$. By the Intermediate Value Theorem there exists a number $c \in [a, b]$ such that $T(c) = 0$; whence $f(c) = c$. ■

The rather trivial example $f(x) = \frac{x}{2}$ shows that a mapping $f : (0, 1] \rightarrow (0, 1]$ need not have a fixed point. The only possible fixed point for such a mapping is the point $0 = f(0)$, but $0 \notin (0, 1]$. Similarly, the mapping $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x + 1$ for each $x \in \mathbb{R}$ cannot have a fixed point. Also, it is very easy to give examples of discontinuous mappings $f : [0, 1] \rightarrow [0, 1]$ which fail to have fixed points.

1.3 The triangle inequality in $\mathbb{R}$.

If $a \in \mathbb{R}$ then the *absolute value* of a is defined to be the ‘distance’ between a and the 0: Thus

$$|a| = \begin{cases} a & \text{if } a \geq 0; \\ -a & \text{if } a < 0. \end{cases}$$

The *triangle inequality* in $\mathbb{R}$ asserts that for any three numbers $a, b, c \in \mathbb{R}$

$$|a + b| \leq |a| + |b|.$$

This fact is totally transparent if one approaches it from the ‘distance’ point of view. Note that upon replacing b with $-b$ and using the fact that $|-b| = |b|$ triangle inequality becomes

$$|a - b| \leq |a| + |b|.$$

Now think of $|a - b|$ as the distance between a and b , and think of $|a|$ (respectively, $|b|$) as the *distance* between a (respectively, b) and 0. There are now only three cases to consider.

1. If both a and b are nonnegative, then clearly

$$|a - b| \leq \max\{a, b\} \leq a + b = |a| + |b|.$$

2. Similarly, if both a and b are nonpositive, then

$$|a - b| \leq \max\{|a|, |b|\} \leq |a| + |b|.$$

3. If one of a or b is positive and the other negative, then

$$|a - b| = |a| + |b|.$$

1.4 The triangle inequality in $\mathbb{R}^n$

It might seem strange to refer to the inequality of the previous section as the 'triangle inequality' since it pertains to points on a line. It is however a very special case of a more general fact that has been known virtually since the inception of rigorous mathematical thought. The triangle inequality involves one of the simplest (and most important) geometrical figures known—the triangle, and one of the most important properties of the triangle as it is understood in euclidean geometry is the fact that the length of no one of its sides exceeds the sum of the lengths of its other two sides.

It is possible to derive the triangle inequality in $\mathbb{R}^n$ from purely geometric principles. Consider the standard euclidean plane (which we shall denote $\mathbb{R}^2$). If A and B are points in $\mathbb{R}^2$ let $|AB|$ denote the distance between A and B . The triangle inequality in $\mathbb{R}^2$ now becomes the statement: For each three points $A, B, C \in \mathbb{R}^2$

$$|AB| \leq |AC| + |BC|.$$

This statement has a rather nice geometric interpretation. Note that if the triangle $\triangle ABC$ has a right angle at C then by the Pythagorean Theorem

$$|AB|^2 = |AC|^2 + |BC|^2.$$

In this case one obviously has

$$|AB| \leq \sqrt{|AC|^2 + |BC|^2} \leq \sqrt{(|AC| + |BC|)^2} = |AC| + |BC|.$$

One can now proceed to the general case by simply drawing an arbitrary triangle and carefully 'dropping perpendiculars'. There are only a few cases to consider, one of which is illustrated by Exercise 1.5.

Now consider the general n -dimensional euclidean space $\mathbb{R}^n$. This is the space whose points consist of all ordered n -tuples $(x_1, x_2, \dots, x_n)$ of real numbers, with the distance between two such points $\mathbf{x} = (x_1, x_2, \dots, x_n)$ and $\mathbf{y} = (y_1, y_2, \dots, y_n)$ taken to be

$$d(\mathbf{x}, \mathbf{y}) = \left(\sum_{i=1}^n |x_i - y_i|^2 \right)^{1/2}.$$

Lifting the triangle inequality from $\mathbb{R}^2$ to $\mathbb{R}^n$ is not difficult since each three points of $\mathbb{R}^n$ lie in a two-dimensional subset (plane) of $\mathbb{R}^n$ which is itself (in terms of distances between points) a copy of $\mathbb{R}^2$. So if one has the triangle inequality in $\mathbb{R}^2$ the triangle inequality in $\mathbb{R}^n$ comes free.

There is an elegant algebraic approach to the triangle inequality in $\mathbb{R}^n$ as well. In explicit terms the triangle inequality in $\mathbb{R}^n$ asserts that for any three n -tuples, $\mathbf{x} = (x_1, x_2, \dots, x_n)$, $\mathbf{y} = (y_1, y_2, \dots, y_n)$, $\mathbf{z} = (z_1, z_2, \dots, z_n)$:

$$\left(\sum_{i=1}^n |x_i - y_i|^2 \right)^{1/2} \leq \left(\sum_{i=1}^n |x_i - z_i|^2 \right)^{1/2} + \left(\sum_{i=1}^n |z_i - y_i|^2 \right)^{1/2}.$$

Some notation will facilitate the proof. For $\mathbf{x} = (x_1, x_2, \dots, x_n)$ and $\mathbf{y} = (y_1, y_2, \dots, y_n)$ introduce the *inner product*

$$\langle \mathbf{x}, \mathbf{y} \rangle = \sum_{i=1}^n x_i y_i,$$

and the *norm*

$$\|\mathbf{x}\| = \left(\sum_{i=1}^n x_i^2 \right)^{1/2}.$$

Then $\langle \mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{y}, \mathbf{x} \rangle$, and $\langle \mathbf{x}, \mathbf{x} \rangle = \|\mathbf{x}\|^2 \geq 0$. In particular for any real number t ,

$$\langle \mathbf{x} + t\mathbf{y}, \mathbf{x} + t\mathbf{y} \rangle \geq 0,$$

and a simple calculation (using linearity in the inner product factors) yields

$$\langle \mathbf{x}, \mathbf{x} \rangle + 2t\langle \mathbf{x}, \mathbf{y} \rangle + t^2\langle \mathbf{y}, \mathbf{y} \rangle \geq 0.$$

If $\mathbf{y} \neq 0$ one can set $t = -\langle \mathbf{x}, \mathbf{y} \rangle / \langle \mathbf{y}, \mathbf{y} \rangle$ and obtain

$$\langle \mathbf{x}, \mathbf{y} \rangle^2 \leq \langle \mathbf{x}, \mathbf{x} \rangle \langle \mathbf{y}, \mathbf{y} \rangle,$$

from which

$$|\langle \mathbf{x}, \mathbf{y} \rangle| \leq \|\mathbf{x}\| \|\mathbf{y}\|.$$

This is the well-known *Cauchy-Schwarz inequality*, and it in turn implies

$$\begin{aligned} \|\mathbf{x} + \mathbf{y}\|^2 &= \langle \mathbf{x} + \mathbf{y}, \mathbf{x} + \mathbf{y} \rangle \\ &= \|\mathbf{x}\|^2 + 2\langle \mathbf{x}, \mathbf{y} \rangle + \|\mathbf{y}\|^2 \\ &\leq \|\mathbf{x}\|^2 + 2\|\mathbf{x}\| \|\mathbf{y}\| + \|\mathbf{y}\|^2 \\ &= (\|\mathbf{x}\| + \|\mathbf{y}\|)^2. \end{aligned}$$

This is the same as $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$ which, on replacing $\mathbf{x}$ with $\mathbf{x} - \mathbf{z}$ and $\mathbf{y}$ with $\mathbf{z} - \mathbf{y}$, becomes the triangle inequality; that is,

$$d(\mathbf{x}, \mathbf{y}) \leq d(\mathbf{x}, \mathbf{z}) + d(\mathbf{z}, \mathbf{y}).$$

1.5 Brouwer's Fixed Point Theorem

In Section 1.2 we saw quite easily that a continuous mapping $f : [a, b] \rightarrow [a, b]$ always has at least one fixed point. It is natural to ask whether one extend this fact to $\mathbb{R}^n$ and, if so, how? The answer is that such an extension is indeed possible, and a very natural one at that, but it is by no means easy.

The notion of continuity in $\mathbb{R}^n$ offers no problem. Merely mimic the definition in $\mathbb{R}$.

Definition 1.2 *If S is a subset of $\mathbb{R}^n$ and if $f : S \rightarrow \mathbb{R}^n$, then f is said to be continuous if for each $\mathbf{a} \in S$,*

$$\lim_{\mathbf{x} \rightarrow \mathbf{a}} f(\mathbf{x}) = f(\mathbf{a}).$$

In precise terms, this means the following: For each $\mathbf{a} \in S$ and each $\varepsilon > 0$ there exists a number $\delta > 0$ (depending on both $\mathbf{a}$ and ε) such that if $\mathbf{x} \in S$ and if $d(\mathbf{x}, \mathbf{a}) < \delta$, then $d(f(\mathbf{x}), f(\mathbf{a})) < \varepsilon$.

An appropriate analogue for the closed interval $[a, b]$ is equally at hand. The interval $[a, b]$ is precisely the set of points of $\mathbb{R}$ whose distance from the midpoint $\frac{a+b}{2}$ does not exceed $\left| \frac{a-b}{2} \right|$. Thus if we let $m = \frac{a+b}{2}$ and $r = \left| \frac{a-b}{2} \right|$, then

$$[a, b] = \{x \in \mathbb{R} : d(m, x) \leq r\}.$$

The analogue of such a set in $\mathbb{R}^n$ is a closed ball centered at a point $\mathbf{m} \in \mathbb{R}^n$ of radius $r \geq 0$:

$$B(\mathbf{m}; r) = \{\mathbf{x} \in \mathbb{R}^n : d(\mathbf{m}, \mathbf{x}) = \|\mathbf{m} - \mathbf{x}\| \leq r\}.$$

Note also that an interval has an algebraic structure. Indeed, we have

$$[a, b] = \{ta + (1-t)b : 0 \leq t \leq 1\}.$$

We are now in a position to state one of the most famous and useful 'fixed point theorems' ever proved.

Theorem 1.4 (*Brouwer's Fixed Point Theorem*) *Let B be closed ball in $\mathbb{R}^n$. Then any continuous mapping $f : B \rightarrow B$ has at least one fixed point.*

Regarding the proof of Brouwer's Theorem, difficulties arise even in the case $n = 2$. We would like to show that if B is a closed ball in $\mathbb{R}^2$, for example, the unit ball:

$$B_1 = \{(x_1, x_2) \in \mathbb{R}^2 : x_1^2 + x_2^2 \leq 1\},$$

then any continuous mapping $f : B_1 \rightarrow B_1$ has a fixed point. The trick in the case $n = 1$ was the application of the Intermediate Value Theorem. Unfortunately it is not obvious how to formulate an appropriate analogue of this theorem in $\mathbb{R}^n$.

There are several ways around this, and for excellent elementary discussions we refer to [57] and [143]. At the same time we shall revisit Brouwer's theorem in Chapter 7 and give detailed proofs of the cases $n = 2$ and $n = 3$, and these in turn will point the way to the general proof.

Brouwer's theorem has a long history. Ideas leading to the proof of Brouwer's theorem were discovered by Henri Poincaré as early as 1886. Brouwer himself proved the theorem for $n = 3$ in 1909. In 1910 Hadamard gave the first proof for arbitrary n , and Brouwer gave another proof in 1912. However in 1904 a result which is equivalent to Brouwer's theorem was published by P. Bohl. For further historical facts see, for example, [28], [106].

Exercises

Exercise 1.1 *Prove Step 1 in the proof of Theorem 1.2.*

Exercise 1.2 *Show that a continuous mapping $f : [0, 1] \rightarrow [0, 1]$ which satisfies $f(f(x)) = x$ for each $x \in [0, 1]$, and for which $f(x) \neq x$ for at least one $x \in [0, 1]$, must have exactly one fixed point.*

Exercise 1.3 *Does a continuous mapping $f : \mathbb{R} \rightarrow \mathbb{R}$ which satisfies $f(f(x)) = x$ for each $x \in \mathbb{R}$ necessarily have a fixed point?*

Exercise 1.4 *Describe a continuous mapping $f : [0, 1] \rightarrow [0, 1]$ for which*

$$f(f(x)) = x \text{ and } f(x) \neq x$$

for more than one $x \in [0, 1]$.

Exercise 1.5 *Suppose $A, B, C \in \mathbb{R}^2$ and suppose the line passing through C which is perpendicular to the line $\ell(A, B)$ joining A and B intersects $\ell(A, B)$ in a point between A and B . Give a geometric proof that $|AB| \leq |AC| + |BC|$.*

Chapter 2

Metric Spaces

A *space* is distinguished from a set by possessing attributes not possessed by a mere collection of elements, and a subspace is a subset of a space which is assumed to have inherited such defining attributes. A metric (distance) space suggests that given two points of the space there should be a real number that measures the distance between them. Accordingly, to discuss a ‘metric’ it is natural to begin with a pair (M, d) where M is a set and $d : M \times M \rightarrow \mathbb{R}^+$ is a mapping of the cartesian product $M \times M$ into the nonnegative reals $\mathbb{R}^+$. If $d(x, y)$ is thought of as the distance between two points $x, y \in M$ it is natural to assume that d satisfies for each $x, y \in M$:

$$(i) \ d(x, y) = 0 \Leftrightarrow x = y; \text{ and}$$

$$(ii) \ d(x, y) = d(y, x).$$

A pair (M, d) satisfying the above assumptions is called a *semimetric space*. These assumptions are in some sense minimal when one thinks of a distance. The semimetric spaces form a subclass of the important class of metric spaces defined below, yet it is doubtful whether semimetric spaces themselves offer sufficient structure to yield very deep results. However even at this point a number of useful concepts can be introduced, surely one of the most important being the concept of “limit”. We begin by recalling that a sequence $\{r_n\}$ of real numbers is said to *converge* to a number r (written $\lim_{n \rightarrow \infty} r_n = r$) if for each $\varepsilon > 0$ there exists an integer N such that $|r_n - r| < \varepsilon$ whenever $n \geq N$.

Definition 2.1 A sequence $\{x_n\}$ in a semimetric space M is said to converge to a point $x \in M$ if $\lim_{n \rightarrow \infty} d(x_n, x) = 0$. Thus for each $\varepsilon > 0$ there exists an integer $N \in \mathbb{N}$ such that

$$n \geq N \Rightarrow d(x_n, x) \leq \varepsilon.$$

In this case x is said to be the limit of the sequence $\{x_n\}$ and we simply write $\lim_{n \rightarrow \infty} x_n = x$.

Immediately something undesirable happens because, in general, there is nothing to assure that in a semimetric space the limit of a given sequence is unique. This defect can be remedied, however, by adding another fairly mild assumption.

Definition 2.2 *The distance d of a semimetric space (M, d) is said to be continuous if for each $p, q \in M$ the conditions $\lim_{n \rightarrow \infty} p_n = p$ and $\lim_{n \rightarrow \infty} q_n = q \Rightarrow \lim_{n \rightarrow \infty} d(p_n, q_n) = d(p, q)$.*

Owing to the complete flexibility one has in assigning distances between points in a semimetric space it is clear that many bizarre examples of such spaces exist, and the preponderance of so many strange spaces diminishes the likelihood of really interesting theorems. In looking for an appropriate assumption to add, mathematicians long ago turned to the obvious model—the euclidean spaces—and added the fundamental inequality known to hold there—the triangle inequality.

Definition 2.3 *A semimetric space (M, d) is called a metric space if it satisfies:*

(iii) *(The triangle inequality) For each three points $x, y, z \in M$,*

$$d(x, y) \leq d(x, z) + d(z, y).$$

The most natural metric space is, of course, the real line $\mathbb{R}$ with the absolute value metric: $d(x, y) = |x - y|$ for $x, y \in \mathbb{R}$. We have already verified the triangle inequality in Section 1.3. However even $\mathbb{R}$ has other interesting metrics. For example, identify $\mathbb{R}$ with the x -axis in $\mathbb{R}^2$ and let S be the circle in $\mathbb{R}^2$ with center $(0, 1)$ and radius 1. Draw a line from $(0, 1)$ to each point x on the x -axis and let x' be the point where this line intersects S . Now for two points x, y on the x -axis define $d_S(x, y)$ to be the length of the shortest circular arc on S which joins x' and y' . Then d_S is a metric on $\mathbb{R}$ with two interesting properties. First, $d_S(x, y) < \pi$ for each $x, y \in \mathbb{R}$; thus d_S is bounded. On the other hand, as with the absolute value metric, if $x, y, z \in \mathbb{R}$ satisfy $x < y < z$, then $d_S(x, y) + d_S(y, z) = d_S(x, z)$. In particular, for a sequence $\{x_n\}$ in $\mathbb{R}$, $\lim_{n \rightarrow \infty} |x_n - x| = 0 \Leftrightarrow \lim_{n \rightarrow \infty} d_S(x_n, x) = 0$. (Note that another way to describe this metric is to take $d_S(x, y) = |\tan^{-1} x - \tan^{-1} y|$ for $x, y \in \mathbb{R}$.)

Having defined metric spaces¹, what should it mean to say that two metric spaces are the same? Since the fundamental notion is distance, it makes sense to say that two metric spaces are the same if there is a (necessarily one-to-one) distance preserving mapping from one *onto* the other. Such mappings are called isometries.

¹These spaces were introduced by M. Fréchet in his thesis *Sur Quelques Points Du Calcul Fonctionnel* (Rendiconti del Circolo Matematico di Palermo, 22(1906), pp. 1–74) and called by him spaces of class (E). However Fréchet and his immediate successors did not develop the theory. The term *metric space* was introduced by F. Hausdorff (*Grundzüge der Mengenlehre*, Leipzig, 1914).

Definition 2.4 Suppose (M, d) and (N, ρ) are metric spaces. A mapping $T : M \rightarrow N$ is said to be an *isometry* if $\rho(T(x), T(y)) = d(x, y)$ for each $x, y \in M$. If T is surjective (onto), then we say that M and N are *isometric*. A surjective isometry $T : M \rightarrow M$ is called a *motion of M* .

2.1 The metric topology

A *topology* on a set X is any family $\mathcal{F}$ of subsets of X which satisfies the following simple axioms:

- (1) $\emptyset$ and X are in $\mathcal{F}$.
 - (2) The union of any subcollection of $\mathcal{F}$ is a member of $\mathcal{F}$.
 - (3) The intersection of any finite subcollection of $\mathcal{F}$ is a member of $\mathcal{F}$.
- Together the pair $(X, \mathcal{F})$ is called a *topological space*.

A subset U of X is said to be an *open* set if $U \in \mathcal{F}$. A *closed* set in X is a set whose complement is open. Thus $B \subseteq X$ is closed if $X \setminus B \in \mathcal{F}$, where

$$X \setminus B = \{x \in X : x \notin B\}.$$

If $(X, \mathcal{F})$ is a topological space then it is clear from the definition (and very elementary properties of sets) that:

- (1') $\emptyset$ and X are closed sets.
- (2') The intersection of any subcollection of closed sets is a closed set.
- (3') The union of any finite subcollection of closed sets is a closed set.

Most topological spaces, and especially those which arise naturally in the study of analysis, satisfy an additional assumption. A topological space X is said to be *Hausdorff* if given any two points $x, y \in X$ there are open sets U and V in X such that $x \in U$, $y \in V$, and $U \cap V = \emptyset$. A sequence $\{x_n\}$ of elements of a topological space X is said to *converge* to $x \in X$ (written $\lim_{n \rightarrow \infty} x_n = x$) if given any open set U containing x there is an integer N such that for $n \geq N$, $x_n \in U$. The assumption that the space is Hausdorff assures that limits of sequences are always unique.

There are two natural ways of introducing the metric topology in a metric space (M, d) . For $x \in M$ and $r > 0$ let

$$U(x; r) = \{y \in M : d(x, y) < r\}.$$

$U(x; r)$ is called the *open ball* centered at x of radius r . The *metric topology* on a metric space M is the topology obtained by taking as open sets the collection of all sets $\mathcal{F}$ in M which have the property $S \in \mathcal{F}$ provided each point $x \in S$ is the center of some open ball $U(x; r)$ (for $r > 0$) which also lies in S . It is easy

to check that $\mathcal{F}$ is indeed a topology², that $\mathcal{F}$ is Hausdorff, and that with this topology the topological notion of limit is consistent with the one defined in the preceding section: $\lim_{n \rightarrow \infty} x_n = x$ if given any $r > 0$ there exists an integer N such that if $n \geq N$, $d(x_n, x) < r$.

This gives rise to an important characterization of closed sets in a metric space.

Theorem 2.1 *A subset B of a metric space M is closed if and only if*

$$\{x_n\} \subseteq B \text{ and } \lim_{n \rightarrow \infty} x_n = x \Rightarrow x \in B. \quad (*)$$

Proof. First assume that B is closed, and suppose $\{x_n\} \subseteq B$ with $\lim_{n \rightarrow \infty} x_n = x$. Since B is closed $M \setminus B$ is open, so if $x \notin B$ there is an open ball $U = U(x; r)$ such that $U \subseteq M \setminus B$. Since $\lim_{n \rightarrow \infty} x_n = x$ there exists an integer N such that $x_n \in U(x; r)$ if $n \geq N$. Hence $x_n \notin B$ for sufficiently large n and this is a contradiction.

Now assume B is not closed and assume that $(*)$ holds. This is the same as saying that $M \setminus B$ is not open. Thus there must be some point $x \in M \setminus B$ which has the property $U(x; r)$ is not contained in $M \setminus B$ for any positive number r . But if $U(x; r)$ is not contained in $M \setminus B$ then $U(x; r) \cap B \neq \emptyset$. Since this is true for any $r > 0$ it must be the case that for each $n = 1, 2, \dots$ there is a point $x_n \in U\left(x; \frac{1}{n}\right) \cap B$. Thus the sequence $\{x_n\}$ lies in B . On the other hand, if $r > 0$ then one can choose $N \in \mathbb{N}$ so that $\frac{1}{N} < r$ and conclude that for $n \geq N$, $d(x_n, x) < r$. Therefore $\lim_{n \rightarrow \infty} x_n = x$, and since $x \notin B$ this contradicts $(*)$. ■

Another efficient way of introducing the metric topology in a metric space is to first define ‘closed sets’. Let (M, d) be a metric space. Call a point $x \in M$ a *limit point* of $S \subseteq M$ if there exists a sequence $\{x_n\}$ in S such that $\lim_{n \rightarrow \infty} x_n = x$. Now define *closed sets* in M to be precisely those sets which contain all of their limit points, and take as *open sets* those sets whose complements are closed. In view of the preceding theorem the topology obtained in this way is indeed the metric topology.

If S is a subset of a topological space X then the closure $\overline{S}$ of S is defined to be the intersection of all closed subsets of X which contain S . It is easy to see that a set S in a topological space is closed if and only if $\overline{S} = S$. Another easy consequence of the preceding theorem is the following.

Theorem 2.2 *If B is a subset of a metric space, then $x \in \overline{B}$ if and only if there exists a sequence $\{x_n\} \subseteq B$ such that $\lim_{n \rightarrow \infty} x_n = x$.*

²As required, $\emptyset$ is an open set in this topology. The assertion “If $x \in \emptyset$, then there is an open ball centered at x which lies in $\emptyset$ ” is true since its hypothesis is vacuous. This is a subtle fact of logic.

A subset S of a topological space $(X, \mathcal{F})$ is said to be *compact* if whenever S is contained in the union of a collection $\mathcal{U}$ of sets of $\mathcal{F}$ (i.e., when S has an *open cover* $\mathcal{U}$) it is the case that some finite subcollection of $\mathcal{U}$ contains S . This definition applies to metric spaces as well, but in the case of metric spaces there is a characterization of compactness that is quite useful.

Theorem 2.3 *A subset S of a metric space M is compact if and only if any sequence $\{x_n\}$ of points of S has a subsequence $\{x_{n_k}\}$ which converges to a point of S .*

Proof. Suppose S is a compact subset of a metric space M and let $\{x_n\}$ be a sequence of points of S . Suppose no point of S is the limit of any subsequence of $\{x_n\}$. This means that given any $x \in S$ there is an integer $N_x \in \mathbb{N}$ and a number $r_x > 0$ such that for $n \geq N_x$, $x_n \notin U(x; r_x)$. Since $S \subseteq \bigcup_{x \in S} U(x; r_x)$ and the family $\{U(x; r_x)\}$ consists of open sets, there must exist a finite subcollection $\{U(x_i; r_{x_i}) : i = 1, \dots, k\}$ such that $S \subseteq \bigcup_{i=1}^k U(x_i; r_{x_i})$. However this implies that if $N = \max\{N_1, \dots, N_k\}$ then $x_N \notin S$. Since this is a contradiction we conclude that some subsequence of $\{x_n\}$ must converge to some point of S . ■

In particular, a closed interval $[a, b]$ in $\mathbb{R}$, indeed any closed ball in $\mathbb{R}^n$, is compact in the sense of Theorem 2.3. In elementary analysis this fact is known as the Bolzano-Weierstrass Theorem. As noted earlier, for a closed interval this is a relatively easy consequence of Proposition 1.1 of Chapter 1.

The previous two theorems yield an important fact: *A subset of a compact metric space is itself compact if and only if it is closed.*

It is important to note that the sequential characterization of compactness is not true for general topological spaces, although it becomes true if ‘sequence’ is replaced with ‘net’ and ‘subsequence’ with ‘subnet’. This is discussed further in the Appendix.

Finally, there is another important fact about compactness that will be used repeatedly in what follows, both for metric spaces and in more general settings. We state and prove the result for metric spaces here, but we shall later invoke the fact that it holds for any topological space. The general result is proved in the same way upon replacing sequential convergence with net convergence.

Theorem 2.4 *Let M be a compact metric space, and let $f : M \rightarrow \mathbb{R}$ be a continuous mapping. Then there is a point $x_0 \in M$ such that*

$$f(x_0) = \inf\{f(x) : x \in M\}.$$

Proof. It is easy to see that $\{f(x) : x \in M\}$ is bounded below, so $m = \inf\{f(x) : x \in M\}$ exists. By the definition of infimum (inf) for each positive integer n there exists $x_n \in M$ such that $m \leq f(x_n) \leq m + \frac{1}{n}$. By compactness of

M the sequence $\{x_n\}$ has a subsequence $\{x_{n_k}\}$ which converges, say, to $x_0 \in M$. Since f is continuous, $\lim_{k \rightarrow \infty} f(x_{n_k}) = f(x_0)$. But it must be the case that

$$m \leq \lim_{k \rightarrow \infty} f(x_{n_k}) = f(x_0) \leq \lim_{k \rightarrow \infty} \left(m + \frac{1}{n_k} \right) = m.$$

Therefore $f(x_0) = m$. ■

Definition 2.5 A metric space (M, d) is said to be *totally bounded* if given $\varepsilon > 0$ there exists a finite set of points $\{x_1, x_2, \dots, x_n\} \subseteq M$, called an ε -net, such that given any $x \in M$ there exists $i \in \{1, 2, \dots, n\}$ for which $d(x, x_i) \leq \varepsilon$.

It is easy to see that compact spaces are totally bounded because if M is compact, then the union of the family $\{U(x; \varepsilon)\}_{x \in M}$ of open sets contains M ; hence some finite subcollection of this family also must contain M .

A subset K of a metric space (or for that manner, any topological space) is said to be *precompact* if its closure $\overline{K}$ is compact. It is useful to know that precompactness is equivalent to total boundedness in many metric spaces. In fact one implication always holds.

Theorem 2.5 If a subspace K of a metric space (M, d) is precompact then it is totally bounded (and in particular bounded).

Proof. Suppose K is precompact and let $\varepsilon > 0$. Since $\overline{K}$ is compact, hence totally bounded, there exists a finite set $\{x_1, x_2, \dots, x_n\} \subseteq \overline{K}$ such that if $x \in \overline{K}$ then for some $1 \leq i \leq n$, $d(x_i, x) \leq \varepsilon/2$. However since each $x_i \in \overline{K}$, for each i there exists $x'_i \in K$ such that $d(x_i, x'_i) < \varepsilon/2$. It follows that for each $x \in K$, $d(x, x'_i) < \varepsilon$ for some $1 \leq i \leq n$, so $\{x'_1, x'_2, \dots, x'_n\}$ is the desired ε -net in K . ■

Sometimes the converse of the above is true as well, a fact we take up later (Theorem 2.12). In the meantime we turn to two other important facts about compact metric spaces.

Theorem 2.6 Let (M, d) and (N, ρ) be metric spaces with M compact. Suppose $T: M \rightarrow N$ is continuous. Then $T(M)$ is compact.

Proof. In view of Theorem 2.3 it only needs to be shown that any sequence $\{y_n\} \subseteq T(M)$ has a subsequence which converges to a point of $T(M)$. Suppose $y_n = T(x_n)$, $n = 1, 2, \dots$. Since M is compact $\{x_n\}$ has a subsequence $\{x_{n_k}\}$ which converges to a point $x \in M$. Since T is continuous, $\lim_{k \rightarrow \infty} y_{n_k} = \lim_{k \rightarrow \infty} T(x_{n_k}) = T(x) \in T(M)$. ■

We conclude this section with another important fact about continuous mappings defined on compact spaces. A mapping $T: (M, d) \rightarrow (N, \rho)$ is said to be *uniformly continuous* if given $\varepsilon > 0$ there exists $\delta > 0$ such that $\rho(T(x), T(y)) < \varepsilon$ whenever $d(x, y) < \delta$.

Theorem 2.7 *Let (M, d) and (N, ρ) be metric spaces with M compact. Suppose $T : M \rightarrow N$ is continuous. Then T is uniformly continuous.*

Proof. Suppose T is not uniformly continuous. Then there exists $\varepsilon > 0$ such that for each $n \in \mathbb{N}$ there exist $x_n, y_n \in M$ such that $d(x_n, y_n) < 1/n$ while $\rho(T(x_n), T(y_n)) \geq \varepsilon$. Since M is compact it is possible to select subsequences $\{x_{n_k}\}$ of $\{x_n\}$ and $\{y_{n_k}\}$ of $\{y_n\}$ for which

$$\lim_{k \rightarrow \infty} x_{n_k} = x \text{ and } \lim_{k \rightarrow \infty} y_{n_k} = y.$$

Since $\lim_{k \rightarrow \infty} d(x_{n_k}, y_{n_k}) \leq \lim_{k \rightarrow \infty} 1/n_k = 0$ it must be the case that $x = y$, and since T is continuous,

$$\lim_{k \rightarrow \infty} T(x_{n_k}) = T(x) = T(y) = \lim_{k \rightarrow \infty} T(y_{n_k}).$$

However, this contradicts the fact that $\rho(T(x_{n_k}), T(y_{n_k})) \geq \varepsilon$ for each k . ■

2.2 Examples of metric spaces

In this section we collect some examples of metric spaces which commonly arise in the study of functional analysis. The first is quite trivial, but it (or some variation) is useful to keep in mind when thinking of counterexamples.

Example 2.1 (*The discrete space*) *Let S be any set and define for each $x, y \in S$*

$$d(x, y) = \begin{cases} 0 & \text{if } x = y; \\ 1 & \text{if } x \neq y. \end{cases}$$

The absolute value metric in $\mathbb{R}$ extends in several natural ways to the vector space $\mathbb{R}^n$ consisting of all n -tuples of real numbers. Perhaps the nicest geometrically is the extension which gives the euclidean spaces $\mathbb{R}_2^n$. For these spaces the distance $d_2(\mathbf{x}, \mathbf{y})$ between two points $\mathbf{x} = (x_1, x_2, \dots, x_n)$ and $\mathbf{y} = (y_1, y_2, \dots, y_n)$ in $\mathbb{R}^n$ is taken to be $\left(\sum_{i=1}^n |x_i - y_i|^2 \right)^{1/2}$. When we think of $\mathbb{R}^n$ as a metric space without further qualification, the underlying metric will always be the euclidean metric.

The euclidean metric on $\mathbb{R}^n$ is only the beginning. A number of other metrics can be described where the underlying space is $\mathbb{R}^n$. We begin by describing three of the most common. Each of these provides $\mathbb{R}^n$ with the same *metric topology* since in each instance metric convergence is equivalent to coordinate-wise convergence.

Example 2.2 ($\mathbb{R}_1^n$) *This is the space whose elements consist of all n -tuples $(x_1, x_2, \dots, x_n)$ of real numbers, with the distance $d_1(\mathbf{x}, \mathbf{y})$ between two elements $\mathbf{x} = (x_1, x_2, \dots, x_n)$ and $\mathbf{y} = (y_1, y_2, \dots, y_n)$ taken as*

$$d_1(\mathbf{x}, \mathbf{y}) = \sum_{i=1}^n |x_i - y_i|.$$

All the metric properties of this space are easy to verify. In fact, the triangle inequality for this space follows immediately from the triangle inequality in $\mathbb{R}$ with the absolute value metric.

Example 2.3 ($\mathbb{R}_\infty^n$) *This is the space whose elements consist of all n -tuples $(x_1, x_2, \dots, x_n)$ of real numbers, with the distance $d_\infty(\mathbf{x}, \mathbf{y})$ between two elements $\mathbf{x} = (x_1, x_2, \dots, x_n)$ and $\mathbf{y} = (y_1, y_2, \dots, y_n)$ taken as*

$$d_\infty(\mathbf{x}, \mathbf{y}) = \max_{1 \leq i \leq n} |x_i - y_i|.$$

The triangle inequality is easy to prove in this case as well because given $\mathbf{x}$, $\mathbf{y}$ and $\mathbf{z} = (z_1, z_2, \dots, z_n)$ one has for each $i = 1, 2, \dots, n$,

$$|x_i - y_i| \leq |x_i - z_i| + |z_i - y_i| \leq \max_{1 \leq i \leq n} |x_i - z_i| + \max_{1 \leq i \leq n} |z_i - y_i|.$$

Hence

$$\max_{1 \leq i \leq n} |x_i - y_i| \leq \max_{1 \leq i \leq n} |x_i - z_i| + \max_{1 \leq i \leq n} |z_i - y_i|.$$

Example 2.4 ($\mathbb{R}_p^n$, $1 < p < \infty$) *This is the space whose elements consist of all n -tuples $(x_1, x_2, \dots, x_n)$ of real numbers, with the distance $d_p(\mathbf{x}, \mathbf{y})$ between two elements $\mathbf{x} = (x_1, x_2, \dots, x_n)$ and $\mathbf{y} = (y_1, y_2, \dots, y_n)$ taken as*

$$d_p(\mathbf{x}, \mathbf{y}) = \left(\sum_{i=1}^n |x_i - y_i|^p \right)^{1/p}.$$

Having already shown that $\mathbb{R}^n$ satisfies the triangle inequality for each of the metrics d_1 , d_2 , and d_∞ it is intuitively clear that the triangle inequality should hold in all the intermediate cases d_p , $1 < p < \infty$. We omit the details (which involve extensions of Cauchy-Schwarz inequality).

Another important class of metric spaces is obtained by extending the $\mathbb{R}_2^n$ to infinite dimensions in a way that preserves the euclidean geometry.

Example 2.5 (*The Hilbert space ℓ_2*) *This is the space whose elements consist of all square summable sequences $(x_1, x_2, \dots)$ of real numbers (i.e., each of the sequences for which $\sum_{i=1}^{\infty} x_i^2$ converges), with the distance $d_2(\mathbf{x}, \mathbf{y})$ between two such sequences $\mathbf{x} = (x_1, x_2, \dots)$ and $\mathbf{y} = (y_1, y_2, \dots)$ taken as $d_2(\mathbf{x}, \mathbf{y}) = \left(\sum_{i=1}^{\infty} (x_i - y_i)^2 \right)^{1/2}$.*

For $\mathbf{x} = (x_1, x_2, \dots)$, $\mathbf{y} = (y_1, y_2, \dots) \in \ell_2$ introduce the *inner product*

$$\langle \mathbf{x}, \mathbf{y} \rangle = \sum_{i=1}^{\infty} x_i y_i,$$

and the *norm*

$$\|\mathbf{x}\| = \left(\sum_{i=1}^{\infty} x_i^2 \right)^{1/2}.$$

By the Cauchy-Schwarz inequality,

$$\left| \sum_{i=1}^n x_i y_i \right| \leq \left(\sum_{i=1}^n x_i^2 \right)^{1/2} \left(\sum_{i=1}^n y_i^2 \right)^{1/2} \leq \|\mathbf{x}\| \|\mathbf{y}\|,$$

and by letting $n \rightarrow \infty$ it follows that $\langle \mathbf{x}, \mathbf{y} \rangle$ is well defined. In turn, this proves that the series $\sum_{i=1}^{\infty} (x_i - y_i)^2$ converges, so $d_2(\mathbf{x}, \mathbf{y})$ is also well defined.

What about the triangle inequality in ℓ_2 ? Intuitively, the quickest way to see that the triangle inequality holds is to realize that any three (nonlinear) points in ℓ_2 lie in a two-dimensional hyperplane (translate of a two-dimensional subspace) of ℓ_2 which is metrically identically to $\mathbb{R}_2^2$. To be a little more rigorous, given three points $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \ell_2$ one can assume $\mathbf{z} = \mathbf{0}$ via the translation $T(\mathbf{u}) = (\mathbf{u} - \mathbf{z})$. A rotation can then be used to place $\mathbf{x}$ on the line determined by the unit vector $\mathbf{e}_1 = (1, 0, 0, 0, \dots)$ and another rotation will place $\mathbf{y}$ in the plane determined by $\mathbf{e}_1$ and $\mathbf{e}_2 = (0, 1, 0, 0, \dots)$. The subspace of ℓ_2 spanned by $\mathbf{e}_1$ and $\mathbf{e}_2$ is clearly identically to $\mathbb{R}_2^2$.

Of course, the above analysis requires some understanding of the geometry of ℓ_2 which may seem to be begging the question. Another approach is to extend the Schwarz inequality to infinite sums and follow the analytic approach used in $\mathbb{R}_2^n$. This leads to $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$ which as before, on replacing $\mathbf{x}$ with $\mathbf{x} - \mathbf{z}$ and $\mathbf{y}$ with $\mathbf{z} - \mathbf{y}$, becomes

$$d_2(\mathbf{x}, \mathbf{y}) \leq d_2(\mathbf{x}, \mathbf{z}) + d_2(\mathbf{z}, \mathbf{y}).$$

Example 2.5 describes a space which is infinite-dimensional yet it has the property that each of its two-dimensional subspaces is a euclidean plane and thus rich in geometric structure. Angles are meaningful, the law of cosines holds, and so forth. We return to these facts in Section 6.3 of Chapter 6.

The other metrics on $\mathbb{R}^n$ described above extend to infinite dimensions quite naturally as well.

Example 2.6 (ℓ_1) *This is the space whose elements consist of all absolutely convergent sequences $(x_1, x_2, \dots)$ of real numbers (i.e., each of the series $\sum_{i=1}^{\infty} |x_i|$ converges), with the distance $d_1(\mathbf{x}, \mathbf{y})$ between two such sequences $\mathbf{x} = (x_1, x_2, \dots)$ and $\mathbf{y} = (y_1, y_2, \dots)$ taken as*

$$d_1(\mathbf{x}, \mathbf{y}) = \sum_{i=1}^{\infty} |x_i - y_i|.$$

The next example will be studied extensively in Chapter 4 because despite the fact that it is infinite-dimensional it shares many nice geometric properties with the real line $\mathbb{R}$.

Example 2.7 (ℓ_∞) *This is the space whose elements consist of all bounded sequences $(x_1, x_2, \dots)$ of real numbers, with the distance $d_\infty(\mathbf{x}, \mathbf{y})$ between two such sequences $\mathbf{x} = (x_1, x_2, \dots)$ and $\mathbf{y} = (y_1, y_2, \dots)$ taken as*

$$d_\infty(\mathbf{x}, \mathbf{y}) = \sup_{1 \leq i < \infty} |x_i - y_i|.$$

In each of the two preceding cases the triangle inequality follows as in the finite-dimensional cases.

There is of course an infinite-dimensional analog of $\mathbb{R}_p^n$.

Example 2.8 (ℓ_p , $1 < p < \infty$) *This is the space whose elements consist of all p -summable sequences $(x_1, x_2, \dots)$ of real numbers (i.e., each of the series $\sum_{i=1}^{\infty} |x_i|^p$ converges), with the distance $d_p(\mathbf{x}, \mathbf{y})$ between two elements $\mathbf{x} = (x_1, x_2, \dots)$ and $\mathbf{y} = (y_1, y_2, \dots)$ taken as*

$$d_p(\mathbf{x}, \mathbf{y}) = \left(\sum_{i=1}^{\infty} |x_i - y_i|^p \right)^{1/p}.$$

Now we turn to some slightly more complicated spaces.

Example 2.9 (*The space $C[0, 1]$*) *This is the space consisting of all continuous real-valued functions defined on the closed unit interval $[0, 1]$, with the distance $d(f, g)$ between two such functions f, g taken as*

$$d(f, g) = \sup\{|f(t) - g(t)| : t \in [0, 1]\}.$$

The metric axioms for $C[0, 1]$ are easy to verify. In particular, the distance $d(f, g)$ is always defined because the function $f - g$ is itself continuous on $[0, 1]$ and hence by Theorem 1.2 bounded. The triangle inequality is derived from the fact that for any $f, g, h \in C[0, 1]$ and any $t \in [0, 1]$ the triangle inequality in $\mathbb{R}$ yields

$$|f(t) - g(t)| \leq |f(t) - h(t)| + |h(t) - g(t)|.$$

The triangle inequality for $C[0, 1]$ now follows from the fact that

$$\sup_{t \in [0, 1]} \{|f(t) - h(t)| + |h(t) - g(t)|\} \leq \sup_{t \in [0, 1]} |f(t) - h(t)| + \sup_{t \in [0, 1]} |h(t) - g(t)|.$$

In view of the fact that $[0, 1]$ is compact, if $f \in C[0, 1]$, then f is, in fact, uniformly continuous.

Example 2.10 (*The space $B[0, 1]$*) This space consists of all bounded real-valued functions defined on $[0, 1]$ with the distance $d(f, g)$ for $f, g \in B[0, 1]$ taken as above:

$$d(f, g) = \sup\{|f(t) - g(t)| : t \in [0, 1]\}.$$

Clearly the space $C[0, 1]$ is a subspace (in fact, a *closed* subspace) of $B[0, 1]$. Indeed here we can think of 'subspace' in both the metric and the algebraic sense.

There are other ways of assigning a distance to the collection of all continuous functions defined on $[0, 1]$.

Example 2.11 Let X be the space consisting of all continuous real-valued functions defined on the closed unit interval $[0, 1]$, with the distance $d(f, g)$ between two such functions f, g taken as

$$d(f, g) = \int_0^1 |f(t) - g(t)| dt.$$

It is easy to see that (X, d) is metric space using standard properties of the integral of continuous functions, such as the fact that for a function f which is continuous on $[0, 1]$:

$$\int_0^1 |f(t)| dt = 0 \Leftrightarrow f(t) = 0 \text{ for each } t \in [0, 1].$$

Example 2.12 Let X be the space consisting of all continuous real-valued functions defined on the closed unit interval $[0, 1]$, with the distance $d(f, g)$ between two such functions f, g taken as

$$d(f, g) = \left(\int_0^1 (f(t) - g(t))^2 dt \right)^{\frac{1}{2}}.$$

Perhaps surprisingly, the above space has geometric properties similar to those of ℓ_2 .

In a more abstract vein there are also standard ways of constructing new metric spaces out of given ones.

Example 2.13 (*The bounded metric d'*) Let (M, d) be a metric space, and define the metric space (M, d') by taking for $x, y \in M$

$$d'(x, y) = \frac{d(x, y)}{1 + d(x, y)}.$$

Clearly $0 \leq d'(x, y) \leq 1$. Also $d'(x, y) = 0 \Leftrightarrow x = y$, and $d'(x, y) = d'(y, x)$. There are two ways to see that the triangle inequality holds. One way is the direct computation:

$$\begin{aligned} d'(x, z) &= d(x, z)/[1 + d(x, z)] \\ &= 1 - 1/[1 + d(x, z)] \\ &\leq 1 - 1/[1 + d(x, y) + d(y, z)] \\ &= d(x, y)/[1 + d(x, y) + d(y, z)] + d(y, z)/[1 + d(x, y) + d(y, z)] \\ &\leq d'(x, y) + d'(y, z). \end{aligned}$$

Another approach is to consider the function $f : [0, \infty) \rightarrow [0, \infty)$ defined by $f(t) = \frac{t}{1+t}$. Then $f(0) = 0$, and for $t > 0$, $f'(t) = \frac{1}{(1+t)^2} > 0$ and $f''(t) = \frac{-2}{(1+t)^3} < 0$. Thus f is increasing and concave downward. This shows that Example 2.13 is a special case of the following:

Example 2.14 (*The metric transform ϕ*) Let (M, d) be a metric space, and define the metric space (M, d_ϕ) by taking for $x, y \in M$

$$d_\phi(x, y) = \phi(d(x, y)),$$

where $\phi : [0, \infty) \rightarrow [0, \infty)$ is increasing, concave downward, and satisfies $\phi(0) = 0$.

Example 2.15 (*The Hausdorff metric*) Let (M, d) be a metric space and let $\mathcal{M}$ denote the family of all nonempty bounded closed subsets of M . For $A \in \mathcal{M}$ and $\varepsilon > 0$ define the ε -neighborhood of A to be the set

$$N_\varepsilon(A) = \{x \in M : \text{dist}(x, A) < \varepsilon\}.$$

where $\text{dist}(x, A) = \inf_{y \in A} d(x, y)$. Now for $A, B \in \mathcal{M}$ set

$$H(A, B) = \inf\{\varepsilon > 0 : A \subseteq N_\varepsilon(B) \text{ and } B \subseteq N_\varepsilon(A)\}.$$

Then $(\mathcal{M}, H)$ is a metric space, and H is called the Hausdorff metric on $\mathcal{M}$.

To see that H is a metric, first observe that if $x \in A$ then $\inf_{y \in A} d(x, y) = 0$ so $x \in N_\varepsilon(A)$ for each $\varepsilon > 0$. Therefore $A \subseteq N_\varepsilon(A)$ for each $\varepsilon > 0$ and $H(A, A) = 0$. On the other hand, if $x \in N_\varepsilon(A)$ for each $\varepsilon > 0$ then for each $n \in \mathbb{N}$ there exists $y_n \in A$ such that $d(x, y_n) < \frac{1}{n}$. It follows that $\lim_{n \rightarrow \infty} y_n = x$, so $x \in \overline{A} = A$. From this it follows that if $H(A, B) = 0$ then $A = \overline{B}$. Since it is obvious from the definition that for each $A, B \in \mathcal{M}$, $H(A, B) = H(B, A)$ and $H(A, B) \geq 0$, H is a metric if it can be shown that the triangle inequality holds.

Now let $A, B, C \in \mathcal{M}$, let $\sigma = H(A, C)$ and $\mu = H(C, B)$, and let $\rho > 0$. Since $A \subseteq N_{\sigma + \rho/4}(C)$, if $a \in A$ there exists $c \in C$ such that $d(a, c) \leq \sigma + \frac{\rho}{2}$.

Also, since $C \subseteq N_{\mu + \rho/4}(B)$, there exists $b \in B$ such that $d(c, b) \leq \mu + \frac{\rho}{2}$. Therefore for any $a \in A$ there exists $b \in B$ such that

$$d(a, b) \leq d(a, c) + d(c, b) \leq \sigma + \mu + \rho.$$

This proves that $A \subseteq N_{\sigma + \mu + \rho}(B)$ for every $\rho > 0$. Hence

$$\inf\{\varepsilon > 0 : A \subseteq N_\varepsilon(B)\} \leq \sigma + \mu.$$

Interchanging the roles of A and B in the above argument yields

$$\inf\{\varepsilon > 0 : B \subseteq N_\varepsilon(A)\} \leq \sigma + \mu.$$

Therefore

$$H(A, B) \leq \sigma + \mu = H(A, C) + H(C, B).$$

■

A natural metric space which exhibits some of the anomalies of ‘metric convexity’ which will be discussed later is the two-dimensional spherical space $S_{2,r}$. (This example extends analogously to its n -dimensional counterpart, but $S_{2,r}$ is easier to visualize.)

Example 2.16 *The elements of $S_{2,r}$ are the collection of all ordered triples (x_1, x_2, x_3) of real numbers which satisfy $\sum_{i=1}^3 x_i^2 = r^2$, $r > 0$. Given a pair $x = (x_1, x_2, x_3)$ and $y = (y_1, y_2, y_3)$ of elements $S_{2,r}$ define the distance $\rho(x, y)$ to be the smallest non-negative number for which*

$$\cos\left(\frac{\rho(x, y)}{r}\right) = \frac{1}{r^2} \left(\sum_{i=1}^3 x_i y_i \right).$$

The above definition assigns as the distance between two points on the surface of the ordinary sphere of radius r in $\mathbb{R}^3$ the length of the shortest arc lying on the sphere which joins them. Since such an arc always lies in the plane which passes through the two points and the origin, in order to see that this formula gives the arc-length distance it suffices to note that if (x_1, x_2, x_3) and (y_1, y_2, y_3) are points of a sphere centered at $0 \in \mathbb{R}^3$ with radius $r > 0$ and if $\theta \in [0, \pi]$ is the angle between the vectors determined by these points, then $\theta = s/r$ where s is the length of the arc subtended by the angle θ . Hence $\cos(s/r) = \cos \theta$. On the other hand, from the ‘dot product’ of elementary vector analysis,

$$\sum_{i=1}^3 x_i y_i = r^2 \cos \theta.$$

The proof of the triangle inequality for this example also involves a generalization of the Cauchy-Schwarz inequality. Details may be found in [11].

We conclude this section with another abstract example. The motions (Definition 2.4) of a metric space M form a group under functional composition, and it is possible to assign a metric to this group as follows.

Example 2.17 (*The group of motions*) Let (M, d) be a metric and let $\mathcal{M}$ denote the group of all motions of M . Fix $p \in M$, and for $\Phi, \Psi \in \mathcal{M}$ define

$$\rho_p(\Phi, \Psi) = \sup\{d(\Phi(x), \Psi(x))e^{-d(p, x)} : x \in M\}.$$

2.3 Completeness

Probably the first person to recognize the truly fundamental role completeness plays in those spaces which usually arise in analysis was the celebrated Polish mathematician Stefan Banach [7]. Indeed, in recognition of its importance Banach took it as an axiom in considering what are now known as Banach spaces. It is, however, an entirely metric condition based on the notion of Cauchy sequences.

Definition 2.6 A sequence $\{x_n\}$ in a metric space (M, d) is said to be a *Cauchy sequence* if for each $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that if $m, n \geq N$, then $d(x_n, x_m) < \varepsilon$.

Definition 2.7 A metric space (M, d) is said to be *complete* if each Cauchy sequence $\{x_n\}$ in M has a limit.

Complete metric spaces are abundant. Each of the metric spaces of the Examples 2.1 through 2.10 of the previous section are complete. Indeed, if the underlying space (M, d) is complete, the spaces of Examples 2.13, 2.14 and 2.15 are complete as well. However some of these facts are not entirely obvious. Before turning to specific examples, we establish two important general facts.

A subset S of a metric space M is said to be *bounded* if S is contained in some ball $B(x; r)$ of M .

Theorem 2.8 A Cauchy sequence in a metric space is always bounded.

Proof. Let $\{x_n\}$ be a Cauchy sequence in a metric space (M, d) . Taking $\varepsilon = 1$ in the definition of Cauchy sequence it is possible to obtain an integer N such that if $n \geq N$, then $d(x_N, x_n) < 1$. Since the set $\{d(x_i, x_j) : 1 \leq i, j \leq N\}$ is finite, it has a largest member, say d . It now follows from the triangle inequality that for any two integers $m, n \in \mathbb{N}$, $d(x_m, x_n) \leq d + 1$. In particular, $\{x_n\} \subseteq B(x_1; d + 1)$. ■

Another easy fact is the following. The proof is left as an exercise.

Theorem 2.9 If $\{x_n\}$ is a Cauchy sequence in a metric space M and if $\{x_n\}$ has a subsequence $\{x_{n_k}\}$ which converges to $x \in M$, then $\lim_{n \rightarrow \infty} x_n = x$.

The *diameter* of a nonempty bounded subset D of a metric space is the number

$$\text{diam}(D) = \sup\{d(x, y) : x, y \in D\}.$$

Note that we have $\text{diam}(\overline{D}) = \text{diam}(D)$.

Theorem 2.10 (*Cantor's Intersection Theorem*) *A metric space (M, d) is complete if and only if given any descending sequence $\{D_n\}$ of nonempty bounded closed subsets of M ,*

$$\lim_{n \rightarrow \infty} \text{diam}(D_n) = 0 \Rightarrow \bigcap_{n=1}^{\infty} D_n \neq \emptyset.$$

Proof. Suppose M is complete and let $\{D_n\}$ be a descending sequence of nonempty bounded closed subsets of M for which $\lim_{n \rightarrow \infty} \text{diam}(D_n) = 0$. For each n select $x_n \in D_n$. Then given $\varepsilon > 0$ there exists an integer $N \in \mathbb{N}$ such that $n \geq N \Rightarrow \text{diam}(D_n) < \varepsilon$. If $m, n \geq N$, both x_n and x_m are in D_N , then $d(x_n, x_m) \leq \varepsilon$. This proves that $\{x_n\}$ is a Cauchy sequence. Since M is complete there exists $x \in M$ such that $\lim_{n \rightarrow \infty} x_n = x$. At the same time this shows that $x \in \overline{D}_n = D_n$ if $n \geq N$. Since the sequence $\{D_n\}$ is descending, we conclude

$$x \in \bigcap_{n=1}^{\infty} D_n.$$

To prove the converse, assume the condition holds and suppose $\{x_n\} \subseteq M$ is a Cauchy sequence. For each n let

$$D_n = \{x_n, x_{n+1}, \dots\}.$$

Then $\{\overline{D}_n\}$ is a descending sequence of nonempty closed subsets of M for which

$$\lim_{n \rightarrow \infty} \text{diam}(\overline{D}_n) = 0.$$

By assumption, there exists a point $x \in \bigcap_{n=1}^{\infty} \overline{D}_n$. Now let $\varepsilon > 0$ and choose $N \in \mathbb{N}$ so large that $n \geq N \Rightarrow \text{diam}(\overline{D}_n) < \varepsilon$. Then clearly for such n it must be the case that $d(x, x_n) \leq \varepsilon$. Hence $\lim_{n \rightarrow \infty} x_n = x$, proving that M is complete. ■

Remark 2.1 *Note that in the statement of the above theorem, the condition $\bigcap_{n=1}^{\infty} D_n \neq \emptyset$ could have been replaced with the condition $\bigcap_{n=1}^{\infty} D_n$ consists of exactly one point.*

Here is a variant of Cantor's Theorem due to Ascoli.

Theorem 2.11 *A metric space M is complete if and only if the intersection of any descending sequence of closed balls in M having radii tending to 0 consists of exactly one point.*

As an application of Cantor's Theorem we have a partial converse of Theorem 2.5.

Theorem 2.12 *If a subspace K of a complete metric space (M, d) is totally bounded, then it is precompact.*

Proof. (This proof is admittedly tedious and perhaps best understood by drawing pictures to illustrate the first two or three steps.) In view of Theorem 2.3 it need only be shown that any sequence $\{x_n\}$ in $\overline{K}$ has a convergent subsequence. Its limit will necessarily lie in $\overline{K}$ proving that $\overline{K}$ is compact. Note also that if an infinite number of terms of $\{x_n\}$ are the same then we could select a subsequence of $\{x_n\}$ which is constant and therefore converges trivially. So by throwing away some terms if necessary, we may assume that each two terms of $\{x_n\}$ are distinct.

Since K is precompact there exist points $\{z_{1,1}, z_{2,1}, \dots, z_{n_1,1}\} \subseteq K$ such that each point of K lies within distance at most 1 from at least one of these points, that is,

$$\overline{K} \subseteq \bigcup_{i=1}^{n_1} B(z_{i,1}; 1).$$

In particular, for one of the $z_{i,1}$'s, say z_1 , an infinite number of terms of $\{x_n\}$ lie in $B_1 = B(z_1; 1)$. Pick one of them, say x_{n_1} . Similarly there is a finite set $\{z_{1,2}, z_{2,2}, \dots, z_{n_2,2}\} \subseteq K$ such that

$$\overline{K} \subseteq \bigcup_{i=1}^{n_2} B\left(z_{i,2}; \frac{1}{2}\right).$$

Since an infinite number of terms of $\{x_n\}_{n > n_1}$ lie in B_1 , for one of the $z_{i,2}$'s, say z_2 , an infinite number of terms of $\{x_n\}_{n > n_1}$ lie in $B_1 \cap B_2$ where $B_2 = B\left(z_2; \frac{1}{2}\right)$. Pick one of them, say x_{n_2} . At this point, notice that $\text{diam}(B_1 \cap B_2) \leq 1$. Proceed by induction. Suppose we have obtained points $\{x_{n_1}, \dots, x_{n_k}\}$ and $\{z_1, \dots, z_k\}$ such that for each $1 \leq i \leq k$, $x_{n_i} \in \bigcap_{j=1}^i B_j$, where $B_i = B\left(z_i; \frac{1}{i}\right)$, and such that an infinite number of terms of $\{x_n\}_{n > n_k}$ lie in $\bigcap_{j=1}^k B_j$. Apply total boundedness to proceed to the next step. There is a finite set

$\{z_{1,k+1}, z_{2,k+1}, \dots, z_{n_{k+1},k+1}\} \subseteq K$ such that

$$\overline{K} \subseteq \bigcup_{i=1}^{n_{k+1}} B\left(z_{i,k+1}; \frac{1}{k+1}\right).$$

In particular, for one of the $z_{i,k}$'s, say z_{k+1} , an infinite number of terms of $\{x_n\}_{n>n_k}$ lie in $\bigcap_{j=1}^{k+1} B_j$ where $B_{k+1} = B\left(z_{k+1}; \frac{1}{k+1}\right)$. Pick one of them, say $x_{n_{k+1}}$. Now let $D_k = \bigcap_{j=1}^k B_j$, $k = 1, 2, \dots$. Then $\{D_k\}$ is a descending sequence of closed sets with the property that $\text{diam}(D_k) \leq \frac{2}{k}$, $k = 1, 2, \dots$. By Cantor's intersection theorem,

$$\bigcap_{k=1}^{\infty} D_k = \{z\}$$

for some $z \in M$. Since $x_{n_k} \in D_k$ it follows that $\lim_{k \rightarrow \infty} x_{n_k} = z$. ■

The following general fact about completeness also is quite useful. We omit the very easy proof.

Proposition 2.1 *Every closed subspace of a complete metric space is itself complete.*

Here is another interesting fact about completeness. A subspace S of a metric space (M, d) is said to be dense in M if the closure of S is all of M (i.e., $\overline{S} = M$). Any metric space M is isometric with a dense subset of a complete metric space called the *completion* of M . (The use of the article 'the' can be justified by showing that any two completions of M must be isometric.) One way to see this is to consider the space M_C of equivalence classes of all Cauchy sequences in M , where two Cauchy sequences $\{x_n\}$ and $\{z_n\}$ are said to be equivalent (written $\{x_n\} \sim \{z_n\}$) if $\lim_{n \rightarrow \infty} d(x_n, z_n) = 0$. Let

$$[\{x_n\}] = \{\{z_n\} \subseteq M : \{z_n\} \sim \{x_n\}\}$$

and for $x^* = [\{x_n\}]$, $y^* = [\{y_n\}]$, set

$$d^*(x^*, y^*) = \lim_{n \rightarrow \infty} d(x_n, y_n).$$

It is easily verified that (i) the above limit does indeed exist; (ii) the space (M_C, d^*) is complete; and (iii) M is isometric to the subspace of (M_C, d^*) consisting of all equivalence classes of the form $[\{x\}]$, $x \in M$.

We now look at some specific examples.

Example 1. Completeness of the space $\mathbb{R}$ of real numbers (with usual distance $d(x, y) = |x - y|$) is a fundamental consequence of the sup axiom. Suppose $\{x_n\}$ is a Cauchy sequence of real numbers and define the two sequences $\{u_n\}$ and $\{v_n\}$ as follows:

$$\begin{aligned} u_n &= \sup\{x_k : k \geq n\}; \\ v_n &= \inf\{x_k : k \geq n\}. \end{aligned}$$

Then $\{u_n\}$ is monotone decreasing and bounded below (by $\inf\{x_n : n \geq 1\}$), while $\{v_n\}$ is monotone increasing and bounded above (by $\sup\{x_n : n \geq 1\}$). Therefore both $\lim_{n \rightarrow \infty} u_n = u$ and $\lim_{n \rightarrow \infty} v_n = v$ exist. (In standard notation, we would write $u = \lim_{n \rightarrow \infty} \sup x$ and $v = \lim_{n \rightarrow \infty} \inf x_n$.) Also, since $\{x_n\}$ is a Cauchy sequence, it is easy to see that

$$\lim_{n \rightarrow \infty} |u_n - v_n| = \lim_{n \rightarrow \infty} u_n - v_n = 0,$$

so $u = v$. Now let $\varepsilon > 0$ and choose $N \in \mathbb{N}$ so that if $n \geq N$ then both $|u_n - v| < \varepsilon/2$ and $|v_n - v| < \varepsilon/2$. Since $v_n \leq x_n \leq u_n$, both x_n and v lie in the interval $[v_n, u_n]$, and if $n \geq N$ this interval has length less than ε . Therefore if $n \geq N$, $|x_n - v| < \varepsilon$. ■

Example 2. Completeness of $\mathbb{R}^n$ follows almost immediately from completeness of $\mathbb{R}$ although some awkwardness in notation arises. Indeed, if $\{\mathbf{x}_i\}$ is a Cauchy sequence in $\mathbb{R}^n$ then each element $\mathbf{x}_i$ is itself an n -tuple:

$$\mathbf{x}_i = (x_1^i, x_2^i, \dots, x_n^i).$$

Now let $\varepsilon > 0$ and choose N so that if $i, j \geq N$, $d(\mathbf{x}_i, \mathbf{x}_j) < \varepsilon$. Then for each $k = 1, 2, \dots, n$:

$$\left| x_k^i - x_k^j \right| \leq \left(\sum_{k=1}^n (x_k^i - x_k^j)^2 \right)^{1/2} < \varepsilon,$$

so for each such k , $\{x_k^i\}_{i=1}^\infty$ is a Cauchy sequence as well. By the completeness of $\mathbb{R}$ there exists for each such k a number x_k such that $\lim_{i \rightarrow \infty} x_k^i = x_k$. Now let $\mathbf{x} = (x_1, x_2, \dots, x_n)$. For each $k = 1, 2, \dots, n$ choose $N_k \in \mathbb{N}$ so that $i \geq N_k$ implies

$$|x_k^i - x_k| < \frac{\varepsilon}{\sqrt{n}}.$$

Then if $i \geq \max\{N_1, N_2, \dots, N_n\}$,

$$d(\mathbf{x}_i, \mathbf{x}) = \left(\sum_{k=1}^n (x_k^i - x_k)^2 \right)^{1/2} < \left(n \left(\frac{\varepsilon^2}{n} \right) \right)^{1/2} = \varepsilon.$$

The above argument extends to ℓ_2 although a little more work is needed. The argument begins as above. If $\{\mathbf{x}_i\}$ is a Cauchy sequence in ℓ_2 then each element $\mathbf{x}_i$ is itself a sequence of real numbers:

$$\mathbf{x}_i = (x_1^i, x_2^i, \dots).$$

Now let $\varepsilon > 0$ and choose N so that if $i, j \geq N$, $d(\mathbf{x}_i, \mathbf{x}_j) < \varepsilon$. Then for each $k = 1, 2, \dots$:

$$|x_k^i - x_k^j| \leq \left(\sum_{n=1}^{\infty} (x_n^i - x_n^j)^2 \right)^{1/2} < \varepsilon$$

so for each such k , $\{x_k^i\}_{i=1}^{\infty}$ is a Cauchy sequence as well. By the completeness of $\mathbb{R}$ there exists for each such k a number x_k such that $\lim_{i \rightarrow \infty} x_k^i = x_k$. Let $\mathbf{x} = (x_1, x_2, \dots)$.

Now there are two things to show. (1) $\mathbf{x} \in \ell_2$, that is, $\sum_{n=1}^{\infty} x_n^2 < \infty$; and (2) $\lim_{i \rightarrow \infty} \mathbf{x}_i = \mathbf{x}$. To this end, let $\varepsilon > 0$ and choose $N \in \mathbb{N}$ so that if $i, j \geq N$ then

$$(d(\mathbf{x}_i, \mathbf{x}_j))^2 = \sum_{k=1}^{\infty} (x_k^i - x_k^j)^2 < \varepsilon.$$

Let $M \in \mathbb{N}$ and rewrite the above in the form

$$\sum_{k=1}^{\infty} (x_k^i - x_k^j)^2 = \sum_{k=1}^M (x_k^i - x_k^j)^2 + \sum_{k=M+1}^{\infty} (x_k^i - x_k^j)^2 < \varepsilon.$$

This in particular means that $\sum_{k=1}^M (x_k^i - x_k^j)^2 < \varepsilon$. Now fix j and let $i \rightarrow \infty$ to obtain

$$\sum_{k=1}^M (x_k - x_k^j)^2 \leq \varepsilon.$$

Since this is true for any M we can let $M \rightarrow \infty$ and conclude

$$\sum_{k=1}^{\infty} (x_k - x_k^j)^2 \leq \varepsilon.$$

The above, along with the fact that $\sum_{k=1}^{\infty} (x_k^j)^2$ converges, implies that $\sum_{k=1}^{\infty} x_k^2$ converges. This proves that $\mathbf{x} \in \ell_2$. At the same time we conclude that $j \geq N$ implies

$$d(\mathbf{x}, \mathbf{x}_j) \leq \varepsilon.$$

Thus $\lim_{j \rightarrow \infty} \mathbf{x}_j = \mathbf{x}$.

Example 3. We now turn to the space $C[0, 1]$. Since this space consists of all continuous real-valued functions defined on $[0, 1]$, with the distance $d(f, g)$ between two such functions f, g taken as

$$d(f, g) = \sup\{|f(t) - g(t)| : t \in [0, 1]\},$$

it is immediate that if $\{f_n\}$ is a Cauchy sequence in $C[0, 1]$ then for each $t \in [0, 1]$ $\{f_n(t)\}$ is a Cauchy sequence as well. Since $\{f_n(t)\} \subseteq \mathbb{R}$ for each t , $\lim_{n \rightarrow \infty} f_n(t)$ exists. Let $f(t) = \lim_{n \rightarrow \infty} f_n(t)$, $t \in [0, 1]$. We need to show that $f \in C[0, 1]$ and that $\lim_{n \rightarrow \infty} d(f_n, f) = 0$.

Assume it is already known that $\lim_{n \rightarrow \infty} d(f_n, f) = 0$. To see that f is continuous let $t_0 \in [0, 1]$ and let $\varepsilon > 0$. Since $\lim_{n \rightarrow \infty} d(f_n, f) = 0$ there exists an integer $N \in \mathbb{N}$ such that

$$\sup_{t \in [0, 1]} |f_N(t) - f(t)| \leq \frac{\varepsilon}{3}.$$

Also, since f_N is continuous at t_0 there is a number $\delta > 0$ such that if $t \in [0, 1]$ and if $|t - t_0| < \delta$ then

$$|f_N(t) - f_N(t_0)| < \frac{\varepsilon}{3}.$$

Hence if $|t - t_0| < \delta$,

$$|f(t) - f(t_0)| \leq |f(t) - f_N(t)| + |f_N(t) - f_N(t_0)| + |f_N(t_0) - f(t_0)| < \varepsilon.$$

This proves continuity of f . To see that $\lim_{n \rightarrow \infty} d(f_n, f) = 0$, let $\varepsilon > 0$ and observe that since $\{f_n\}$ is a Cauchy sequence there is an integer N such that if $m, n \geq N$ then

$$\sup_{t \in [0, 1]} |f_n(t) - f_m(t)| \leq \varepsilon,$$

that is,

$$f_n(t) - \varepsilon \leq f_m(t) \leq f_n(t) + \varepsilon.$$

Letting $m \rightarrow \infty$ we see that for any $t \in [0, 1]$ and $n \geq N$,

$$f_n(t) - \varepsilon \leq f(t) \leq f_n(t) + \varepsilon;$$

hence

$$|f(t) - f_n(t)| \leq \varepsilon,$$

from which $d(f_n, f) \leq \varepsilon$. Since $\varepsilon > 0$ is arbitrary we conclude $\lim_{n \rightarrow \infty} d(f_n, f) = 0$.

2.4 Separability and connectedness

Most, but not all, of the metric spaces described in this chapter share another characteristic whose presence is often helpful to know.

Definition 2.8 *A metric space M is said to be separable if it possesses a countable dense subset.*

Thus if M is separable, then there exists a sequence $\{u_1, u_2, \dots\} \subseteq M$ such that if $A = \{u_i : i = 1, 2, \dots\}$, then $\overline{A} = M$. This concept will not play a large role in what follows so we only make a few quick comments here. However, one of the deeper results of Chapter 5 (Theorem 5.12) depends very heavily on separability.

First, the rational numbers form a countable dense subset of the reals $\mathbb{R}$; hence $\mathbb{R}$ is separable. More generally, the space ℓ_1 of Example 2.6, all of the spaces ℓ_p ($1 < p < \infty$) of Example 2.8, and the space $C[0, 1]$ of Example 2.9 are separable. The space ℓ_∞ is not.

For the ℓ_p spaces ($1 \leq p < \infty$) a countable dense subset is obtained by taking the collection of all sequences which have rational coordinates and for which *all but a finite number of coordinates are 0*. The latter restriction is essential to assure that the dense subset is in fact countable.

For $C[0, 1]$ the situation is more intricate. For given $n \in \mathbb{N}$ let P_n denote the set of all functions defined as follows. If $x = k/n$, $k = 0, 1, \dots, n$, let $f(x)$ be a rational number, and for $(k-1)/n \leq x \leq k/n$ let f be the function whose graph is a line joining the point $((k-1)/n, f((k-1)/n))$ to the point $(k/n, f(k/n))$. Then the set $P = \bigcup_{n \geq 1} P_n$ is countable, and moreover, $\overline{P} = C[0, 1]$. The proof

of the latter fact hinges on the fact that any function in $C[0, 1]$ is uniformly continuous. Armed with this fact, let $f \in C[0, 1]$, let $\varepsilon > 0$, and choose $\delta > 0$ so that $|f(u) - f(v)| < \varepsilon/5$ whenever $|u - v| < \delta$. Now choose n so that $1/n < \delta$. For each $k = 0, 1, \dots, n$ it is possible to choose a rational number w_k so that $|f(k/n) - w_k| < \varepsilon/5$. Now let $g \in P_n$ be the function for which $g(k/n) = w_k$. Then if $x \in [k/n, (k+1)/n]$,

$$\begin{aligned} |f(x) - g(x)| &\leq |f(x) - f(k/n)| + |f(k/n) - g(k/n)| + |g(k/n) - g(x)| \\ &\leq 2\varepsilon/5 + |g(k/n) - g((k+1)/n)| \\ &\leq 2\varepsilon/5 + |f(k/n) - f((k+1)/n)| + 2\varepsilon/5 \\ &\leq \varepsilon. \end{aligned}$$

Having established the above, it follows that given any $f \in C[0, 1]$ there exists $g \in P$ such that $d(f, g) \leq \varepsilon$. This implies $\overline{P} = C[0, 1]$.

What we have just observed is that there is a countable family of *piecewise linear* continuous functions which is dense in $C[0, 1]$. It is also the case that the family of all polynomial functions with rational coefficients is dense in $C[0, 1]$. This collection is countable as well.

In fact, the space $C[0, 1]$ is more than just separable. It is universal with respect to separability in the following profound sense!

Theorem 2.13 *Every separable metric space is isometric with a subset of $C[0, 1]$.*

It is possible to give an elementary (but quite detailed) proof of this fact; see, for example, [145], p. 192.

The following is an easy consequence of the fact that compact metric spaces are totally bounded.

Theorem 2.14 *Every compact metric space (M, d) is separable.*

Proof. Let (M, d) be a compact metric space and for each $n \in \mathbb{N}$, let P_n be a subset of M which is maximal with respect to the property: if $x, y \in P_n$ then $d(x, y) \geq 1/n$ (a simple induction argument suffices for this). Since M is compact each of the sets P_n is finite, so $P = \bigcup_{i=1}^{\infty} P_i$ is countable. Also by maximality of P_n , if $x \in M$ it must be the case that there exists $p \in P_n$ such that $d(x, p) < 1/n$. Now let $\varepsilon > 0$ and choose n so that $1/n < \varepsilon$. It follows that if $x \in M$ there exists $p \in P$ such that $d(x, p) < 1/n < \varepsilon$, and this proves that $\bar{P} = M$. ■

Finally, we mention another interesting fact about separability. If (M, d) is separable then the space of all nonempty bounded closed subsets of M endowed with the Hausdorff metric need not be separable. However, if only the collection of all nonempty compact subsets is considered then the resulting space is separable (Exercise 2.23).

Another basic topological concept which arises often is that of connectedness.

Definition 2.9 *A metric space (M, d) is said to be connected if M is not the union of two disjoint nonempty open subsets of M .*

The above definition immediately leads to the conclusion that a metric space is connected if and only if the only subsets of M that are both open and closed are the empty set and M itself.

Obviously the real line $\mathbb{R}$ is connected, as is any interval and ray in $\mathbb{R}$. Indeed, the Intermediate Value Theorem (Theorem 1.1 of Chapter 1) rests on the connectedness of intervals in $\mathbb{R}$. In fact, a more general version of that theorem holds.

Theorem 2.15 (*Intermediate Value Theorem*) *Let M be a connected metric space and let $f : M \rightarrow \mathbb{R}$ be continuous. Suppose $x, y \in M$ and suppose c is any point which lies between $f(x)$ and $f(y)$. Then there is a point $z \in M$ such that $f(z) = c$.*

The proof is left as an exercise.

2.5 Metric convexity and convexity structures

If (M, d) is a metric space and if $p, q \in M$ then a point $r \in M$ is said to be *metrically between* p and q if $p \neq r \neq q$ and $d(p, q) = d(p, r) + d(r, q)$. A metric space is said to be *metrically convex* if given any two points $p, q \in M$ there exists at least one point $r \in M$ such that r is metrically between p and q .

Metric convexity is a fundamental concept in the axiomatic study of the geometry of metric spaces. However, in its most general form, it fails to satisfy one of the basic properties of convexity in the algebraic sense (in a linear space); namely, if A and B are two metrically convex subsets of a metric space M , then *it need not be the case that $A \cap B$ is metrically convex*. To see this it is sufficient only to consider the ordinary unit circle S in $\mathbb{R}^2$ where the distance between two points is taken to be the length of the shortest arc joining them. The upper half circle and lower half circle of S are each metrically convex but the intersection of these two sets is easily seen to be a set consisting of just two antipodal points.

A subset S of a metric space M is called a *metric segment* with endpoints (or joining) $p, q \in S$ if there exists a closed interval $[a, b]$ in the real line $\mathbb{R}$ and an *isometry* φ which maps $[a, b]$ onto S with $\varphi(a) = p$ and $\varphi(b) = q$. If a metric space M has the property that each two of its points are the endpoints of a metric segment, then clearly M is metrically convex, and it is natural to ask when the converse is true. The answer is surprisingly often. This is a fundamental result in the theory due to Karl Menger, one of the pioneers in the study of abstract metric spaces. For a proof see, for example, [12], [67].

Theorem 2.16 *If M is a complete and metrically convex metric space, then each two points of M are the endpoints of at least one metric segment of M .*

We shall be concerned almost entirely with complete metric spaces; hence with the stronger notion of convexity assured by Menger's Theorem whenever the concept arises. For reasons that will be clear later, the usefulness of metric convexity in a fixed point context is hindered by the fact that the family of all convex subsets of a metrically convex metric space need not be stable under intersections. A quick way to remedy this defect is to turn to a more abstract notion of convexity.

A family $\mathcal{C}$ of subsets of a set X is called an (abstract) *convexity structure* if

(1) Both $\emptyset$ and X are in $\mathcal{C}$.

(2) $\mathcal{C}$ is stable under intersections; that is, if $\{D_\alpha\}_{\alpha \in I}$ is any nonempty subfamily of $\mathcal{C}$ then $\bigcap_{\alpha \in I} D_\alpha \in \mathcal{C}$.

(3) $\mathcal{C}$ is stable for nested unions; that is, if $\{D_\alpha\}_{\alpha \in I}$ is any nonempty subfamily of $\mathcal{C}$ which is totally ordered by set inclusion, then $\bigcup_{\alpha \in I} D_\alpha \in \mathcal{C}$.

Unfortunately, for our purposes, there is even a problem with this abstract formulation. In the study of metric fixed point theory closed balls turn out to play a fundamental role. A subset A of a bounded metric space M will be said to be an *admissible* subset of M if A can be written as the intersection of a family of closed balls centered at points of M . The family $\mathcal{A}(M)$ of all admissible subsets of M enters into the study of metric fixed point theory in a very natural way and is therefore the obvious candidate for the needed underlying convexity structure. However, while this family satisfies assumptions (1) and (2), in general, property (3) does not hold. This fact cannot be rectified even if the collection is enlarged to include all ball intersections rather than just intersections of *closed* balls. Nonetheless, the family $\mathcal{A}(M)$ will arise frequently in the ensuing chapters. Thus we shall exchange property (3) with the assumption that the convexity structure under consideration contains all the closed balls of M .

The structure $\mathcal{A}(M)$ has one very distinctive advantage when compared with metric convexity. Any subset A of a metric space is contained in the set

$$\text{cov}(A) = \bigcap \{B : B \text{ is a closed ball in } M \text{ and } B \supseteq A\}.$$

Moreover $\text{cov}(A) \in \mathcal{A}(M)$. This provides a nice analog for the concept of 'closed convex hull' in functional analysis, with the family $\mathcal{A}(M)$ replacing the family of all closed (algebraically) convex sets.

This analogy does not quite work with metric convexity. It is natural to define the *metric convex hull* of a subset A of a convex metric space M to be a set which is closed, metrically convex, contains A , and for which none of its proper subsets has those properties. With this definition the existence of a unique metric convex hull for each subset A of a complete convex metric space *which has unique metric segments* is immediate. Merely take the 'metric convex hull' of A to be the intersection of all closed convex sets which contain A . However, problems arise in the general case since, as we have observed, the intersection of two metrically convex sets need not be metrically convex. On the other hand, if M is any compact metric space and if $A \subseteq M$, then A does have a metric convex hull, although it need not be unique. This is a classical result proved by Karl Menger in 1931. It rests on the following lemma.

Lemma 2.1 *Let $C_1 \supseteq C_2 \supseteq \cdots$ be a descending sequence of nonempty closed metrically convex subsets of a compact metric space (M, d) . Then $\bigcap_{n=1}^{\infty} C_n$ is nonempty and metrically convex.*

Proof. The fact that the intersection is nonempty is immediate from compactness. Suppose $x, y \in \bigcap_{n=1}^{\infty} C_n$ with $x \neq y$. Then in each of the sets C_n there exists a point z_n such that

$$d(x, z_n) = d(y, z_n) = \frac{1}{2}d(x, y).$$

(This uses the fact that x and y are actually joined by a metric segment which lies in C_n .) By compactness of M the sequence $\{z_n\}$ has a subsequence $\{z_{n_k}\}$ which converges to a point $z \in M$ and since each of the sets C_n is closed, $z \in \bigcap_{n=1}^{\infty} C_n$. Since the metric d is continuous,

$$d(x, z) = d(y, z) = \frac{1}{2}d(x, y).$$

This proves that $\bigcap_{n=1}^{\infty} C_n$ is metrically convex. ■

Theorem 2.17 *If A is any nonempty subset of a compact and metrically convex metric space M , then A has a metric convex hull.*

Proof. Let Σ denote the collection of all closed metrically convex subsets of M which contain A . Observe that $\Sigma \neq \emptyset$ since $M \in \Sigma$. Since M is compact, for each $n \in \mathbb{N}$ there is a finite collection $\mathcal{U}_n = \{U_{n(1)}, U_{n(2)}, \dots, U_{n(k_n)}\}$ of open balls of radius $1/n$ which covers M . For each $K \in \Sigma$ there is a well-defined integer $\mu_n(K)$ ($\mu_n(K) \leq k(n)$) such that K intersects $\mu_n(K)$ members of $\mathcal{U}_n$. Choose $C_1 \in \Sigma$ so that

$$\mu_1(C_1) = \inf\{\mu_1(C) : C \in \Sigma\}.$$

Having defined C_n choose $C_{n+1} \subseteq C_n$ so that

$$\mu_{n+1}(C_{n+1}) = \inf\{\mu_{n+1}(C) : C \in \Sigma \text{ and } C \subseteq C_n\}.$$

Then clearly $C_1 \supseteq C_2 \supseteq \dots$ and $A \subseteq C = \bigcap_{n=1}^{\infty} C_n$. Also, by Lemma 2.1, C is metrically convex. Suppose some $H \in \Sigma$ is a proper subset of C . Then there exists a point $c \in C \setminus H$ and in turn there exists an integer n for which the open ball $U(c; 1/n) \cap C \subset C \setminus H$. In particular, some member U of $\mathcal{U}_{2n}$ lies in $U(c; 1/n)$; hence

$$\mu_{2n}(H) < \mu_{2n}(C) \leq \mu_{2n}(C_{2n}),$$

contradicting the definition of $\mu_{2n}(C_{2n})$. ■

It is possible to give a much quicker proof of the above theorem using Zorn's Lemma. (See the Appendix.) The proof just given is Menger's original and it predates the discovery of Zorn's Lemma. In fact, with Zorn's Lemma it is quite easy to prove the following.

Theorem 2.18 *Let M be a metrically convex metric space, and suppose the intersection of every descending chain of closed metrically convex subsets of M is itself metrically convex. Then every nonempty subset A of M has a metric convex hull.*

Proof. As above, let Σ denote the family of all closed metrically convex subsets of M which contain A , and order Σ by set inclusion. Then $\Sigma \neq \emptyset$ since $M \in \Sigma$, and by assumption every descending chain in $(\Sigma, \supseteq)$ is bounded below by its intersection. By Zorn's lemma Σ has a minimal element, completing the proof. ■

Exercises

Exercise 2.1 Show that if a semimetric space (M, d) has continuous distance function, then

$$\lim_{n \rightarrow \infty} p_n = p \text{ and } \lim_{n \rightarrow \infty} p_n = q \Rightarrow p = q.$$

Exercise 2.2 Suppose (M, d) is a semimetric space and suppose $\{x_n\}$ is a sequence in M which converges to $x \in M$. Is $\{x_n\}$ necessarily a Cauchy sequence? What if d is continuous?

Exercise 2.3 Give an example of two different metric spaces, each of which is isometric with a subspace of the other.

Exercise 2.4 Suppose (M, d) is a metric space and suppose $\{x_n\}$ is a sequence in M which converges to $x \in M$. Show that $\{x_n\}$ is a Cauchy sequence.

Exercise 2.5 Let M be the real unit interval $[0, 1]$ and for $x, y \in M$ define $d(x, y) = |x - y|^2$. Show that (M, d) is a semimetric space with continuous distance which is not a metric space.

Exercise 2.6 Let S be any set of nonnegative real numbers which contains 0. Show that there is a metric space (M, d) such that given $r \in S$ there exist $x, y \in M$ for which $d(x, y) \in S$.

Exercise 2.7 Complete the proof of Theorem 2.3, that is, show that if any sequence $\{x_n\}$ of points of S has a subsequence $\{x_{n_k}\}$ which converges to a point of S , then S is compact.

Exercise 2.8 Show that if X is a compact metric space, and $f : X \rightarrow \mathbb{R}$ is a continuous function, then there is a point $x_1 \in X$ such that

$$f(x_1) = \sup\{f(x) : x \in X\}.$$

Exercise 2.9 Suppose A is a precompact subset of a metric space (M, d) and suppose $T : A \rightarrow M$ is continuous. Then is T uniformly continuous?

Exercise 2.10 Show that (M, d_ϕ) is a metric space.

Exercise 2.11 Show that $(\mathcal{M}, \rho_p)$ is a metric space.

Exercise 2.12 Show that if (M, d) is compact, then $(\mathcal{M}, \rho_p)$ is compact (Example 2.17).

Exercise 2.13 Suppose a sequence $\{x_n\}$ in a metric space (M, d) satisfies the following condition. For each $p \in \mathbb{N}$ and each $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that if $n \geq N$, then $d(x_n, x_{n+p}) < \varepsilon$. Is $\{x_n\}$ necessarily a Cauchy sequence?

Exercise 2.14 Let $\{x_n\}$ be a sequence in a metric space for which

$$\sum_{i=1}^{\infty} d(x_i, x_{i+1}) < \infty.$$

Show that $\{x_n\}$ is a Cauchy sequence. Is the converse true?

Exercise 2.15 Prove Theorem 2.11.

Exercise 2.16 Show that (M_C, d^*) is a metric space.

Exercise 2.17 Show that (M_C, d^*) is complete.

Exercise 2.18 Show that (M, d^*) is dense in (M_C, d^*) .

Exercise 2.19 Show that the space of Example 2.11 is not complete.

Exercise 2.20 Show that if M is complete then the space $(\mathcal{M}, H)$ of Example 2.15 is complete.

Exercise 2.21 Show that the metric space consisting of all irrational numbers is separable.

Exercise 2.22 Show that any subspace of a separable metric space is separable.

Exercise 2.23 Show that the space of all nonempty compact subsets of a separable metric space M endowed with the Hausdorff metric is separable.

Exercise 2.24 Let (M, d) be a complete metric space and $\phi : M \rightarrow [0, \infty)$ an arbitrary nonnegative function. Assume that

$$\inf\{\phi(x) + \phi(y) : d(x, y) \geq \varepsilon\} > 0$$

for any $\varepsilon > 0$. Prove that each sequence $\{x_n\}$ in M such that $\phi(x_n) \rightarrow 0$, converges to one and the same point $x \in M$.

Exercise 2.25 Let (M, d) be an arbitrary metric space and let $A \subset M$ be a compact set. Let $\phi : M \rightarrow [0, \infty)$ be an arbitrary nonnegative function such that

$$\inf\{\phi(x) : d(x, A) \geq \varepsilon\} > 0$$

for any $\varepsilon > 0$. Prove that each sequence $\{x_n\}$ in M such that $\phi(x_n) \rightarrow 0$, contains a subsequence which converges to some point $x \in A$.

Chapter 3

Metric Contraction Principles

3.1 Banach's Contraction Principle

Banach's Contraction Mapping Principle is remarkable in its simplicity, yet it is perhaps the most widely applied fixed point theorem in all of analysis. This is because the contractive condition on the mapping is simple and easy to test, because it requires only a complete metric space for its setting, because it provides a constructive algorithm, and because it finds almost canonical applications in the theory of differential and integral equations. Although the basic idea was known to others earlier, the principle first appeared in explicit form in Banach's 1922 thesis [7], where it was used to establish the existence of a solution to an integral equation. The underlying space was taken to be the space $C[0, 1]$ of Example 2.9 of Chapter 2.

Let (M, d) be a metric space. A mapping $T : M \rightarrow M$ is said to be *lipschitzian* if there is a constant $k \geq 0$ such that for all $x, y \in M$

$$d(T(x), T(y)) \leq k d(x, y). \quad (3.1)$$

The smallest number k for which Eq. (3.1) holds is called the *Lipschitz constant* of T .

Definition 3.1 *A lipschitzian mapping $T : M \rightarrow M$ with Lipschitz constant $k < 1$ is said to be a contraction mapping.*

Theorem 3.1 (*Banach's Contraction Mapping Principle*) *Let (M, d) be a complete metric space and let $T : M \rightarrow M$ be a contraction mapping. Then T has a unique fixed point x_0 , and for each $x \in M$, $\lim_{n \rightarrow \infty} T^n(x) = x_0$. Moreover,*

$$d(T^n(x), x_0) \leq \frac{k^n}{1 - k} d(x, T(x)).$$

Proof. Suppose T has Lipschitz constant $k \in (0, 1)$. Then for each $x \in M$,

$$d(T(x), T^2(x)) \leq k d(x, T(x)).$$

Adding $d(x, T(x))$ to both sides of the above gives

$$d(x, T(x)) + d(T(x), T^2(x)) \leq d(x, T(x)) + k d(x, T(x)),$$

which can be rewritten as

$$d(x, T(x)) - k d(x, T(x)) \leq d(x, T(x)) - d(T(x), T^2(x)).$$

This, in turn, is equivalent to

$$d(x, T(x)) \leq (1 - k)^{-1} [d(x, T(x)) - d(T(x), T^2(x))].$$

Now define the function $\varphi : M \rightarrow \mathbb{R}^+$ by setting $\varphi(x) = (1 - k)^{-1} d(x, T(x))$, for $x \in M$. Thus

$$d(x, T(x)) \leq \varphi(x) - \varphi(T(x)), \quad x \in M.$$

Therefore if $x \in M$ and $m, n \in \mathbb{N}$ with $n < m$,

$$d(T^n(x), T^{m+1}(x)) \leq \sum_{i=n}^m d(T^i(x), T^{i+1}(x)) \leq \varphi(T^n(x)) - \varphi(T^{m+1}(x)).$$

(Note that the last inequality comes from cancellation in the telescoping sum.) In particular, by taking $n = 1$ and letting $m \rightarrow \infty$ we conclude that

$$\sum_{i=1}^{\infty} d(T^i(x), T^{i+1}(x)) \leq \varphi(T(x)) < \infty.$$

This implies that $\{T^n(x)\}$ is a Cauchy sequence (Chapter 2, Exercise 2.14). Since M is complete there exists $x_0 \in M$ such that

$$\lim_{n \rightarrow \infty} T^n(x) = x_0$$

and since T is continuous

$$x_0 = \lim_{n \rightarrow \infty} T^n(x) = \lim_{n \rightarrow \infty} T^{n+1}(x) = T(x_0).$$

Thus x_0 is a fixed point of T . In order to complete the proof of the existence part of the theorem we need to show that x_0 is the only fixed point. Indeed, let y be a fixed point of T . Then from the previous result,

$$x_0 = \lim_{n \rightarrow \infty} T^n(y) = y.$$

Returning to the inequality

$$d(T^n(x), T^{m+1}(x)) \leq \varphi(T^n(x)) - \varphi(T^{m+1}(x)),$$

upon letting $m \rightarrow \infty$ we see that

$$d(T^n(x), x_0) \leq \varphi(T^n(x)) = (1 - k)^{-1} d(T^n(x), T^{n+1}(x)).$$

Since $(1 - k)^{-1} d(T^n(x), T^{n+1}(x)) \leq \frac{k^n}{1 - k} d(x, T(x))$, we obtain

$$d(T^n(x), x_0) \leq \frac{k^n}{1 - k} d(x, T(x)).$$

■

While the preceding estimate on the rate of convergence of $\{T^n(x)\}$ is sharp, in practical situations it might be necessary also to approximate the values of $T^n(x)$. Here a natural question arises. If one replaces the sequence $\{T^n(x)\}$ with $\{y_n\}$ where $y_0 = x$ and y_{n+1} is 'approximately' $T(y_n)$, then under what conditions will it still be the case that $\lim_{n \rightarrow \infty} y_n = x_0$? The following response to this question was given by Ostrowski in [129]. (For a recent generalization of this result as well as related literature, see [76].)

Theorem 3.2 *Let (M, d) be a complete metric space, let $T : M \rightarrow M$ be a contraction mapping with Lipschitz constant $k \in (0, 1)$, and suppose $x_0 \in M$ is the fixed point of T . Let $\{\varepsilon_n\}$ be a sequence of positive numbers for which $\lim_{n \rightarrow \infty} \varepsilon_n = 0$, let $y_0 \in M$, and suppose $\{y_n\} \subseteq M$ satisfies*

$$d(y_{n+1}, T(y_n)) \leq \varepsilon_n.$$

Then $\lim_{n \rightarrow \infty} y_n = x_0$.

Proof. Let $y_0 = x$ and observe that

$$\begin{aligned} d(T^{m+1}(x), y_{m+1}) &\leq d(T(T^m(x)), T(y_m)) + d(T(y_m), y_{m+1}) \\ &\leq kd(T^m(x), y_m) + \varepsilon_m, \end{aligned}$$

which implies

$$d(T^{m+1}(x), y_{m+1}) \leq \sum_{i=0}^m k^{m-i} \varepsilon_i.$$

Thus

$$\begin{aligned} d(y_{m+1}, x_0) &\leq d(y_{m+1}, T^{m+1}(x)) + d(T^{m+1}(x), x_0) \\ &\leq \sum_{i=0}^m k^{m-i} \varepsilon_i + d(T^{m+1}(x), x_0). \end{aligned}$$

Now let $\varepsilon > 0$. Since $\lim_{n \rightarrow \infty} \varepsilon_n = 0$ there exists $N \in \mathbb{N}$ such that for $m \geq N$, $\varepsilon_m \leq \varepsilon$. Thus

$$\begin{aligned} \sum_{i=0}^m k^{m-i} \varepsilon_i &= \sum_{i=0}^N k^{m-i} \varepsilon_i + \sum_{i=N+1}^m k^{m-i} \varepsilon_i \\ &\leq k^{m-N} \sum_{i=0}^N k^{N-i} \varepsilon_i + \varepsilon \sum_{i=N+1}^m k^{m-i}. \end{aligned}$$

Hence

$$\lim_{m \rightarrow \infty} \sum_{i=0}^m k^{m-i} \varepsilon_i \leq \varepsilon \left(\frac{k^{N+1}}{1-k} \right).$$

Since $\varepsilon > 0$ is arbitrary, and since $\lim_{m \rightarrow \infty} d(T^{m+1}(x), x_0) = 0$, we conclude that $\lim_{m \rightarrow \infty} y_{m+1} = x_0$. ■

The contraction principle also has a trivial, yet somewhat surprising, extension. In this theorem the mapping T is not even assumed to be continuous!

Theorem 3.3 *Suppose (M, d) is a complete metric space and suppose $T : M \rightarrow M$ is a mapping for which T^N is a contraction mapping for some positive integer N . Then T has a unique fixed point.*

Proof. By Banach's Theorem T^N has a unique fixed point x . Hence

$$T^{N+1}(x) = T(T^N(x)) = T(x),$$

so $T(x)$ is also a fixed point of T^N . Since the fixed point of T^N is unique, it must be the case that $T(x) = x$. Also, if $T(y) = y$, then $T^N(y) = y$, proving (again by uniqueness) that $y = x$. ■

The following is another extension of Banach's principle. We omit the proof since a more general result is taken up in the next section. However, the result itself is a nice exercise.

Theorem 3.4 *Let M be a metric space that has two metrics d and ρ , and suppose $\rho(x, y) \leq d(x, y)$ for each $x, y \in M$. Suppose (M, ρ) is complete, and suppose $T : M \rightarrow M$ is a continuous mapping with respect to the metric ρ and a contraction mapping with respect to the metric d . Then T has a unique fixed point in M .*

In considering lipschitzian mappings an obvious question that arises immediately is whether it is possible to weaken the contraction assumption even a little bit and still obtain the existence of fixed points. In a broad sense the answer is no and here is an example. Begin with the complete metric space

$C[0, 1]$ and consider the closed subspace M of $C[0, 1]$ consisting of those mappings $f \in C[0, 1]$ for which $f(1) = 1$. Since M is a closed subspace of a complete metric space, M is itself complete. Now define $T : M \rightarrow M$ by taking $T(f)$ to be the function in M obtained by setting

$$T(f)(t) = tf(t) \quad t \in [0, 1].$$

If $f, g \in M$ then $|T(f) - T(g)| \in C[0, 1]$, so by the Maximum Value Theorem $|T(f) - T(g)|$ attains its maximum value at some point $t_0 \in [0, 1]$. We then have

$$d(T(f), T(g)) = \sup_{t \in [0, 1]} |T(f)(t) - T(g)(t)| = t_0 |f(t_0) - g(t_0)| \leq d(f, g).$$

But if $f \neq g$ it must be the case that $f(t) \neq g(t)$ for some $t \in [0, 1]$ and since $f(1) = g(1) = 1$, this in turn implies $t_0 < 1$. Therefore, if $f, g \in M$ and $f \neq g$,

$$d(T(f), T(g)) < d(f, g).$$

Now suppose $T(f) = f$ for $f \in M$. This implies that for each $t \in [0, 1]$, $f(t) = tf(t)$. This implies that $f(t) = 0$ for all $t \in [0, 1)$. On the other hand, $f(1) = 1$. This contradicts the assumption that f is continuous, so T can have no fixed point in M . Therefore the Banach Contraction Mapping Principle does not even extend to the following slightly more general class of mappings.

Definition 3.2 A mapping $T : M \rightarrow M$ is said to be contractive if

$$d(T(x), T(y)) < d(x, y)$$

for each $x, y \in M$ with $x \neq y$.

The next obvious question is whether there is a meaningful fixed point result for the contractive mappings. Now the answer is yes, but the class of spaces to which it applies is much more restrictive.

Theorem 3.5 Let (M, d) be a compact metric space and let $T : M \rightarrow M$ be a contractive mapping. Then T has a unique fixed point x_0 , and moreover, for each $x \in M$, $\lim_{n \rightarrow \infty} T^n(x) = x_0$.

Proof. The existence of a fixed point for T is easy. Introduce the mapping $\psi : M \rightarrow \mathbb{R}^+$ by setting

$$\psi(x) = d(x, T(x)), \quad x \in M.$$

Then ψ is continuous and bounded below, so ψ assumes its minimum value at some point $x_0 \in M$. Since $x_0 \neq T(x_0)$ implies

$$\psi(T(x_0)) = d(T(x_0), T^2(x_0)) < d(x_0, T(x_0)) = \psi(x_0),$$

it must be the case that $x_0 = T(x_0)$.

Now let $x \in M$ and consider the sequence $\{d(T^n(x), x_0)\}$. If $T^n(x) \neq x_0$,

$$d(T^{n+1}(x), x_0) = d(T^{n+1}(x), T(x_0)) < d(T^n(x), x_0),$$

so $\{d(T^n(x), x_0)\}$ is strictly decreasing (until perhaps it reaches x_0). Consequently the limit

$$r = \lim_{n \rightarrow \infty} d(T^n(x), x_0)$$

exists and $r \geq 0$. Also, since M is compact, the sequence $\{T^n(x)\}$ has a convergent subsequence $\{T^{n_k}(x)\}$, say $\lim_{k \rightarrow \infty} T^{n_k}(x) = z \in M$. Since $\{T^n(x)\}$ is decreasing,

$$r = d(z, x_0) = \lim_{k \rightarrow \infty} d(T^{n_k}(x), x_0) = \lim_{k \rightarrow \infty} d(T^{n_k+1}(x), x_0) = d(T(z), x_0).$$

But if $z \neq x_0$ then $d(T(z), x_0) = d(T(z), T(x_0)) < d(z, x_0)$. This proves that any convergent subsequence of $\{T^n(x)\}$ converges to x_0 , so it must be the case that $\lim_{n \rightarrow \infty} T^n(x) = x_0$. ■

3.2 Further extensions of Banach's Principle

The strength of the Contraction Mapping Principle lies in the fact that the underlying space is quite general (complete metric) while the conclusion is very strong. The fixed point is unique and sequences of iterates of the original mapping always converge to this fixed point. It seems that any theorem which should claim to be a genuine extension of the contraction principle should offer just as much, and indeed, many such extensions appear in the literature. The extensions we present in this section are easy to describe and include many of the others. They are in no way definitive, however.

The first extension we take up, which is due to Geraghty [63], was inspired by an earlier theorem of Rakotch [134]. We then consider extensions due to F. Browder and others.

Let S denote the class of those functions $\alpha : \mathbb{R}^+ \rightarrow [0, 1)$ which satisfy the simple condition $\alpha(t_n) \rightarrow 1 \Rightarrow t_n \rightarrow 0$.

Theorem 3.6 *Let (M, d) be a complete metric space, let $T : M \rightarrow M$, and suppose there exists $\alpha \in S$ such that for each $x, y \in M$,*

$$d(T(x), T(y)) \leq \alpha(d(x, y))d(x, y).$$

Then T has a unique fixed point $z \in M$, and $\{T^n(x)\}$ converges to z , for each $x \in M$.

Proof. Fix $x \in M$ and let $x_n = T^n(x)$, $n = 1, 2, \dots$. We break the argument into two steps, each of which illustrates something more.

Step 1. $\lim_{n \rightarrow \infty} d(x_n, x_{n+1}) = 0$.

Proof. Since T is contractive the sequence $\{d(x_n, x_{n+1})\}$ is monotone decreasing and bounded below so $\lim_{n \rightarrow \infty} d(x_n, x_{n+1}) = r \geq 0$. Assume $r \neq 0$. Then by the contractive condition

$$\frac{d(x_{n+1}, x_{n+2})}{d(x_n, x_{n+1})} \leq \alpha(d(x_n, x_{n+1})), \quad n = 1, 2, \dots$$

Letting $n \rightarrow \infty$ we see that $1 \leq \lim_{n \rightarrow \infty} \alpha(d(x_n, x_{n+1}))$, and since $\alpha \in S$ this in turn implies $r = 0$. This contradiction establishes Step 1.

Step 2. $\{x_n\}$ is a Cauchy sequence.

Proof. Assume $\limsup_{m, n \rightarrow \infty} d(x_n, x_m) > 0$. By the triangle inequality

$$d(x_n, x_m) \leq d(x_n, x_{n+1}) + d(x_{n+1}, x_{m+1}) + d(x_{m+1}, x_m),$$

so by the contractive condition

$$d(x_n, x_m) \leq (1 - \alpha(d(x_n, x_m)))^{-1} [d(x_n, x_{n+1}) + d(x_{m+1}, x_m)].$$

Under the assumption $\limsup_{m, n \rightarrow \infty} d(x_n, x_m) > 0$ Step 1 now implies

$$\limsup_{m, n \rightarrow \infty} (1 - \alpha(d(x_n, x_m)))^{-1} = +\infty,$$

from which

$$\limsup_{m, n \rightarrow \infty} \alpha(d(x_n, x_m)) = 1.$$

But since $\alpha \in S$ this implies $\limsup_{m, n \rightarrow \infty} d(x_n, x_m) = 0$, which is again a contradiction.

Proof of Theorem 3.6 completed. Let $x \in M$. Since M is complete and since $\{T^n(x)\}$ is a Cauchy sequence, $\lim_{n \rightarrow \infty} T^n(x) = z \in M$, and since T is continuous, $T(z) = z$. Uniqueness of z follows from the contractive condition on T . ■

For the next result we let S' denote the collection of all monotone decreasing mappings $\psi : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ for which $0 < \psi(r) < r$ and for which ψ is continuous from the right (i.e., $r_j \downarrow r \geq 0 \Rightarrow \psi(r_j) \rightarrow \psi(r)$). This extension of Banach's Principle is due to Browder [25].

Theorem 3.7 *Let (M, d) be a bounded complete metric space, let $T : M \rightarrow M$ be continuous, and suppose there exists $\psi \in \mathcal{S}'$ such that for each $x, y \in M$,*

$$d(T(x), T(y)) \leq \psi(d(x, y)). \quad (3.2)$$

Then T has a unique fixed point z , and $\{T^n(x)\}$ converges to z , for each $x \in M$.

Proof. This theorem is actually a special case of the previous theorem. First introduce the function $\bar{\psi} : \mathbb{R} \rightarrow [0, 1)$ by setting $\bar{\psi}(0) = \psi(0)$ and

$$\bar{\psi}(t) = \frac{\psi(t)}{t} \text{ for } t > 0.$$

To see that $\bar{\psi}$ is in the class $\mathcal{S}$ suppose $\bar{\psi}(t_n) \rightarrow 1$. Then $\{t_n\}$ must be bounded (otherwise, $\lim_{n \rightarrow \infty} \inf \bar{\psi}(t_n) = 0$). By passing to a subsequence we may assume that $t_n \rightarrow t_0$ and we may assume further that either $t_n \uparrow t_0$ or $t_n \downarrow t_0$. But since ψ is continuous from the right, the assumption $t_n \downarrow t_0$ leads to $\psi(t_0) = t_0$, and since $\psi(r) < r$ for $r > 0$ this in turn implies $t_0 = 0$, that is, $t_n \rightarrow 0$. On the other hand, if $t_n \uparrow t_0$ we have, by monotonicity of ψ ,

$$\frac{\psi(t_0)}{t_n} \geq \frac{\psi(t_n)}{t_n} \rightarrow 1,$$

from which $\frac{\psi(t_0)}{t_0} \geq 1$ and this again implies $t_0 = 0$. So, any convergent subsequence of the *original* sequence $\{t_n\}$ must converge to 0. It follows that $t_n \rightarrow 0$, proving that $\bar{\psi}$ is in the class $\mathcal{S}$. Since

$$d(T(x), T(y)) \leq \bar{\psi}(d(x, y))d(x, y), \quad x, y \in M,$$

the proof is complete. ■

Subsequently, Boyd and Wong [17] obtained a more general result. In this theorem it is assumed that $\psi : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ is upper semicontinuous from the right (i.e., $r_j \downarrow r \geq 0 \Rightarrow \limsup_{j \rightarrow \infty} \psi(r_j) \leq \psi(r)$).

Theorem 3.8 *Let M be a complete metric space and suppose $f : M \rightarrow M$ satisfies*

$$d(f(x), f(y)) \leq \psi(d(x, y)) \text{ for each } x, y \in M,$$

where $\psi : \mathbb{R}^+ \rightarrow [0, \infty)$ is upper semi-continuous from the right and satisfies $0 \leq \psi(t) < t$ for $t > 0$. Then f has a unique fixed point, $\bar{x}$, and $\{f^n(x)\}$ converges to $\bar{x}$ for each $x \in M$.

Proof. Fix $x \in M$ and let $x_n = T^n(x)$, $n = 1, 2, \dots$. We break the argument into two steps.

Step 1. $\lim_{n \rightarrow \infty} d(x_n, x_{n+1}) = 0$.

Proof. Since T is contractive the sequence $\{d(x_n, x_{n+1})\}$ is monotone decreasing and bounded below so $\lim_{n \rightarrow \infty} d(x_n, x_{n+1}) = r \geq 0$. Assume $r > 0$. Then

$$d(x_{n+1}, x_{n+2}) \leq \psi(d(x_n, x_{n+1})) \Rightarrow r \leq \psi(r) \Rightarrow r = 0.$$

Step 2. $\{x_n\}$ is a Cauchy sequence.

Proof. Suppose not. Then there exists $\varepsilon > 0$ such that for any $k \in \mathbb{N}$, there exist $m_k > n_k \geq k$, such that

$$d(x_{m_k}, x_{n_k}) \geq \varepsilon. \quad (3.3)$$

Furthermore, assume that for each k , m_k is the smallest number greater than n_k for which (3.3) holds. In view of Step 1 there exists k_0 such that $k \geq k_0 \Rightarrow d(x_k, x_{k+1}) \leq \varepsilon$. For such k we have

$$\begin{aligned} \varepsilon &\leq d(x_{m_k}, x_{n_k}) \leq d(x_{m_k}, x_{m_k-1}) + d(x_{m_k-1}, x_{n_k}) \\ &\leq d(x_{m_k}, x_{m_k-1}) + \varepsilon \leq d(x_k, x_{k-1}) + \varepsilon. \end{aligned}$$

This proves $\lim_{k \rightarrow \infty} d(x_{m_k}, x_{n_k}) = \varepsilon$. On the other hand,

$$\begin{aligned} d(x_{m_k}, x_{n_k}) &\leq d(x_{m_k}, x_{m_k+1}) + d(x_{m_k+1}, x_{n_k+1}) + d(x_{n_k+1}, x_{n_k}) \\ &\leq 2d(x_k, x_{k-1}) + \psi(d(x_{m_k}, x_{n_k})). \end{aligned}$$

It follows that $\varepsilon \leq \psi(\varepsilon)$ — a contradiction.

The proof is completed by observing that since $\{T^n(x)\}$ is a Cauchy sequence and M is complete, $\lim_{n \rightarrow \infty} T^n(x) = z \in M$. Since T is continuous, $T(z) = z$. Uniqueness of T follows from the contractive condition. ■

Remark 3.1 *Boyd and Wong also showed in [17] that if the space M is metrically convex, then the upper semicontinuity assumption on ψ can be dropped. Matkowski has extended this fact even further in [117] by showing that it suffices to assume that ψ is continuous at 0 and that there exists a sequence $t_n \downarrow 0$ for which $\psi(t_n) < t_n$.*

Since it is the explicit control over the error term that contributes so much to the widespread usefulness of Banach's principle, the following variant of the Boyd-Wong theorem due to Browder [25] is also of interest.

Theorem 3.9 *Let X be a complete metric space and M a bounded closed subset of X . Suppose $f : M \rightarrow M$ satisfies*

$$d(f(x), f(y)) \leq \psi(d(x, y)) \text{ for each } x, y \in M,$$

where $\psi : [0, \infty) \rightarrow [0, \infty)$ is monotone nondecreasing and continuous from the right, such that $\psi(t) < t$ for all $t > 0$. Then there is a unique element $\bar{x} \in M$ such that $\{f^n(x)\}$ converges to $\bar{x}$ for each $x \in M$. Moreover, if d_0 is the diameter of M , then

$$d(f^n(x), \bar{x}) \leq \psi^n(d_0)$$

and $\psi^n(d_0) \rightarrow 0$ as $n \rightarrow \infty$.

Another variant is due to Matkowski [118]. In this result the continuity condition on ψ is replaced with another condition.

Theorem 3.10 *Let M be a complete metric space and suppose $f : M \rightarrow M$ satisfies*

$$d(f(x), f(y)) \leq \psi(d(x, y)) \text{ for each } x, y \in M,$$

where $\psi : (0, \infty) \rightarrow (0, \infty)$ is monotone nondecreasing and satisfies $\lim_{n \rightarrow \infty} \psi^n(t) = 0$ for $t > 0$. Then f has a unique fixed point $\bar{x}$, and $\lim_{n \rightarrow \infty} d(f^n(x), \bar{x}) = 0$ for every $x \in M$.

Proof. Fix $x \in M$ and let $x_n = T^n(x)$, $n = 1, 2, \dots$. As before, we break the argument into two steps.

Step 1. $\lim_{n \rightarrow \infty} d(x_n, x_{n+1}) = 0$.

Proof. $0 \leq \limsup_{n \rightarrow \infty} d(x_n, x_{n+1}) \leq \limsup_{n \rightarrow \infty} \psi^n(d(x, x_1)) = 0$.

Step 2. $\{x_n\}$ is a Cauchy sequence.

Proof. Since $\psi^n(t) \rightarrow 0$ for $t > 0$, $\psi(\epsilon) < \epsilon$ for any $\epsilon > 0$. In view of Step 1, given any $\epsilon > 0$ it is possible to choose n so that

$$d(x_{n+1}, x_n) \leq \epsilon - \psi(\epsilon).$$

Now let

$$K(x_n, \epsilon) = \{x \in M : d(x, x_n) \leq \epsilon\}.$$

Then if $z \in K(x_n, \epsilon)$,

$$\begin{aligned} d(T(z), x_n) &\leq d(T(z), T(x_n)) + d(T(x_n), x_n) \\ &\leq \psi(d(z, x_n)) + d(x_{n+1}, x_n) \\ &\leq \psi(\epsilon) + (\epsilon - \psi(\epsilon)) = \epsilon. \end{aligned}$$

Therefore $T : K(x_n, \epsilon) \rightarrow K(x_n, \epsilon)$ and it follows that $d(x_m, x_n) \leq \epsilon$ for all $m \geq n$. This completes Step 2.

The conclusion of the proof follows as in Theorem 3.8. ■

The key step in proving the existence of a fixed point in each of the previous results involved showing that given $x \in M$, $\{T^n(x)\}$ is a Cauchy sequence (and then invoking continuity of T). It is possible to carry this idea much further. First we need some notation. For any mapping $F : M \rightarrow M$ we use $k(F)$ to denote the Lipschitz constant of F ; thus

$$k(F) = \sup \left\{ \frac{d(F(x), F(y))}{d(x, y)} : x, y \in M, x \neq y \right\}.$$

Note in particular that if $F, G : M \rightarrow M$ are two Lipschitzian mappings, then

$$k(F \circ G) \leq k(F)k(G).$$

Now let (M, d) be a complete metric space and suppose $T : M \rightarrow M$ is Lipschitzian. Fix $x \in M$, and let $x_n = T^n(x)$, $n = 1, 2, \dots$. By the triangle inequality, if $m > n$, then

$$d(x_n, x_m) \leq \sum_{i=n}^{m-1} d(x_i, x_{i+1}).$$

Consequently $\{x_n\}$ is a Cauchy sequence if $\sum_{i=1}^{\infty} d(x_i, x_{i+1}) < \infty$, and since

$$d(x_i, x_{i+1}) = d(T^i(x), T^{i+1}(x)) \leq k(T^i)d(x, T(x))$$

it follows that

$$\sum_{i=1}^{\infty} k(T^i) < \infty \Rightarrow \sum_{i=1}^{\infty} d(x_i, x_{i+1}) < \infty.$$

Also

$$k(T^{m+n}) \leq k(T^n)k(T^m).$$

Specifically $[k(T^n)]^{1/n} \leq k(T)$ and so

$$k_{\infty}(T) = \limsup_{n \rightarrow \infty} [k(T^n)]^{1/n}$$

exists. This is all we really need at this point. However it is possible to say more. Replacing T with T^p in $[k(T^n)]^{1/n} \leq k(T)$ and taking the p th root of both sides gives

$$[k(T^{pn})]^{1/pn} \leq [k(T^p)]^{1/p}, \quad p = 1, 2, \dots$$

Also, a simple calculation shows that

$$\lim_{n \rightarrow \infty} \frac{[k(T^n)]^{1/n}}{[k(T^{n+1})]^{1/n+1}} = 1.$$

These two facts lead to the conclusion:

$$k_{\infty}(T) = \lim_{n \rightarrow \infty} [k(T^n)]^{1/n} = \inf \{ [k(T^n)]^{1/n} : n = 1, 2, \dots \}.$$

Now, by the Root Test for convergence of series, if $k_{\infty}(T) < 1$ then $\sum_{i=1}^{\infty} k(T^i) < \infty$. This leads to the following theorem.

Theorem 3.11 *Let M be a complete metric space and let $T : M \rightarrow M$ be a continuous mapping for which $k_{\infty}(T) < 1$. Then T has a unique fixed point $z \in M$, and for each $x \in M$ the sequence $\{T^n(x)\}$ converges to z .*

Proof. In view of what was shown just prior to the statement of the theorem, all that remains is to show that the fixed point of T is unique. However, this follows from the fact that if $k_{\infty}(T) < 1$, then for n sufficiently large, $k(T^n) < 1$. ■

There is a point to the previous development. One might ask whether *qualitatively*, Theorem 3.11 is stronger than Banach's contraction principle, and indeed the answer is no. To see this we introduce the concept of equivalent metrics.

Two metrics, ρ and d on a space M are said to be *equivalent* if there exist positive numbers a and b such that for each $x, y \in M$

$$a d(x, y) \leq \rho(x, y) \leq b d(x, y).$$

From this it follows that if $T : M \rightarrow M$ then

$$k_d(T) < \infty \Leftrightarrow k_{\rho}(T) < \infty.$$

Moreover,

$$d(T(x), T(y)) \leq \frac{1}{a} \rho(T(x), T(y)) \leq \frac{1}{a} k_{\rho}(T) \rho(x, y) \leq \frac{b}{a} k_{\rho}(T) d(x, y).$$

This implies $k_d(T) \leq \frac{b}{a} k_{\rho}(T)$. Similarly,

$$\rho(T(x), T(y)) \leq b d(T(x), T(y)) \leq b k_d(T) d(x, y) \leq \frac{b}{a} k_d(T) \rho(x, y);$$

hence $k_{\rho}(T) \leq \frac{b}{a} k_d(T)$. Consequently,

$$\frac{a}{b} k_{\rho}(T) \leq k_d(T) \leq \frac{b}{a} k_{\rho}(T),$$

and since $\lim_{n \rightarrow \infty} \left(\frac{b}{a}\right)^{1/n} = \lim_{n \rightarrow \infty} \left(\frac{a}{b}\right)^{1/n} = 1$ we conclude

$$\lim_{n \rightarrow \infty} [k_d(T^n)]^{1/n} = \lim_{n \rightarrow \infty} [k_\rho(T^n)]^{1/n}.$$

Hence $(k_d)_\infty(T) = (k_\rho)_\infty(T)$, that is, $k_\infty(T)$ is the same for all equivalent metrics on M .

Now suppose (M, d) is a complete metric space and let $T : M \rightarrow M$ be a mapping for which $k_\infty(T) < 1$. Then if $\lambda \in \left[0, \frac{1}{k_\infty(T)}\right)$,

$$\lambda^n d(T^n(x), T^n(y)) \leq \lambda^n k(T^n) d(x, y)$$

with

$$\lim_{n \rightarrow \infty} [\lambda^n k(T^n)]^{1/n} = \lambda k_\infty(T) < 1.$$

Therefore,

$$r_\lambda(x, y) = \sum_{n=0}^{\infty} \lambda^n d(T^n(x), T^n(y)) < \infty,$$

and moreover

$$d(x, y) \leq r_\lambda(x, y) \leq \left[\sum_{n=0}^{\infty} \lambda^n k(T^n) \right] d(x, y).$$

This proves that r_λ and d are equivalent metrics on M .

Finally, for $x, y \in M$

$$\begin{aligned} r_\lambda(T(x), T(y)) &= \sum_{n=0}^{\infty} \lambda^n d(T^{n+1}(x), T^{n+1}(y)) \\ &= (1/\lambda) \sum_{n=0}^{\infty} \lambda^{n+1} d(T^{n+1}(x), T^{n+1}(y)) \\ &= (1/\lambda) [r_\lambda(x, y) - d(x, y)] \\ &\leq (1/\lambda) r_\lambda(x, y). \end{aligned}$$

Thus $k_{r_\lambda}(T) \leq \frac{1}{\lambda} < 1$. This proves that T is a contraction mapping on the metric space (M, r_λ) . Also, since r_λ and d are equivalent metrics, (M, r_λ) is a complete space. It is now possible to invoke Banach's original theorem to

conclude that T has a unique fixed point x_0 (with $\lim_{n \rightarrow \infty} d(T^n(x), x_0) = 0$ for each $x \in M$).

The preceding ideas lead to the following variant of Theorem 3.4.

Theorem 3.12 *Suppose (M, d) is a metric space and suppose $T : M \rightarrow M$ is a mapping for which $(k_d)_\infty(T) < 1$. Suppose also that T is continuous relative to a metric ρ on M for which (M, ρ) is complete, and suppose $\rho(x, y) \leq d(x, y)$ for each $x, y \in M$. Then T has a unique fixed point x_0 , and moreover $\lim_{n \rightarrow \infty} T^n(x) = x_0$ for each $x \in M$.*

Proof. There exists a metric d' on M which is equivalent to d and such that $T : (M, d') \rightarrow (M, d')$ is a contraction mapping. Therefore, for each $x \in M$, $\{T^n(x)\}$ is a Cauchy sequence in (M, d') ; hence $\{T^n(x)\}$ is Cauchy in (M, d) and, since $\rho \leq d$, $\{T^n(x)\}$ is in fact Cauchy in (M, ρ) as well. Therefore, relative to ρ , $\lim_{n \rightarrow \infty} T^n(x) = x_0$ for some $x_0 \in M$ and the conclusion follows because T is continuous on (M, ρ) . ■

We now turn to a principle of a different kind. In this result, the contractive condition is imposed only at the first step. This paves the way for the results of the next section.

Theorem 3.13 *Suppose M is a complete metric space and suppose $T : M \rightarrow M$ is a continuous mapping which satisfies for some $\varphi : M \rightarrow \mathbb{R}^+$,*

$$d(x, T(x)) \leq \varphi(x) - \varphi(T(x)), \quad x \in M. \quad (*)$$

Then $\{T^n(x)\}$ converges to a fixed point of T for each $x \in M$.

Proof. This is a piece of the argument used in the proof of Banach's contraction principle. The condition $(*)$ implies that $\{\varphi(T^n(x))\}$ is monotone decreasing and hence $\lim_{n \rightarrow \infty} \varphi(T^n(x)) = r \geq 0$. By the triangle inequality, if $m, n \in \mathbb{N}$ and $m > n$ then

$$d(T^n(x), T^m(x)) \leq \sum_{i=n}^{m-1} d(T^i(x), T^{i+1}(x)) \leq \varphi(T^n(x)) - \varphi(T^m(x))$$

so $\lim_{m, n \rightarrow \infty} d(T^n(x), T^m(x)) = 0$. Since M is complete there exists $z \in M$ such that $\lim_{n \rightarrow \infty} T^n(x) = z$ and by continuity of T , $z = T(z)$. ■

Remark 3.2 *In the above result, one can obtain an estimate on the rate of convergence of $\{T^n(x)\}$ by referring back to the inequality*

$$\sum_{i=n}^{m-1} d(T^i(x), T^{i+1}(x)) \leq \varphi(T^n(x)) - \varphi(T^m(x)).$$

This yields

$$d(T^n(x), T^m(x)) \leq \varphi(T^n(x)) - \varphi(T^m(x)) \leq \varphi(T^n(x)),$$

and if $T(z) = z$, upon letting $m \rightarrow \infty$ we get

$$d(T^n(x), z) \leq \varphi(T^n(x)).$$

3.3 The Caristi-Ekeland Principle

In view of the fact that continuity of T was essential to the proof of Theorem 3.13 it is remarkable to find that the result remains true without such an assumption if it is assumed that $\varphi : M \rightarrow \mathbb{R}$ is *lower semicontinuous (l.s.c.)*, that is, for any sequence $\{x_n\} \subset M$, if $\lim_{n \rightarrow \infty} x_n = x$ and $\lim_{n \rightarrow \infty} \varphi(x_n) = r$, then $\varphi(x) \leq r$. This new fact can be formulated in two equivalent ways.

Theorem 3.14 (E) (Ekeland, 1974) Let (M, d) be a complete metric space and $\varphi : M \rightarrow \mathbb{R}^+$ l.s.c. Define:

$$x \leq y \Leftrightarrow d(x, y) \leq \varphi(x) - \varphi(y), \quad x, y \in M.$$

Then $(M, \leq)$ has a maximal element.

Theorem 3.15 (C) (Caristi, 1975) Let M and φ be as above. Suppose $g : M \rightarrow M$ satisfies:

$$d(x, g(x)) \leq \varphi(x) - \varphi(g(x)), \quad x \in M.$$

Then g has a fixed point.

(E) $\Rightarrow$ (C). Proof. With M, φ as above, and g as in Theorem (C), define $x \leq y \Leftrightarrow d(x, y) \leq \varphi(x) - \varphi(y)$, $x, y \in M$. By (E) there exists $\bar{x} \in M$ such that $\bar{x}$ is maximal in $(M, \leq)$. But $d(\bar{x}, g(\bar{x})) \leq \varphi(\bar{x}) - \varphi(g(\bar{x})) \Rightarrow \bar{x} \leq g(\bar{x})$. Hence by maximality, $\bar{x} = g(\bar{x})$.

(C) $\Rightarrow$ (E). Proof. Assume (E) is false. Then $\forall x \in M \exists g(x) \in M$ such that $x < g(x)$. It follows that

$$d(x, g(x)) \leq \varphi(x) - \varphi(g(x)), \quad x \in M. \quad (*)$$

By (C) g must have a fixed point $\bar{x}$. But by assumption, $\bar{x} < g(\bar{x})$ —a contradiction.

Those familiar with the logical foundations of mathematics might note that the proof of implication (C) $\Rightarrow$ (E) uses the Axiom of Choice, whereas the proof that (E) $\Rightarrow$ (C) does not. In fact, these two theorems are equivalent *only* if one

assumes (as we do) the Axiom of Choice. (A proof of (C) can be given within the basic axioms of Zermelo-Fraenkel set theory [116]. At the same time it is known that (E) is roughly equivalent to the Countable Axiom of Choice [32].)

We first prove Caristi's theorem using the principle of transfinite induction, and then we prove a more general version of Caristi's theorem using a Zorn's lemma argument. We do this merely to illustrate the methods involved, so nothing is lost in the overall development if the transfinite induction argument is omitted. Also, more constructive proofs could have been given in each instance.

The transfinite induction argument given below is a simplified version (due to Chi Song Wong) of Caristi's original proof.

Proof of Caristi's Theorem. Let $x_0 \in M$, let Ω denote the smallest uncountable ordinal, let $\beta \in \Omega$, and suppose for each $\alpha \in \Omega$ with $\alpha < \beta$, $x_\alpha \in M$ has been defined so that

- (1) $\mu \leq \alpha \Rightarrow d(x_\mu, x_\alpha) \leq \varphi(x_\mu) - \varphi(x_\alpha)$;
- (2) $\mu < \alpha \Rightarrow x_{\mu+1} = g(x_\mu)$.

We need to define x_β and show that (1), (2) hold for $\alpha \leq \beta$.

Case 1. $\beta = \alpha + 1$. Define $x_\beta = g(x_\alpha)$. Then (2) holds by definition. Also from (*)

$$d(x_\alpha, x_{\alpha+1}) \leq \varphi(x_\alpha) - \varphi(x_{\alpha+1}).$$

If $\mu \leq \alpha$, then by assumption

$$d(x_\mu, x_\alpha) \leq \varphi(x_\mu) - \varphi(x_\alpha).$$

Hence

$$\begin{aligned} d(x_\mu, x_{\alpha+1}) &\leq d(x_\mu, x_\alpha) + d(x_\alpha, x_{\alpha+1}) \\ &\leq \varphi(x_\mu) - \varphi(x_\alpha) + \varphi(x_\alpha) - \varphi(x_{\alpha+1}) \\ &= \varphi(x_\mu) - \varphi(x_{\alpha+1}). \end{aligned}$$

This establishes (1).

Case 2. β is a limit ordinal. First note that (1) implies $\{\varphi(x_\alpha)\}_{\alpha < \beta}$ is a nonincreasing net of real numbers. Since φ is bounded below there exists $r \in \mathbb{R}$ such that

$$\lim_{\alpha \rightarrow \beta^-} \varphi(x_\alpha) = \inf_{\alpha < \beta} \varphi(x_\alpha) = r.$$

Let $\varepsilon > 0$. Then there exists $\alpha_0 < \beta$ such that if $\alpha_0 < \alpha < \beta$ then

$$\varphi(x_\alpha) < r + \varepsilon.$$

In particular, if $\alpha_0 < \mu < \alpha < \beta$, then

$$d(x_\mu, x_\alpha) \leq \varphi(x_\mu) - \varphi(x_\alpha) < \varepsilon.$$

This proves that $\{x_\alpha\}_{\alpha < \beta}$ is a Cauchy net. Since M is complete there exists $x \in M$ such that $\lim_{\alpha \rightarrow \beta^-} x_\alpha = x$. Since $\lim_{\alpha \rightarrow \beta^-} \varphi(x_\alpha) = r$ and φ is lower semicontinuous, $\varphi(x) \leq r$. Now define $x_\beta = x$. Then if $\mu < \beta$, the definition of x_β and lower semicontinuity of φ give

$$\begin{aligned} d(x_\mu, x_\beta) &= \lim_{\alpha \rightarrow \beta^-} d(x_\mu, x_\alpha) \\ &\leq \lim_{\alpha \rightarrow \beta^-} (\varphi(x_\mu) - \varphi(x_\alpha)) \\ &= \varphi(x_\mu) - r \\ &\leq \varphi(x_\mu) - \varphi(x) \\ &= \varphi(x_\mu) - \varphi(x_\beta). \end{aligned}$$

Thus (1) holds for β , and (2) holds vacuously when β is a limit ordinal.

So $\{x_\alpha\}$ is defined for all $\alpha \in \Omega$. Since $\{\varphi(x_\alpha)\}$ is nonincreasing and Ω is uncountable there exists $\alpha_0 \in \Omega$ such that $\{\varphi(x_\alpha)\}_{\alpha \geq \alpha_0}$ is constant. In particular,

$$\varphi(x_{\alpha+1}) = \varphi(x_\alpha),$$

from which

$$d(x_{\alpha_0}, g(x_{\alpha_0})) = d(x_{\alpha_0}, x_{\alpha_0+1}) \leq \varphi(x_{\alpha_0}) - \varphi(x_{\alpha_0+1}) = 0;$$

thus $g(x_{\alpha_0}) = x_{\alpha_0}$. ■

Next we illustrate a Zorn's lemma approach. The following theorem reduces to Caristi's theorem in the case that $M = Y$, f is the identity mapping, and $c = 1$. Here we need another definition. A mapping f of a subset A of metric space M into a metric space N is said to be *closed* if it has a closed graph; thus $f : A \rightarrow N$ is closed if for $\{x_n\} \subseteq A$ the conditions $\lim_{n \rightarrow \infty} x_n = x$ and $\lim_{n \rightarrow \infty} f(x_n) = y$ imply $x \in A$ and $f(x) = y$. Applications of this result to nonlinear operator theory may be found in [48].

Theorem 3.16 *Let M and Y be complete metric spaces and let $g : M \rightarrow M$ be an arbitrary mapping. Suppose there exist a closed mapping $f : M \rightarrow Y$, a lower semicontinuous mapping $\varphi : f(M) \rightarrow \mathbb{R}^+$, and a constant $c > 0$ such that for each $x \in M$,*

$$\max\{d(x, g(x)), c d(f(x), f(g(x)))\} \leq \varphi(f(x)) - \varphi(f(g(x))). \quad (**)$$

Then there exists $x \in M$ such that $g(x) = x$.

Proof. Introduce the partial order $\geq$ in M as follows. For $x, y \in M$ say that $y \geq x \Leftrightarrow$

$$\max\{d(x, y), c d(f(x), f(y))\} \leq \varphi(f(x)) - \varphi(f(y)).$$

Let $\{x_\alpha\}_{\alpha \in I}$ be any chain in $(M, \geq)$, and for $\alpha, \beta \in I$ set $\beta \geq \alpha \Leftrightarrow x_\beta \geq x_\alpha$. Then $\{\varphi(f(x_\alpha))\}_{\alpha \in I}$ is a nonincreasing net in $\mathbb{R}^+$ so there exists $r \geq 0$ such that

$$\lim_{\alpha} \varphi(f(x_\alpha)) = r.$$

Let $\varepsilon > 0$. Then there exists $\alpha_0 \in I$ such that $\alpha \geq \alpha_0$ implies

$$r \leq \varphi(f(x_\alpha)) \leq r + \varepsilon$$

and so for $\beta \geq \alpha \geq \alpha_0$,

$$\max\{d(x_\alpha, x_\beta), c d(f(x_\alpha), f(x_\beta))\} \leq \varphi(f(x_\alpha)) - \varphi(f(x_\beta)) \leq \varepsilon.$$

Thus $\{f(x_\alpha)\}$ is a Cauchy net in Y while at the same time $\{x_\alpha\}$ is a Cauchy net in M . It follows that there exists $\bar{y} \in Y$ and $\bar{x} \in M$ such that $\lim_{\alpha} f(x_\alpha) = \bar{y}$ and $\lim_{\alpha} x_\alpha = \bar{x}$. Since f is a closed mapping, $f(\bar{x}) = \bar{y}$, and lower semicontinuity of φ yields $\varphi(f(\bar{x})) \leq r$. Moreover, if $\alpha, \beta \in I$ with $\beta \geq \alpha$, then

$$\max\{d(x_\alpha, x_\beta), c d(f(x_\alpha), f(x_\beta))\} \leq \varphi(f(x_\alpha)) - \varphi(f(x_\beta)) \leq \varphi(f(x_\alpha)) - r$$

so taking the limit with respect to β yields

$$\max\{d(x_\alpha, \bar{x}), c d(f(x_\alpha), f(\bar{x}))\} \leq \varphi(f(x_\alpha)) - \varphi(f(\bar{x})).$$

This proves that $\bar{x} \geq x_\alpha$ for each $\alpha \in I$.

Having thus shown that every chain in $(M, \geq)$ has an upper bound we can appeal to Zorn's lemma to conclude that $(M, \geq)$ has a maximal element, say x . But by (**) $g(x) \geq x$; hence $x = g(x)$. ■

3.4 Equivalents of the Caristi-Ekeland Principle

There is a vast literature on this subject and a complete summary of the topic would be beyond the scope of this treatment. The following is an example of a seemingly more general formulation of Caristi's theorem which actually reduces to a simple application of the original result.

Theorem 3.17 *Let M be a complete metric space and suppose $\varphi : M \rightarrow \mathbb{R}$ is lower semicontinuous and bounded below. Suppose $\xi : \mathbb{R} \rightarrow \mathbb{R}$ is continuous, strictly increasing, concave downward, and vanishes at 0, and suppose $g : M \rightarrow M$ satisfies:*

$$\xi(d(x, g(x))) \leq \varphi(x) - \varphi(g(x)), \quad x \in M.$$

Then g has a fixed point.

Proof. Observe that $\xi \circ d$ is a metric on M (Example 2.14 of Chapter 2). Suppose $\{x_n\}$ is a Cauchy sequence in $(M, \xi \circ d)$. Since ξ is strictly increasing and vanishes at 0, $\lim_{m,n \rightarrow \infty} \xi(d(x_n, x_m)) = 0$ implies $\lim_{m,n \rightarrow \infty} d(x_n, x_m) = 0$. Since M is complete there exists $x \in M$ such that $\lim_{n \rightarrow \infty} d(x_n, x) = 0$. By continuity of ξ this in turn implies $\lim_{n \rightarrow \infty} \xi(d(x_n, x)) = 0$. Therefore $\{x_n\}$ converges to x in $(M, \xi \circ d)$, proving that $(M, \xi \circ d)$ is itself complete. It is now possible to apply Caristi's theorem directly to the space $(M, \xi \circ d)$ to obtain a fixed point x for g , completing the proof. ■

Corollary 3.1 *Let M be a complete metric space and suppose $\varphi : M \rightarrow \mathbb{R}$ is lower semicontinuous and bounded below. Suppose for some $r \in (0, 1]$, $g : M \rightarrow M$ satisfies:*

$$d(x, g(x))^r \leq \varphi(x) - \varphi(g(x)), \quad x \in M.$$

Then g has a fixed point.

Proof. Take $\xi(t) = t^r$, $t \in \mathbb{R}$, and apply the previous theorem. (It is shown in [4] that Corollary 3.1 is false if $r > 1$.) ■

It is also trivial to see that Caristi's theorem extends to set valued mappings.

Theorem 3.18 *Let M be a complete metric space and let $\varphi : M \rightarrow \mathbb{R}$ be lower semicontinuous and bounded below. Suppose $G : M \rightarrow 2^M$ satisfies: For each $x \in M$ there exists $u \in G(x)$ such that*

$$d(x, u) \leq \varphi(x) - \varphi(u).$$

Then there exists $x \in M$ such that $x \in G(x)$.

We now take up one of the more recent developments due to Oettli and Théra [127]. This approach provides yet another proof of Caristi's theorem, one which uses neither transfinite induction nor Zorn's lemma.

Let (M, d) be a complete metric space and let $f : M \times M \rightarrow (-\infty, \infty]$ be an extended real-valued function which is lower semicontinuous in its second variable and satisfies

$$f(v, v) = 0 \text{ for all } v \in M; \quad (3.4)$$

$$f(u, v) \leq f(u, w) + f(w, v) \text{ for all } u, v, w \in M. \quad (3.5)$$

Assume also that there exists $v_0 \in M$ such that

$$\inf_{v \in M} f(v_0, v) > -\infty. \quad (3.6)$$

Let

$$S_0 = \{v \in M : f(v_0, v) + d(v_0, v) \leq 0\}. \quad (3.7)$$

Observe that $S_0 \neq \emptyset$ since by (3.4), $v_0 \in S_0$.

Theorem 3.19 *Assume M , f and S_0 satisfy the assumptions of the preceding paragraph. Let $S \subseteq M$ have the property: For every $\bar{v} \in S_0 \setminus S$ there exists $v \in M$ such that $\bar{v} \neq v$ and*

$$f(\bar{v}, v) + d(\bar{v}, v) \leq 0. \quad (3.8)$$

Then there exists $v^ \in S_0 \cap S$.*

Before proving this theorem we show how it implies Theorem 3.18. Suppose $G : M \rightarrow 2^M$ and suppose for each $\bar{v} \in M$ there exists $v \in G(\bar{v})$ such that

$$d(\bar{v}, v) \leq \varphi(\bar{v}) - \varphi(v), \quad (3.9)$$

where $\varphi : M \rightarrow \mathbb{R}$ is lower semicontinuous and bounded below. Define $f : M \times M \rightarrow \mathbb{R}$ by setting

$$f(u, v) = \varphi(v) - \varphi(u), \quad u, v \in M.$$

Thus (3.9) is equivalent to (3.8). Fix $v_0 \in M$. Since φ is bounded below,

$$\inf_{v \in M} f(v_0, v) = \inf_{v \in M} (\varphi(v) - \varphi(v_0)) > -\infty.$$

Now let $S_0 = \{v \in M : f(v_0, v) + d(v_0, v) \leq 0\}$ and $S = \{\bar{v} \in M : \bar{v} \in G(\bar{v})\}$. By assumption if $\bar{v} \in S_0 \setminus S$ there exists $v \in G(\bar{v})$ (necessarily $\bar{v} \neq v$) such that

$$d(\bar{v}, v) \leq \varphi(\bar{v}) - \varphi(v) = -f(\bar{v}, v).$$

Thus (3.8) holds. So by Theorem 3.19 there exists $v^* \in S_0 \cap S$; thus $v^* \in G(v^*)$.

Proof of Theorem 3.19. This is a simple induction procedure. For each $x \in M$ define the set

$$S(x) = \{v \in M : f(x, v) + d(x, v) \leq 0\},$$

and define

$$\gamma(x) = \inf_{v \in S(x)} f(x, v).$$

Then clearly $x \in S(x)$ and $\gamma(x) \leq 0$ for each $x \in M$. Now let $S_0 = S(v_0)$. By assumption $\gamma(v_0) = \gamma_0 = \inf_{v \in M} f(v_0, v) > -\infty$. Let $n \in \mathbb{N}$ and suppose v_{n-1} has been defined with $\gamma(v_{n-1}) > -\infty$. Choose $v_n \in S(v_{n-1})$ so that

$$f(v_{n-1}, v_n) \leq \gamma(v_{n-1}) + \frac{1}{n}.$$

Now let $v \in S(v_n)$. Then

$$f(v_n, v) + d(v_n, v) \leq 0.$$

Since $v_n \in S(v_{n-1})$

$$f(v_{n-1}, v_n) + d(v_{n-1}, v_n) \leq 0.$$

The triangle inequality and (3.5) now give

$$f(v_{n-1}, v) + d(v_{n-1}, v) \leq 0.$$

This proves $v \in S(v_{n-1})$, that is, $S(v_n) \subseteq S(v_{n-1})$. Therefore,

$$\begin{aligned} \gamma(v_n) = \inf_{v \in S(v_n)} f(v_n, v) &\geq \inf_{v \in S(v_n)} (f(v_{n-1}, v) - f(v_{n-1}, v_n)) \\ &\geq \inf_{v \in S(v_{n-1})} f(v_{n-1}, v) - f(v_{n-1}, v_n) \\ &= \gamma(v_{n-1}) - f(v_{n-1}, v_n) \geq -\frac{1}{n}. \end{aligned}$$

Thus if $v \in S(v_n)$ then $d(v_n, v) \leq -f(v_n, v) \leq -\gamma(v_n) \leq \frac{1}{n}$. Since $v_n \in S(v_n)$ this implies that the diameters of the sets $S(v_n)$ tend to zero and in turn that the sequence $\{v_n\}$ is Cauchy. Also, since f is lower semicontinuous in its second variable, each of the sets $S(v_n)$ is closed. Therefore there exists $v^* \in \bigcap_{n=0}^{\infty} S(v_n)$; indeed $\bigcap_{n=0}^{\infty} S(v_n) = \{v^*\}$. We claim $v^* \in S$. If not, then by assumption there would exist $v \neq v^*$ for which $f(v^*, v) + d(v^*, v) \leq 0$. Since $v^* \in \bigcap_{n=0}^{\infty} S(v_n)$ we have

$$f(v_n, v^*) + d(v_n, v^*) \leq 0 \text{ for all } n.$$

Combined with (3.5) this yields

$$f(v_n, v) + d(v_n, v) \leq 0 \text{ for all } n$$

and this in turn implies $v \in \bigcap_{n=0}^{\infty} S(v_n)$. However this contradicts $v \neq v^*$. Thus $v^* \in S$, completing the proof. ■

3.5 Set-valued contractions

Banach's Contraction Mapping Principle extends nicely to set-valued mappings, a fact first noticed by S. Nadler [125]. The key idea is the following: If A and B are nonempty closed bounded subsets of a metric space and if $x \in A$, then given $\varepsilon > 0$ there must exist a point $y \in B$ such that

$$d(x, y) \leq H(A, B) + \varepsilon,$$

where $H(A, B)$ denotes the Hausdorff distance between A and B . This is because the definition of Hausdorff distance (recall Example 2.15 of Chapter 2) assures that for any $\mu > 0$

$$A \subseteq N_{\rho+\mu}(B),$$

where $\rho = H(A, B)$.

Theorem 3.20 *Let (M, d) be a complete metric space, and let $\mathcal{M}$ be the collection of all nonempty bounded closed subsets of M endowed with the Hausdorff metric H . Suppose $T : M \rightarrow \mathcal{M}$ is a contraction mapping in the sense that for some $k < 1$:*

$$H(T(x), T(y)) \leq k d(x, y), \quad x, y \in M.$$

Then there exists a point $x \in M$ such that $x \in T(x)$.

Proof. Select $x_0 \in M$ and $x_1 \in T(x_0)$. By the observation immediately preceding the statement of the theorem there must exist $x_2 \in T(x_1)$ such that

$$d(x_1, x_2) \leq H(T(x_0), T(x_1)) + k.$$

Similarly, there exists $x_3 \in T(x_2)$ such that

$$d(x_2, x_3) \leq H(T(x_1), T(x_2)) + k^2.$$

Proceed by induction to obtain a sequence $\{x_n\}$ in M with the property that for each $i \in \mathbb{N}$, $x_{i+1} \in T(x_i)$ and for which

$$\begin{aligned} d(x_i, x_{i+1}) &\leq H(T(x_{i-1}), T(x_i)) + k^i \\ &\leq k d(x_{i-1}, x_i) + k^i \\ &\leq k[H(T(x_{i-2}), T(x_{i-1})) + k^{i-1}] + k^i \\ &\leq k^2 d(x_{i-2}, x_{i-1}) + 2k^i \\ &\leq \dots \\ &\leq k^i d(x_0, x_1) + ik^i. \end{aligned}$$

Therefore,

$$\sum_{i=0}^{\infty} d(x_i, x_{i+1}) \leq d(x_0, x_1) \left(\sum_{i=0}^{\infty} k^i \right) + \sum_{i=0}^{\infty} ik^i < \infty.$$

This proves that $\{x_n\}$ is a Cauchy sequence, so since M is complete there exists $x \in M$ such that $\lim_{n \rightarrow \infty} x_n = x$. Also, since T is continuous, $\lim_{n \rightarrow \infty} H(T(x_n), T(x)) = 0$. Since $x_n \in T(x_{n-1})$,

$$\lim_{n \rightarrow \infty} \text{dist}(x_n, T(x)) = \lim_{n \rightarrow \infty} \inf\{d(x_n, y) : y \in T(x)\} = 0.$$

This implies that

$$\text{dist}(x, T(x)) = \inf\{d(x, y) : y \in T(x)\} = 0,$$

and since $T(x)$ is closed it must be the case that $x \in T(x)$. ■

Two things are worth noting about the preceding theorem. First, in contrast to Banach's Theorem, it is not asserted that the fixed point x is unique. Indeed, it need not be. Also, since M is complete, $(\mathcal{M}, H)$ is complete as well, but this fact is not needed in the proof.

There is an interesting stability result due to T. C. Lim that holds for set-valued contractions (hence ordinary contractions as well). We begin with a technical lemma.

Lemma 3.1 *Suppose M and $\mathcal{M}$ are as in the preceding theorem, and let $T_i : M \rightarrow \mathcal{M}$, $i = 1, 2$, be two contraction mappings each having Lipschitz constant $k < 1$. Then if $F(T_1)$ and $F(T_2)$ denote the respective fixed point sets of T_1 and T_2 ,*

$$H(F(T_1), F(T_2)) \leq (1 - k)^{-1} \sup_{x \in M} H(T_1(x), T_2(x)).$$

Proof. Let $\varepsilon > 0$, choose $c > 0$ so that $c \sum_{n=1}^{\infty} nk^n < 1$, and set

$$\varepsilon_1 = \frac{c\varepsilon}{(1 - k)}.$$

Select $x_0 \in F(T_1)$, and then select $x_1 \in T_2(x_0)$ so that

$$d(x_0, x_1) \leq H(T_1(x_0), T_2(x_0)) + \varepsilon.$$

Since $H(T_2(x_1), T_2(x_0)) \leq k d(x_1, x_0)$ there exists $x_2 \in T_2(x_1)$ such that

$$d(x_2, x_1) \leq k d(x_1, x_0) + k\varepsilon_1.$$

Now define $\{x_n\}$ inductively so that $x_{n+1} \in T_2(x_n)$ and

$$d(x_{n+1}, x_n) \leq k d(x_n, x_{n-1}) + k^n \varepsilon_1, \quad n = 1, 2, \dots$$

Then

$$\begin{aligned} d(x_{n+1}, x_n) &\leq k d(x_n, x_{n-1}) + k^n \varepsilon_1 \\ &\leq k(k d(x_{n-1}, x_{n-2}) + k^{n-1} \varepsilon_1) + k^n \varepsilon_1 \\ &= k^2 d(x_{n-1}, x_{n-2}) + 2k^n \varepsilon_1. \end{aligned}$$

Continuing in this fashion we obtain

$$d(x_{n+1}, x_n) \leq k^n d(x_1, x_0) + nk^n \varepsilon_1.$$

Therefore,

$$\sum_{n=m}^{\infty} d(x_{n+1}, x_n) \leq k^m (1 - k)^{-1} d(x_1, x_0) + \sum_{n=m}^{\infty} nk^n \varepsilon_1.$$

Since the right side of the above tends to 0 as $m \rightarrow \infty$, this proves that $\{x_n\}$ is a Cauchy sequence with limit, say z . Since T_2 is continuous, $\lim_{n \rightarrow \infty} H(T_2(x_n), T_2(z)) = 0$. Also, since $x_{n+1} \in T_2(x_n)$ it must be the case that $z \in T_2(z)$, that is, $z \in F(T_2)$. Furthermore,

$$\begin{aligned} d(x_0, z) &\leq \sum_{n=0}^{\infty} d(x_{n+1}, x_n) \\ &\leq (1-k)^{-1}d(x_1, x_0) + \sum_{n=1}^{\infty} nk^n \varepsilon_1 \\ &< (1-k)^{-1}(d(x_1, x_0) + \varepsilon) \\ &\leq (1-k)^{-1}(H(T_1(x_0), T_2(x_0)) + 2\varepsilon). \end{aligned}$$

Reversing the roles of T_1 and T_2 and repeating the argument just given leads to the conclusion that for each $y_0 \in F(T_2)$ there exist $y_1 \in T_1(y_0)$ and $w \in F(T_1)$ such that

$$d(y_0, w) \leq (1-k)^{-1}(H(T_1(y_0), T_2(y_0)) + 2\varepsilon).$$

Since $\varepsilon > 0$ is arbitrary, the conclusion follows. ■

Theorem 3.21 *Suppose M and $\mathcal{M}$ are as in the preceding theorem, and let $T_i : M \rightarrow \mathcal{M}$, $i = 1, 2, \dots$ be a sequence of contraction mappings each having Lipschitz constant $k < 1$. If $\lim_{n \rightarrow \infty} H(T_n(x), T_0(x)) = 0$ uniformly for $x \in M$, then $\lim_{n \rightarrow \infty} H(F(T_n), F(T_0)) = 0$.*

Proof. Let $\varepsilon > 0$. Since $\lim_{n \rightarrow \infty} H(T_n(x), T_0(x)) = 0$ uniformly it is possible to choose $N \in \mathbb{N}$ so that for $n \geq N$, $\sup_{x \in M} H(T_n(x), T_0(x)) < (1-k)\varepsilon$. By the lemma, $H(F(T_n), F(T_0)) < \varepsilon$ for all such n . ■

3.6 Generalized contractions

A thorough search of the mathematical literature would yield numerous extensions of Banach's Theorem in which the contractive condition involves iterates of the mapping T under consideration and/or the relationship between $d(T(x), T(y))$ and the five remaining distances determined by $\{x, y, T(x), T(y)\}$ for any x, y in the domain of T . Many, but by no means all, of these results are subsumed by the approach of this section.

Again we assume (M, d) is a metric space with $T : M \rightarrow M$. For $x \in M$ we adopt the notation

$$O(x) = \{x, T(x), T^2(x), \dots\},$$

and we use $O(x, y)$ to denote $O(x) \cup O(y)$ for $x, y \in M$. Let ϕ be a *contractive gauge function* on M . By this we mean $\phi : M \rightarrow \mathbb{R}^+$ is continuous, nondecreasing, and satisfies $\phi(s) < s$ for $s > 0$.

As we have observed before, if $\phi(x) = kx$ for $x \in M$ and fixed $k \in [0, 1]$, then a mapping $T : M \rightarrow M$ satisfying

$$d(T(x), T(y)) \leq \phi(d(x, y))$$

for all $x, y \in M$ is a contraction mapping in the sense of Banach, and thus if M is complete T has a unique fixed point $z \in M$. It is a far-reaching extension of this approach inspired by Felix Browder that we take up here. We begin with the following fact due to Walter [157].

Theorem 3.22 *Let M be a bounded complete metric space and suppose $T : M \rightarrow M$ satisfies the following condition. For each $x \in M$ there exists $n(x) \in \mathbb{N}$ such that for all $n \geq n(x)$ and $y \in M$,*

$$d(T^n(x), T^n(y)) \leq \phi(\text{diam}(O(x, y))). \quad (3.10)$$

Then there exists $z \in M$ such that $\lim_{k \rightarrow \infty} T^k(x) = z$ for each $x \in M$.

Proof. This is essentially the proof of Browder [27]. There are four steps. For $x \in M$ we use the notation $x^k = T^k(x)$, $k = 0, 1, 2, \dots$.

(1) If $m = \max\{n(x), n(y)\}$, then $\text{diam}(O(x^m, y^m)) \leq \phi(\text{diam}(O(x, y)))$.

Proof. Suppose $n \geq m$ and $r \geq 0$. Then any two elements (u, v) of $O(x^m, y^m)$ are of one of the forms: (x^n, y^{n+r}) , (x^{n+r}, y^n) , (x^n, x^{n+r}) , or (y^n, y^{n+r}) . In the first case we have

$$d(u, v) = d(T^n(x), T^n(y^r)) \leq \phi(\text{diam}(O(x, y^r))) \leq \phi(\text{diam}(O(x, y))).$$

The other three cases follow by similar inequalities.

Next we define sequences $\{k(i)\} \subset \mathbb{N}$ and $A_i \subset M$ by

$$k(0) = 0; \quad k(i+1) = k(i) + \max\{n(x^{k(i)}), n(y^{k(i)})\}$$

and

$$A_i = O(x^{k(i)}, y^{k(i)}), \quad i = 0, 1, 2, \dots$$

(2) $\text{diam}(A_{i+1}) \leq \phi(\text{diam}(A_i))$ for $i = 0, 1, 2, \dots$

Proof. For $i = 0$ this is just a restatement of (1). Now let i be arbitrary and let $\xi = x^{k(i)}$, $\eta = y^{k(i)}$, and $\mu = \max\{n(x^{k(i)}), n(y^{k(i)})\}$. Applying Case (1) we have

$$\text{diam}(O(\xi^\mu, \eta^\mu)) \leq \phi(\text{diam } O(\xi, \eta)). \quad (3.11)$$

However $\xi^\mu = x^{k(i)+\mu} = x^{k(i)+\max\{n(x^{k(i)}), n(y^{k(i)})\}} = x^{k(i+1)}$. Similarly, $\eta^\mu = y^{k(i)+\mu} = y^{k(i+1)}$. Thus $\text{diam}(O(\xi^\mu, \eta^\mu)) = \text{diam}(A_{i+1})$ and (3.11) coincides with (2).

Now let $a_i = \text{diam}(A_i)$.

$$(3) \lim_{i \rightarrow \infty} a_i = 0.$$

Proof. From (2) and the fact that ϕ is decreasing we have $a_{i+1} \leq \phi(a_i) \leq a_i$. Thus $\{a_i\}$ is decreasing so there exists $a \geq 0$ such that $\lim_{i \rightarrow \infty} a_i = a \leq \phi(a)$. Since $\phi(a) < a$ if $a > 0$, it must be the case that $a = 0$.

(4) We have shown that $\lim_{i \rightarrow \infty} \text{diam}(A_i) = \lim_{i \rightarrow \infty} \text{diam}(O(x^{k(i)}, y^{k(i)})) = 0$. This clearly implies $\lim_{k \rightarrow \infty} \text{diam}(O(x^k, y^k)) = 0$. This implies that both $\{x^k\}$ and $\{y^k\}$ are Cauchy sequences and have the same limit, say $z \in M$. Since $y \in M$ is arbitrary, we conclude that, in fact, $\lim_{k \rightarrow \infty} x^k = z$ for each $x \in M$. ■

Corollary 3.2 *If in addition to the assumptions of Theorem 3.22 T is continuous, then $T(z) = z$.*

By strengthening the assumption (3.10) to require that $n(x) = 1$ for all $x \in M$ it is possible to conclude that T has a fixed point without assuming continuity.

Theorem 3.23 *Let M be a bounded complete metric space and suppose $T : M \rightarrow M$ satisfies the following condition. For each $x, y \in M$,*

$$d(T(x), T(y)) \leq \phi(\text{diam}(O(x, y))). \quad (3.12)$$

Then T has a unique fixed point $z \in M$ and $\lim_{k \rightarrow \infty} T^k(x) = z$ for each $x \in M$.

Proof. By Theorem 3.22 there exists $z \in M$ such that $\lim_{k \rightarrow \infty} T^k(x) = z$ for each $x \in M$. Assume $z \neq T(z)$. Then $\text{diam}(O(z)) = \alpha > 0$. From this it is possible to select two sequences $\{p(k)\}$ and $\{q(k)\}$ such that $0 \leq p(k) < q(k)$ for which $\lim_{k \rightarrow \infty} d(z^{p(k)}, z^{q(k)}) = \alpha$. Since $\lim_{k \rightarrow \infty} z^k = z$ there exists $k_0 \in \mathbb{N}$ such that for $k, l \geq k_0$, $d(z^k, z^l) \leq \alpha/2$. Hence for some p with $0 \leq p < k_0$, it must be the case that $p(k) \equiv p$ for infinitely many k . Therefore there is a subsequence $\{r(k)\}$ of $\{q(k)\}$ such that $\lim_{k \rightarrow \infty} d(z^p, z^{r(k)}) = \alpha$. If $r(k) = q$ infinitely often, then $d(z^p, z^q) = \alpha$. Otherwise there exists a subsequence $\{s(k)\}$ of $\{r(k)\}$ with $s(k) \rightarrow \infty$ as $k \rightarrow \infty$ and this implies $d(z^p, z) = \alpha$. In any case, there exist $p, q \geq 0$ such that $d(z^p, z^q) = \alpha$. If $p, q \geq 1$ then (3.12) implies

$$d(z^p, z^q) = d(T(z^{p-1}), T(z^{q-1})) \leq \phi(\text{diam}(O(z^{p-1}, z^{q-1}))) \leq \phi(\text{diam}(O(z))).$$

Since $\alpha = d(z^p, z^q) = \text{diam}(O(z))$ this gives $\alpha \leq \phi(\alpha)$.

On the other hand, if $d(z, z^q) = \alpha$, then since $\lim_{k \rightarrow \infty} z^k = z$

$$\alpha = \lim_{k \rightarrow \infty} d(z^k, z^q) \leq \phi(\text{diam}(O(z^{k-1}, z^{q-1}))) \leq \phi(\alpha).$$

In either case we have a contradiction since $\phi(\alpha) < \alpha$ if $\alpha > 0$. Hence $\alpha = 0$, that is, $O(z) = \{z\}$. ■

Exercises

Exercise 3.1 Suppose M is a compact metric space and suppose $T : M \rightarrow M$ is a mapping of M into M which satisfies $d(T(x), T(y)) \geq d(x, y)$. Is T necessarily an isometry?

Exercise 3.2 Let (M, d) be a complete metric space, let $\mathcal{M}$ denote the family of all nonempty bounded closed subsets of M , and let $\mathcal{C}$ denote the subfamily of $\mathcal{M}$ consisting of all compact sets. Let $T_i : M \rightarrow M$, $i = 1, \dots, n$, be a family of contractions. Show that there exists a unique nonempty compact subset X of M such that

$$X = \bigcup_{i=1}^n T_i(X).$$

Hint: Consider the mapping $T^* : \mathcal{C} \rightarrow \mathcal{C}$ defined by

$$T^*(X) = \bigcup_{i=1}^n T_i(X).$$

Show that T^* is a contraction.

Application. Consider the real line $\mathbb{R}$ and the two contractions

$$T_1(x) = \frac{1}{3}x; \quad T_2(x) = \frac{1}{3}x + \frac{2}{3}.$$

Show that the compact X which satisfies $X = T_1(X) \cup T_2(X)$ is the well-known Cantor set (which is described in standard topological texts).

Exercise 3.3 Let (M, d) be a complete metric space and $T : M \rightarrow M$. Assume that for each $\varepsilon > 0$, there exists $\delta > 0$ such that

$$d(x, Tx) < \delta \Rightarrow T(B(x, \varepsilon)) \subset B(x, \varepsilon).$$

Prove that if $d(T^n(x), T^{n+1}(x)) \rightarrow 0$, for some $x \in M$, then the sequence $\{T^n(x)\}$ converges to a fixed point of T .

Note: The mapping T is not necessarily continuous.

Exercise 3.4 Let (M, d) be a complete metric space, and X an arbitrary metric space. Let $T : M \times X \rightarrow M$ be continuous in each variable separately. Assume also that there exists a positive number $k < 1$ such that

$$d(T(x, a), T(y, a)) \leq kd(x, y)$$

for any $a \in X$ and any $x, y \in M$. Therefore, for any $a \in X$, the mapping $x \mapsto T(x, a)$ has a unique fixed point a^* . Show that the mapping $a \mapsto a^*$ is continuous.

Exercise 3.5 Let (M, d) be a complete metric space, and $T_n : M \rightarrow M$ a sequence of Lipschitzian mappings. Assume that each T_n has a fixed point x_n and the sequence $\{T_n\}$ converges pointwise to a mapping T . Assume also that the sequence $\{L(T_n)\}$ of their Lipschitz constants is bounded (by a positive number K). Show that

1. T is Lipschitzian and $L(T) \leq K$.
2. If $\{x_n\}$ converges to x_0 , then x_0 is a fixed point of T .
3. If $K < 1$, then $\{x_n\}$ converges to the unique fixed point of T .

Show also that the boundedness condition of $\{L(T_n)\}$ cannot be relaxed.

Exercise 3.6 Let (M, d) be a complete metric space and $T : M \rightarrow M$ be such that

$$d(T(x), T(y)) \geq kd(x, y)$$

for some $k > 1$ and any $x, y \in M$, and assume that $T(M) = M$. Prove that T is one-to-one (or injective). Moreover, prove that T has a unique fixed point x_0 , with $\lim_{n \rightarrow \infty} T^{-n}(x) \rightarrow x_0$ for any $x \in M$.

Exercise 3.7 Let (M, d) be a complete metric space and $T : M \rightarrow M$ a continuous mapping. Assume there exist an integer n and a positive number $k < \frac{1}{2}$ such that

$$d(T(x), T(y)) \leq k(d(x, T^n(z)) + d(y, T^n(z)))$$

for all $x, y, z \in M$. Prove that T has a unique fixed point.

Exercise 3.8 A metric space (M, d) is called ε -chainable if for any $x, y \in M$, there exist $x_0 = x, x_1, \dots, x_n, x_{n+1} = y$ in M such that $d(x_i, x_{i+1}) < \varepsilon$ for $0 \leq i \leq n$. Let $T : M \rightarrow M$ be a map. Assume that (M, d) is complete and ε -chainable, for some $\varepsilon > 0$, and there exists a positive number $k < 1$ such that

$$d(T(x), T(y)) \leq kd(x, y)$$

whenever $d(x, y) < \varepsilon$. Show that T has a fixed point.

Exercise 3.9 Let (M, d) be an arbitrary metric space and $T : M \rightarrow M$ a mapping which satisfies:

$$d(T(x), T(y)) < d(x, y)$$

whenever $x \neq y$. Assume that for some $x \in M$, the sequence $\{T^n(x)\}$ has a subsequence which converges to a point $z \in M$. Show that z is a fixed point of T .

Exercise 3.10 Let (M, d) be an arbitrary metric space and $T : M \rightarrow M$ a continuous mapping. Assume that for some $x \in M$, the orbit $\{T^n(x)\}$ contains a convergent subsequence $\{T^{n_i}(x)\}$. Show that if $d(T^{n_i}(x), T^{n_i+1}(x)) \rightarrow 0$ then T has a fixed point.

Exercise 3.11 (Volterra Equation) Consider the integral equation

$$f(x) = \lambda \int_a^x K(x, t) f(t) dt + \phi(x)$$

for a fixed real number λ , where $K(x, t)$ is continuous on $[a, b] \times [a, b]$. Consider the metric space $C[a, b]$ of continuous real-valued functions defined on $[a, b]$. Consider the mapping $T : C[a, b] \rightarrow C[a, b]$ defined by

$$(T(f))(x) = \lambda \int_a^x K(x, t) f(t) dt + \phi(x).$$

For $f_1, f_2 \in C[a, b]$, show that

$$d(T^n(f_1), T^n(f_2)) \leq |\lambda|^n M^n \frac{(b-a)^n}{n!} d(f_1, f_2),$$

where

$$M = \max\{|K(x, t)|; (x, t) \in [a, b] \times [a, b]\}.$$

Then show that the above equation has a unique solution $f(x)$.

Note: Recall

$$d(f_1, f_2) = \max\{|f_1(x) - f_2(x)|; x \in [a, b]\}.$$

for $f_1, f_2 \in C[a, b]$.

Chapter 4

Hyperconvex Spaces

4.1 Introduction

We motivate the study of hyperconvex spaces with a simple observation about the real line $\mathbb{R}^1$.

Proposition 4.1 *Let $\{I_\alpha\}_{\alpha \in A}$ be a family of bounded closed intervals of $\mathbb{R}^1$ each two of which intersect. Then*

$$\bigcap_{\alpha \in A} I_\alpha \neq \emptyset.$$

Proof. Suppose the intersection is empty. Then by compactness there exist

$$\{I_1, \dots, I_{N+1}\} \subseteq \{I_\alpha\}$$

such that

$$I = \bigcap_{i=1}^N I_i \neq \emptyset \text{ while } \bigcap_{i=1}^{N+1} I_i = \emptyset.$$

Thus I and I_{N+1} are disjoint closed intervals in $\mathbb{R}^1$. Select any point x that lies *strictly between* them. (This is possible because the complement of $I \cup I_{N+1}$ is an open set.) Then by the fact that any *two* members of the original family intersect,

$$x \in I_i \cap I_{N+1}, \quad i = 1, \dots, N.$$

This implies

$$x \in \bigcap_{i=1}^{N+1} I_i,$$

which is a contradiction. ■

We note in passing that the above fact is a special case of a famous result in geometry known as Helly's Theorem. Although it will not be needed in the sequel, we pause to state this result and to give an outline of a very elegant proof which is quite easy to visualize in $\mathbb{R}^2$.

Theorem 4.1 (*Helly's Theorem*) *Let $\mathcal{F} = \{K_\alpha\}_{\alpha \in A}$ be a family of bounded closed convex sets in $\mathbb{R}^n$, and suppose each subcollection of $\mathcal{F}$ consisting of $n+1$ members intersect. Then*

$$\bigcap_{\alpha \in A} K_\alpha \neq \emptyset.$$

Proof. By induction. Suppose the theorem is true in $\mathbb{R}^{n-1}$ and false in $\mathbb{R}^n$. Then there exists a family $\mathcal{F} = \{K_\alpha\}_{\alpha \in A}$ of bounded closed convex sets in $\mathbb{R}^n$ having the property that each $n+1$ members of $\mathcal{F}$ intersect while

$$\bigcap_{\alpha \in A} K_\alpha = \emptyset.$$

The basic line of argument follows the one given for Proposition 4.1. By compactness there must exist

$$\{K_1, \dots, K_{N+1}\} \subseteq \mathcal{F},$$

for which

$$K = \bigcap_{i=1}^N K_i \neq \emptyset \text{ while } \bigcap_{i=1}^{N+1} K_i = \emptyset.$$

Now let L be any hyperplane of $\mathbb{R}^n$ (a translate of an $n-1$ -dimensional subspace of $\mathbb{R}^n$) which separates K and K_{N+1} . (To produce such an L , let $\mathbf{x}$ and $\mathbf{y}$ be points of K and K_{N+1} respectively, such that

$$d(\mathbf{x}, \mathbf{y}) = \inf \{d(\mathbf{u}, \mathbf{v}) : \mathbf{u} \in K, \mathbf{v} \in K_{N+1}\}$$

and let L be the hyperplane through the midpoint $(\mathbf{x} + \mathbf{y})/2$ of the segment joining $\mathbf{x}$ and $\mathbf{y}$ which is orthogonal to the segment joining $\mathbf{x}$ and $\mathbf{y}$.) Now consider the family $\{K_i \cap L\}_{i=1}^N$. Since the intersection of any n members of $\{K_i\}_{i=1}^N$ also intersects K_{N+1} the intersection of any n members of $\{K_i\}_{i=1}^N$ must also intersect L . Hence the intersection of any n members of $\{K_i \cap L\}_{i=1}^N$ is nonempty. By the inductive hypothesis

$$\bigcap_{i=1}^N (K_i \cap L) = \left(\bigcap_{i=1}^N K_i \right) \cap L = K \cap L \neq \emptyset.$$

This is a contradiction. ■

The argument of Proposition 4.1 extends to ℓ_∞ (Example 2.7 of Chapter 2).

Theorem 4.2 Suppose $\{B_\alpha\}_{\alpha \in A}$ is a family of closed balls in ℓ_∞ each two of which intersect. Then

$$\bigcap_{\alpha \in A} B_\alpha \neq \emptyset.$$

Proof. Let

$$L_n = \{\mathbf{x} = (x_1, x_2, \dots) \in \ell_\infty : x_i = 0 \text{ if } i \neq n\}.$$

Then L_n is an isometric copy of $\mathbb{R}^1$ and $\{B_\alpha \cap L_n\}$ is a family of closed intervals of L_n each two of which intersect. Thus for each $n \in \mathbb{N}$,

$$\bigcap_{\alpha \in A} (L_n \cap B_\alpha) \neq \emptyset.$$

Select

$$x_n \in \bigcap_{\alpha \in A} (L_n \cap B_\alpha) \quad n = 1, 2, \dots$$

Then if $\mathbf{x} = (x_1, x_2, \dots)$ it follows that

$$\mathbf{x} \in \bigcap_{\alpha \in A} B_\alpha.$$

■

Another simple observation about ℓ_∞ :

Proposition 4.2 Suppose $B(\mathbf{x}_\alpha; r_\alpha)$ and $B(\mathbf{x}_\beta; r_\beta)$ are two closed balls in ℓ_∞ . Then

$$B(\mathbf{x}_\alpha; r_\alpha) \cap B(\mathbf{x}_\beta; r_\beta) \neq \emptyset$$

if and only if

$$d_\infty(\mathbf{x}_\alpha, \mathbf{x}_\beta) \leq r_\alpha + r_\beta.$$

Proof. One implication is completely trivial. If $\mathbf{z} \in B(\mathbf{x}_\alpha; r_\alpha) \cap B(\mathbf{x}_\beta; r_\beta)$, then

$$d_\infty(\mathbf{x}_\alpha, \mathbf{x}_\beta) \leq d_\infty(\mathbf{x}_\alpha, \mathbf{z}) + d_\infty(\mathbf{z}, \mathbf{x}_\beta) \leq r_\alpha + r_\beta.$$

Now suppose

$$d_\infty(\mathbf{x}_\alpha, \mathbf{x}_\beta) \leq r_\alpha + r_\beta.$$

Then if $\mathbf{x}_\alpha = (x_1, x_2, \dots)$ and $\mathbf{x}_\beta = (y_1, y_2, \dots)$ it must be the case that

$$|x_i - y_i| \leq r_\alpha + r_\beta \quad i = 1, 2, \dots$$

Let $z_i = \frac{r_\beta x_i + r_\alpha y_i}{r_\alpha + r_\beta}$, $i = 1, 2, \dots$. Then $|x_i - z_i| \leq r_\alpha$ and $|z_i - y_i| \leq r_\beta$; where

$$\begin{aligned} d_\infty(\mathbf{x}_\alpha, \mathbf{z}) &\leq r_\alpha; \\ d_\infty(\mathbf{x}_\beta, \mathbf{z}) &\leq r_\beta, \end{aligned}$$

that is,

$$\mathbf{z} \in B(\mathbf{x}_\alpha; r_\alpha) \cap B(\mathbf{x}_\beta; r_\beta).$$

■

Combining the above with Theorem 4.2, we obtain

Theorem 4.3 Suppose $\{B(\mathbf{x}_\alpha; r_\alpha)\}_{\alpha \in A}$ is any family of closed balls in ℓ_∞ each two of which satisfy

$$d_\infty(\mathbf{x}_\alpha, \mathbf{x}_\beta) \leq r_\alpha + r_\beta.$$

Then $\bigcap_{\alpha \in A} B(\mathbf{x}_\alpha; r_\alpha) \neq \emptyset$.

Therefore, the space ℓ_∞ provides a nontrivial example of a metric space which is *hyperconvex*. The general definition is given in the next section.

Before turning to more abstract considerations we take up another property of $\mathbb{R}^1$. Because this fact is a little deeper we give a detailed proof here and repeat the essence of the argument in a more general setting in the next section.

Recall that a subset L of a metric space is a *metric segment* if L is isometric with some real line interval $[a, b] \subseteq \mathbb{R}^1$. A *nonexpansive retraction* of a metric space (M, d) onto one of its subspaces S is a nonexpansive mapping R of M onto S (i.e., $d(R(x), R(y)) \leq d(x, y)$ for each $x, y \in M$) which leaves each point of S fixed. In this case S is said to be a *nonexpansive retract* of M .

Proposition 4.3 Suppose L is a metric segment in a metric space (M, d) . Then there is a nonexpansive retraction R of M onto L . Equivalently, L is a nonexpansive retract of M .

Proof. Consider the pairs (T_F, F) where F is a subspace of M for which $L \subseteq F \subseteq M$ and $T_F : F \rightarrow L$ is nonexpansive with $T_F(y) = y$ for each $y \in L$. Let $\mathcal{N}$ denote the collection of all such pairs, and partially order $\mathcal{N}$ as follows:

$$(T_F, F) \prec (T_G, G)$$

if and only if $F \subseteq G$ and $T_G(z) = T_F(z)$ for each $z \in F$. Note that $\mathcal{N}$ is nonempty since $(L, I) \in \mathcal{N}$, where I denotes the identity mapping on L .

If $\{(T_{F_\alpha}, F_\alpha)\}_{\alpha \in A}$ is a chain in $\mathcal{N}$ relative to this partial order, take $G = \bigcup_{\alpha} F_\alpha$, and define $T_G : G \rightarrow L$ by setting $T_G(x) = T_{F_\alpha}(x)$ for any α such that

$x \in F_\alpha$. Then T_G is well defined since if $x \in F_\alpha \cap F_\beta$ then, since either $F_\alpha \subseteq F_\beta$ or $F_\beta \subseteq F_\alpha$, it must be the case that $T_{F_\alpha}(x) = T_{F_\beta}(x)$. With this definition we see that

$$(T_{F_\alpha}, F_\alpha) \prec (T_G, G) \text{ for each } \alpha \in A.$$

Therefore, each chain in $(\mathcal{N}, \prec)$ is bounded above, so by Zorn's Lemma $(\mathcal{N}, \prec)$ has a maximal element, say (T_H, H) . All we need establish now is the following:

Assertion. $H = M$.

Proof. Suppose not. Then there exists $z \in M$ such that $z \notin H$. Let $H_1 = H \cup \{z\}$. We wish to define an extension T_{H_1} of T_H such that $T_{H_1} : H_1 \rightarrow L$ and T_{H_1} is nonexpansive. To this end, consider the following family of closed subintervals of L :

$$\mathcal{F} = \left\{ I(x) \cap L; x \in H \right\},$$

where $I(x) = [T_H(x) - d(x, z), T_H(x) + d(x, z)]$. Suppose $x_1, x_2 \in H$ and label them so that $T_H(x_1) \leq T_H(x_2)$. Then

$$T_H(x_2) - T_H(x_1) = |T_H(x_1) - T_H(x_2)| \leq d(x_1, x_2) \leq d(x_1, z) + d(x_2, z).$$

It follows that

$$T_H(x_1) + d(x_1, z) \geq T_H(x_2) - d(x_2, z)$$

and consequently

$$I(x_1) \cap I(x_2) \neq \emptyset.$$

Therefore, each two intervals in the family $\mathcal{F}$ intersect, so by Proposition 4.1

$$\bigcap_{x \in H} \left\{ [T_H(x) - d(x, z), T_H(x) + d(x, z)] \cap L \right\} \neq \emptyset.$$

Select any point $w \in \bigcap \mathcal{F}$, define $T_{H_1}(z) = w$, and let $T_{H_1}(x) = T_H(x)$ for all $x \in H$. Then since

$$T_{H_1}(z) \in [T_H(x) - d(x, z), T_H(x) + d(x, z)]$$

for each $x \in H$, we conclude that

$$|T_{H_1}(z) - T_{H_1}(x)| = |T_{H_1}(z) - T_H(x)| \leq d(x, z).$$

Hence T_{H_1} is nonexpansive. This implies

$$(T_H, H) \prec (T_{H_1}, H_1)$$

with $H_1 \neq H$, contradicting maximality of (T_H, H) and completing the proof. ■

Now let $\{B(x_\alpha; r_\alpha)\}_{\alpha \in A}$ be a family of closed balls in ℓ_∞ and suppose

$$D = \bigcap_{\alpha \in A} B(x_\alpha; r_\alpha) \neq \emptyset.$$

As before, let

$$L_n = \{x = (x_1, x_2, \dots) \in \ell_\infty : x_i = 0 \text{ for all } i \neq n\}.$$

Then $D_n = L_n \cap D$ is a nonempty closed interval which lies in an isometric copy of the real line $\mathbb{R}^1$.

It is easy to extend Proposition 4.3 to the set $D = \bigcap_n D_n$. To this end, suppose (an isometric copy of) D is contained in a metric space (M, d) . Then by Proposition 4.3, for each $n \in \mathbb{N}$ there exists a nonexpansive retraction $R_n : M \rightarrow D_n$. Thus

$$|R_n(x) - R_n(y)| \leq d(x, y)$$

for each $x, y \in M$. Define $R : M \rightarrow D$ by setting

$$R(x) = (R_1(x), R_2(x), \dots).$$

Then

$$d_\infty(R(x), R(y)) = \sup_n |R_n(x) - R_n(y)| \leq d(x, y).$$

Also, since $R_n(x_n) = x_n$ if $x_n \in D_n$, $R(x) = x$ for each $x \in D$. Therefore, R is a nonexpansive retraction of M onto D and we have the following.

Theorem 4.4 *Let $\{B_\alpha\}$ be a family of closed balls in ℓ_∞ and suppose $D = \bigcap_\alpha B_\alpha \neq \emptyset$. Let M be any metric space which contains (an isometric copy of) $\overset{\alpha}{D}$. Then there exists a nonexpansive retraction R of M onto D .*

Finally, we turn to another elementary property of $\mathbb{R}^1$. If $\{I_\alpha\}$ is a family of bounded closed intervals of $\mathbb{R}^1$ for which $I = \bigcap_\alpha I_\alpha \neq \emptyset$. Then I is itself a closed interval, say $I = [a, b]$, and if $m = \frac{a+b}{2}$ and $d = |a-b|$, then $I = \left[m - \frac{d}{2}, m + \frac{d}{2}\right]$. Thus I coincides with the closed ball $B\left(m; \frac{d}{2}\right)$. This fact extends to ℓ_∞ as follows.

Theorem 4.5 *Let $\{B_\alpha\}$ be a family of closed balls in ℓ_∞ and suppose $D = \bigcap_\alpha B_\alpha \neq \emptyset$. Let $d = \text{diam}(D)$. Then there exists a point $z \in D$ such that*

$$D \subseteq B\left(z; \frac{d}{2}\right).$$

Proof. Again let

$$L_n = \{\mathbf{x} = (x_1, x_2, \dots) \in \ell_\infty : x_i = 0 \text{ for all } i \neq n\}$$

and use the fact that $\{B_\alpha \cap L_n\}$ is a family of closed subintervals in (an isometric copy of) $\mathbb{R}^1$. Thus for each n , $D_n = \bigcap_\alpha (B_\alpha \cap L_n)$ is a nonempty closed interval in L_n . Moreover, if

$$d = \sup\{\text{diam}(D_n) : n = 1, 2, \dots\}$$

and if z_n is the midpoint of the interval D_n , then

$$D_n \subseteq \left[z_n - \frac{d}{2}, z_n + \frac{d}{2} \right].$$

Therefore, if $\mathbf{z} = (z_1, z_2, \dots)$, then $\mathbf{z} \in D$, and if $\mathbf{x} \in D$,

$$d_\infty(\mathbf{z}, \mathbf{x}) = \sup_n |z_n - x_n| \leq \frac{d}{2}.$$

This proves that $D \subseteq B\left(\mathbf{z}; \frac{d}{2}\right)$. ■

Remark 4.1 *All of the preceding arguments extend routinely to wider classes of spaces; for example, to the space $B_\infty[0, 1]$ of all bounded real-valued functions defined on $[0, 1]$. (Again, distance is defined in the usual way: $d_\infty(f, g) = \sup_{t \in [0, 1]} |f(t) - g(t)|$.)*

4.2 Hyperconvexity

The preceding observations should serve to motivate the following more general discussion. Let (X, ρ) and (M, d) be metric spaces. A mapping $f : X \rightarrow M$ is said to be *nonexpansive* if for each $x, y \in X$

$$d(f(x), f(y)) \leq \rho(x, y).$$

A metric space M is said to be *injective* if it has the following extension property: Whenever Y is a subspace of a metric space X and $f : Y \rightarrow M$ is nonexpansive, then f has a nonexpansive extension $\tilde{f} : X \rightarrow M$. This fact has several nice consequences. For example, suppose M is injective and suppose M is a subspace of a metric space X . Then since the identity mapping $I : M \rightarrow M$ is nonexpansive then I can be extended to a nonexpansive mapping $R : X \rightarrow M$. Since R is a retraction of X onto M we have the following.

Theorem 4.6 *An injective metric space is a nonexpansive retract of any metric space in which it is metrically embedded.*

In light of the above it is clear that an injective metric space must be complete because it is a nonexpansive retract (hence a closed subspace) of its own completion.

It might seem that a class of spaces with such nice properties would be small, but, in fact, the opposite is true. We have already seen that the spaces $\mathbb{R}_\infty^n$ and ℓ_∞ provide explicit examples of spaces which satisfy the conclusion of Theorem 4.6. We shall show that these spaces are, in fact, injective.

The *injective envelope* of a metric space M is an injective metric space $\tilde{M}$ which satisfies (i) $\tilde{M}$ contains an isometric copy of M and (ii) no proper injective subspace of $\tilde{M}$ contains an isometric copy of M . Thus $\tilde{M}$ is isometric with some subspace of any injective metric space which contains an isometric copy of M . It is known that *every* metric space has an injective envelope. This well-known fact, due to J. R. Isbell [75], will be discussed in detail in the last section of this chapter.

The injective metric spaces have been given an explicit metric characterization by Aronszajn and Panitchpakdi [3].

Definition 4.1 *A metric space (M, d) is called hyperconvex if for any indexed class of closed balls $B(x_i; r_i)$, $i \in I$, of M which satisfy*

$$d(x_i, x_j) \leq r_i + r_j \quad i, j \in I,$$

it is necessarily the case that $\bigcap_{i \in I} B(x_i; r_i) \neq \emptyset$.

Aronszajn and Panitchpakdi proved that a metric space is injective if and only if it is hyperconvex. This fact, which in light of the discussion of the preceding section is perhaps not too surprising, is taken up in the next section.

The term ‘hyperconvex’ suggests that the definition has something to do with convexity. This is indeed true.

It is easy to see that a hyperconvex metric space M is metrically convex. Indeed if $x, y \in M$ with $x \neq y$ let $r_1 = \alpha d(x, y)$ and $r_2 = (1 - \alpha)d(x, y)$, for any $\alpha \in [0, 1]$. Then $d(x, y) = r_1 + r_2$, so by hyperconvexity $B(x; r_1) \cap B(y; r_2) \neq \emptyset$. Let z be in this intersection. Then we have

$$d(x, z) \leq \alpha d(x, y) \text{ and } d(y, z) \leq (1 - \alpha)d(x, y).$$

In fact, the triangle inequality will imply

$$d(x, z) = \alpha d(x, y) \text{ and } d(y, z) = (1 - \alpha)d(x, y).$$

Now suppose M is a metric space and suppose $\{B(x_i; r_i)\}_{i \in I}$ is a family of closed balls in a metric space each two of which intersect. Then if $z \in B(x_i; r_i) \cap B(x_j; r_j)$ for $i, j \in I$,

$$d(x_i, x_j) \leq d(x_i, z) + d(z, x_j) \leq r_i + r_j.$$

Therefore, if the space is hyperconvex,

$$\bigcap_{i \in I} B(x_i; r_i) \neq \emptyset.$$

In particular, hyperconvex spaces have the *binary ball intersection property* (any family of closed balls, each two of which intersect must have nonempty intersection).

Next suppose M is a complete metric space which has the binary ball intersection property, and suppose M is metrically convex. If $\{B(x_i; r_i)\}_{i \in I}$ is a family of closed balls in M for which

$$d(x_i, x_j) \leq r_i + r_j,$$

then there is a metric segment joining x_i and x_j and clearly some point of this segment must lie in $B(x_i; r_i) \cap B(x_j; r_j)$. Therefore, $\bigcap_{i \in I} B(x_i; r_i) \neq \emptyset$ by the binary ball intersection property.

We conclude, therefore, that for a complete metric space M the following are equivalent.

- (1) M is hyperconvex.
- (2) M is metrically convex and has the binary ball intersection property.

Completeness is not actually necessary for the above. Indeed, in the next section, we will prove that any metric space which satisfies the binary ball intersection property is complete.

At this point we introduce some notation which will be used throughout the remainder of this chapter and in later ones as well.

For a subset A of a metric space (M, d) , set:

$$\begin{aligned} r_x(A) &= \sup\{d(x, y) : y \in A\}, & x \in M; \\ r_M(A) &= \inf\{r_x(A) : x \in M\}; \\ r(A) &= \inf\{r_x(A) : x \in A\}; \\ \text{diam}(A) &= \sup\{d(x, y) : x, y \in A\}; \\ C_M(A) &= \{x \in M : r_x(A) = r(A)\}; \\ C(A) &= \{x \in A : r_x(A) = r(A)\}; \\ \text{cov}(A) &= \bigcap\{B_i : B_i \text{ is a closed ball and } B_i \supseteq A\}. \end{aligned}$$

$r_M(A)$ is called the *radius* of A (relative to M), $\text{diam}(A)$ is called the *diameter* of A , $C_M(A)$ is called the *center* of A (relative to M), $r(A)$ is called the *Chebyshev radius* of A , $C(A)$ is called the *Chebyshev center* of A , and $\text{cov}(A)$ is called the *cover* of A .

4.3 Properties of hyperconvex spaces

We begin with a useful fact.

Proposition 4.4 *Any hyperconvex metric space is complete.*

Proof. Let $\{x_n\}$ be a Cauchy sequence in a hyperconvex space M and for each $n \in \mathbb{N}$ set $\rho_n = \sup\{d(x_n, x_m) : m \geq n\}$. Then for $m \geq n$, $d(x_n, x_m) \leq \rho_n \leq \rho_n + \rho_m$. Thus by hyperconvexity there exists a point z in the intersection

$$\bigcap_{n=1}^{\infty} B(x_n; \rho_n).$$

Since $\lim_{n \rightarrow \infty} \rho_n = 0$, then $\lim_{n \rightarrow \infty} x_n = z$. ■

Remark 4.2 *The above proof suggests that the conclusion is valid if we assume that the collection of balls satisfies a kind of compactness property. Indeed, if M is a metric space such that any family of closed balls has a nonempty intersection provided any finitely many of them have a nonempty intersection, then M is complete.*

The above proposition tells us even more about the structure of hyperconvex spaces. Since a hyperconvex space is metrically convex, each two points of such a space are joined by at least one metric segment. In topological terms this assures that hyperconvex spaces are always pathwise connected.

We now prove a technical lemma.

Lemma 4.1 *Suppose A is a bounded subset of a hyperconvex metric space M . Then:*

1. $\text{cov}(A) = \bigcap \{B(x; r_x(A)) : x \in M\}$.
2. $r_x(\text{cov}(A)) = r_x(A)$, for any $x \in M$.
3. $r_M(\text{cov}(A)) = r_M(A)$.
4. $r_M(A) = \frac{1}{2} \text{diam}(A)$.
5. $\text{diam}(\text{cov}(A)) = \text{diam}(A)$.

Proof. 1. Since $B(x; r_x(A))$ contains A for each $x \in M$ it must be the case that

$$\text{cov}(A) \subseteq \bigcap \{B(x; r_x(A)) : x \in M\}.$$

On the other hand, if $A \subseteq B(x; r)$, then $r_x(A) \leq r$ so $B(x; r_x(A)) \subseteq B(x; r)$. Hence

$$\cap \{B(z; r_z(A)) : z \in M\} \subseteq B(x; r).$$

This clearly implies

$$\text{cov}(A) = \cap \{B(x; r_x(A)) : x \in M\}.$$

2. By 1 $r_x(\text{cov}(A)) = \sup \left\{ d(x, y) : y \in \bigcap_{z \in M} B(z; r_z(A)) \right\}$. In particular, $y \in \text{cov}(A)$ implies $y \in B(x; r_x(A))$, for any $x \in M$. Hence $d(x, y) \leq r_x(A)$. This proves $r_x(\text{cov}(A)) \leq r_x(A)$. The reverse inequality is obvious since $A \subseteq \text{cov}(A)$.

3. This is immediate from the definition of r .

4. Let $\delta = \text{diam}(A)$ and consider the family $\left\{ B\left(a; \frac{\delta}{2}\right) : a \in A \right\}$. If $a, b \in A$ then $d(a, b) \leq \delta = \frac{\delta}{2} + \frac{\delta}{2}$ so by hyperconvexity

$$\bigcap_{a \in A} B\left(a; \frac{\delta}{2}\right) \neq \emptyset.$$

If x is any point in this intersection then $d(x, a) \leq \frac{\delta}{2}$ so $r_x(A) \leq \frac{\delta}{2}$. On the other hand $d(a, b) \leq d(a, z) + d(z, b)$ for any $a, b \in A$ and $z \in M$ so $\delta \leq 2r_z(A)$ from which $\delta \leq 2r_M(A)$. Therefore, $\delta \leq 2r_M(A) \leq 2r_x(A) \leq \delta$, proving $r_M(A) = \frac{\delta}{2}$.

5. Using 3 and 4, $\text{diam}(A) = 2r_M(A) = 2r_M(\text{cov}(A)) = \text{diam}(\text{cov}(A))$. ■

Remark 4.3 Note that we have

$$r_M(A) \leq r(A).$$

In general, we do not have the equality. But if $A = \text{cov}(A)$, that is, A is an intersection of balls in M , then we have

$$r_M(A) = r(A) = \frac{1}{2} \text{diam}(A).$$

We are now ready to prove the following.

Theorem 4.7 (Aronszajn and Panitchpakdi) *A metric space is injective if and only if it is hyperconvex.*

Proof. Assume (M, d) is injective. In order to prove that M is hyperconvex we need only to show that M is metrically convex and has the binary intersection property. Let $x, y \in M$ with $x \neq y$. Let Y be the metric subspace of M consisting of the points $\{x, y\}$ and let (X, ρ) be the metric space consisting of the points $\{x, y, w\}$ (w an arbitrary element) with distance

$$\rho(x, y) = d(x, y); \quad \rho(x, w) = \rho(y, w) = \frac{1}{2}d(x, y).$$

The identity mapping $I : Y \rightarrow M$ is nonexpansive, so by injectivity of M it has a nonexpansive extension $\tilde{I} : X \rightarrow M$. Therefore,

$$d(x, y) \leq d(x, \tilde{I}(w)) + d(y, \tilde{I}(w)) \leq \rho(x, w) + \rho(y, w) = d(x, y).$$

Since

$$d(x, \tilde{I}(w)) = d(\tilde{I}(x), \tilde{I}(w)) \leq \rho(x, w) = \frac{1}{2}d(x, y),$$

and

$$d(y, \tilde{I}(w)) \leq \frac{1}{2}d(x, y),$$

it must be the case that

$$d(x, \tilde{I}(w)) = d(y, \tilde{I}(w)) = \frac{1}{2}d(x, y),$$

and this proves that M is metrically convex.

Now we show that M has the binary ball intersection property. Suppose $\{B_i(x_i; r_i)\}_{i \in I}$ is a family of closed balls of M each two of which have nonempty intersection. Let Y be the subspace of M consisting of the points $\{x_i : i \in I\}$ and let $I_d : Y \rightarrow M$ be the identity mapping. As before, let $X = Y \cup \{w\}$, where w is an arbitrary element (of course, not in Y), and define the distance on X by extending d to

$$d(x_j, w) = \inf\{r : B(x_j; r) \supseteq B(x_i; r_i) \text{ for some } i \in I\},$$

$j = 1, 2, \dots$. To see that this is a metric one only need to check that for $j, k \in I$,

$$d(x_j, x_k) \leq d(x_j, w) + d(x_k, w).$$

This is where the binary ball intersection property enters in. Note that since $B(x_j; r_j) \supseteq B(x_j; r_j)$, $d(x_j, w) \leq r_j$ for each $j \in I$. Also, either $d(x_j, w) = r_j$ or for some $i \neq j$, $B(x_i; r_i) \subseteq B(x_j; d(x_j, w))$. In either case, given $j, k \in I$ there exist $i, n \in I$ such that

$$B(x_i; r_i) \subseteq B(x_j; d(x_j, w)) \text{ and } B(x_n; r_n) \subseteq B(x_k; d(x_k, w)).$$

Since $B(x_i; r_i) \cap B(x_k; r_k) \neq \emptyset$, it must be the case that

$$B(x_j; d(x_j, w)) \cap B(x_k; d(x_k, w)) \neq \emptyset,$$

from which $d(x_j, x_k) \leq d(x_j, w) + d(x_k, w)$.

Finally, since M is injective I_d has a nonexpansive extension $\tilde{I}_d : X \rightarrow M$. Then we have $\tilde{I}_d(w) \in M$ and

$$d(x_j, \tilde{I}_d(w)) \leq d(x_j, w) \leq r_j \text{ for each } j \in I.$$

Therefore, $\tilde{I}_d(w) \in \bigcap_{i \in I} B(x_i; r_i)$.

We now prove the reverse implication. This is a reworking of the proof of Proposition 4.3. Suppose (M, d) is hyperconvex, let Y be a metric space with $T : Y \rightarrow M$ nonexpansive, and suppose Y is a subspace of a metric space X . Consider the family $\mathcal{N}$ defined as follows: $\mathcal{N}$ is the collection of all pairs (T_F, F) where F is a subspace of X for which $Y \subseteq F \subseteq X$ and $T_F : F \rightarrow M$ is nonexpansive with $T_F(y) = T(y)$ for each $y \in Y$.

The family $\mathcal{N}$ can be partially ordered by setting

$$(T_F, F) \prec (T_G, G)$$

if $F \subseteq G$ and $T_G(z) = T_F(z)$ for each $z \in F$. Also, if $\{(T_{F_\alpha}, F_\alpha)\}_{\alpha \in A}$ is a chain in $\mathcal{N}$ relative to this order, take $G = \bigcup_{\alpha \in A} F_\alpha$ and define $T_G : G \rightarrow M$ by taking $T_G(x) = T_{F_\alpha}(x)$ for any α such that $x \in F_\alpha$. Then clearly

$$(T_{F_\alpha}, F_\alpha) \prec (T_G, G) \text{ for each } \alpha \in A,$$

so each chain in $\mathcal{N}$ is bounded above. By Zorn's Lemma $\mathcal{N}$ has a maximal element, say (T_H, H) .

We assert that $H = X$. If not, then there exists $z \in X$ such that $z \notin H$. Now let $H_1 = H \cup \{z\}$. We want to define a nonexpansive extension of T_H on H_1 . Consider the family of balls $\{B(T_H(x); d(x, z))\}_{x \in H}$. Note that for $x_1, x_2 \in H$,

$$d(T_H(x_1), T_H(x_2)) \leq d(x_1, x_2) \leq d(x_1, z) + d(x_2, z).$$

Since M is hyperconvex it follows that

$$\bigcap_{x \in H} B(T_H(x); d(x, z)) \neq \emptyset.$$

Select $w \in \bigcap_{x \in H} B(T_H(x); d(x, z))$ and define $T_{H_1} : H_1 \rightarrow M$ by taking $T_{H_1}(x) = T_H(x)$ if $x \in H$ and setting $T_{H_1}(z) = w$. Since

$$T_{H_1}(z) \in \bigcap_{x \in H} B(T_H(x); d(x, z)),$$

it follows that

$$d(T_{H_1}(x), T_{H_1}(z)) \leq d(x, z) \text{ for each } x \in H.$$

Therefore, T_{H_1} is nonexpansive. This implies $(T_H, H) \prec (T_{H_1}, H_1)$ with $H_1 \neq H$, contradicting the maximality of (T_H, H) . $\blacksquare$

4.4 A fixed point theorem

In view of the intimate relationship which exists between the nonexpansive mappings and hyperconvexity via injectivity, it is perhaps not surprising that hyperconvex spaces should admit an interesting fixed point theory for such mappings.

We begin with a basic definition.

Definition 4.2 *A bounded subset D of a metric space is admissible if $D = \text{cov}(D)$.*

The collection of all admissible subsets of a metric space M will be denoted by $\mathcal{A}(M)$. Note that a set is admissible if and only if it can be written as the intersection of a family of closed balls in M . For this reason, the family $\mathcal{A}(M)$ is closed under arbitrary intersections. The admissible subsets of a hyperconvex space are interesting for another reason.

Proposition 4.5 *Suppose (M, d) is a hyperconvex metric space. Then each set $D \in \mathcal{A}(M)$ is itself hyperconvex.*

Proof. Since D is admissible, we can write $D = \bigcap_{i \in I} B(x_i; r_i)$, $x_i \in M$. Now let $\{B(x_\alpha; r_\alpha)\}_{\alpha \in A}$ be a family of closed balls centered at points $x_\alpha \in D$ for which $d(x_\alpha, x_\beta) \leq r_\alpha + r_\beta$, $\alpha, \beta \in A$. By the hyperconvexity of M ,

$$\bigcap_{\alpha \in A} B(x_\alpha; r_\alpha) \neq \emptyset.$$

Now consider the family of closed balls

$$\{B(x_\alpha; r_\alpha) : \alpha \in A\} \cup \{B(x_i; r_i) : i \in I\}.$$

Then for $\alpha \in A$ and $i, j \in I$,

$$d(x_\alpha, x_i) \leq r_i \leq r_\alpha + r_i$$

and

$$d(x_i, x_j) \leq d(x_i, x_\alpha) + d(x_\alpha, x_j) \leq r_i + r_j,$$

so it again follows from the hyperconvexity of M that

$$\left(\bigcap_{\alpha \in A} B(x_\alpha; r_\alpha) \right) \cap \left(\bigcap_{i \in I} B(x_i; r_i) \right) = \left(\bigcap_{\alpha \in A} B(x_\alpha; r_\alpha) \right) \cap D \neq \emptyset.$$

This proves that D is hyperconvex. ■

Theorem 4.8 *If (M, d) is a bounded hyperconvex metric space and if $T : M \rightarrow M$ is nonexpansive, then the fixed point set $\text{Fix}(T)$ of T in M is nonempty and hyperconvex.*

Proof. We first show that $\text{Fix}(T) \neq \emptyset$. This is another standard Zorn's Lemma approach. Let

$$\mathcal{F} = \{D \in \mathcal{A}(M) : D \neq \emptyset \text{ and } T : D \rightarrow D.\}$$

Then $\mathcal{F} \neq \emptyset$ since $M \in \mathcal{F}$ and $\mathcal{F}$ is partially ordered by reverse set-inclusion. ($A \geq B \Leftrightarrow A \subseteq B$).

First note that if $\mathcal{C}$ is a chain in $\mathcal{F}$ then $\bigcap \mathcal{C} \in \mathcal{A}(M)$. To see that $\mathcal{C} \in \mathcal{F}$ we need only to show that $\bigcap \mathcal{C} \neq \emptyset$. To see that this is true apply the definition of hyperconvexity. First index $\mathcal{C}$, say $\mathcal{C} = \{C_\alpha\}_{\alpha \in A}$. Each set C_α can be written as the intersection of closed balls in M , say $C_\alpha = \bigcap_{i_\alpha \in I_\alpha} B(x_{i_\alpha}; r_{i_\alpha})$. Suppose $C_\alpha \subseteq C_\beta$, and select $z \in C_\alpha$. Then for each $i_\alpha \in I_\alpha, i_\beta \in I_\beta$,

$$d(x_{i_\alpha}, x_{i_\beta}) \leq d(x_{i_\alpha}, z) + d(z, x_{i_\beta}) \leq r_{i_\alpha} + r_{i_\beta}.$$

Therefore by hyperconvexity of M ,

$$\bigcap_{\alpha \in A} C_\alpha = \bigcap_{\alpha \in A} \left(\bigcap_{i_\alpha \in I_\alpha} B(x_{i_\alpha}; r_{i_\alpha}) \right) \neq \emptyset.$$

Since each chain in $\mathcal{F}$ is bounded above (by the intersection of the elements from the chain), $\mathcal{F}$ has a maximal element, which, of course, is a *minimal* element relative to set inclusion. Call this minimal element D . Our strategy is to show that D consists of a single point.

Since

$$T(D) \subseteq \bigcap_{x \in M} B(x; r_x(T(D)))$$

and since $r_x(T(D)) \leq r_x(D)$, it must be the case that

$$\text{cov}(T(D)) \subseteq \bigcap_{x \in M} B(x; r_x(T(D))) \subseteq \bigcap_{x \in M} B(x; r_x(D)) = D.$$

In particular,

$$T(\text{cov}(T(D))) \subseteq T(D) \subseteq \text{cov}(T(D)).$$

This proves that $T : \text{cov}(T(D)) \rightarrow \text{cov}(T(D))$. Therefore, $\text{cov}(T(D)) \in \mathcal{F}$, and, since $\text{cov}(T(D)) \subseteq D$ with D minimal, it must be the case that $D = \text{cov}(T(D))$. Thus

$$D = \bigcap_{x \in M} B(x; r_x(T(D))),$$

which, in turn, implies $r_x(D) \leq r_x(T(D))$ for each $x \in M$; hence

$$r_x(D) = r_x(T(D)), \quad x \in M.$$

Now let $\delta = \text{diam}(D)$ and let

$$C(D) = \bigcap_{x \in D} B\left(x; \frac{\delta}{2}\right).$$

We have already seen that $C(D) \neq \emptyset$ by hyperconvexity $(x, y \in D \Rightarrow d(x, y) \leq \delta = \frac{\delta}{2} + \frac{\delta}{2})$. What is important here is the fact that

$$D_1 = C(D) \cap D \neq \emptyset.$$

To see this we use the fact that D is admissible, say

$$D = \bigcap_{i \in I} B(x_i; r_i)$$

and consider the family

$$\mathcal{M} = \left\{ B(x_i; r_i); i \in I \right\} \cup \left\{ B\left(x; \frac{\delta}{2}\right); x \in D \right\}.$$

We have already seen that $d(x_i, x_j) \leq r_i + r_j$ for $i, j \in I$ and that $d(x, y) \leq \frac{\delta}{2} + \frac{\delta}{2}$ for $x, y \in D$. But also if $i \in I$ and $x \in D$ then by taking $z \in D$ we see that

$$d(x_i, x) \leq d(x_i, z) + d(z, x) \leq r_i + \frac{\delta}{2}.$$

So by hyperconvexity we conclude

$$D_1 = \bigcap \mathcal{M} \neq \emptyset.$$

Now let $z \in D_1$. Since T is nonexpansive, $r_{T(z)}(T(D)) \leq r_z(D)$. However, we have already seen that $r_{T(z)}(T(D)) = r_{T(z)}(D)$, and since $r_z(D) = \frac{\delta}{2}$ we obtain $r_{T(z)}(D) \leq \frac{\delta}{2}$. Since the reverse inequality is obvious, we conclude $r_{T(z)}(D) = \frac{\delta}{2}$ and, since $T(z) \in D$ we have shown $T(z) \in D_1$, that is, $T : D_1 \rightarrow D_1$. Since D_1 is admissible, this proves that $D_1 \in \mathcal{F}$ and so by minimality of D we have $D_1 = D$. But $\text{diam}(D_1) = \frac{\delta}{2}$ and $\text{diam}(D) = \delta$. This can only happen if $\delta = 0$. We conclude, therefore, that D consists of a single point which of course must be a fixed point of T . We have proved the first part of the theorem, namely, that $\text{Fix}(T) \neq \emptyset$.

We now show that $\text{Fix}(T)$ is hyperconvex. Let $\{B(x_i; r_i)\}_{i \in I}$ be a family of closed balls centered at points $x_i \in \text{Fix}(T)$ with the property $d(x_i, x_j) \leq r_i + r_j$. Since M is hyperconvex the set

$$B = \bigcap_{i \in I} B_i(x_i; r_i) \neq \emptyset.$$

Since B is admissible, B is itself hyperconvex. Also, if $z \in B$ then for each $i \in I$,

$$d(T(z), x_i) = d(T(z), T(x_i)) \leq d(z, x_i) \leq r_i.$$

This proves that $T(z) \in B(x_i; r_i)$, that is, $T : B \rightarrow B$. Therefore, we can conclude from what we have shown above that T has a fixed point in B . Thus

$$\left(\bigcap_{i \in I} B_i(x_i; r_i) \right) \cap \text{Fix}(T) \neq \emptyset.$$

This proves that $\text{Fix}(T)$ is hyperconvex. ■

4.5 Intersections of hyperconvex spaces

In general, the intersection of two hyperconvex subspaces of a given hyperconvex space need not be hyperconvex. Easy examples exist even in $\mathbb{R}_\infty^2$. Let H_1 denote the line segment joining $(-1, 0)$ to $(1, 0)$ and let H_2 be the union of the line segment joining $(-1, 0)$ to $(0, 1)$ and the line segment joining $(0, 1)$ to $(1, 0)$. Then both H_1 and H_2 are hyperconvex subsets of the space $\mathbb{R}_\infty^2$ but $H_1 \cap H_2$ consists of two distinct points. On the other hand, the intersection of two admissible subsets of a hyperconvex space is also admissible and again hyperconvex.

There are other situations where the intersection of hyperconvex spaces is again hyperconvex.

Theorem 4.9 *Let (M, d) be a bounded hyperconvex metric space and let $\{H_n\}$ be a descending sequence of nonempty hyperconvex subspaces of M . Then $H = \bigcap_{n=1}^{\infty} H_n$ is nonempty and hyperconvex.*

Proof. We begin by showing that $H \neq \emptyset$, and to do this we introduce a new space.

Let $\mathcal{H} = \prod_{n=1}^{\infty} H_n$ be the metric space consisting of all sequences $x = \{x_n\}$ with $x_i \in H_i$, $i = 1, 2, \dots$, and for $x = \{x_n\}, y = \{y_n\} \in \mathcal{H}$ set

$$d(x, y) = \sup\{d(x_i, y_i) : i = 1, 2, \dots\}.$$

Then $\mathcal{H}$ is bounded and hyperconvex.

Now consider the shift operator $T : \mathcal{H} \rightarrow \mathcal{H}$ defined by

$$T(\{x_n\}) = \{x_{n+1}\}.$$

It is trivial to see that T is nonexpansive, so, by Theorem 4.8, T has at least one fixed point $\{x_n\}$. But this implies

$$x_n = x_{n+1}, \quad n = 1, 2, \dots$$

This, in turn, implies $x_n \equiv x \in \bigcap_{n=1}^{\infty} H_n$.

To prove H is hyperconvex we use what we have just shown. Let $\{x_\alpha\} \subset H$ and $\{r_\alpha\} \subset \mathbb{R}^+$ satisfy

$$d(x_\alpha, x_\beta) \leq r_\alpha + r_\beta.$$

Since $\{x_\alpha\} \subset H_n$ and H_n is hyperconvex, $(\bigcap_\alpha B(x_\alpha; r_\alpha)) \cap H_n$ is a nonempty admissible (hence hyperconvex) subset of H_n for each n . Therefore $\{(\bigcap_\alpha B(x_\alpha; r_\alpha)) \cap H_n\}_{n=1}^{\infty}$ is a descending sequence of nonempty hyperconvex subsets of M , so by what we just proved

$$\bigcap_{n=1}^{\infty} [(\bigcap_\alpha B(x_\alpha; r_\alpha)) \cap H_n] = (\bigcap_\alpha B(x_\alpha; r_\alpha)) \cap H \neq \emptyset.$$

This proves H is hyperconvex. ■

Remark 4.4 *Theorem 4.9 is actually true for arbitrary, rather than countable, descending families. This fact is an indirect consequence of Theorem 5.5 in the next chapter.*

Definition 4.3 *If M is hyperconvex, then a subset $H \subseteq M$ is said to be externally hyperconvex relative to M , if for any collection $\{B(x_\alpha; r_\alpha)\}_{\alpha \in A}$ of closed balls in M (not necessarily centered in H) for which $d(x_\alpha, x_\beta) \leq r_\alpha + r_\beta$ and $\text{dist}(x_\alpha, H) \leq r_\alpha$, it is the case that $H \cap \left(\bigcap_\alpha B(x_\alpha; r_\alpha)\right) \neq \emptyset$.*

Theorem 4.10 *Let M be hyperconvex and let $H \subseteq M$ be externally hyperconvex relative to M . Then $H \cap D$ is hyperconvex for each $D \in \mathcal{A}(M)$.*

Proof. Let $\{B(x_\alpha; r_\alpha)\}$ be a collection of closed balls in M centered in $H \cap D$ for which $d(x_\alpha, x_\beta) \leq r_\alpha + r_\beta$. Suppose $D = \bigcap_{i \in I} B(x_i; r_i)$, and let $z \in H \cap D$. Then $d(z; x_i) \leq r_i$ so $\text{dist}(x_i, H) \leq r_i$. Trivially $\text{dist}(x_\alpha, H) \leq r_\alpha$. Since $d(x_i, x_j) \leq d(x_i, z) + d(x_j, z) \leq r_i + r_j$ and $d(x_i, x_\alpha) \leq r_i \leq r_i + r_\alpha$, the family $\{B(x_\alpha; r_\alpha)\} \cup \{B(x_i; r_i)\}$ satisfies the assumptions of external hyperconvexity for H so

$$(H \cap D) \cap \left(\bigcap_\alpha B(x_\alpha; r_\alpha)\right) = H \cap \left(\bigcap_{i \in I} B(x_i; r_i)\right) \cap \left(\bigcap_\alpha B(x_\alpha; r_\alpha)\right) \neq \emptyset.$$
■

4.6 Approximate fixed points

The fundamental ideas in this section are due to Sine [147]. We begin with some more notation. For any set A in a metric space, let

$$N_\epsilon(A) = \bigcup_{a \in A} B(a; \epsilon).$$

Thus $x \in N_\epsilon(A)$ if and only if there exists $a \in A$ such that $d(x, a) \leq \epsilon$. There is a nice characterization of $N_\epsilon(A)$ if A is an admissible set in a hyperconvex space.

Lemma 4.2 *Suppose M is a hyperconvex metric space and let $A \in \mathcal{A}(M)$, say $A = \bigcap_{i \in I} B(x_i; r_i)$. Then for each $\epsilon > 0$,*

$$N_\epsilon(A) = \bigcap_{i \in I} B(x_i; r_i + \epsilon).$$

Proof. Suppose $y \in N_\epsilon(A)$. Then $d(y, a) \leq \epsilon$ for some $a \in A$. But for each $i \in I$,

$$d(y, x_i) \leq d(a, x_i) + d(a, y) \leq r_i + \epsilon,$$

and this proves $N_\epsilon(A) \subseteq \bigcap_{i \in I} B(x_i; r_i + \epsilon)$. (This implication does not require hyperconvexity.) Now suppose $y \in \bigcap_{i \in I} B(x_i; r_i + \epsilon)$ and let $i \in I$. Hence

$$d(y, x_i) \leq r_i + \epsilon, i \in I.$$

Since A is not empty, we must have

$$d(x_i, x_j) \leq r_i + r_j, i, j \in I.$$

So by hyperconvexity of M

$$A \cap B(y; \epsilon) = \left(\bigcap_{i \in I} B(x_i; r_i) \right) \cap B(y; \epsilon) \neq \emptyset.$$

Consequently, $y \in N_\epsilon(A)$ and the proof is complete. ■

We also need the following:

Lemma 4.3 *Suppose M is a hyperconvex metric space and let $A \in \mathcal{A}(M)$. Then for each $\epsilon > 0$ there is a nonexpansive retraction R of $N_\epsilon(A)$ onto A which has the property $d(x, R(x)) \leq \epsilon$ for each $x \in N_\epsilon(A)$.*

Proof. In particular, assume $A = \bigcap_{i \in I} B(x_i; r_i)$. By Lemma 4.2 we know that $N_\varepsilon(A) \in \mathcal{A}(M)$ and so $N_\varepsilon(A)$ is itself hyperconvex. Consider the family $\mathcal{F}$ of all pairs (D, R_D) , where

$$A \subseteq D \subseteq N_\varepsilon(A)$$

and R_D is a nonexpansive retraction of D onto A for which $d(x, R_D(x)) \leq \varepsilon$ for each $x \in D$. Note that $(A, I_d) \in \mathcal{F}$ so $\mathcal{F} \neq \emptyset$. If one orders $\mathcal{F}$ in the usual way $((D, R_D) \preceq (H, R_H) \text{ if and only if } D \subseteq H \text{ and } R_H \text{ is an extension of } R_D)$ then each chain in $(\mathcal{F}, \preceq)$ is bounded above, so by Zorn's Lemma $\mathcal{F}$ has a maximal element which we again denote (D, R_D) . We need to show that $D = N_\varepsilon(A)$.

Suppose there exists $x \in N_\varepsilon(A)$ such that $x \notin D$, and consider the set

$$C = \left(\bigcap_{w \in D} B(R_D(w); d(w, x)) \right) \cap \left(\bigcap_{i \in I} B(x_i; r_i) \right) \cap B(x; \varepsilon).$$

First we show that $C \neq \emptyset$, and in order to do this we need only to show that each two members of the family of balls used in defining C intersect. (This is because M is hyperconvex and thus has the binary ball intersection property.)

First if $w_1, w_2 \in D$ then

$$d(R_D(w_1), R_D(w_2)) \leq d(w_1, w_2) \leq d(w_1, x) + d(w_2, x).$$

This proves that

$$B(R_D(w_1); d(w_1, x)) \cap B(R_D(w_2); d(w_2, x)) \neq \emptyset,$$

so each two members of the first family intersect. Also, for each $w \in D$, $R_D(w) \in A = \bigcap_{i \in I} B(x_i; r_i)$ so each ball in the first family intersects each ball in the second family. Since

$$x \in N_\varepsilon(A) = \bigcap_{i \in I} B(x_i; r_i + \varepsilon)$$

we know that $B(x; \varepsilon) \cap B(x_i; r_i) \neq \emptyset$ for each $i \in I$. Finally, if $w \in D$, then

$$d(R_D(w), x) \leq d(R_D(w), w) + d(w, x) \leq \varepsilon + d(w, x)$$

and this proves that

$$B(R_D(w); d(w, x)) \cap B(x; \varepsilon) \neq \emptyset.$$

We conclude therefore that $A \supseteq C \neq \emptyset$.

Now let $u \in C$ and define $R' : D \cup \{x\} \rightarrow A$ by setting $R'(w) = R_D(w)$ if $w \in D$ and $R'(x) = u$. Then for $w \in D$,

$$d(R'(x), R'(w)) = d(u, R_D(w)) \leq d(u, w)$$

so R' is nonexpansive. Also $d(R'(x), x) = d(u, x) \leq \varepsilon$. With this we conclude that the pair $(D \cup \{x\}, R')$ contradicts the maximality of (D, R_D) in $(\mathcal{F}, \preceq)$. Therefore, $D = N_\varepsilon(A)$ and the proof is complete. ■

For a mapping $T : M \rightarrow M$ we use $F_\varepsilon(T)$ to denote the set of ε -fixed points of T ; thus

$$F_\varepsilon(T) = \{x \in M : d(x, T(x)) \leq \varepsilon\}.$$

Theorem 4.11 *Suppose M is a hyperconvex metric space and suppose $T : M \rightarrow M$ is nonexpansive. Then for any $\varepsilon > 0$, the set $F_\varepsilon(T)$ is also hyperconvex.*

Proof. Clearly we may suppose $F_\varepsilon(T) \neq \emptyset$. For each i in some index set I , let $x_i \in F_\varepsilon(T)$, and let $r_i \geq 0$ satisfy

$$d(x_i, x_j) \leq r_i + r_j.$$

We need to show that

$$\left(\bigcap_{i \in I} B(x_i; r_i) \right) \cap F_\varepsilon(T) \neq \emptyset.$$

We know that $J = \bigcap_{i \in I} B(x_i; r_i) \neq \emptyset$ by hyperconvexity of M . Also if $x \in J$ then for each $i \in I$,

$$d(x_i, T(x)) \leq d(T(x), T(x_i)) + d(T(x_i), x_i) \leq d(x, x_i) + \varepsilon \leq r_i + \varepsilon.$$

This proves that $T(x) \in N_\varepsilon(J)$. Now, by Lemma 4.3, there is a nonexpansive retraction R of $N_\varepsilon(J)$ onto J for which $d(R(x), x) \leq \varepsilon$ for each $x \in N_\varepsilon(J)$. Also since $R \circ T$ is a nonexpansive mapping of J into J it must have a fixed point by Theorem 4.8. Suppose $R \circ T(x_0) = x_0$ for $x_0 \in J$. Then

$$d(x_0, T(x_0)) = d(R \circ T(x_0), T(x_0)) \leq \varepsilon.$$

and the proof is complete. ■

4.7 Isbell's hyperconvex hull

As we mentioned before, J. R. Isbell [75] showed that every metric space has an injective envelope. The *injective envelope* of a metric space M is an injective metric space $\bar{M}$ that contains an isometric copy of M and which is isometric with a subspace of any hyperconvex metric space which contains an isometric copy of M . In this section, we will discuss Isbell's ideas. (In light of Theorem 4.7 we may use the terms 'injective' and 'hyperconvex' interchangeably.)

Let M be a metric space. For any $x \in M$, define the positive real-valued function $f_x : M \rightarrow [0, \infty)$ by

$$f_x(y) = d(x, y).$$

Let us discuss some of the properties satisfied by these functions.

1. Using the triangle inequality, we get

$$d(x, y) \leq f_a(x) + f_a(y)$$

and

$$f_a(x) \leq d(x, y) + f_a(y)$$

for any $x, y, a \in M$.

2. Let $f : M \rightarrow [0, \infty)$ be such that

$$d(x, y) \leq f(x) + f(y), \text{ for any } x, y \in M,$$

and for some $a \in M$, assume that $f(x) \leq f_a(x)$, for any $x \in M$. In this case, we have $f = f_a$. Indeed, first we have $f(a) \leq f_a(a) = 0$, which implies $f(a) = 0$. Using the above inequality, we get

$$f_a(x) = d(x, a) \leq f(x) + f(a) = f(x)$$

for any $x \in M$. Combined with the assumptions on $f(x)$, we get $f(x) = f_a(x)$, for any $x \in M$. This is a minimality property for the pointwise order.

Using the above properties, Isbell introduced what he called the *injective envelope* of A , denoted $\epsilon(A)$, for any subset A of M . The set $\epsilon(A)$ is the set of all *extremal functions* defined on A , where $f : A \rightarrow [0, \infty)$ is said to be *extremal* if f satisfies

$$d(x, y) \leq f(x) + f(y) \text{ for all } x, y \text{ in } A$$

and, moreover, f is pointwise minimal with respect to this property, that is, if $g : A \rightarrow [0, \infty)$ satisfies

$$d(x, y) \leq g(x) + g(y) \text{ for all } x, y \text{ in } A$$

and $g(x) \leq f(x)$ for all $x \in A$, then it follows that $f = g$. In particular, we have $f_a \in \epsilon(A)$ for any $a \in A$. Consider the mapping $e : A \rightarrow \epsilon(A)$, defined by

$$e(a) = f_a \text{ for all } a \in A.$$

The mapping e is an isometry. In other words,

$$d(e(a), e(b)) = \sup_{x \in A} |f_a(x) - f_b(x)| = \sup_{x \in A} |d(a, x) - d(b, x)| = d(a, b).$$

So, A and $e(A)$ are isometric spaces and we may identify A and $e(A)$ via $a \leftrightarrow e(a)$. Before we give some detailed properties of extremal functions, we will need the following lemma.

Lemma 4.4 *Let A be a nonempty subset of M . Let $r : A \rightarrow [0, \infty)$ be such that*

$$d(x, y) \leq r(x) + r(y)$$

for any $x, y \in A$. Then there exists $R : M \rightarrow [0, \infty)$ which extends r such that

$$d(x, y) \leq R(x) + R(y)$$

for any $x, y \in M$. Moreover, there exists an extremal function f defined on M such that $f(x) \leq R(x)$ for any $x \in M$.

Proof. This is left as an exercise at the end of the chapter.

Let us give some properties of extremal functions.

Proposition 4.6 *The following statements are true.*

1. *If $f \in \epsilon(A)$, then f satisfies*

$$f(x) \leq d(x, y) + f(y) \text{ for all } x, y \text{ in } A.$$

Moreover, we have

$$f(x) = \sup_{y \in A} |f(y) - f_x(y)| = d(f, e(x)).$$

2. *For any $f \in \epsilon(A)$, $\delta > 0$, and $x \in A$, there exists $y \in A$ such that*

$$f(x) + f(y) < d(x, y) + \delta.$$

3. *If A is compact, then $\epsilon(A)$ is compact.*

4. *If s is an extremal function on the metric space $\epsilon(A)$, then $s \circ e$ is extremal on A .*

Proof

1. Assume not. Then there exist $x_0, y_0 \in A$, such that

$$d(x_0, y_0) + f(y_0) < f(x_0).$$

Set

$$g(x) = \begin{cases} f(x) & \text{if } x \neq x_0 \\ d(x_0, y_0) + f(y_0) & \text{if } x = x_0. \end{cases}$$

It is clear that we have $g(x) \leq f(x)$, for any $x \in A$. In particular, we have $g(x_0) < f(x_0)$. Let us show that for any $x, y \in A$, we have

$$d(x, y) \leq g(x) + g(y).$$

If both x and y are different from x_0 , we use the properties of f . So we can assume $x = x_0$ and $y \neq x_0$. We have

$$d(x, y) = d(x_0, y) \leq d(x_0, y_0) + d(y_0, y) \leq d(x_0, y_0) + f(y_0) + f(y),$$

which obviously implies

$$d(x_0, y) \leq g(x_0) + g(y).$$

The minimality behavior of f gives us $f = g$, which is a contradiction. Combining this inequality with the fundamental one (for extremal functions), we get

$$|f(y) - f_x(y)| \leq f(x)$$

for any $y \in A$. The equality holds for $y = x$ which clearly implies

$$f(x) = \sup_{y \in A} |f(y) - f_x(y)| = d(f, e(x)).$$

2. Assume not. Then there exist $x \in A$ and $\delta > 0$ (we may take it less than $f(x)$) such that for any $y \in A$, we have

$$d(x, y) + \delta \leq f(x) + f(y).$$

Set

$$h(z) = \begin{cases} f(z) & \text{if } z \neq x \\ f(x) - \delta & \text{if } z = x. \end{cases}$$

It is easy to check that

$$d(y, z) \leq h(y) + h(z)$$

for any $y, z \in A$. Since $h \leq f$ and $h(x) < f(x)$, we get a contradiction with the minimality of f . This completes the proof of property 2.

3. From the property 1, we get

$$|f(x) - f(y)| \leq d(x, y)$$

for any $x, y \in A$. This implies $\epsilon(A) \subset Lip_1(A)$, where $Lip_1(A)$ is the space of all lipschitzian real-valued functions with Lipschitz constant less than or equal 1. Hence, all extremal functions are equicontinuous. Also, it is quite easy to show that pointwise-limit of extremal functions is an extremal function. Since A is compact, the Arzelà-Ascoli theorem (see Chapter 6, Exercise 6.6) implies that $\epsilon(A)$ is compact.

4. Let s be an extremal function on the metric space $\epsilon(A)$. Note that for any $x, y \in A$, we have

$$d(x, y) = d(f_x, f_y) \leq s(f_x) + s(f_y) = s \circ e(x) + s \circ e(y).$$

Assume that $s \circ e$ is not an extremal function on A . Then there exists an extremal function $h \in \epsilon(A)$ such that $h(x) \leq s \circ e(x)$, for any $x \in A$, and the inequality is strict at some point x_0 . Define the function

$$t(f) = \begin{cases} s(f) & \text{if } f \neq e(x_0) \\ h(x_0) & \text{if } f = e(x_0). \end{cases}$$

Let us show that t satisfies the inequality

$$d(f, g) \leq t(f) + t(g)$$

for any $f, g \in \epsilon(A)$. Since t and s coincide almost everywhere and s is an extremal function, then we only need to prove the above inequality for $g = e(x_0)$ and $f \neq e(x_0)$, that is,

$$d(f, e(x_0)) \leq t(f) + t(e(x_0)).$$

For any $\delta > 0$, there exists $y \in A$ such that $f(x_0) + f(y) < d(x_0, y) + \delta$. If $y = x_0$, then we must have $f(x_0) \leq 1/2\delta$. Hence

$$d(f, e(x_0)) = f(x_0) \leq \frac{1}{2}\delta + t(f) + t(e(x_0)).$$

On the other hand, if $y \neq x_0$ and $f \neq e(x_0)$, then

$$d(f, e(x_0)) + f(y) - \delta = f(x_0) + f(y) - \delta < d(x_0, y)$$

and

$$d(x_0, y) \leq h(x_0) + h(y) \leq h(x_0) + s \circ e(y) = t(e(x_0)) + t(e(y)).$$

Since s is an extremal function, then

$$t(e(y)) = s(e(y)) \leq s(f) + d(f, e(y)) = t(f) + f(y),$$

where we used the fact that f is an extremal function to get $d(f, e(y)) = f(y)$. So we have

$$d(f, e(x_0)) + f(y) - \delta < t(e(x_0)) + t(e(y))$$

and

$$t(e(y)) \leq t(f) + f(y).$$

Adding the above two inequalities, we get

$$d(f, e(x_0)) + f(y) - \delta + t(e(y)) < t(e(x_0)) + t(e(y)) + t(f) + f(y),$$

which leads to

$$d(f, e(x_0)) - \delta < t(e(x_0)) + t(f).$$

Since δ is arbitrary, we get the desired inequality

$$d(f, e(x_0)) \leq t(e(x_0)) + t(f).$$

The link between Isbell's ideas and hyperconvexity is given in the following proposition.

Proposition 4.7 *The following statements are true.*

1. $\epsilon(A)$ is hyperconvex.
2. $\epsilon(A)$ is an injective envelope of A , that is, no proper subset of $\epsilon(A)$ which contains A (metrically) is hyperconvex. Moreover, any hyperconvex metric space H which contains A metrically and is minimal (i.e., any proper subset of H which contains A is not hyperconvex), is isometric to $\epsilon(A)$.

Proof

1. In order to prove that $\epsilon(A)$ is hyperconvex, let $\{f_\alpha\}$ be a family of points in $\epsilon(A)$ and $\{r_\alpha\}$ be a family of positive numbers such that

$$d(f_\alpha, f_\beta) \leq r_\alpha + r_\beta$$

for any α and β . Define the mapping $r : \{f_\alpha\} \rightarrow [0, \infty)$ by $r(f_\alpha) = r_\alpha$. By Lemma 4.4, we extend r to the entire set $\epsilon(A)$ such that

$$d(f, g) \leq r(f) + r(g)$$

for any $f, g \in \epsilon(A)$. Using the same Lemma 4.4, there exists h an extremal function on $\epsilon(A)$ such that $h \leq r$. Using property 4, we know that $h \circ e$ is an extremal function on A , that is, $h \circ e \in \epsilon(A)$. It is easy to see that

$$h \circ e \in \bigcap_{f \in \epsilon(A)} B(f; r(f)) \subset \bigcap_{\alpha} B(f_\alpha; r_\alpha).$$

Indeed, we have

$$h \circ e(x) - f(x) = h \circ e(x) - d(f, e(x)) \leq h(f) \leq r(f)$$

and

$$f(x) - h \circ e(x) = d(f, e(x)) - h \circ e(x) \leq h(f) \leq r(f)$$

for any $x \in A$. Hence

$$d(f, h \circ e) \leq r(f)$$

for any $f \in \epsilon(A)$. The proof of our claim is therefore complete.

2. Recall that A and $e(A)$ are isometric, and let H be a subset of $\epsilon(A)$ such that $e(A) \subset H$. Assume that H is hyperconvex. Since $\epsilon(A)$ is hyperconvex, there exists a nonexpansive retraction $R : \epsilon(A) \rightarrow H$. Let $f \in \epsilon(A)$. We have

$$d(R(f), e(x)) = R(f)(x) \leq d(f, e(x)) = f(x)$$

for any $x \in A$. Since f is an extremal function, we must have $R(f) = f$. This clearly implies that $H = \epsilon(A)$. So no proper subset of $\epsilon(A)$ which contains $e(A)$ is hyperconvex. On the other hand, let $\mathcal{H}$ be a hyperconvex metric space which contains A metrically such that no proper subset of $\mathcal{H}$ which contains A metrically is hyperconvex. Since $\mathcal{H}$ contains A metrically, there exists an isometry $i : A \rightarrow \mathcal{H}$. Using the fundamental extension property of hyperconvex metric spaces, there exists $R_1 : \epsilon(A) \rightarrow \mathcal{H}$ a nonexpansive mapping which extends the mapping i . Since the mapping $e \circ i^{-1} : i(A) \rightarrow e(A) \subset \epsilon(A)$, there exists $R_2 : \mathcal{H} \rightarrow \epsilon(A)$ a nonexpansive mapping which extends the mapping $e \circ i^{-1}$. The composition $R_2 \circ R_1 : \epsilon(A) \rightarrow \epsilon(A)$ is nonexpansive and extends the identity mapping on $e(A)$. The argument described above forces this mapping to be the identity mapping on $\epsilon(A)$. In other words, the two maps R_1 and R_2 are inverse of each other. Since both are nonexpansive, then they are isometries. The proof of property 2 is complete.

So one conclusion from the previous investigation is that if M is any metric space, there is a hyperconvex space, $h(M)$, such that M embeds isometrically in $h(M)$ and the embedding is contained in no hyperconvex proper subset of $h(M)$. The hyperconvex space $h(M)$ is called the *hyperconvex hull* or *injective envelope* of M . Moreover, any two hyperconvex hulls of M are isometric. Whenever M is a subset of a hyperconvex space H , a hyperconvex hull of M , $h(M)$ can be found so that $M \subset h(M) \subset H$ (due to Theorem 5.5 in the next chapter). Thus $h(M)$ is not unique.

Exercises

Exercise 4.1 Show that the space $\mathcal{H}$ in Theorem 4.9 is itself hyperconvex.

Exercise 4.2 Show that if M is hyperconvex and if $D \in \mathcal{A}(M)$, then D is externally hyperconvex relative to M . (See Definition 4.3.)

Exercise 4.3 Verify the last two statements in Remark 4.3.

Exercise 4.4 Prove the statements of Lemma 4.4.

Hint: Use the proof of Theorem 4.7.

Exercise 4.5 Suppose (M, d) is hyperconvex. Show that the metric space (N, ρ) is hyperconvex if and only if the space $(M \times N)_\infty$ is hyperconvex.

Exercise 4.6 Suppose (M, d) is hyperconvex. Show that the space of all nonempty closed subsets of M endowed with the Hausdorff metric is also hyperconvex.

Exercise 4.7 Let H be a hyperconvex metric space and suppose A is an externally hyperconvex subset of H (see Definition 4.3). Show that $N_\varepsilon(A)$ is externally hyperconvex (in H) for each $\varepsilon > 0$.

Exercise 4.8 Let H be a bounded hyperconvex metric space and suppose $\{H_\alpha\}$ is a descending chain of nonempty externally hyperconvex subsets of H . Show that $\bigcap_\alpha H_\alpha$ is nonempty and externally hyperconvex in H .

Exercise 4.9 Let C be the bounded closed convex subset of l_∞ defined by

$$C = \{(x_n); x_n \in [0, 1] \text{ and } \lim_{n \rightarrow \infty} x_n = 0\}.$$

Find a nonexpansive mapping $T : C \rightarrow C$ with no fixed point. This will prove that the conclusion of Theorem 4.8 does not extend to any bounded closed convex subset of l_∞ .

Exercise 4.10 In this exercise, we will show that the conclusion of Theorem 4.8 does not extend to unbounded hyperconvex metric spaces. Consider the mapping $T : l_\infty \rightarrow l_\infty$ defined by

$$T(x_1, x_2, \dots) = (1 - \lambda((x_n)), x_1, x_2, \dots),$$

where λ is a Banach limit. Check that T is an isometry with bounded orbits, and is fixed point free.

Exercise 4.11 Let (M, d) be a metric space and let $\mathcal{B}(M)$ denote the collection of nonempty, bounded subsets of M .

- (i) The Kuratowski measure of noncompactness $\alpha : \mathcal{B}(M) \rightarrow [0, \infty)$ is defined by

$$\alpha(A) = \inf \{ \varepsilon > 0; A \subset \bigcup_{i=1}^{i=n} A_i \text{ with } A_i \in \mathcal{B}(M) \text{ and } \text{diam}(A_i) \leq \varepsilon \}.$$

- (ii) The Hausdorff (or ball) measure of noncompactness $\chi : \mathcal{B}(M) \rightarrow [0, \infty)$ is defined by

$$\chi(A) = \inf \{ r > 0; A \subset \bigcup_{i=1}^{i=N} B(x_i, r) \text{ with } x_i \in M \}.$$

Show that if H is a hyperconvex metric space and A a bounded subset of H , then we have

$$\alpha(A) = 2\chi(A).$$

Exercise 4.12 Let M be any bounded metric space and $\epsilon(M)$ its hyperconvex hull. Show that

$$\chi(\epsilon(M)) = \chi(M).$$

Hint: Consider the space $\lambda_{[a,b]}(M)$ of Lipschitzian real-valued functions defined on M with Lipschitz constant less than λ with values in the interval $[a, b]$. Show that

$$\alpha(\lambda_{[a,b]}(M)) \leq 2\lambda\chi(M).$$

Exercise 4.13 (A Selection Theorem) Let H be hyperconvex, and let $T^* : \rightarrow \mathcal{E}(H)$, where $\mathcal{E}(H)$ denotes the family of all nonempty bounded externally hyperconvex subsets of H . Show that there exists a mapping $T : H \rightarrow H$ for which $T(x) \in T^*(x)$ for each $x \in H$ and for which

$$d(T(x), T(y)) \leq d_H(T^*(x), T^*(y))$$

for each $x, y \in H$. (Recall that d_H is the Hausdorff distance and $\mathcal{E}(H)$ is the family of externally hyperconvex subsets of H .)

Hint: Use the ideas developed in the proof of Theorem 4.7.

Exercise 4.14 Let H be a bounded hyperconvex set, and for $\lambda > 0$, let $\mathcal{F}_\lambda$ denote the family of all λ -lipschitzian functions of H into H . Show that $\mathcal{F}_\lambda$ is hyperconvex (relative to the metric $\rho(f, g) = \sup \{ d(f(u), g(u)) : u, v \in H \}$).

Chapter 5

“Normal” Structures in Metric Spaces

5.1 A fixed point theorem

The behavior of the collection $\mathcal{A}(M)$ of admissible subsets of a hyperconvex metric space M was central to the proof of fixed point theorem for nonexpansive mappings in the previous chapter. Two facts about $\mathcal{A}(M)$ were essential to the proof. (1) The intersection of any descending chain of nonempty sets in $\mathcal{A}(M)$ is nonempty and also in $\mathcal{A}(M)$. (2) For any $D \in \mathcal{A}(M)$, $r(D) = \frac{1}{2} \text{diam}(D)$. The principal observation of this section is the fact that if (1) is assumed to hold for a bounded metric space M , then an assumption much weaker than (2) is sufficient to assure that every nonexpansive mapping $T : M \rightarrow M$ has at least one fixed point. In Chapter 8 we discuss how this fact is frequently realized in Banach spaces.

Let (M, d) be a bounded metric space, and let $\mathcal{A}(M)$ denote the family of all admissible sets in M . (Thus $D \in \mathcal{A}(M) \Leftrightarrow D = \cap \{B : B \text{ is a closed ball in } M \text{ which contains } D\}$.) We now follow the definitions in the hyperconvex case:

$$r_x(D) = \sup\{d(x, y) : y \in D\}, \quad x \in M,$$

and

$$r(D) = \inf\{r_x(D) : x \in D\}.$$

Also we use $r_M(D)$ to denote the number:

$$r_M(D) = \inf\{r_x(D) : x \in M\}.$$

Thus, in general, we have

$$\frac{1}{2} \text{diam}(D) \leq r_M(D) \leq r(D) \leq \text{diam}(D).$$

We have seen that if M is hyperconvex $r(D) = r_M(D) = \frac{1}{2} \text{diam}(D)$.

As before, let

$$C(D) = \{x \in D : r_x(D) = r(D)\}$$

and

$$C_M(D) = \{x \in M : r_x(D) = r_M(D)\}.$$

Also, for $A \subseteq M$ we denote

$$\begin{aligned} \text{cov}(A) &= \cap \{B : B \text{ is a closed ball and } B \supseteq A\} \\ &= \cap \{B : B \in \mathcal{A}(M) \text{ and } B \supseteq A\}. \end{aligned}$$

Definition 5.1 $\mathcal{A}(M)$ is said to be compact if every descending chain of nonempty members of $\mathcal{A}(M)$ has nonempty intersection.

It was noted in the previous chapter that the binary intersection property for closed balls implies completeness. Compactness of $\mathcal{A}(M)$ leads to the same conclusion.

Proposition 5.1 If M is a metric space for which $\mathcal{A}(M)$ is compact, then M is complete.

Proof. Let $\{x_n\}$ be a Cauchy sequence in M and let

$$r_n = \sup\{d(x_i, x_j) : i, j \geq n\}.$$

Since $\{x_n\}$ is Cauchy, $\lim_{n \rightarrow \infty} r_n = 0$. Also if F is a finite subset of $\mathbb{N}$ (the natural numbers), then

$$A_F = \cap \{B(x_i; r_i) : i \in F\} \neq \emptyset$$

since $x_{\sup F} \in A_F$. Compactness of $\mathcal{A}(M)$ implies

$$\cap \{B(x_i; r_i) : i \in \mathbb{N}\} \neq \emptyset.$$

Since $\lim_{n \rightarrow \infty} r_n = 0$, this intersection must be a single point which is the limit of the sequence $\{x_n\}$. ■

Definition 5.2 $\mathcal{A}(M)$ is said to be normal (or have normal structure) if $r(D) < \text{diam}(D)$ whenever $D \in \mathcal{A}(M)$ and $\text{diam}(D) > 0$.

The assumption that $\mathcal{A}(M)$ is normal is equivalent to the following: If $D \in \mathcal{A}(M)$ consists of more than one point then there is a number $r < \text{diam}(D)$ and a point $z \in D$ such that $D \subseteq B(z; r)$.

At first the Definitions 5.1 and 5.2 may seem esoteric. However, we will see later that both hold in many Banach spaces. In fact, a quick examination of convex sets in $\mathbb{R}^2$ shows that the normal structure assumption is typical.

Theorem 5.1 *Suppose (M, d) is a (nonempty) bounded metric space for which $\mathcal{A}(M)$ is compact and normal. Then every nonexpansive $T : M \rightarrow M$ has at least one fixed point.*

Proof. Compactness of $\mathcal{A}(M)$ and an application of Zorn's Lemma as in the proof of Theorem 4.8 of Chapter 4 yields the existence of a set $D \in \mathcal{A}(M)$ which is minimal with respect to being nonempty and mapped into itself by T . Also, $\text{cov}(T(D)) \subseteq D$ and $T : \text{cov}(T(D)) \rightarrow \text{cov}(T(D))$, so minimality of D implies

$$D = \text{cov}(T(D)).$$

Now assume $\text{diam}(D) > 0$ and choose r so that $r(D) < r < \text{diam}(D)$. It follows that the set

$$C = \{x \in D : D \subseteq B(x; r)\} \neq \emptyset.$$

Also one can quickly check that

$$C = \left(\bigcap_{x \in D} B(x; r) \right) \cap D.$$

This proves that $C \in \mathcal{A}(M)$. Now let $z \in C$. Then if $x \in D$

$$d(T(z), T(x)) \leq d(x, z) \leq r.$$

Therefore, $T(x) \in B(T(z); r)$ for every $x \in D$; hence $T(D) \subseteq B(T(z); r)$ from which $\text{cov}(T(D)) \subseteq B(T(z); r)$. But $D = \text{cov}(T(D))$, so $D \subseteq B(T(z); r)$. This proves that $T(z) \in C$. Hence $T : C \rightarrow C$. However if $z, w \in C$ then $d(z, w) \leq r$, so $\text{diam}(C) \leq r < \text{diam}(D)$. This proves that C is a proper subset of D . Since $C \in \mathcal{A}(M)$ and $T : C \rightarrow C$ this contradicts the minimality of D . We thus conclude that $\text{diam}(D) = 0$, so D consists of a single point which must be a fixed point of T . ■

5.2 Structure of the fixed point set

The development of this section will lead to the fact that Theorem 1.1 extends easily to finite commutative families of nonexpansive mappings (and with somewhat more effort to arbitrary commutative families).

The following is the key to the structure of the fixed point sets of nonexpansive mappings (and to the existence of common fixed points for commuting families of such mappings). A subset A of M is said to be a *1-local retract* of M if for each family $\{B_i\}_{i \in I}$ of closed balls centered at A for which

$$\bigcap_{i \in I} B_i \neq \emptyset$$

it is the case that $A \cap \left(\bigcap_{i \in I} B_i \right) \neq \emptyset$. (It is easy to check that every nonexpansive retract of M is a 1-local retract of M .)

We now describe several fundamental properties of 1-local retracts. Most of these results are taken from [90]. We begin with a direct consequence of Definition 5.1.

Proposition 5.2 *If M is a metric space for which $\mathcal{A}(M)$ is compact and if $\{A_n\}$ is a descending sequence of sets in $\mathcal{A}(M)$ for which $\lim_{n \rightarrow \infty} \text{diam}(A_n) = 0$, then $\bigcap_{n=1}^{\infty} A_n = \{z\}$.*

The following technical proposition collects several other properties needed in the sequel. These are analogous to ones already established in the hyperconvex case.

Proposition 5.3 *Let M be a metric space and let A be a nonempty subset of M . Then:*

- (1) $\text{cov}(A) = \bigcap \{B(x; r_x(A)) : x \in A\}$;
- (2) $r_x(A) = r_x(\text{cov}(A))$ for every $x \in M$;
- (3) $r(\text{cov}(A)) \leq r(A)$;
- (4) $\text{diam}(\text{cov}(A)) = \text{diam}(A)$.

Proof.

1. This is immediate since $B(x; r_x(A))$ is the smallest ball centered at x which contains A .

2. Since $A \subseteq \text{cov}(A)$ it follows from the definition of r_x that

$$r_x(A) \leq r_x(\text{cov}(A)).$$

On the other hand, (1) implies $r_x(\text{cov}(A)) \leq r_x(A)$.

3. This is immediate from the definition of r and (2).

4. It obviously suffices to show that $\text{diam}(\text{cov}(A)) \leq \text{diam}(A)$. Let $z \in \text{cov}(A)$. Then $z \in B(x; r_x(A))$ for each $x \in M$. In particular, $d(x, z) \leq r_x(A) \leq \text{diam}(A)$ for each $x \in A$; thus $A \subseteq B(z; \text{diam}(A))$. Hence

$$\text{cov}(A) \subseteq B(z; \text{diam}(A)).$$

It follows that

$$\text{diam}(\text{cov}(A)) \leq \text{diam}(A).$$

■

The next proposition, which shows that in the case of 1-local retracts normality is a hereditary property, is fundamental to what follows.

Proposition 5.4 *Let M be a metric space and suppose $\mathcal{A}(M)$ is compact and normal. Suppose N is a given subset of M which is a 1-local retract of M . Then $\mathcal{A}(N)$ is compact and normal.*

The key to the proof of the above is the following lemma.

Lemma 5.1 *Under the hypothesis of Proposition 5.4, $r(\text{cov}(A)) = r(A)$ for each $A \in \mathcal{A}(N)$.*

Proof. Clearly we may suppose $\text{diam}(A) > 0$. Since Proposition 5.3 implies $r(\text{cov}(A)) \leq r(A)$ we only need to show $r(A) \leq r(\text{cov}(A))$. By assumption $A \in \mathcal{A}(N)$, so A is of the form

$$A = N \cap \left(\bigcap \{B(x_i; r_i) : x_i \in N\} \right).$$

Also,

$$\text{cov}(A) \subseteq \bigcap \{B(x_i; r_i) : x_i \in N\}.$$

Choose $z \in \text{cov}(A)$ and let $r = r_z$. Then

$$z \in S = \left(\bigcap_{x \in A} B(x; r) \right) \cap \left(\bigcap \{B(x_i; r_i) : x_i \in N\} \right).$$

S is a nonempty set belonging to $\mathcal{A}(M)$. Since N is a 1-local retract of M and S is the intersection of balls centered in N , $S \cap N \neq \emptyset$. Let $w \in S \cap N$. Then

$$w \in \left(\bigcap \{B(x_i; r_i) : x_i \in N\} \right) \cap N,$$

that is, $w \in A$. On the other hand, $w \in \bigcap \{B(x; r) : x \in A\}$, so $r_w(A) \leq r$. It follows that

$$r(A) \leq r \leq r_z(\text{cov}(A)).$$

Since z was an arbitrary element of $\text{cov}(A)$ the proof is complete. $\blacksquare$

Proof of Proposition 5.4. The definition of a 1-local retract assures that $\mathcal{A}(N)$ is compact. To see that $\mathcal{A}(N)$ is normal, let $A \in \mathcal{A}(N)$ and recall that $\text{diam}(\text{cov}(A)) = \text{diam}(A)$ by Proposition 5.3. By Lemma 5.1, $r(\text{cov}(A)) = r(A)$. Since $\mathcal{A}(M)$ is normal, $r(\text{cov}(A)) < \text{diam}(\text{cov}(A))$, that is, $r(A) < \text{diam}(A)$. $\blacksquare$

We now have the following ‘structure’ theorem.

Theorem 5.2 *Let M be a bounded metric space for which $\mathcal{A}(M)$ is compact and normal, and let $T : M \rightarrow M$ be nonexpansive. Then the fixed point set F_T of T is a nonempty 1-local retract of M , and consequently $\mathcal{A}(F_T)$ is compact and normal.*

Proof. The fact that $F_T \neq \emptyset$ is immediate from Theorem 5.1. To see that F_T is a 1-local retract of M let $\{B_i\}$ be a family of closed balls centered in F_T for which

$$S = \bigcap_i B_i \neq \emptyset.$$

Then since T is nonexpansive, $T : S \rightarrow S$. Also, since S is admissible, $\mathcal{A}(S)$ is compact and normal. Therefore, again by Theorem 5.1, T has a fixed point in S , that is, $F_T \cap S \neq \emptyset$. This proves that F_T is a 1-local retract of M . The final assertion of the theorem follows from Proposition 5.4.

It is now possible to extend Theorem 5.1 to finite commutative families.

Theorem 5.3 *Let M be a bounded metric space for which $\mathcal{A}(M)$ is compact and normal. Then every finite family Φ of commuting self-mappings of M has a nonempty common fixed point set F_Φ . Moreover, F_Φ is a 1-local retract of M .*

Proof. It suffices to show that $F_\Phi \neq \emptyset$. The fact that F_Φ is a 1-local retract of M can be proved as above. Suppose $\Phi = \{T_1, \dots, T_n\}$. By Theorem 5.2, if F_{T_i} denotes the fixed point set of T_i , $i = 1, \dots, n$, then each family $\mathcal{A}(F_{T_i})$ is compact and normal. Since T_1 and T_2 commute, it is immediate that $T_2 : F_{T_1} \rightarrow F_{T_1}$. Thus $F_{T_1} \cap F_{T_2} \neq \emptyset$. Proceeding step by step one concludes $\bigcap_{i=1}^n F_{T_i} \neq \emptyset$. ■

5.3 Uniform normal structure

We now turn to the following question: Are there normal type conditions, weaker than hyperconvexity, that are sufficient to assure that the intersection of any descending chain of nonempty admissible sets in a metric space is nonempty (and admissible)? If the underlying metric space is complete the answer is yes, but the approach, due to Kulesza and Lim [105], is a little sophisticated. We continue with the notation of the previous section.

Definition 5.3 $\mathcal{A}(M)$ is said to be *uniformly normal* if there exists $c \in (0, 1)$ such that $r(D) \leq c \operatorname{diam}(D)$ for each $D \in \mathcal{A}(M)$.

We shall see later that the family of admissible subsets of any bounded closed convex subset of a uniformly convex Banach space is uniformly normal.

Theorem 5.4 *Suppose (M, d) is a bounded complete metric space for which $\mathcal{A}(M)$ is uniformly normal. Then $\mathcal{A}(M)$ is compact.*

We first establish a more elementary fact. $\mathcal{A}(M)$ is said to be *countably compact* if every descending sequence $\{D_n\}$ of nonempty sets in $\mathcal{A}(M)$ has nonempty intersection.

Lemma 5.2 *If $\mathcal{A}(M)$ is uniformly normal, then $\mathcal{A}(M)$ is countably compact.*

Proof. Let

$$k_0 = \sup\{r(D)/\operatorname{diam}(D) : D \in \mathcal{A}(M) \text{ and } \operatorname{diam}(D) > 0\}.$$

Since M is uniformly normal, $k_0 < 1$. Now fix $k \in (k_0, 1)$, and for $D \in \mathcal{A}(M)$ set

$$A(D) = \{x \in D : d(x, y) \leq k \operatorname{diam}(D) \text{ for each } y \in D\},$$

that is,

$$A(D) = D \cap \left(\bigcap_{y \in D} B(y; k \operatorname{diam}(D)) \right).$$

Clearly $A(D) \neq \emptyset$ and $A(D) \in \mathcal{A}(M)$. Moreover, $A(D)$ is a proper subset of D if $\operatorname{diam}(D) > 0$.

Now let $\{D_n^0\}$ be a descending sequence of nonempty sets in $\mathcal{A}(M)$. We wish to show that $\bigcap_{n=1}^{\infty} D_n^0 \neq \emptyset$. To this end, set

$$\begin{aligned} D_1^1 &= \operatorname{cov} \left(\bigcup_{i=1}^{\infty} A(D_i^0) \right) \\ D_2^1 &= \operatorname{cov} \left(\bigcup_{i=2}^{\infty} A(D_i^0) \right) \\ &\vdots \\ D_n^1 &= \operatorname{cov} \left(\bigcup_{i=n}^{\infty} A(D_i^0) \right). \\ &\dots \end{aligned}$$

We assert that $\operatorname{diam}(D_n^1) \leq k \operatorname{diam}(D_n^0)$. To see this let $x, y \in \bigcup_{i=n}^{\infty} A(D_i^0)$. Then $x \in A(D_p^0)$ and $y \in A(D_q^0)$ with, say, $n \leq p \leq q$. Since $D_q^0 \subseteq D_p^0 \subseteq D_n^0$, the definition of $A(D_p^0)$ implies

$$d(x, y) \leq k \operatorname{diam}(D_p^0) \leq k \operatorname{diam}(D_n^0).$$

This proves that $\operatorname{diam} \left(\bigcup_{i=n}^{\infty} A(D_i^0) \right) \leq k \operatorname{diam}(D_n^0)$. The conclusion now follows from the fact that $\operatorname{diam}(\operatorname{cov}(A)) = \operatorname{diam}(A)$ for any $A \subseteq M$.

We now repeat the above construction in a step-by-step manner to obtain the following:

$$\begin{array}{ccccccc} D_1^0 & \supseteq & D_2^0 & \supseteq & \cdots & \supseteq & D_n^0 \supseteq \cdots \\ \cup & & \cup & & & & \cup \\ D_1^1 & \supseteq & D_2^1 & \supseteq & \cdots & \supseteq & D_n^1 \supseteq \cdots \\ \cup & & \cup & & & & \cup \\ \vdots & & \vdots & & & & \vdots \\ \cup & & \cup & & & & \cup \\ D_1^n & \supseteq & D_2^n & \supseteq & \cdots & \supseteq & D_n^n \supseteq \cdots \end{array}$$

We now invoke the completeness of M . Since $\text{diam}(D_i^n) \leq k^n \text{diam}(D_i^0) \rightarrow 0$ as $n \rightarrow \infty$, the diagonal sequence $\{D_{n+1}^n\}_{n \geq 1}$ has nonempty intersection by Cantor's theorem. We thus have

$$\emptyset \neq \bigcap_{n=1}^{\infty} D_{n+1}^n \subseteq \bigcap_{n=1}^{\infty} D_n^0,$$

completing the proof. ■

For the next step it is necessary to escalate the method of argument. The key idea in the proof of Theorem 5.4 is the following fact. Any uncountable chain of real numbers which is either strictly decreasing or strictly increasing is eventually constant. However, we need a preliminary fact which has to do with ordinal numbers.

Lemma 5.3 *If $\mathcal{A}(M)$ is countably compact but not compact, then there exists an uncountable ordinal Γ and a descending transfinite chain $\{D_\alpha, \alpha < \Gamma\}$ of nonempty members of $\mathcal{A}(M)$ such that*

$$\bigcap_{\alpha < \Gamma} D_\alpha = \emptyset.$$

Proof. Since $\mathcal{A}(M)$ is not compact there exists a descending chain $\{D_i\}_{i \in I}$ of nonempty sets in $\mathcal{A}(M)$ for which $\bigcap_{i \in I} D_i = \emptyset$. Choose an ordinal γ whose cardinality exceeds the cardinality of the index set I . Select $i_1 \in I$ and set $D_1 = D_{i_1}$. Proceed by transfinite induction. Let $\beta < \gamma$. Having defined $D_\alpha = D_{i_\alpha}$ for $\alpha < \beta$ (where $i_\alpha \in I$), consider two cases.

- (1) $\bigcap_{\alpha < \beta} D_\alpha = \emptyset$. In this case there is nothing to prove—take $\Gamma = \beta$.
- (2) $\bigcap_{\alpha < \beta} D_\alpha \neq \emptyset$. In this case there must exist $i \in I$ for which D_i is a proper subset of $\bigcap_{\alpha < \beta} D_\alpha$. (Such a set must exist because $\bigcap_{i \in I} D_i = \emptyset$.) Define $D_\beta = D_i$.

Since the cardinality of γ exceeds that of I case (2) cannot prevail throughout the induction process, so case (1) must hold for some $\beta < \gamma$. Also β must be uncountable because $\mathcal{A}(M)$ is countably compact. The set of all ordinals for which the conclusion of Lemma 5.3 holds is nonempty and thus by well-ordering there is a *smallest such ordinal*, which we again call Γ . ■

A chain $\{D_\alpha\}$ that satisfies the conclusions of Lemma 5.3 for Γ will be called a Γ -chain.

Given a Γ -chain $\{D_\alpha\}$, for each $\alpha \in \Gamma$ let

$$d_\alpha = \text{diam}(D_\alpha).$$

Since $\{d_\alpha\}_{\alpha < \Gamma}$ is nonincreasing, $\lim_{\alpha} d_\alpha = d$ (in fact, for α sufficiently large, $d_\alpha \equiv d$). We call d the *diameter of* $\{D_\alpha\}$ and write

$$d = \text{diam}(\{D_\alpha\}).$$

Finally, we say that a Γ -chain $\{J_\alpha\}$ is a refinement of a Γ -chain $\{D_\alpha\}$ if $J_\alpha \subseteq D_\alpha$ for each $\alpha < \Gamma$. We now have the following.

Lemma 5.4 *Under the hypothesis of Lemma 5.3, there exists a Γ -chain $\{D_\alpha\}$ with the property: If $\{J_\alpha\}$ is a refinement of $\{D_\alpha\}$ then $\text{diam}(\{J_\alpha\}) = \text{diam}(\{D_\alpha\})$.*

Proof. We now know there exists at least one Γ -chain $\{D_\alpha\}$. Define a sequence $\{\mathcal{J}_n\}$ of Γ -chains inductively as follows. Let $\mathcal{J}_1 = \{D_\alpha\}$ and suppose $\mathcal{J}_j$ has been defined. Let

$$\hat{d}_j = \inf\{\delta : \text{there exists a refinement of } \mathcal{J}_j \text{ with diameter } \delta\}.$$

Now let $\mathcal{J}_{j+1} = \{D_\alpha^{j+1}\}$ be a refinement of $\mathcal{J}_j = \{D_\alpha^j\}$ for which

$$\text{diam}(\mathcal{J}_{j+1}) \leq \hat{d}_j + \frac{1}{j+1}.$$

For each $\alpha < \Gamma$ define $D_\alpha^* = \bigcap_{j=1}^{\infty} D_\alpha^j$. By countable compactness of $\mathcal{A}(M)$ each set D_α^* is nonempty (and in $\mathcal{A}(M)$). Therefore $\{D_\alpha^*\}$ is a Γ -chain.

Suppose $\mathcal{J}$ is refinement of $\{D_\alpha^*\}$. Then $\mathcal{J}$ is a refinement of $\mathcal{J}_j$ for each j as well, so

$$\hat{d}_j \leq \text{diam}(\mathcal{J}) \leq \text{diam}(\{D_\alpha^*\}) \leq \text{diam}(\mathcal{J}_{j+1}) \leq \hat{d}_j + \frac{1}{j+1}.$$

Since this holds for each $j = 1, 2, \dots$, we conclude $\text{diam}(\mathcal{J}) = \text{diam}(\{D_\alpha^*\})$. ■

Theorem 5.5 *Suppose (M, d) is a bounded metric space for which $\mathcal{A}(M)$ is countably compact and normal. Then $\mathcal{A}(M)$ is compact.*

Proof. Suppose $\mathcal{A}(M)$ is not compact. Then by Lemma 5.4 there exists an uncountable Γ -chain $\mathcal{I} = \{D_\alpha\}$ in $\mathcal{A}(M)$ whose diameter d is minimal with respect to refinement. Since $\lim_{\alpha < \Gamma} \text{diam}(D_\alpha) = d$ and Γ is uncountable, there exists $\beta \in \Gamma$ such that for all $\alpha > \beta$, $\text{diam}(D_\alpha) = d$. We may assume $d > 0$ since otherwise each of the sets of the chain $\{D_\alpha\}$ would eventually consist of the same point, implying $\bigcap_{\alpha} D_\alpha \neq \emptyset$.

Now let $\alpha > \beta$. Since $\mathcal{A}(M)$ is normal, for each such α there exists $r_\alpha < \text{diam}(D_\alpha) = d$ and $p_\alpha \in D_\alpha$ such that $D_\alpha \subseteq B(p_\alpha; r_\alpha)$.

We now come to a subtle part of the argument. Let

$$I_n = \left[d \left(1 - \frac{1}{n} \right), d \left(1 - \frac{1}{n+1} \right) \right], \quad n = 1, 2, \dots$$

Note that for each $\beta < \alpha < \Gamma$, $r_\alpha \in I_n$ for some n . To see this suppose for each n there exists β_n such that $\alpha > \beta_n \Rightarrow r_\alpha \notin I_n$. Since Γ is uncountable and $\{\beta_n\}$ is countable, there exists $\mu < \Gamma$ such that $\mu > \beta_n$ for each n . But this would imply that r_μ for such μ could not lie in any of the sets I_n —a contradiction. We therefore conclude that for some m the set $S = \{\alpha > \beta : r_\alpha \in I_m\}$ is cofinal in Γ . Let $r = d\left(1 - \frac{1}{m+1}\right)$ and let $J = \cap\{B(p_\alpha; r) : \alpha \in S\}$. Now let $\mu, \nu \in S$, and say $\mu > \nu$. Then $D_\mu \subseteq D_\nu$. Also $r_\nu \in I_m$ so $r_\nu < d\left(1 - \frac{1}{m+1}\right) = r$. Therefore,

$$p_\mu \in D_\mu \subseteq D_\nu \subseteq B(p_\nu; r_\nu) \subseteq B(p_\nu; r).$$

We conclude from this that

$$\{p_\alpha : \alpha \in S\} \subseteq J.$$

Now let $K = \cap\{B(x; r) : x \in J\}$. Since $\{p_\alpha : \alpha \in S\} \subseteq J$ we have $K \subseteq J$. Suppose $y, z \in K$. Then $z \in J$ and $y \in B(z; r)$ so $d(y, z) \leq r$. This implies $\text{diam}(K) \leq r$. Also, if $x \in J$ and $\alpha \in S$ then $x \in B(p_\alpha; r)$, so $p_\alpha \in K$, that is,

$$\{p_\alpha : \alpha \in S\} \subseteq K.$$

Define the chain $\mathcal{J} = \{J_\alpha\}$ by setting $J_\alpha = D_\alpha$ for $\alpha \leq \beta$ and $J_\alpha = D_\alpha \cap K$ for $\beta < \alpha < \Gamma$. Suppose $\beta < \mu < \Gamma$. Then since S is cofinal in Γ there exists $\alpha \in S$ such that $\mu \leq \alpha$. Hence $p_\alpha \in D_\alpha \cap K \subseteq D_\mu \cap K$ and it follows that $J_\alpha \neq \emptyset$ for each α (and clearly $J_\alpha \in \mathcal{A}(M)$). Since $\text{diam}(\mathcal{J}) \leq \text{diam}(K) < d$ and $\mathcal{J}$ is a refinement of $\mathcal{I}$ we have a contradiction. ■

5.4 Uniform relative normal structure

This is a weaker variant of the concept of uniform normal structure that does not seem to imply normal structure. Yet it does assure the existence of fixed points for nonexpansive mappings. This concept was introduced by Sordi in [149].

The admissible sets $\mathcal{A}(M)$ in M are said to be *uniformly relatively normal* if there exists $c \in (0, 1)$ such that for each $D \in \mathcal{A}(M)$ there exists $z_D \in M$ such that:

- (a) $D \subseteq B(z_D; c \text{diam}(D))$;
- (b) if $D \subseteq B(y; c \text{diam}(D))$ for $y \in M$, then $d(z_D, y) \leq c \text{diam}(D)$.

Theorem 5.6 *Let M be a nonempty bounded metric space for which $\mathcal{A}(M)$ is compact and uniformly relatively normal. Then every nonexpansive $T : M \rightarrow M$ has at least one fixed point.*

Proof. By the compactness of $\mathcal{A}(M)$ and Zorn's Lemma there exists $A_0 \in \mathcal{A}(M)$ such that

$$A_0 = \text{cov}(T(A_0)).$$

Let $r = \text{diam}(A_0)$. Then the set

$$A = \{z \in M : A_0 \subseteq B(z; cr)\} = \bigcap_{x \in A_0} B(x; cr)$$

is nonempty ($z_{A_0} \in A$). Moreover if $z \in A$ and $u \in A_0$, then

$$d(T(z), T(u)) \leq d(z, u) \leq cr;$$

hence $T(A_0) \subseteq B(T(z); cr)$, from which

$$A_0 = \text{cov}(T(A_0)) \subseteq B(T(z); cr).$$

Thus $T : A \rightarrow A$. Now let

$$H = \left(\bigcap_{z \in A} B(z; cr) \right) \cap A,$$

and let

$$A'_1 = \cap \{D \in \mathcal{A}(M) : T(D) \subseteq D \text{ and } D \supseteq H\}.$$

Clearly, $T : A'_1 \rightarrow A'_1$. We show next that $\text{diam}(A'_1) \leq cr$. To see this, let

$$F = \left(\bigcap_{z \in A'_1} B(z; cr) \right) \cap A'_1.$$

Since $H \subseteq A'_1$, then we must have $F \supseteq H$. Indeed, let $x \in H$. Then $x \in A'_1$. Since $x \in H$, we have

$$x \in \bigcap_{z \in A} B(z; cr) \subseteq \bigcap_{z \in A'_1} B(z; cr),$$

where the second assertion is true because $A'_1 \subseteq A$. Now let $z \in F$ and set

$$G = B(T(z); cr) \cap A'_1.$$

Then if $w \in H$, $d(w, u) \leq cr$ for each $u \in A'_1$; hence $d(w, T(z)) \leq cr$. Therefore, $H \subseteq G$. Also, if $x \in G$ then $d(T(x), T(z)) \leq d(x, z) \leq cr$ and thus $T(x) \in G$, that is, $T : G \rightarrow G$. It follows that $A'_1 \subseteq G$; hence $A'_1 = G$. Therefore $B(T(z); cr) \supseteq A'_1$ proving $T(z) \in F$ and, in turn, proving $F = A'_1$. Hence $\text{diam}(A'_1) = \text{diam}(F) \leq cr$. Notice also that since $A'_1 \subseteq A$, $A_0 \subseteq B(x; cr)$ for each $x \in A'_1$.

Now apply Zorn's Lemma again to obtain a set $A_1 \in \mathcal{A}(M)$ such that $A_1 \subseteq A'_1$ and such that $\text{cov}(T(A_1)) = A_1$. Proceeding in this way it is possible to construct a sequence $\{A_n\} \subseteq \mathcal{A}(M)$ such that

- (i) $T : A_n \rightarrow A_n$;
- (ii) $\text{diam}(A_n) \leq rc^n$;
- (iii) $d(u, v) \leq rc^n$ if $u \in A_n, v \in A_{n-1}$.

The argument is completed by selecting $u_n \in A_n$ for each n and observing that the sequence $\{u_n\}$ is a Cauchy sequence which converges, necessarily, to a fixed point of T . ■

5.5 Quasi-normal structure

We now turn to the question of whether the assumption that $\mathcal{A}(M)$ is normal in Theorem 5.1 can be weakened. There is an obvious candidate for such a weakening. As before, we assume M is a bounded metric space.

Definition 5.4 *The admissible structure $\mathcal{A}(M)$ of M is said to be quasi-normal if for each $D \in \mathcal{A}(M)$ with $\text{diam}(D) > 0$ there exists a point $p \in D$ such that for each $x \in D$:*

$$d(p, x) < \text{diam}(D).$$

Obviously, the following implications hold:

$$\text{uniformly normal} \Rightarrow \text{normal} \Rightarrow \text{quasi-normal}.$$

The metric definition of quasi-normal was inspired by its Banach space counterpart which we take up later. While the concept is well understood in Banach spaces its significance in metric spaces is less clear. However the question of whether the condition is strong enough to imply the existence of fixed points for nonexpansive mappings can be quickly settled in the negative. (The answer is unknown for Banach spaces.)

Example 5.1 *Let $\mathbb{R}^+$ denote the nonnegative reals, and for $x, y \in \mathbb{R}^+$ introduce the metric*

$$d(x, y) = \frac{|x - y|}{1 + |x - y|}.$$

Then if $M = (\mathbb{R}^+, d)$, $\mathcal{A}(M)$ is both compact and quasi-normal. However, the mapping $T : M \rightarrow M$ defined by $T(x) = x + 1$ is nonexpansive (in fact, isometric) and fixed point free.

Proof. We observed in Chapter 2 that d is a metric. Now suppose $\mathfrak{S}$ is a family of balls in M with the finite intersection property. If $B(x; r) \in \mathfrak{S} \Rightarrow r \geq 1$ then $\cap \mathfrak{S} \supseteq M \neq \emptyset$. Otherwise there exists $B(x; r) \in \mathfrak{S}$ with $r < 1$. In this case $B(x; r)$ is a compact subset of $\mathbb{R}^+$ and $\{B \cap B(x; r) : B \in \mathfrak{S}\}$ has the finite

intersection property, from which $\cap \mathfrak{S} \neq \emptyset$. It follows that $\mathcal{A}(M)$ is compact. To see that $\mathcal{A}(M)$ is quasi-normal note that the closed balls in M (and hence their intersections) correspond to closed intervals in $\mathbb{R}^+$. If such an interval is bounded then its midpoint will be a nondiametral point of the interval relative to the metric d . On the other hand, if D is unbounded in $\mathbb{R}^+$ then relative to d , $\text{diam}(D) = 1$ while $d(x, y) < 1$ for each $x, y \in D$. ■

The concept of quasi-normality does have fixed point theoretic implications for a special class of mappings.

Definition 5.5 *A mapping $T : M \rightarrow M$ is said to be a Kannan nonexpansive mapping if for each $x, y \in M$,*

$$d(T(x), T(y)) \leq \frac{1}{2} [d(x, T(x)) + d(y, T(y))].$$

Note that mappings of the above type may or may not be nonexpansive in the usual sense. In fact, the Kannan condition does not even imply continuity of the mapping.

Kannan nonexpansive mappings are notable for two reasons. On the one hand they (and their contractive analogs) have inspired a branch of fixed point theory devoted exclusively to the study of increasingly complex generalizations of contractive type conditions. This is an esoteric branch of the theory which seems to be of limited interest. However, it is the case that Kannan mappings are inextricably tied to the concept of quasi-normality. This fact, observed in [127], may be of wider interest.

Theorem 5.7 *Suppose (M, d) is a nonempty bounded metric space. If $\mathcal{A}(M)$ is (countably) compact and quasi-normal, then any Kannan nonexpansive mapping $T : M \rightarrow M$ has a unique fixed point. Conversely, if $\mathcal{A}(M)$ is not quasi-normal, then there exists a nonempty $D \in \mathcal{A}(M)$ and a Kannan nonexpansive $T : D \rightarrow D$ which is fixed point free.*

The easy part of the above theorem is the converse. If $\mathcal{A}(M)$ is not quasi-normal then there exists $D \in \mathcal{A}(M)$ for which $\text{diam}(D) > 0$ but which has the property that for each $x \in D$ there exists $T(x) \in D$ such that $d(x, T(x)) = \text{diam}(D)$. From this we have for each $x, y \in D$,

$$d(T(x), T(y)) \leq \text{diam}(D) = \frac{1}{2} [d(x, T(x)) + d(y, T(y))].$$

We now turn to the main part of the theorem, beginning with the following lemma.

Lemma 5.5 *For $r \geq 0$, assume $A_r = \{x \in M : d(x, T(x)) \leq r\} \neq \emptyset$, and let $C_r = \text{cov}(T(A_r))$. Then $C_r \subseteq A_r$ and $\text{diam}(C_r) \leq r$.*

Proof. Fix $x \in C_r$. If $d(x, T(x)) = 0$ the theorem follows, so we suppose $d(x, T(x)) > 0$. Let $s = \sup\{d(T(z), T(x)) : z \in A_r\}$. Then $B(T(x); s) \supseteq T(A_r)$, where $B(T(x); s) \supseteq \text{cov}(T(A_r)) = C_r$. Since $x \in C_r$ we have $d(x, T(x)) \leq s$. By the definition of s , for each $\varepsilon > 0$ there exists $z \in A_r$ such that $d(T(x), T(z)) \geq s - \varepsilon$. We now have

$$d(x, T(x)) - \varepsilon \leq s - \varepsilon \leq d(T(x), T(z)) \leq \frac{1}{2} [d(x, T(x)) + d(z, T(z))].$$

This immediately implies $d(x, T(x)) \leq d(z, T(z)) + 2\varepsilon \leq r + 2\varepsilon$, that is, $C_r \subseteq A_r$, since $\varepsilon > 0$ was arbitrary.

To complete the proof of the lemma, observe that

$$\text{diam}(C_r) = \text{diam}(\text{cov}(T(A_r))) = \text{diam}(T(A_r)).$$

But $\text{diam}(T(A_r)) = \sup\{d(T(x), T(y)) : x, y \in A_r\} \leq \frac{1}{2}[r + r] = r$. ■

Proof of Theorem 5.7. Set

$$r_0 = \inf\{d(x, T(x)) : x \in M\}.$$

To see that $A_{r_0} \neq \emptyset$ notice that if $\{r_n\}$ is a sequence strictly decreasing to r_0 then each of the sets

$$C_{r_1} \supseteq C_{r_2} \supseteq \cdots \supseteq C_{r_n} \supseteq \cdots$$

is nonempty and in $\mathcal{A}(M)$. Thus

$$A_{r_0} = \bigcap_{n=1}^{\infty} A_{r_n} \supseteq \bigcap_{n=1}^{\infty} C_{r_n} \neq \emptyset.$$

Now suppose $r_0 > 0$. Since $\mathcal{A}(M)$ is quasi normal there exists $z \in C_{r_0}$ such that $d(z, y) < \text{diam}(C_{r_0}) \leq r_0$ for each $y \in C_{r_0}$. But $T(z) \in T(A_{r_0}) \subseteq C_{r_0}$, so this implies $d(z, T(z)) < r_0$ and this contradicts the definition of r_0 . ■

The foregoing discussion raises a basic question about nonexpansive mappings.

Definition 5.6 *We say that a metric space M has the FPP for nonexpansive mappings if for any nonempty $D \in \mathcal{A}(M)$ and any nonexpansive $T : D \rightarrow D$ it is always the case that T has at least one fixed point.*

Problem. Is there a geometric condition which lies between normal and quasi-normal which is both necessary and sufficient for the FPP?

Related questions are abundant. For example:

Problem. If M has the FPP, is $\mathcal{A}(M)$ necessarily compact? Is $\mathcal{A}(M)$ necessarily quasi-normal?

5.6 Stability and normal structure

In this section we look at mappings which do not increase 'large' distances. Such mappings do not need to be continuous, and our objective is not to seek fixed points but rather to determine what can be said about minimal displacement for such mappings.

Definition 5.7 Let M be a metric space. A mapping $T : M \rightarrow M$ is said to be h -nonexpansive for $h > 0$ if $d(T(x), T(y)) \leq \max\{d(x, y), h\}$ for $x, y \in M$.

Our principal result is the following: The surprising aspect of the conclusion of this theorem is the fact that h -nonexpansive mappings have minimal displacement *strictly less than* h in the presence of normality.

Theorem 5.8 Let (M, d) be a metric space for which $\mathcal{A}(M)$ is compact and normal, and suppose $T : M \rightarrow M$ is h -nonexpansive for $h > 0$. Then there exists $z \in M$ such that $d(z, T(z)) < h$.

Proof. By a standard Zorn's Lemma argument there exists $D \in \mathcal{A}(M)$ which is minimal with respect to being nonempty and invariant under T . Then clearly $D = \text{cov}(T(D))$. Let $r = r(D)$ and let $C = C(D)$. If $\text{diam}(D) = 0$ then D consists of a single point which is fixed under T and there is nothing to prove. So assume $\text{diam}(D) > 0$ and let $z \in C$. Observe that if $r < h$, then $d(z, T(z)) \leq r < h$ and again we are finished. So we assume $h \leq r$ and show that this leads to a contradiction. Let $x \in D$. If $d(z, x) \geq h$ then $d(T(z), T(x)) \leq d(z, x) \leq r$. On the other hand, if $d(z, x) < h$, then $d(T(z), T(x)) < h \leq r$. So in either case $T(x) \in B(T(z); r)$. Thus $T(D) \subseteq B(T(z); r)$, from which $D = \text{cov}(T(D)) \subseteq B(T(z); r)$, and this, in turn, implies $T(z) \in C$. Therefore, $T : C \rightarrow C$. But since $C \in \mathcal{A}(M)$ this contradicts the minimality of D . ■

If $\mathcal{A}(M)$ is *uniformly normal*, it is possible to say a little more. The proof is only a slight modification of the one just given.

Theorem 5.9 Let (M, d) be a metric space for which $\mathcal{A}(M)$ is compact and uniformly normal with normality constant $c \in (0, 1)$, and suppose $T : M \rightarrow M$ is h -nonexpansive. Then there exists $z \in M$ such that $d(z, T(z)) \leq ch$.

Proof. Again assume $D \in \mathcal{A}(M)$ is minimal with respect to being nonempty and invariant under T . Then clearly $D = \text{cov}(T(D))$. Let $r = r(D)$ and let $C = C(D)$. If $\text{diam}(D) = 0$ then D consists of a single point which is fixed under T and there is nothing to prove. So assume $\text{diam}(D) > 0$ and let $z \in C$. Next observe that if $\text{diam}(D) \leq h$, then $d(z, T(z)) \leq r = c \text{diam}(D) \leq ch$ and again we are finished. So we assume $h < \text{diam}(D)$ and show that this leads to a contradiction. To this end let $h' = \max\{h, r\}$ and let

$$C' = \{z \in D : D \subseteq B(z; h')\}.$$

Then $C' \in \mathcal{A}(M)$ and since $C \subseteq C'$, $C' \neq \emptyset$. Let $z \in C'$ and let $x \in D$. If $d(z, x) \geq h'$, then $d(z, x) \geq h$ and $d(T(z), T(x)) \leq d(z, x) \leq h'$ so in this case $T(x) \in B(T(z); h')$. On the other hand, if $d(z, x) < h'$, then $d(T(z), T(x)) \leq h < h'$ and again $T(x) \in B(T(z); h')$. Thus $T(D) \subseteq B(T(z); h')$, from which $D = \text{cov}(T(D)) \subseteq B(T(z); h')$, and this, in turn, implies $T(z) \in C'$. Therefore, $T : C' \rightarrow C'$. But since $C' \in \mathcal{A}(M)$ is nonempty and since C' is a proper subset of D (because $\text{diam}(C') \leq h' < \text{diam}(D)$), this contradicts the minimality of D . ■

Corollary 5.1 *Let (M, d) be a hyperconvex metric space and suppose $T : M \rightarrow M$ is h -nonexpansive for $h > 0$. Then there exists $z \in M$ such that*

$$d(z, T(z)) \leq \frac{h}{2}.$$

Proof. As we saw in Chapter 4, for a hyperconvex space M the admissible sets $\mathcal{A}(M)$ are compact and uniformly normal for $c = 1/2$. ■

5.7 Ultrametric spaces

Ultrametric spaces and hyperconvex metric spaces share many common properties, yet they are quite different in very distinctive ways. The most striking similarity has to do with the injective extension property; the most striking difference is likely the fact that while hyperconvex metric spaces are always metrically convex, ultrametric spaces never are.

Ultrametric spaces are metric spaces for which a much stronger version of the triangle inequality holds.

Definition 5.8 *A metric space (M, d) is an ultrametric space if, in addition to the usual metric axioms, the following property holds for each $x, y, z \in M$:*

$$d(x, z) \leq \max\{d(x, y), d(y, z)\}.$$

It is immediate from the definition that if $d(x, y) \neq d(y, z)$ then, in fact, $d(x, z) = \max\{d(x, y), d(y, z)\}$. Therefore, each three points of an ultrametric space represent vertices of an isosceles triangle. This leads to another anomaly.

(1) If $B(a; r_1)$ and $B(b; r_2)$ are two closed balls in an ultrametric space, with $r_1 \leq r_2$, then either $B(a; r_1) \cap B(b; r_2) = \emptyset$ or $B(a; r_1) \subseteq B(b; r_2)$. In particular, if $a \in B(b; r_2)$, then $B(a; r_1) \subseteq B(b; r_2)$.

Ultrametries arise naturally in the study of non-archimedean analysis; in particular, in the study of normed vector spaces over non-archimedean valuation fields ([122], [126], [139]). Here we discuss a few results which hold in the most general setting. Most of these results are taken from [131] and [130].

We begin with a characterization of ultrametric spaces in terms of nonexpansive extensions found in [132]. (Actually, a more general version of this theorem

is proved in [132] in which the 'distance' values lie in linearly ordered sets with smallest elements rather than in $\mathbb{R}$.)

An ultrametric space M is said to be *spherically complete* if every chain of closed balls in M has nonempty intersection. As a consequence of (1) the admissible sets $\mathcal{A}(M)$ of M coincide with the closed balls of M . Therefore, the assertion that M is spherically complete implies $\mathcal{A}(M)$ is compact in previous terminology.

The following is now an immediate consequence of (1).

(2) If $\mathcal{F}$ is a family of closed balls in a spherically complete ultrametric space and if each two members of $\mathcal{F}$ intersect, then $\cap \mathcal{F} \neq \emptyset$.

Another consequence of (1).

(3) If $B(a; r_1), B(b; r_2)$ are closed balls in an ultrametric space, if $B(a; r_1) \subseteq B(b; r_2)$, and if $b \notin B(a; r_1)$, then $d(b, a) = d(b, z)$ for each $z \in B(a; r_1)$.

Proof. Since $b \notin B(a; r_1)$ and $z \in B(a; r_1)$, $d(b, a) > d(z, a)$. Hence $d(b, a) = d(b, z)$. ■

Since the diameter of a closed ball in an ultrametric space can never exceed its radius, $\mathcal{A}(M)$ is never normal for an ultrametric space M .

An ultrametric space (M, d) is said to have the *extension property* (EP) if given any ultrametric space (X, ρ) and any subspace Y of X , every nonexpansive mapping $f : Y \rightarrow M$ has a nonexpansive extension $F : X \rightarrow M$.

Theorem 5.10 *An ultrametric space is spherically complete if and only if it has the extension property.*

Proof. ($\Rightarrow$) Suppose (M, d) is spherically complete, let (X, ρ) be an ultrametric space, let Y be a subspace of X , and suppose $f : Y \rightarrow M$ is nonexpansive. Let $z \in X$ with $z \notin Y$ and let $Y' = Y \cup \{z\}$. We first show that f has a nonexpansive extension $F : Y' \rightarrow M$.

Let $\mathcal{F} = \{B(f(y); \rho(y, z)) : y \in Y\}$. We assert that each two members of $\mathcal{F}$ intersect. Indeed, suppose $y_1, y_2 \in Y$ with $\rho(y_1, z) \leq \rho(y_2, z)$. Then $z \in B(y_1; \rho(y_1, z)) \cap B(y_2; \rho(y_2, z))$ so by (1)

$$B(y_1; \rho(y_1, z)) \subseteq B(y_2; \rho(y_2, z)).$$

Therefore, $d(f(y_1), f(y_2)) \leq \rho(y_1, y_2) \leq \rho(y_2, z)$, so $f(y_1) \in B(f(y_2); \rho(y_2, z))$. By (2) $\cap \mathcal{F} \neq \emptyset$, so let $p \in \cap \mathcal{F}$ and define $F(z) = p$. Then if $F(y) = f(y)$ for each $y \in Y$, $d(F(z), F(y)) \leq \rho(z, y)$ and F is a nonexpansive extension of f to Y' . The proof of this implication is now completed by using Zorn's Lemma as in the extension theorem of Aronszajn and Panitchpakdi (Theorem 4.7 of Chapter 4).

($\Leftarrow$) Now assume (M, d) has the extension property but is not spherically complete. Then there exists a decreasing family $\{B(x_i; r_i)\}_{i \in I}$ of closed balls in M for which $\bigcap_{i \in I} B(x_i; r_i) = \emptyset$. Let $M' = M \cup \{p\}$ where $p \notin M$ and define a metric ρ on M' as follows. Set $\rho(x, y) = d(x, y)$ if $x, y \in M$; otherwise for $x \in M$ set $\rho(p, x) = d(x, x_j)$, where $x \notin B(x_j; r_j)$. By assumption, such $j \in I$ must exist. To see that ρ is well defined, notice that if $x \notin B(x_k; r_k)$ then, since these balls are nested, $d(x_j, x_k) < d(x, x_j)$. Thus $d(x, x_j) = d(x, x_k)$.

By the extension property, the identity mapping on M has an extension $F : M' \rightarrow M$. Also, if $x_i \notin B(x_j; r_j)$ it must be the case that $B(x_j; r_j) \subseteq B(x_i; r_i)$. Thus

$$d(F(p), x_i) = d(F(p), F(x_i)) \leq \rho(p, x_i) = d(x_i, x_j) \leq r_i,$$

and $F(p) \in \bigcap_{i \in I} B(x_i; r_i)$. This contradicts the original assumption and completes the proof. $\blacksquare$

The following 'fixed point result' is attributed to Petalas and Vidalis [130].

Theorem 5.11 *Suppose (M, d) is a spherically complete ultrametric space and suppose $T : M \rightarrow M$ is nonexpansive. Then either T has a fixed point, or there exists a ball $B = B(x; r)$ in M such that (i) $T : B \rightarrow B$ and (ii) $d(z, T(z)) = r$ for each $z \in B$.*

Proof. Consider the family $\mathcal{F} = \{B(x; d(x, T(x))) : x \in M\}$ ordered by set inclusion and let $\mathcal{A}$ be a descending chain in $\mathcal{F}$. Since M is spherically complete, $\bigcap \mathcal{A} = B \neq \emptyset$. Let $b \in B$ and $B(a; d(a, T(a))) \in \mathcal{A}$. Then if $x \in B(b; d(b, T(b)))$,

$$d(x, b) \leq d(b, T(b)) \leq \max\{d(a, b), d(a, T(a)), d(T(a), T(b))\} = d(a, T(a)).$$

Thus

$$d(x, a) \leq \max\{d(a, b), d(x, b)\} \leq d(a, T(a)).$$

This proves that $B(b; d(b, T(b))) \subseteq B(a; d(a, T(a)))$; hence each descending chain in $\mathcal{F}$ is bounded below by a member of $\mathcal{F}$. It follows from Zorn's Lemma that $\mathcal{F}$ has a minimal element, say $B(z; d(z, T(z)))$. For any $x \in B(z; d(z, T(z)))$, we have

$$d(x, T(x)) \leq \max\{d(x, z), d(z, T(z)), d(T(z), T(x))\} = d(z, T(z)).$$

Since $B(x; d(x, T(x))) \cap B(z; d(z, T(z))) \neq \emptyset$, it follows (from (1)) that

$$B(x; d(x, T(x))) \subseteq B(z; d(z, T(z)))$$

so by minimality, $B(x; d(x, T(x))) = B(z; d(z, T(z)))$. If $d(x, T(x)) < d(z, T(z))$ then

$$d(x, z) = d(z, T(z)) > d(x, T(x)).$$

However, this would imply $z \notin B(x; d(x, T(x)))$, contradicting $B(x; d(x, T(x))) = B(z; d(z, T(z)))$. Therefore, it must be the case that

$$d(x, T(x)) = d(z, T(z)) \text{ for each } x \in B(x; d(x, T(x))).$$

■

Corollary 5.2 *Under the assumptions of the theorem, if $T : M \rightarrow M$ is contractive then T has a fixed point.*

Proof. If $B = B(z; d(z, T(z)))$ satisfies alternative (ii), then

$$T(z) \in B(z; d(z, T(z)))$$

and $d(T(z), T^2(z)) = d(z, T(z))$. Since T is contractive this, in turn, implies $z = T(z)$. ■

Remarks. Ultrametrics arise in the study of non-archimedean analysis and in particular in the study of Banach spaces over non-archimedean valuation fields.

Definition 5.9 *A (real) valuation on a field K is a mapping $|\cdot| : K \rightarrow \mathbb{R}$ which satisfies for each $a, b \in K$:*

- (i) $|a| \geq 0$;
- (ii) $|a| = 0 \Leftrightarrow a = 0$;
- (iii) $|ab| = |a| \cdot |b|$;
- (iv) $|a + b| \leq |a| + |b|$.

In addition, $|1| = 1$; $|-a| = |a|$; and $|a^{-1}| = |a|^{-1}$ if $a \neq 0$.

The most obvious valuation is just 'absolute value' on the field of reals $\mathbb{R}$. If K is the field of complex numbers the usual valuation on K is given by $|z| = \sqrt{x^2 + y^2}$ for $z = x + iy \in K$.

A less obvious example arises as follows: Let Q be the field of rational numbers and let p be a fixed prime number. Each $a \in Q$ has a unique factorization of the form $a = (\alpha/\beta)p^n$, where α and β are relatively prime to p . The *p-adic valuation* on Q is given by setting $|a| = p^{-n}$, $|0| = 0$.

A valuation $|\cdot|$ is said to be *archimedean* if there exists an integer n in the underlying field K for which $|n| > 1$. If $|n| \leq 1$ for each integer n in K , then the valuation is said to be *non-archimedean*. It can be shown that a valuation $|\cdot|$ is non-archimedean if and only if the inequality in (iv) is strengthened to

$$|a + b| \leq \max\{|a|, |b|\}.$$

It is also known that a valuation on Q is non-archimedean if and only if it is equivalent to p -adic valuation for some prime p . (Two valuations $|\cdot|_1$ and $|\cdot|_2$ on K are said to be equivalent if and only if for each $a \in K$, $|a|_1 < 1 \Leftrightarrow |a|_2 < 1$.)

Given a valuation field K the function $d(a, b) = |a - b|$, $a, b \in K$, always defines a metric on K . If the valuation is non-archimedean then the metric is an ultrametric. The p -adic numbers arise as the completion of the space $(Q, |\cdot|)$, where $|\cdot|$ is the p -adic valuation on Q .

If a norm on a normed space X over a valuation field K satisfies the inequality

$$\|x + y\| \leq \max \{ \|x\|, \|y\| \}$$

for each $x, y \in X$, then it can be shown that, in fact, the valuation on K is non-archimedean.

5.8 Fixed point set structure—separable case

This chapter concludes with a rather deep theorem due to R. E. Bruck [29]. We have already seen that if M is bounded and hyperconvex then the fixed point set $F(T)$ of a nonexpansive mapping $T : M \rightarrow M$ is always hyperconvex and thus a nonexpansive retract of M . We have also seen that if $\mathcal{A}(M)$ is merely assumed to be compact and normal, then $F(T)$ is only known to be a 1-local retract of M . This raises the question of whether hyperconvexity is necessary for the stronger conclusion. A complete answer to this question is unknown, but it follows from the result proved below that in many instances $F(T)$ is indeed a nonexpansive retract of the domain of T .

Properties of $\mathcal{A}(M)$ do not play a direct role in the development of this section, and to be fully appreciated the result proved here should be recast in a Banach space framework. We shall do this later. However, the proof given below illustrates the generality of the metric approach.

Some ideas will be needed here which have not yet been discussed. We consider a metric space M and the family M^M of all mappings of M into M . One can think of M^M as the product space

$$\prod_{x \in M} M_x,$$

where each of the spaces M_x coincides with M itself. The usual product topology on M^M then coincides with the topology of pointwise convergence on M . To be precise, a net $\{f_\alpha\} \subseteq M^M$ converges to $f \in M^M$ if and only if

$$\lim_{\alpha} f_\alpha(x) = f(x) \text{ for each } x \in M.$$

In particular, if $S \subseteq M^M$, then $f \in \bar{S}$ (the closure of S in the product topology) if and only if $f = \lim_{\alpha} f_\alpha$ for some net $\{f_\alpha\} \subseteq S$. Hoping to avoid confusion, we shall reserve this notation to denote closure in M^M , and we shall use $\text{cl } K$ to denote the closure of a subset $K \subseteq M$ in the metric of M .

In the following theorem we consider a semigroup S of (nonexpansive) mappings in M^M . To say that S is a semigroup simply means that the composition $f \circ g \in S$ whenever $f, g \in S$. A *left ideal* J of S is a subset of S with the property $s \circ j \in J$ for each $j \in J$ and $s \in S$. We use $F(S)$ to denote the common fixed point set of S , that is,

$$F(S) = \{x \in M : s(x) = x \text{ for each } s \in S\}.$$

Also, for $x \in M$,

$$S(x) = \{s(x) : s \in S\}.$$

Recall that a subset $A \subseteq M$ is said to be dense in M if $\text{cl}(A) = M$, and that M is separable if it contains a countable dense subset. Separability is essential to the validity of the approach given here.

Theorem 5.12 *Let M be a separable complete metric space and let S be a semigroup of nonexpansive self-mappings of M . Then there exists in $\bar{S}$ a (nonexpansive) retraction r of M onto $F(S)$ if and only if one of the following two equivalent conditions holds:*

- (i) *Each nonempty closed S -invariant subset of M contains a fixed point of S .*
- (ii) *Whenever $x \in M$ then $\text{cl } S(x) \cap F(S) \neq \emptyset$.*

Proof. We begin with the easy implication. Suppose r is any retraction of M onto $F(S)$, suppose H is a nonempty closed S -invariant subset of M and let $f \in \bar{S}$. Then for some net $\{f_\alpha\} \subseteq S$, $\lim_\alpha f_\alpha(x) = f(x)$ for each $x \in M$. Since H is closed, $f(x) \in H$ for each $x \in H$, so H is also invariant under $\bar{S}$. Since $r \in \bar{S}$, $r(H) \subseteq H$. But r is a retraction of M onto $F(S)$, so this implies $r(H) \subseteq F(S)$. This proves $F(S) \cap H \supseteq r(H) \neq \emptyset$, so (i) holds.

Now observe that in order to prove there exists in $\bar{S}$ a retraction r of M onto $F(S)$ (which is necessarily nonexpansive because S is a nonexpansive semigroup) it suffices to show that $\bar{S}$ contains a one-element left ideal $\{e\}$. From this the conclusion is easy because then one has $s \circ e = e$ for each $s \in \bar{S}$. This means that $s \circ e(x) = e(x)$ for each $x \in M$ and $s \in \bar{S}$. In particular, the range of e is contained in $F(\bar{S})$ (and, of course, $F(\bar{S}) = F(S)$). At the same time $e(x) = x \Rightarrow s(x) = x$ so $F(e) \subseteq F(S)$. But since $e \in \bar{S}$, $F(\bar{S}) = F(S) \subseteq F(e)$. This proves that the range of e and $F(S)$ coincide with $F(e)$, proving e is a retraction of M onto $F(S)$.

We now embark on the task of proving that (ii) implies $\bar{S}$ contains a one-element left ideal. (We shall prove the equivalence of (i) and (ii) later.)

Note that since S is a semigroup of nonexpansive mappings it must also be the case that $\bar{S}$ is as well. We begin by introducing a metric ρ on $\bar{S}$ relative to which $\bar{S}$ is complete. A one-element left ideal of $\bar{S}$ is then constructed via Cantor's theorem as the intersection of a descending sequence of closed left ideals whose diameters converge to 0.

The metric ρ is defined as follows: Let $\{p_1, p_2, \dots\}$ be a countable dense subset of M and for $s, t \in \bar{S}$ set

$$\rho(s, t) = \sum_{i=1}^{\infty} \frac{1}{2^i} \left(\frac{d(s(p_i), t(p_i))}{1 + d(s(p_i), t(p_i))} \right).$$

To see that ρ is complete, let $\{s_n\}$ be a Cauchy sequence in $(\bar{S}, \rho)$. Then

$$\frac{d(s_n(p_i), s_m(p_i))}{1 + d(s_n(p_i), s_m(p_i))} \leq 2^i \rho(s_n, s_m),$$

so for each i , $\{s_n(p_i)\}$ is a Cauchy sequence in M . Let $s(p_i) = \lim_{n \rightarrow \infty} s_n(p_i)$. (M is complete.) Now let $\varepsilon > 0$ and choose $N \in \mathbb{N}$ so that $\sum_{i=N+1}^{\infty} 2^{-i} < \frac{\varepsilon}{2}$. Then

$$\rho(s_n, s) < \sum_{i=1}^N \frac{1}{2^i} \left(\frac{d(s_n(p_i), s(p_i))}{1 + d(s_n(p_i), s(p_i))} \right) + \frac{\varepsilon}{2}.$$

But for each $i = 1, \dots, N$, there exists n_i such that $n \geq n_i \Rightarrow d(s_n(p_i), s(p_i)) < \frac{\varepsilon}{2N}$. It follows that if $n \geq \max\{n_i : i = 1, \dots, N\}$, then $\rho(s_n, s) < \varepsilon$, proving that ρ is complete.

Now for $u \in F(S)$ and $k > 0$, let $N_k(u) = B\left(u; \frac{1}{k}\right)$. Then obviously $s : N_k(u) \rightarrow N_k(u)$ for each $s \in \bar{S}$.

Assertion. *If J is any closed left ideal of $\bar{S}$, if $x \in M$, and if k is any positive integer, then there exists a closed left ideal $J' \subseteq J$ and an element $u \in F(S)$ such that $J'(x) \subseteq N_k(u)$.*

Proof of the Assertion. Since J is a left ideal of $\bar{S}$, $s \circ j(x) \in J(x)$, so $J(x)$ is $\bar{S}$ -invariant. By continuity $\text{cl } J(x)$ is $\bar{S}$ -invariant as well. By (ii) there exists $u \in \text{cl } J(x) \cap F(\bar{S})$. In particular, $N_k(u) \cap J(x) \neq \emptyset$; hence $J' \neq \emptyset$, where

$$J' = \{j \in J : j(x) \in N_k(u)\}.$$

Also, if $s \in \bar{S}$ and $j \in J'$, then $s \circ j(x) \in J(x)$ and because

$$d(s \circ j(x), u) = d(s \circ j(x), s(u)) \leq d(j(x), u),$$

$s \circ j(x) \in N_k(u)$. This proves that $s \circ j \in J'$; hence that J' is a left ideal of $\bar{S}$. Also, J' is closed since J is closed and $N_k(u)$ is closed. (Suppose $\{j'_\alpha\}$ is a net in J' which converges to j' . Then $j' \in J$ and $\lim_{\alpha} j'_\alpha(x) = j'(x)$. But $j'_\alpha(x) \in N_k(u)$ for each α and since $N_k(u)$ is closed this implies $j'(x) \in N_k(u)$. Hence $j' \in J'$.) This proves the assertion.

Returning to the proof of the theorem, let $\{n_i\}_{i=1}^\infty$ be a sequence of positive integers which has the property that every positive integer appears in the sequence infinitely often. Define a sequence $\{J_k\}$ of left ideals of $\overline{S}$ as follows. Let $J_0 = \overline{S}$ and proceed by induction. Having defined J_{k-1} , invoke the Assertion to choose a closed left ideal J_k of $\overline{S}$ with $J_k \subseteq J_{k-1}$ and an element $u_k \in F(S)$ such that

$$J_k(p_{n_k}) \subseteq N_k(u_k).$$

Now fix $i \in \mathbb{N}$. Then for infinitely many k , $i = n_k$, and for such k

$$J_k(p_i) \subseteq N_k(u_k).$$

It follows that $\text{diam}(J_k(p_i)) \leq \frac{2}{k}$ for infinitely many k . Since the ideals $\{J_n\}$ are descending, for each fixed i the sequence $\{\text{diam}(J_n(p_i))\}$ is nonincreasing; consequently, $\lim_{n \rightarrow \infty} \text{diam}(J_n(p_i)) = 0$ for each i . Since for $s, t \in J_n$,

$$\rho(s, t) = \sum_{i=1}^{\infty} \frac{1}{2^i} \left(\frac{d(s(p_i), t(p_i))}{1 + d(s(p_i), t(p_i))} \right) \leq \text{diam}(J_n(p_i)),$$

this, in turn, implies that $\lim_{n \rightarrow \infty} \text{diam}(J_n) = 0$ in $(\overline{S}, \rho)$. Since $(\overline{S}, \rho)$ is complete we invoke Cantor's theorem to conclude that $\bigcap_{n=1}^{\infty} J_n = \{e\}$, where e is a left ideal of $\overline{S}$. As we saw at the outset e must be a retraction of M onto $F(S)$.

It remains only to show that (i) $\Leftrightarrow$ (ii). To see that (ii) $\Rightarrow$ (i), let H be a nonempty closed S -invariant subset of M . Then if $x \in H$, $\text{cl } S(x) \subseteq H$. But $\text{cl } S(x) \cap F(S) \neq \emptyset$. Hence $H \cap F(S) \neq \emptyset$. To see that (i) $\Rightarrow$ (ii), let $x \in M$ and note that $\text{cl } S(x)$ is closed and S -invariant; hence by (i) $S(x) \cap F(S) \neq \emptyset$. ■

Exercises

Exercise 5.1 Show that if M is compact then $(\mathcal{K}(M), H)$ is compact.

Exercise 5.2 If M is compact, is the metric space $(\mathcal{A}(M), H)$ a compact metric space (where here $\mathcal{A}(M)$ denotes the collection of all nonempty admissible subsets of M)?

Exercise 5.3 If M is a compact metric space, is $\mathcal{A}(\mathcal{A}(M))$ compact in the sense of Definition 5.1 (relative to the Hausdorff metric on $\mathcal{A}(M)$)?

Exercise 5.4 Show that if M is compact and metrically convex, then $(\mathcal{K}(M), H)$ is metrically convex. Is $(\mathcal{A}(M), H)$ metrically convex?

Exercise 5.5 Show that $(\mathcal{K}(M), H)$ in the previous exercise need not be compact if M is not compact.

Hint: Take M to be the subspace of ℓ_∞ consisting of all sequences $\{x_n\}$ which converge to 0. Consider

$$A = \{ \{x_n\} \in M : x_n = 1 + 1/n \text{ for an odd number of } n \text{ and } 0 \text{ elsewhere} \};$$

$$B = \{ \{x_n\} \in M : x_n = 1 + 1/n \text{ for an even number of } n \text{ and } 0 \text{ elsewhere} \}.$$

We now introduce a new definition. Let M be a metric space, and for $\epsilon > 0$ let

$$k_\epsilon = \sup \{ r(D) / \text{diam}(D) : D \in \mathcal{A}(M) \text{ and } \text{diam}(D) \geq \epsilon \}.$$

We say that $\mathcal{A}(M)$ is *quasi-uniformly normal* if $k_\epsilon < 1$ for each $\epsilon > 0$.

Exercise 5.6 Show that if M is complete and $\mathcal{A}(M)$ quasi-uniformly normal, then $\mathcal{A}(M)$ is compact.

Hint: Use the ideas developed in the proof of Theorem 5.4.

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Part II

Banach Spaces

Chapter 6

Banach Spaces: Introduction

6.1 The definition

In this chapter we collect information that will be needed for an understanding of the remainder of the text. Although details are included in some cases, many of the fundamental principles of functional analysis are merely stated without proof.

The spaces considered henceforth arise when an algebraic structure is combined in a special way with a metric structure. Many (indeed, most) of the examples of metric spaces we have already given have an inherent algebraic structure; coordinatewise addition and scalar multiplication in the cases of $\mathbb{R}^n$ and ℓ_∞ , functional addition and scalar multiplication in the case of $C[0, 1]$, $B[0, 1]$, and so on. An abstraction of these ideas gives rise to the body of spaces, both abstract and explicit, that invariably arise in the study of analysis. Many of the fixed point results in subsequent chapters will rely heavily on this added algebraic structure.

Let S denote any nonempty set that contains with each of its elements x and each real number λ a unique element $\lambda \cdot x$, written λx , called a *scalar multiple* of x . (One could also include complex numbers λ as well, but we restrict ourselves here to the real case.) Also assume that for each two elements $x, y \in S$ there exists a unique element $x + y \in S$ called the *sum* of x and y . The system $\{S, \cdot, +\}$ is called a *linear space* or a *vector space* (over $\mathbb{R}$) if the following conditions are satisfied. Here, $x, y, z \in S$; $\lambda, \mu, 1 \in \mathbb{R}$.

$$(1) \quad x + y = y + x;$$

$$(2) \quad x + (y + z) = (x + y) + z;$$

$$(3) \lambda(x + y) = \lambda x + \lambda y;$$

$$(4) x + y = x + z \text{ implies } y = z;$$

$$(5) (\lambda + \mu)x = \lambda x + \mu x;$$

$$(6) \lambda(\mu x) = (\lambda\mu)x;$$

$$(7) 1x = x.$$

Axioms (4) and (7) immediately give rise to a unique element $\theta \in S$ for which $x + \theta = x$ for each $x \in S$. Following custom we denote $\theta = 0$, that is, the zero element of the linear space shares its notation with $0 \in \mathbb{R}$. (Of course, $\mathbb{R}$ is itself a linear space over $\mathbb{R}$.)

Also for $x, y \in S$, we write $x - y = x + (-1)y$.

A *norm* on a linear space S is a mapping $\|\cdot\| : S \rightarrow \mathbb{R}^+$ which satisfies for each $x, y \in S$; $\lambda \in \mathbb{R}$:

$$(8) \|x\| = 0 \text{ if and only if } x = 0;$$

$$(9) \|\lambda x\| = |\lambda| \|x\|;$$

$$(10) \|x + y\| \leq \|x\| + \|y\|.$$

A linear space with a norm (called a *normed linear space*) carries a natural metric, namely, the distance $d : S \times S \rightarrow \mathbb{R}^+$ defined by taking

$$d(x, y) = \|x - y\|.$$

Taking $\lambda = -1$ in (9) and combining this with (10) immediately gives

$$\|x - y\| \leq \|x\| + \|y\|,$$

and the triangle inequality for d quickly follows from this.

Definition 6.1 A Banach space is a normed linear space $(S, \|\cdot\|)$ which is complete relative to the metric d defined above.

A linear space X has many norms. In general the space may be complete relative to one norm and not complete relative to another unless the two norms

are equivalent. Two norms $\|\cdot\|_1$ and $\|\cdot\|_2$ on X are said to be *equivalent* if there exist a and b (with $0 < a \leq b$) such that

$$a \|x\|_1 \leq \|x\|_2 \leq b \|x\|_1$$

for any vector $x \in X$.

Many of the metric spaces we have already discussed are, in fact, examples of Banach spaces. These include the spaces $\mathbb{R}^n$, $\mathbb{R}_p^n$ ($1 \leq p \leq \infty$), ℓ_p ($1 \leq p \leq \infty$), $C[0, 1]$, and $B[0, 1]$. In each instance a norm is defined by taking $\|x\| = d(x, 0)$.

Here are three additional examples.

Example 6.1 (c_0) *This is the space of all sequences $x = \{x_1, x_2, \dots\}$ of real numbers for which $\lim_{n \rightarrow \infty} x_n = 0$, with*

$$\|x\| = \sup_{1 \leq n < \infty} |x_n|.$$

Example 6.2 (c) *This is the space of all convergent sequences $x = \{x_1, x_2, \dots\}$ of real numbers, with*

$$\|x\| = \sup_{1 \leq n < \infty} |x_n|.$$

Example 6.3 $\ell_\infty(M)$ *This is the space of all bounded real-valued functions $f : M \rightarrow \mathbb{R}$ where M is a complete metric space and*

$$\|f\| = \sup_{x \in M} |f(x)|.$$

The completeness of $\ell_\infty(M)$ follows from the fact that if $\{f_n\}$ is a Cauchy sequence in $\ell_\infty(M)$ then $\{f_n(x)\}$ is a Cauchy sequence in M for each $x \in M$. The function defined by $f(x) = \lim_{n \rightarrow \infty} f_n(x)$, $x \in M$, exists since M is complete. It is quite easy to show that $\lim_{n \rightarrow \infty} \|f_n - f\| = 0$.

We will make use of the following embedding result later.

Proposition 6.1 *Any complete metric space M is isometric to a closed subset of a Banach space.*

Proof. Fix a point $x_0 \in M$ and for arbitrary $x \in M$ consider the function $f_x \in \ell_\infty(M)$ defined by

$$f_x(y) = d(y, x) - d(y, x_0), \quad y \in M.$$

First observe that since $|f_x(y)| \leq d(x, x_0)$, f_x is bounded. We claim that the mapping $F : M \rightarrow \ell_\infty(M)$ defined by taking $F(x) = f_x$ is an isometry. To see this take x_1 and x_2 in M and consider $\|f_{x_1} - f_{x_2}\|$. For $y \in M$ we have

$$\begin{aligned} |f_{x_1}(y) - f_{x_2}(y)| &= |d(y, x_1) - d(y, x_0) - d(y, x_2) + d(y, x_0)| \\ &= |d(y, x_1) - d(y, x_2)| \\ &\leq d(x_1, x_2). \end{aligned}$$

This proves $\|f_{x_1} - f_{x_2}\| \leq d(x_1, x_2)$. Conversely, upon taking $y = x_2$ we see that

$$|f_{x_1}(y) - f_{x_2}(y)| = |d(x_2, x_1) - d(x_2, x_0) - d(x_2, x_2) + d(x_2, x_0)| = d(x_1, x_2),$$

so $\|f_{x_1} - f_{x_2}\| \geq d(x_1, x_2)$.

Finally notice that $F(M)$ is a closed subset of $\ell_\infty(M)$. To see this suppose that $\lim_{n \rightarrow \infty} \|f_{x_n} - g\| = 0$. Then $\{f_{x_n}\}$ is a Cauchy sequence in $\ell_\infty(M)$ and it follows that $\{x_n\}$ is a Cauchy sequence in M . (Notice as above that $\|f_{x_n} - f_{x_m}\| = d(x_n, x_m)$.) Since M is complete $\lim_{n \rightarrow \infty} x_n = x \in M$, from which $\lim_{n \rightarrow \infty} f_{x_n} = f_x = g$. ■

Remark 6.1 Since the functions f_x defined above are also continuous, it follows that the isometry F maps M onto a subset of $C(M)$, the closed subspace of $\ell_\infty(M)$ consisting of all bounded real-valued continuous functions $f : M \rightarrow \mathbb{R}$. This fact has further implications since the space $\overline{C(M)}$ is always complete regardless of whether or not M is complete. Since $\overline{F(M)}$ is closed in $C(M)$, $\overline{F(M)}$ is itself complete. This proves that every metric space is isometric with a dense subspace of a complete metric space. This space, which is unique up to isometries, is called the completion of M . (See the discussion in Chapter 2, Section 2.3.) In particular, the completion of any normed linear space is a Banach space.

We conclude with an intricate example due to R. C. James. We will allude to many interesting properties of this space later (but without giving details).

Example 6.4 (The James space J) Let X denote the collection of all $x = (x_1, x_2, x_3, \dots) \in c_0$ for which

$$\|x\|_J = \sup \left[\sum_{i=1}^n (x_{p_i} - x_{p_{i+1}})^2 + (x_{p_{n+1}} - x_{p_1})^2 \right]^{1/2} < \infty,$$

where the supremum is taken over all positive integers n and all finite increasing sequences $(p_1, \dots, p_{n+1})$. The James space J is the completion of X with respect to the norm $\|x\|_J$.

6.2 Convexity

Definition 6.2 A subset K of a normed linear space is said to be convex if $\lambda x + (1 - \lambda)y \in K$ for each $x, y \in K$ and each scalar $\lambda \in [0, 1]$.

The next fact follows from a routine induction argument.

Proposition 6.2 A subset K of a normed linear space is convex if and only if $\sum_{i=1}^n \lambda_i x_i \in K$ for any finite set $\{x_1, x_2, \dots, x_n\} \subseteq K$ and any scalars $\lambda_i \geq 0$ for which $\sum_{i=1}^n \lambda_i = 1$.

The family of all convex subsets of a Banach space forms a *convexity structure*, in the sense that the family of such sets is closed under arbitrary intersections, and under unions of ascending chains. The family of all *closed* convex subsets of a Banach space is also closed under arbitrary intersections, but it fails the latter property. However, this class is more relevant to the study of fixed point theory. Since the assumption of completeness will be essential to much of what we do, we shall accordingly confine our attention to Banach spaces (as opposed to normed linear spaces).

We shall use $(X, \|\cdot\|)$ to denote a Banach space, and simply write X if only one norm on X is under consideration. In keeping with previous notation, we shall use $B(x; r)$ to denote the closed ball centered at $x \in X$ of radius $r \geq 0$:

$$B(x; r) = \{y \in X : \|x - y\| \leq r\}.$$

The ball $B(0; 1)$ is called the *unit ball* in X and is often denoted B_X . The boundary $\partial B(0; r)$ of $B(0; r)$ is the sphere

$$S_X = S(0; r) = \{x \in X : \|x\| = r\}.$$

For a subset A of a metric space X the concept

$$\text{cov}(A) = \cap \{B : B \text{ is a closed ball in } X \text{ and } B \supseteq A\}$$

has been already introduced. Related to this is another important concept.

Definition 6.3 For a subset A of a Banach space, the set

$$\text{conv}(A) = \cap \{K \subseteq X : K \text{ is convex and } K \supseteq A\}$$

is called the *convex hull* of A . The set

$$\overline{\text{conv}}(A) = \cap \{K \subseteq X : K \text{ is closed, convex and } K \supseteq A\}$$

is called the *convex closure* (or *closed convex hull*) of A .

Proposition 6.3 *For any set $A \subseteq X$*

$$\overline{\text{conv}}(A) = \overline{\text{conv}(\overline{A})}.$$

In general, $\overline{\text{conv}}(A) \neq \text{conv}(\overline{A})$. However, the following is true.

Proposition 6.4 *If $A \subseteq X$ is compact, then $\overline{\text{conv}}(A)$ is compact.*

A proof of the above fact will be given later in the section on condensing mappings (Chapter 7, Section 7.8).

Finally, since $A \subseteq \text{conv}(A) \subseteq \overline{\text{conv}}(A) \subseteq \text{cov}(A)$, and since we have already shown that in an arbitrary bounded metric space $\text{diam}(\text{cov}(A)) = \text{diam}(A)$, we have the following.

Proposition 6.5 *If $A \subseteq X$ is bounded, then*

$$\text{diam}(A) = \text{diam}(\text{conv}(A)) = \text{diam}(\overline{\text{conv}}(A)).$$

6.3 ℓ_2 revisited

There is some euclidean geometry inherited by ℓ_2 that will be needed in the next section. Let $x = (x_1, x_2, \dots)$, $y = (y_1, y_2, \dots) \in \ell_2$, and let

$$\langle x, y \rangle = \sum_{i=1}^{\infty} x_i y_i.$$

As in $\mathbb{R}^n$, the number $\langle x, y \rangle$ is called the *inner product* of x and y , and the fact that it is well defined follows from the Schwarz inequality, which now takes the form:

$$\langle x, y \rangle \leq \|x\| \|y\|.$$

Properties of the Inner Product: The following algebraic properties of the inner product follow instantly from the definition. For $x, y, z \in \ell_2$ and $\lambda \in \mathbb{R}$:

$$(i) \quad \langle x, x \rangle = \|x\|^2 \geq 0,$$

$$(ii) \quad \langle x, x \rangle = 0 \Leftrightarrow x = 0,$$

$$(iii) \quad \langle x, y \rangle = \langle y, x \rangle,$$

$$(iv) \quad \langle \lambda x, y \rangle = \lambda \langle x, y \rangle,$$

$$(v) \langle x, y + z \rangle = \langle x, y \rangle + \langle x, z \rangle.$$

Although there is nothing that obviously suggests this in the definition, a direct calculation shows that the inner product in ℓ_2 can be expressed explicitly in terms of the norm via the *Polarization Identity*:

$$(vi) \langle x, y \rangle = \frac{1}{4} \left[\|x + y\|^2 - \|x - y\|^2 \right].$$

A similar direct computation also establishes the *Parallelogram Law* :

$$(vii) \|x + y\|^2 + \|x - y\|^2 = 2 \left[\|x\|^2 + \|y\|^2 \right].$$

Keep in mind that only the real case is being considered here. If the scalar field is taken to be the complex numbers, then the inner product takes the form

$$\langle x, y \rangle = \sum_{i=1}^{\infty} x_i \bar{y}_i,$$

where $\bar{y}_i$ denotes the complex conjugate of y_i . In this case (iii) becomes $\langle x, y \rangle = \overline{\langle y, x \rangle}$.

The procedure of deriving an inner product from a norm is reversible. A linear space X in which an inner product, that is, a mapping $\langle \cdot, \cdot \rangle : X \times X \rightarrow \mathbb{R}$, can be defined which satisfies properties (i)–(v) is called an *inner product space*. Any inner product space X can be assigned a norm by taking

$$\|x\| = \sqrt{\langle x, x \rangle}, \quad x \in X,$$

and moreover it can be seen, again by a direct computation, that this norm satisfies the Parallelogram Law. The converse is true as well. If a normed linear space X satisfies the Parallelogram Law, then there exists an inner product on X which $\|x\| = \sqrt{\langle x, x \rangle}$, $x \in X$. The Polarization Identity suggests how to define this inner product. Thus if a normed linear space satisfies either the Polarization Identity or the Parallelogram Law, then it necessarily is an inner product space.

If an inner product space $(X, \|\cdot\|)$ is complete relative to its norm metric, then X is called a *Hilbert space*. In particular, ℓ_2 is a Hilbert space. There are many others, the most well known among these being the space $L_2[0, 1]$ of Lebesgue measurable functions on $[0, 1]$. (See the comments at the end of Section 6.5.)

In the study of calculus in $\mathbb{R}^n$ the inner product is $\langle x, y \rangle$ is called the dot product and denoted $x \cdot y$. Consider the well-known formula from calculus:

$$\cos \theta = \frac{\langle x, y \rangle}{\|x\| \|y\|},$$

where θ denotes the angle between the vectors x and y . By the Law of Cosines,

$$\|x - y\|^2 = \|x\|^2 + \|y\|^2 - 2\|x\|\|y\|\cos\theta.$$

Therefore,

$$2\langle x, y \rangle = \|x\|^2 + \|y\|^2 - \|x - y\|^2.$$

Replacing y with $-y$ and multiplying both sides by -1 gives

$$2\langle x, y \rangle = -\|x\|^2 - \|y\|^2 + \|x + y\|^2.$$

Note that upon adding the two preceding equations one obtains the Polarization Identity. This illustrates that the Polarization Identity is merely the Law of Cosines in another guise.

One useful consequence of the Parallelogram Law is the fact that if K is a closed convex subset of a Hilbert space H and if $z \in H \setminus K$, then there is a unique point Pz of K which is nearest to z . To see this, suppose, without loss of generality, that $z = 0 \notin K$, let $r = \inf\{\|x\| : x \in K\}$, and suppose $\{x_n\} \subset K$ satisfies $\lim_{n \rightarrow \infty} \|x_n\| = r$. Because

$$r \leq \left\| \frac{x_n + x_m}{2} \right\| \leq \frac{\|x_n\| + \|x_m\|}{2},$$

it must be the case that $\lim_{n, m \rightarrow \infty} \left\| \frac{x_n + x_m}{2} \right\| = r$. Thus by the Parallelogram Law,

$$\|x_n - x_m\|^2 = 2 \left[\|x_n\|^2 + \|x_m\|^2 \right] - \|x_n + x_m\|^2.$$

Since the right-hand side of the above equality tends to zero as $m, n \rightarrow \infty$, $\{x_n\}$ is a Cauchy sequence, and since K is closed there exists $x \in K$ such that $\lim_{n \rightarrow \infty} x_n = x$. Clearly $\|x\| = r$. If there were two such points, say $x, y \in K$, such that $\|x\| = \|y\| = r$ with $x \neq y$, then since K is convex $\frac{1}{2}(x + y) \in K$. Again, by the Parallelogram Law,

$$r^2 = \left\| \frac{x + y}{2} \right\|^2 = \frac{1}{2} \left[\|x\|^2 + \|y\|^2 \right] - \left\| \frac{x - y}{2} \right\|^2 = r^2 - \left\| \frac{x - y}{2} \right\|^2,$$

which is a contradiction. Therefore, the point of K having smallest norm is unique.

We now take a closer look at the mapping P which associates with each point $z \in H$ the point $P(z)$ of K which is nearest to z . The following result holds only in a Hilbert space.

Theorem 6.1 *Let K be a closed convex set in a Hilbert space H . Let $x, y \in H$, and suppose Px and P_y are points of K which are, respectively, nearest to x and y . Then*

$$\|Px - P_y\| \leq \|x - y\|.$$

The following will facilitate the proof.

Lemma 6.1 *Let $x \in K$ with K as above in Theorem 6.1, and let $p \in H$, $x \neq p$. Then x is the point of K nearest to p if and only if for all $z \in K$,*

$$\xi = \langle z - x, x - p \rangle \geq 0.$$

Proof. In view of the Law of Cosines this is intuitively clear. The lemma just asserts that the angle between the vector from p to x and the vector from x to z is at least $\frac{\pi}{2}$. A detailed proof using the Polarization Identity follows.

Suppose x is the point of K nearest to p and let $z \in K$. Since K is convex,

$$\lambda z + (1 - \lambda)x \in K \text{ for all } 0 \leq \lambda \leq 1.$$

Now

$$\begin{aligned} \lambda^2 \|x - z\|^2 + 2\lambda\xi &= \langle \lambda(x - z), \lambda(x - z) \rangle - 2\langle \lambda(x - z), x - p \rangle \\ &= \langle \lambda(x - z), [\lambda(x - z) - 2(x - p)] \rangle \\ &= \frac{1}{4} \left[\|2\lambda(x - z) - 2(x - p)\|^2 - \|2(x - p)\|^2 \right] \\ &= \left[\|p - (\lambda z + (1 - \lambda)x)\|^2 - \|x - p\|^2 \right] \\ &\geq 0. \end{aligned}$$

Thus $\lambda \|x - z\|^2 + 2\xi \geq 0$ for $\lambda \in (0, 1]$ and upon letting $\lambda \rightarrow 0^+$ we conclude $\xi \geq 0$.

For the converse, suppose $\xi \geq 0$. Then

$$\begin{aligned} \|z - p\|^2 - \|p - x\|^2 &= \langle z - p, z - p \rangle - \langle p - x, p - x \rangle \\ &= \langle z, z \rangle - 2\langle z, p \rangle + 2\langle p, x \rangle - \langle x, x \rangle \\ &= \langle z, z \rangle - 2\langle z, x \rangle + \langle x, x \rangle + 2\langle z, x \rangle \\ &\quad - 2\langle p, z \rangle + 2\langle p, x \rangle - 2\langle x, x \rangle \\ &= \langle z - x, z - x \rangle + 2 \left[\langle p, x \rangle + \langle z, x \rangle - \langle p, z \rangle - \langle x, x \rangle \right] \\ &= \|z - x\|^2 + 2\langle z - x, x - p \rangle \\ &\geq 0. \end{aligned}$$

■

Proof of Theorem 6.1. Let $A = \langle Px - P_y, P_y - y \rangle$ and $B = \langle P_y - Px, Px - x \rangle$. By Lemma 6.1, $A \geq 0$ and $B \geq 0$. Write $B = \langle Px - P_y, x - Px \rangle$. Then

$$\begin{aligned} A + B &= \langle 2(Px - P_y), x - y + P_y - Px \rangle \\ &= 2[\langle Px - P_y, x - y \rangle - \|Px - P_y\|^2]. \end{aligned}$$

Since $A + B \geq 0$, this gives

$$\|Px - Py\|^2 \leq \langle Px - Py, x - y \rangle.$$

But the Schwarz inequality implies

$$\langle Px - Py, x - y \rangle \leq \|Px - Py\| \|x - y\|.$$

Combining the two previous inequalities leads to the desired conclusion:

$$\|Px - Py\| \leq \|x - y\|.$$

■

6.4 The modulus of convexity

Interest in the modulus of convexity arose from a careful study of geometric properties of the ℓ_2 space. Indeed, let x and y be any vectors in ℓ_2 . The Parallelogram Law implies

$$\|x + y\|^2 = 2\left[\|x\|^2 + \|y\|^2\right] - \|x - y\|^2.$$

So if we assume that x and y are in the unit ball and bounded away from each other, meaning that $\|x - y\| \geq \varepsilon > 0$, then

$$\|x + y\|^2 \leq 4 - \varepsilon^2,$$

which implies

$$\left\| \frac{x + y}{2} \right\| \leq \sqrt{1 - \frac{\varepsilon^2}{2}}.$$

Thus the midpoint of the segment joining x and y is uniformly bounded away from the surface of the unit ball by a factor depending on ε . This property is known as *uniform convexity*.

Definition 6.4 *The modulus of convexity of a Banach space X is the function $\delta_X : [0, 2] \rightarrow [0, 1]$ defined by*

$$\delta_X(\varepsilon) = \inf \left\{ 1 - \left\| \frac{x + y}{2} \right\| : \|x\| \leq 1, \|y\| \leq 1, \|x - y\| \geq \varepsilon \right\}. \quad (6.1)$$

In working within a specific space we generally drop the subscript X from δ_X and simply write δ . Note that for a given space the number $\delta(\varepsilon)$ is the largest number for which the following inequality holds:

$$\left. \begin{array}{l} \|x\| \leq 1 \\ \|y\| \leq 1 \\ \|x - y\| \geq \varepsilon \end{array} \right\} \Rightarrow \left\| \frac{x + y}{2} \right\| \leq 1 - \delta(\varepsilon). \quad (6.2)$$

One very useful observation is the rather trivial fact that the above inequality is equivalent to the following: If $x, y, p \in X$, $R > 0$, and $r \in [0, 2R]$, then

$$\left. \begin{array}{l} \|x - p\| \leq R \\ \|y - p\| \leq R \\ \|x - y\| \geq r \end{array} \right\} \Rightarrow \left\| \frac{x + y}{2} \right\| \leq \left(1 - \delta \left(\frac{r}{R} \right) \right) R.$$

Related to the modulus of convexity is another important function.

Definition 6.5 *The characteristic (or coefficient) of convexity of a Banach space X is the number*

$$\varepsilon_0 = \varepsilon_0(X) = \sup\{\varepsilon \geq 0 : \delta(\varepsilon) = 0\}.$$

The following is very important, but we will not give the proof here. (For a proof see, e.g., [67].)

Proposition 6.6 *Let X be a Banach space with modulus of convexity δ and characteristic of convexity ε_0 . Then δ is continuous on $[0, 2)$ and strictly increasing on $[\varepsilon_0, 2]$.*

The function δ can actually be *discontinuous* at $\varepsilon = 2$. However, it is always the case that

$$\lim_{\varepsilon \rightarrow 2^-} \delta(\varepsilon) = 1 - \frac{\varepsilon_0}{2}.$$

In general, it is quite difficult to calculate the modulus of convexity of a Banach space, but for ℓ_2 the situation is clear.

Example 6.5 *We have seen that the Parallelogram Law holds for ℓ_2 . Thus in ℓ_2 , if $\|x\| = \|y\| = 1$ and $\|x - y\| = \varepsilon \in [0, 2]$, then by what we have seen above,*

$$\delta_2(\varepsilon) \geq 1 - \left\| \frac{x + y}{2} \right\| \geq 1 - \sqrt{1 - \left(\frac{\varepsilon}{2} \right)^2}.$$

A straightforward calculation shows that the reverse inequality holds (see the proof of Theorem 6.2 in the next section), so for ℓ_2 we have

$$\delta_2(\varepsilon) = 1 - \sqrt{1 - \left(\frac{\varepsilon}{2} \right)^2}.$$

An analogous formula, which we take up in more detail in the next section, holds for the modulus of convexity δ_p for the spaces $X = \ell_p$ (or L_p) ($2 < p < \infty$):

$$\delta_p(\varepsilon) = 1 - \left(1 - \left(\frac{\varepsilon}{2} \right)^p \right)^{1/p}.$$

The L_p spaces are a natural prototype of spaces which satisfy the following more general condition.

Definition 6.6 A Banach space X with modulus of convexity δ is said to be uniformly convex if $\delta(\varepsilon) > 0$ for each $\varepsilon \in (0, 2]$, or equivalently, if $\varepsilon_0(X) = 0$. If $\varepsilon_0(X) < 2$, then X is said to be uniformly nonsquare.

Every nonempty closed convex subset of a uniformly convex space contains a unique element of minimal norm (see Exercise 6.10). In fact, a Banach space may satisfy this conclusion yet fail to be uniformly convex.

Definition 6.7 A Banach space X is said to be strictly convex (or have strictly convex norm) if for $x, y, z \in X$

$$\|x - z\| = \|y - z\| = \frac{1}{2} \|x - y\| \Rightarrow z = \frac{x + y}{2}.$$

The properties in this section are purely geometric. Later we shall combine some of these ideas with (weak) topological properties of the space to classify various categories of Banach spaces.

6.5 Uniform convexity of the ℓ_p spaces

We begin with the following elementary lemma. The key ingredient is a fundamental inequality from analysis known as Hölder's inequality, which asserts that if $p, q \in \mathbb{R}^+$ and if $1/p + 1/q = 1$, then for $x_i, y_i \in \mathbb{R}$, $i = 1, \dots, n$,

$$\sum_{i=1}^n x_i y_i \leq \left(\sum_{i=1}^n |x_i|^p \right)^{1/p} \left(\sum_{i=1}^n |y_i|^q \right)^{1/q}.$$

Lemma 6.2 If $2 \leq p < \infty$ and if $a, b \in \mathbb{R}^+$, then

- (i) $a^p + b^p \leq (a^2 + b^2)^{p/2}$;
- (ii) $(a^2 + b^2)^{p/2} \leq 2^{(p-2)/2} (a^p + b^p)$.

Proof. Both inequalities hold if either a or b is 0, so we may assume $a \neq 0 \neq b$. Since

$$\frac{a^2}{a^2 + b^2} \leq 1 \quad \text{and} \quad \frac{b^2}{a^2 + b^2} \leq 1$$

and $p/2 \geq 1$, we have

$$\begin{aligned} \frac{a^p}{(a^2 + b^2)^{p/2}} + \frac{b^p}{(a^2 + b^2)^{p/2}} &= \left[\frac{a^2}{a^2 + b^2} \right]^{p/2} + \left[\frac{b^2}{a^2 + b^2} \right]^{p/2} \\ &\leq \frac{a^2}{a^2 + b^2} + \frac{b^2}{a^2 + b^2} = 1. \end{aligned}$$

Thus (i) holds. Note that part (ii) is trivial if $p = 2$, so assume $p > 2$, and set $p' = p/2 > 1$ and $q' = p'(p' - 1) = p/(p - 2)$. Then $1/p' + 1/q' = 1$, so by Hölder's inequality,

$$\begin{aligned} a^2 + b^2 &\leq \left[(a^2)^{p'} + (b^2)^{p'} \right]^{1/p'} \left[1^{q'} + 1^{q'} \right]^{1/q'} \\ &= [a^p + b^p]^{2/p} [2^{(p-2)/p}]. \end{aligned}$$

This leads to

$$(a^2 + b^2)^{p/2} \leq 2^{(p-2)/2} (a^p + b^p).$$

■

Theorem 6.2 *The modulus of convexity δ_p of ℓ_p , ($2 < p < \infty$), satisfies*

$$\delta_p(\varepsilon) = 1 - \left(1 - \left(\frac{\varepsilon}{2} \right)^p \right)^{1/p}, \quad \varepsilon \in (0, 2).$$

Proof. First observe that if $u, v \in \mathbb{R}$ then by Lemma 6.2

$$|u + v|^p + |u - v|^p \leq \left[|u + v|^2 + |u - v|^2 \right]^{p/2}$$

and

$$\left[|u|^2 + |v|^2 \right]^{p/2} \leq 2^{(p-2)/2} [|u|^p + |v|^p].$$

Consequently,

$$\begin{aligned} |u + v|^p + |u - v|^p &\leq \left[|u + v|^2 + |u - v|^2 \right]^{p/2} \\ &= \left[2|u|^2 + 2|v|^2 \right]^{p/2} \\ &\leq 2^{p-1} [|u|^p + |v|^p]. \end{aligned}$$

Therefore, if $\mathbf{u} = (u_1, u_2, \dots)$, $\mathbf{v} = (v_1, v_2, \dots) \in \ell_p$

$$\sum_{i=1}^{\infty} |u_i + v_i|^p + \sum_{i=1}^{\infty} |u_i - v_i|^p \leq 2^{p-1} \left[\sum_{i=1}^{\infty} |u_i|^p + \sum_{i=1}^{\infty} |v_i|^p \right].$$

In other words,

$$\|\mathbf{u} + \mathbf{v}\|^p + \|\mathbf{u} - \mathbf{v}\|^p \leq 2^{p-1} [\|\mathbf{u}\|^p + \|\mathbf{v}\|^p]$$

for each $\mathbf{u}, \mathbf{v} \in \ell_p$.

Now let $\varepsilon \in (0, 2)$ and suppose $\mathbf{u}, \mathbf{v} \in \ell_p$ satisfy $\|\mathbf{u}\| \leq 1$, $\|\mathbf{v}\| \leq 1$, and $\|\mathbf{u} - \mathbf{v}\| \geq \varepsilon$. Then

$$\begin{aligned} \left\| \frac{\mathbf{u} + \mathbf{v}}{2} \right\|^p &\leq 2^{p-1} \left[\left\| \frac{\mathbf{u}}{2} \right\|^p + \left\| \frac{\mathbf{v}}{2} \right\|^p \right] - \left\| \frac{\mathbf{u} - \mathbf{v}}{2} \right\|^p \\ &\leq 1 - \left(\frac{\varepsilon}{2} \right)^p. \end{aligned}$$

From this it follows that

$$\delta_p(\varepsilon) \geq 1 - \left(1 - \left(\frac{\varepsilon}{2}\right)^p\right)^{1/p}.$$

For the reverse inequality, let $\varepsilon \in (0, 2)$, let $\mathbf{u} = (u_1, u_2) \in \ell_p^{(2)}$ satisfy

$$u_1 = \left(1 - \left(\frac{\varepsilon}{2}\right)^p\right)^{1/p} \quad \text{and} \quad u_2 = \frac{\varepsilon}{2},$$

and take $\mathbf{v} = (u_1, -u_2)$. Then $\|\mathbf{u}\| = \|\mathbf{v}\| = 1$. Moreover, it is easy to see that $\|\mathbf{u} - \mathbf{v}\| = \varepsilon$, while

$$\left\|\frac{\mathbf{u} + \mathbf{v}}{2}\right\| = \left(1 - \left(\frac{\varepsilon}{2}\right)^p\right)^{1/p}.$$

■

An implicit formula for δ_p is also known for the spaces $X = \ell_p$ (or L_p) ($1 < p < 2$). In 1956 Hanner showed that for $1 < p < 2$ the following inequality holds:

$$|||x| - |y|||^p + |||x| + |y|||^p \leq \|x - y\|^p + \|x + y\|^p \quad x, y \in \ell_p$$

and from this he obtained the precise implicit formula

$$\left|1 - \delta_p(\varepsilon) + \frac{\varepsilon}{2}\right|^p + \left|1 - \delta_p(\varepsilon) - \frac{\varepsilon}{2}\right|^p = 2.$$

Therefore, for all the spaces ℓ_p ($1 < p < \infty$) it is the case that $\delta_p(\varepsilon) > 0$ whenever $\varepsilon > 0$.

So far we have concentrated on the sequential ℓ_p and said nothing about their functional analogs, the important class of L_p spaces.

Let $(X, \mathcal{S}, \mu)$ be a finite measure space. This means that X is a set, $\mathcal{S}$ is a σ -algebra of subsets of X , and $\mu : \mathcal{S} \rightarrow \mathbb{R}$ is a finite measure. (We consider only the real case.) In particular $X \in \mathcal{S}$, the complement of any set in $\mathcal{S}$ is also in $\mathcal{S}$, and $\mathcal{S}$ is closed under countable unions. The measure μ is a real-valued non-negative function defined on $\mathcal{S}$ for which $\mu(\emptyset) = 0$ and

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu(A_n)$$

for every disjoint countable family $\{A_n\}$ in $\mathcal{S}$. A mapping $f : X \rightarrow \mathbb{R}$ is said to be μ -measurable if $f^{-1}(U) \in \mathcal{S}$ for every open set $U \subseteq \mathbb{R}$.

For each p , $1 \leq p < \infty$, the space $L_p(\mu) = L_p(X, \mathcal{S}, \mu)$ is the space of equivalence classes of μ -measurable functions f for which $\int_X |f|^p d\mu < \infty$. Two

elements $f, g \in L_p(\mu)$ are said to be equivalent if $\int_X |f - g|^p d\mu = 0$, and the norm of an element $f \in L_p(\mu)$ is given by

$$\|f\|_p = \left(\int_X |f|^p d\mu \right)^{1/p}.$$

The norm for $L_\infty(\mu)$ is a little more technical. $L_\infty(\mu)$ consists of those μ -measurable functions f for which

$$\|f\|_{L_\infty} = \text{ess sup}\{|f(\omega)| : \omega \in X\} = \inf_{E \in \mathcal{N}} \sup_{X \setminus E} |f| < \infty,$$

where $\mathcal{N} = \{E \in \mathcal{S} : \mu(E) = 0\}$.

It is possible to prove that the spaces $L_p(\mu)$ are uniformly convex for $1 < p < \infty$ by making minor adjustments in the argument for the ℓ_p case. If $u, v \in L_p(\mu)$ then the line of argument used in Theorem 6.2 leads to the fact that for almost all $t \in X$

$$|u(t) + v(t)|^p + |u(t) - v(t)|^p \leq 2^{p-1} (|u(t)|^p + |v(t)|^p).$$

On integrating both sides of the above we obtain

$$\|u + v\|^p + \|u - v\|^p \leq 2^{p-1} (\|u\|^p + \|v\|^p)$$

for each $u, v \in L_p(\mu)$. The argument can now be completed as before. Thus we have the following.

Theorem 6.3 *The spaces $L_p(\mu)$, $1 < p < \infty$, are uniformly convex.*

Of course, the ℓ_p spaces can be thought of as a special subclass of the $L_p(X, \mathcal{S}, \mu)$ spaces with $X = \mathbb{N}$ and μ the counting measure on bounded subsets S of $\mathbb{N}$.

Among the L_p spaces, the case $p = 2$ is of particular interest. If μ is taken to be Lebesgue measure on $[0, 1]$ then $L_2[0, 1]$ is the space consisting of all measurable functions f with $|f|^2$ summable, identifying those functions which differ only on a set of measure zero. This space, is in, fact metrically identical to ℓ_2 . The correspondence is obtained by assigning to each $f \in L_2$ the sequence $(a_1, a_2, \dots)$ of its Fourier coefficients. Then the corresponding series $\sum_{n=1}^{\infty} a_n^2$ converges. On the other hand, if the series $\sum_{n=1}^{\infty} a_n^2$ converges, then it can be shown that $a_1, a_2, \dots$, are the Fourier coefficients of some $f \in L_2$. It follows that the spaces L_2 and ℓ_2 are isometric and share the same metric properties.

6.6 The dual space: Hahn-Banach Theorem

If X is a Banach space, then a mapping $f : X \rightarrow \mathbb{R}$ is called a *linear functional* if for each $x, y \in X$ and $\alpha, \beta \in \mathbb{R}$,

$$f(\alpha x + \beta y) = \alpha f(x) + \beta f(y).$$

The space of all *continuous* linear functionals on X is denoted X^* . Since $\lim_{n \rightarrow \infty} x_n = x$ if and only if $\lim_{n \rightarrow \infty} (x_n - x) = 0$, if a linear functional is continuous at 0 then it is continuous on the whole space. Also, if f is a linear functional and if f is *bounded*, in the sense that there exists a constant c such that for each $x \in X$

$$|f(x)| \leq c \|x\|,$$

then clearly f is continuous at 0. The converse is true as well.

Proposition 6.7 *A continuous linear functional f on a Banach (or normed) space X is always bounded.*

Proof. Assume f is not bounded. Then for each $n \in \mathbb{N}$ there exists $x_n \in X$ such that

$$f(x_n) > n \|x_n\| \quad n = 1, 2, \dots$$

Since $f(0) = 0$ this, in turn, implies $x_n \neq 0$. Let $y_n = \frac{x_n}{n \|x_n\|}$, $n = 1, 2, \dots$.

Then $\lim_{n \rightarrow \infty} y_n = 0$. On the other hand, $f(y_n) = f\left(\frac{x_n}{n \|x_n\|}\right) > 1$, and this implies f is not continuous at 0. ■

In view of Proposition 6.7, for each $f \in X^*$ there is a number $\|f\|_* \in \mathbb{R}^+$ such that

$$\|f\|_* = \sup\{|f(x)| : x \in S_X\},$$

where, as usual, $S_X = \{x \in X : \|x\| = 1\}$. It is a fairly straightforward matter to verify that the space $(X^*, \|\cdot\|_*)$, called the *dual* (or *conjugate*) space of X , is, in fact, a normed space. Although it is a little less obvious, $(X^*, \|\cdot\|_*)$ is also complete.

Proposition 6.8 *If X is a normed space, then its dual space $(X^*, \|\cdot\|_*)$ is always a Banach space.*

Proof. Let $\{f_n\} \subseteq X^*$ be a Cauchy sequence. Then given $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that if $m, n \geq N$, $\|f_n - f_m\|_* \leq \varepsilon$. Thus for $m, n \geq N$

$$\|f_n - f_m\|_* = \sup\{|(f_n - f_m)(x)| : x \in S_X\},$$

from which, for each $x \in X$

$$|f_n(x) - f_m(x)| = |(f_n - f_m)(x)| \leq \|f_n - f_m\|_* \|x\| \leq \varepsilon \|x\|.$$

So for each such x , $\{f_n(x)\}$ is a Cauchy sequence in $\mathbb{R}$ and has a limit which we denote $f(x)$. Also if $x, y \in S_X$,

$$f(\alpha x + \beta y) = \lim_{n \rightarrow \infty} f_n(\alpha x + \beta y) = \alpha \lim_{n \rightarrow \infty} f_n(x) + \beta \lim_{n \rightarrow \infty} f_n(y),$$

which implies $f(\alpha x + \beta y) = \alpha f(x) + \beta f(y)$. Therefore, f is a *linear functional*. To see that $f \in X^*$ we only need to show that f is bounded. We have already observed that every Cauchy sequence is bounded. Therefore, there exists $M \in \mathbb{R}$ such that

$$M = \sup\{\|f_n\|_* : n \in \mathbb{N}\}.$$

Let $\varepsilon > 0$ and choose N as above. Then for each $x \in X$ and $m, n \geq N$,

$$|f_n(x)| \leq |f_n(x) - f_m(x)| + |f_m(x)| \leq (\varepsilon + M) \|x\|.$$

Letting $n \rightarrow \infty$ gives $|f(x)| \leq (\varepsilon + M) \|x\|$, so f is bounded, hence continuous. Finally, for $n \geq N$,

$$\|f - f_n\|_* = \sup\{|f(x) - f_n(x)| : x \in S_X\} \leq \varepsilon.$$

This proves $\lim_{n \rightarrow \infty} f_n = f$ in $(X^*, \|\cdot\|_*)$. ■

Proposition 6.8 offers an easy way to see that certain normed spaces are complete. For example, once it is known that ℓ_1 is complete and that $(\ell_1)^* = \ell_\infty$ (Section 6.8), then completeness of ℓ_∞ follows.

We now state, without proof, the Hahn-Banach Theorem and one of its many useful consequences. The Hahn-Banach Theorem is usually formulated in a slightly more abstract setting and the standard proof, which utilizes Zorn's Lemma, can be found in many texts.

Theorem 6.4 (Hahn-Banach) *Let X be a Banach space and let H be a linear subspace of X . Then given any continuous linear functional $f \in H^*$ there is a continuous linear functional $\tilde{f} \in X^*$ such that*

- (i) $\tilde{f}(x) = f(x)$ for each $x \in H$; and
- (ii) $\|\tilde{f}\|_* \leq \|f\|_*$.

It is important to note that the norm $\|f\|_*$ in the above statement is relative to the dual space H^* .

Corollary 6.1 *Let X be a normed space and let $x \in X$, $x \neq 0$. Then there exists $f_x \in X^*$ such that $\|f_x\| = 1$ and for which $f_x(x) = \|x\|$.*

6.7 The weak and weak* topologies

In our discussion of Tychonoff's Theorem in the Appendix we describe the product topology in terms of 'net' convergence. This is also a convenient way to describe the weak and weak* topologies. Let X be a Banach space. We shall say that a net $\{x_\alpha\}$ *converges to* $x \in X$ *in the weak topology* (or *converges weakly to* x) if $\lim_\alpha f(x_\alpha) = f(x)$ for each $f \in X^*$. When this is true we write

$$w\text{-}\lim_\alpha x_\alpha = x.$$

A subset K of X is *weakly closed* if it is closed in the weak topology, that is, if it contains the weak limit of each of its weakly convergent nets. The *weakly open* sets are now taken as those sets whose complements are weakly closed. The resulting topology on X is called the *weak topology* on X . Sets which are compact in this topology are said to be *weakly compact*. It can be shown that the weak topology is the weakest topology for which all the functionals $f \in X^*$ are, in fact, continuous.

It is important to know that the weak topology on a Banach space is a Hausdorff topology, and that weak limits are unique. This is because the functionals in X^* *separate points* in X , that is, given any two points $x, y \in X$ with $x \neq y$ there exists an $f \in X^*$ such that $f(x) \neq f(y)$. This is another consequence of the Hahn-Banach Theorem.

For $x \in X$ and $f \in X^*$ define $i(x)(f) = f(x)$. It is easy to see that $i(x) \in X^{**}$ and that, in fact, the mapping $i : X \rightarrow X^{**}$ is an isometric isomorphism, called the *canonical embedding* of X into X^{**} . If $i(X) = X^{**}$ then X is said to be *reflexive*.

Now observe that if X^* is the dual of a Banach space then it too has a weak topology determined by its dual space X^{**} . Thus $\{f_\alpha\} \subseteq X^*$ converges weakly to $f \in X^*$ if $\lim_\alpha F(f_\alpha) = F(f)$ for each $F \in X^{**}$. However, as we have seen above, there is a natural isomorphism between X and a subspace, say $i(X)$, of X^{**} . If it is the case that $\lim_\alpha x^{**}(f_\alpha) = x^{**}(f)$ for each $x^{**} \in i(X)$ then $\{f_\alpha\}$ is said to converge to f in the weak* topology. When this is the case we write

$$w^*\text{-}\lim_\alpha f_\alpha = f.$$

Note that one always has the following implication:

$$w\text{-}\lim_\alpha f_\alpha = f \Rightarrow w^*\text{-}\lim_\alpha f_\alpha = f.$$

However, the converse only holds if X (hence X^*) is reflexive.

A subset K of X^* is said to be *weak* closed* if it contains the weak* limit of each of its weak* converging sequences. The *weak* open* sets are now taken as those sets whose complements are weak* closed. The resulting topology on X^* is called the *weak* topology* on X^* . It is important to notice that this topology depends on X as well as X^* . It is possible for two different Banach spaces to have the same dual space X^* ; consequently X^* could have two different weak*

topologies, one generated by each of its preduals.

Another concept which will be useful for us in the subsequent sections is the adjoint of operators.

Definition 6.8 *Let X and Y be two Banach spaces. For any operator (continuous linear mapping) $T : X \rightarrow Y$, define the adjoint operator $T^* : Y^* \rightarrow X^*$ defined by*

$$T^*(y^*) = y^* \circ T$$

or

$$T^*(y^*)(x) = y^*(T(x))$$

for any $x \in X$.

The adjoint operator has some fundamental properties which play a crucial role in the study of Banach spaces. In particular, we have

$$\|T^*\| = \|T\|,$$

where $\|T\|$ is defined by $\|T\| = \sup\{\|T(x)\| : x \in S_X\}$.

Another common notation for the weak topology is $\sigma(X, X^*)$. This suggests the topology on X induced by X^* . In this notation $\sigma(X^*, X^{**})$ denotes the weak topology on X^* , whereas $\sigma(X^*, X)$ denotes the weak* topology on X^* .

We conclude this section with another consequence of the Hahn-Banach Theorem. It will be used later, for example, in the proof of Theorem 9.12, Chapter 9.

Theorem 6.5 (Basic Separation Theorem) *Let A and B be disjoint convex subsets of a Banach space X .*

(a) *If A is open, then there exists $x^* \in X^*$ and a number $r \in \mathbb{R}$ such that*

$$x^*(x) > r \text{ if } x \in A \text{ and } x^*(x) \leq r \text{ if } x \in B.$$

(b) *If B is compact (or, more generally, weakly compact) and A is closed, then there exist $x^* \in X^*$ and $z \in B$ such that*

$$x^*(z) = \|z\| = \max\{x^*(x) : x \in B\} < \inf\{x^*(x) : x \in A\}.$$

6.8 The spaces c , c_0 , ℓ_1 and ℓ_∞

The space c is the Banach space of all sequences of real numbers which converge and c_0 is the subspace of c consisting of those sequences converging to 0. In each case $\|x_n\| = \sup\{|x_n| : n = 1, 2, \dots\}$ and thus the following subspace inclusions hold: $c_0 \subseteq c \subseteq \ell_\infty$.

A sequence $\{x_n\}$ in a (real) Banach space is called a (Schauder) *basis* for X if for each $x \in X$ there is a unique sequence $\{a_n\}$ of real numbers such that

$$x = \lim_{n \rightarrow \infty} \sum_{i=1}^n a_i x_i = \sum_{i=1}^{\infty} a_i x_i.$$

The first three classical sequence spaces listed in the section heading all have bases while ℓ_∞ , since it is not separable, does not. In fact, the classical unit vectors $\{e^i\}$, where

$$e^i = (0, \dots, 1, 0, \dots)$$

and 1 appears in the i th position and 0's appear elsewhere can be shown to be a basis for both c_0 and ℓ_1 .

Among other things we show in this section that the dual spaces of both c and c_0 are isometrically isomorphic to ℓ_1 . Moreover $(\ell_1)^* = \ell_\infty$. These facts lead to three interesting questions. If $\{u^k\}_{k=1}^\infty$ is a bounded sequence in ℓ_1 when does $\{u^k\}$ converge weakly; when does $\{u^k\}$ converge in the weak* topology regarding ℓ_1 as the dual of c ; when does $\{u^k\}$ converge in the weak* topology regarding ℓ_1 as the dual of c_0 ?

The answer to these questions should help to sort out subtle differences in the concepts, and the answer to the first question is a little surprising. A bounded sequence $\{u^k\}$ in ℓ_1 converges weakly to $u \in \ell_1$ if and only if $\lim_{k \rightarrow \infty} \|u^k - u\| = 0$. Spaces which have this property are said to have the *Schur property*. Thus the weakly compact subsets of such spaces coincide with the norm compact subsets. (Note, however, that ℓ_1 is *not* finite-dimensional.) We discuss the Schur property of ℓ_1 in detail later in Section 6.10.

The latter two questions are a little less complicated. First we show that (up to isomorphism) $c^* = c_0^* = \ell_1$.

Let $y = (y_1, y_2, \dots) \in \ell_1$ and $(x_1, x_2, \dots) \in c_0$, and define

$$y(x) = \sum_{i=1}^{\infty} x_i y_i.$$

This defines a *linear* functional. Also,

$$|y(x)| \leq \sum_{i=1}^{\infty} |x_i y_i| \leq \|x\|_\infty \sum_{i=1}^{\infty} |y_i| = \|x\|_\infty \|y\|_1,$$

so y is a bounded (continuous) linear functional with $\|y\|_* \leq \|y\|_1$. Now let $\varepsilon > 0$ and choose N so that

$$\sum_{i=1}^N |y_i| > \|y\|_1 - \varepsilon.$$

Let

$$x = (\operatorname{sgn} y_1, \operatorname{sgn} y_2, \dots, \operatorname{sgn} y_N, 0, \dots) \in c_0,$$

where “sgn” denotes the function which assigns to a number 1, -1, or 0 according to whether the number, respectively, is positive, negative, or zero. Then

$$y(x) = \sum_{i=1}^N |y_i| > \|y\|_1 - \varepsilon$$

with $\|x\|_\infty = 1$. It follows that $\|y\|_* \geq \|y\|_1$; thus $\|y\|_* = \|y\|_1$.

On the other hand, if $f \in c_0^*$ and $x = (x_1, x_2, \dots) \in c_0$, then since the vectors $\{e^i\}$ form a basis for c_0 we have $x = \lim_{n \rightarrow \infty} \sum_{i=1}^n x_i e^i$. This leads to

$$f(x) = f\left(\lim_{n \rightarrow \infty} \sum_{i=1}^n x_i e^i\right) = \lim_{n \rightarrow \infty} \sum_{i=1}^n x_i f(e^i) = \sum_{i=1}^{\infty} x_i f(e^i).$$

In particular, $\sum_{i=1}^{\infty} |x_i f(e^i)|$ exists, that is,

$$y = (f(e^1), f(e^2), \dots) \in \ell_1,$$

and moreover $\|y\|_1 \leq \|f\|_*$. At the same time, $\|f\|_* \geq \|y\|_1$ because

$$|f(x)| = \left| \sum_{i=1}^{\infty} x_i f(e^i) \right| \leq \|x\|_\infty \sum_{i=1}^{\infty} |f(e^i)| = \|x\|_\infty \|y\|_1.$$

Therefore, the mapping $\varphi : c_0^* \rightarrow \ell_1$ defined by

$$\varphi(f) = (f(e^1), f(e^2), \dots)$$

is an isometry of c_0^* onto ℓ_1 .

A subtlety arises in showing that ℓ_1 is also isomorphic to c^* because the vectors $\{e^i\}$ do not form a basis for c . Suppose $x = (x_1, x_2, \dots) \in c$. Then $\lim_{n \rightarrow \infty} x_n = L \in \mathbb{R}$ and

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n (x_i - L)e^i = \sum_{i=1}^{\infty} (x_i - L)e^i = (x_1 - L, x_2 - L, \dots) \in c_0.$$

Moreover, $\sum_{i=1}^{\infty} (x_i - L)e^i = x - L \cdot e$, where $e = (1, 1, \dots)$. Thus $x = L \cdot e + \sum_{i=1}^{\infty} (x_i - L)e^i$. It follows that the sequence $\{e^i\}$ along with the vector e provides a basis for c . Moreover it can be shown that the mapping $\Phi : c^* \rightarrow \ell_1$ defined by

$$\Phi(f) = (f(e) - \sum_{i=1}^{\infty} f(e^i), f(e^1), f(e^2), \dots)$$

is an isometry of c^* onto ℓ_1 . [Note that the convergence of $\sum_{i=1}^{\infty} f(e^i)$ is assured because $f \in c^*$ implies $f \in c_0^*$; hence $\varphi(f) = (f(e^1), f(e^2), \dots) \in \ell_1$.]

Since the basis vectors $e^i \in c_0 \subset c$, $i = 1, 2, \dots$, if $x = (x_1, x_2, \dots) \in \ell_1$, then the function defined by $e^i(x) = x_i$ can be thought of as an element of ℓ_1^* which is identified under the natural isomorphism with an element of c_0 (or c). Thus a *necessary* condition for a bounded sequence in ℓ_1 to converge in the weak* topology induced on ℓ_1 by either c_0 or c is that it converge coordinatewise. In the case of $\ell_1 = c_0^*$, however, coordinatewise convergence is also a *sufficient* condition for weak* convergence of bounded sequences in ℓ_1 . This is because $\{e^i\}$ is a basis for c_0 .

We conclude by remarking that reasoning similar to the above shows that $\ell_1^* = \ell_\infty$ and that a bounded sequence in ℓ_∞ converges in the weak* topology induced by ℓ_1 if and only if it converges coordinatewise to an element of ℓ_∞ . Similarly, weak convergence of a bounded sequence in ℓ_1 implies coordinatewise convergence. Also a bounded sequence $\{x^n\}$ in c_0 converges weakly if and only if it converges coordinatewise to $x \in c_0$, while a bounded sequence $\{x^n\}$ in c converges weakly to $x \in c$ if and only if it converges coordinatewise to x and $\lim_{n \rightarrow \infty} \lim_{i \rightarrow \infty} x_i^n = \lim_{i \rightarrow \infty} x_i$.

6.9 Some more general facts

We now collect for later reference some well-known properties of the weak and weak* topologies. Despite the fact that proofs of these results can be found in any standard functional analysis text we include selected details.

We begin with yet another consequence of the Hahn-Banach Theorem.

Proposition 6.9 *A convex subset K of a Banach space is weakly closed if and only if it is closed.*

The above leads to the following.

Proposition 6.10 *If a subset K of a Banach space is weakly compact, then $\overline{\text{conv}}(K)$ is also weakly compact.*

Neither of the above facts holds for the weak* topology. However the following holds for both topologies.

Proposition 6.11 *If $K \subseteq X$ is weakly compact (or weak* compact if X is a dual space), then K is bounded.*

This is a consequence of the following classical result, which is also known as the *uniform boundedness principle*.

Proposition 6.12 (*Banach Steinhaus Theorem*) *Let X be a Banach space. Then a subset S of X^* is bounded in norm if and only if $\{f(x) : x \in X\}$ is bounded for each $f \in S$.*

The Banach Steinhaus Theorem can be used to prove Proposition 6.11 by identifying K with its natural imbedding in X^{**} and invoking the fact that a continuous real-valued function defined on a compact set is bounded.

The following fact holds only for the weak* topology (except, of course, in reflexive spaces).

Proposition 6.13 (*Alaoglu's Theorem*) *The unit ball B_{X^*} (hence any ball) in a dual space X^* is always compact in the weak* topology.*

Proof. This follows nicely from Tychonoff's Theorem. Let

$$B_{X^*} = \{f \in X^* : \|f\|_* \leq 1\}$$

and recall that for a net $\{f_\alpha\} \subset B_{X^*}$, $\text{weak}^* \lim f_\alpha = 0 \Leftrightarrow \lim_\alpha f_\alpha(x) = 0$ for all $x \in B_X$. Consider the product space $\prod_{x \in B_X} I_x$, where $I_x = [-1, 1]$ for each $x \in X$. Then for each $f \in B_{X^*}$, $f(x) \in I_x$ for each $x \in B_X$. Thus $f \in \prod_{x \in B_X} I_x$, that is,

$$B_{X^*} \subset \prod_{x \in B_X} I_x.$$

Since the weak* topology on B_{X^*} is just the topology of coordinatewise convergence, it coincides with the product topology on $\prod_{x \in B_X} I_x$, and since $\prod_{x \in B_X} I_x$ is compact in the product topology by Tychonoff's Theorem (each of the sets I_x is compact), it only needs to be shown that B_{X^*} is a closed subset of $\prod_{x \in B_X} I_x$.

To see this, let $\{f_\alpha\}$ be a net in B_{X^*} which converges in the product topology to $f \in \prod_{x \in B_X} I_x$. We need to show that $f \in B_{X^*}$. However, since $\{f_\alpha\}$ converges to f in the product topology, $\lim_\alpha f_\alpha(x) = f(x)$ for each $x \in B_X$. Clearly this implies f is linear. Moreover, since $|f_\alpha(x)| \leq |f_\alpha| \leq 1$, it follows that $\lim_\alpha |f_\alpha(x)| \leq 1$, from which $|f(x)| \leq 1$ for each $x \in B_X$. Thus $f \in B_{X^*}$. ■

Proposition 6.14 *A Banach space X is reflexive if and only if its unit ball B_X is compact in the weak topology.*

The following is a very deep fact. It says that in the weak topology compactness is equivalent to sequential compactness. This fact holds also for the weak* topology on X^* if X is separable, because in this case the weak* topology is metrizable, but it does not hold in general for the weak* topology.

Proposition 6.15 (*Eberlein-Smulian Theorem*) *For any weakly closed subset A of a Banach space the following are equivalent:*

- (a) *Each sequence $\{x_n\}$ in A has a subsequence which converges weakly to a point of A .*
- (b) *Each net $\{x_\alpha\}$ in A has a subnet which converges weakly to a point of A .*
- (c) *A is weakly compact.*

Finally, we list several properties which characterize reflexivity. Each is at times useful. For example, property (g) offers a quick way to confirm the fact that uniformly convex Banach spaces are always reflexive.

Proposition 6.16 *For a Banach space X the following are equivalent.*

- (a) *X is reflexive.*
- (b) *X^* is reflexive.*
- (c) *B_X is weakly compact in X .*
- (d) *Any bounded sequence in X has a weakly convergent subsequence.*
- (e) *For any $f \in X^*$ there exists $x \in B_X$ such that $f(x) = \|f\|$.*
- (f) *For any bounded closed convex subset K of X and any $f \in X^*$ there exists $x \in K$ such that $f(x) = \sup\{f(y) : y \in K\}$.*
- (g) *If $\{K_n\}$ is any descending sequence of nonempty bounded closed convex subsets of X , then $\bigcap_{n=1}^{\infty} K_n \neq \emptyset$.*

Condition (b) is immediate from the definition, (c) is a consequence of the Hahn-Banach Theorem, and (d) follows from the Eberlein-Smulian Theorem. The characterizations (e) and (f) are due to R. C. James, and (g) is due to Smulian.

6.10 The Schur property and ℓ_1

We return to the Schur property for ℓ_1 , and here we include the details because the argument we give has an entirely metric flavor. Aside from an application of Alaoglu's Theorem, the key ingredients are the Baire Category Theorem and the fact that the weak* topology on X^* is metrizable if X is separable.

To establish the Baire Category Theorem we need some definitions.

Definition 6.9 *If S is dense in some open subset U of a topological space T (that is, $\bar{S} \supseteq U$), then S is said to be somewhere dense in T . Otherwise, S is said to be nowhere dense in T .*

Definition 6.10 *A subset S of a topological space T is said to be of the first category in T if it is the union of a countable family of nowhere dense sets in T . Otherwise S is said to be of the second category in T .*

Theorem 6.6 (Baire Category Theorem) *Any complete metric space M is of the second category in itself.*

It is the following reformulation of the Baire Category Theorem that we will actually use later.

Corollary 6.2 *If M is a complete metric space and if $M = \bigcup_{i=1}^{\infty} A_i$, then for some n , $\bar{A}_n$ contains an open set (i.e., $\bar{A}_n$ has nonempty interior).*

Proof. By the Baire Category Theorem some A_n is not nowhere dense. This means that $\bar{A}_n$ contains an open set. ■

The Baire Category Theorem is an easy consequence of the following lemma. Here we use $U(a; r)$ to denote the *open* ball centered at $a \in M$ with radius $r > 0$; as usual, $B(a; r)$ will denote the corresponding *closed* ball.

Lemma 6.3 *Let $\{G_n\}$ be a sequence of dense open sets in a complete metric space M . Then $\bigcap_{n=1}^{\infty} G_n$ is also dense in M .*

Proof. Let U be an open ball in M . Because $\bar{G}_1 = M$, $G_1 \cap U$ is a nonempty open set, there exist $a_1 \in G_1 \cap U$ and $r_1 > 0$ such that $B(a_1; r_1) \subseteq G_1 \cap U$. Similarly, $G_2 \cap U(a_1; r_1)$ is nonempty and open, so there exist $a_2 \in G_2 \cap U(a_1; r_1)$ and $r_2 > 0$ with $r_2 \leq \frac{1}{2}r_1$ such that $B(a_2; r_2) \subseteq G_2 \cap U(a_1; r_1)$. Continue in this way to obtain

$$B(a_n; r_n) \subseteq G_n \cap U(a_{n-1}; r_{n-1}), \quad n = 1, 2, \dots,$$

with $r_n \leq \frac{1}{2}r_{n-1} \rightarrow 0$ as $n \rightarrow \infty$. Observe that $m > n \Rightarrow$

$$B(a_1; r_1) \supseteq \cdots \supseteq B(a_n; r_n) \supseteq \cdots \supseteq B(a_m; r_m).$$

Thus $d(a_n, a_m) \leq r_n$ and it follows that $\{a_n\}$ is a Cauchy sequence. Since M is complete there exists $a \in M$ such that $\lim_{n \rightarrow \infty} a_n = a$. But for any given n , $a_m \in B(a_n; r_n)$ for all $n < m$. Hence $a = \lim_{n \rightarrow \infty} a_n \in B(a_n; r_n)$. Consequently,

$$a \in \bigcap_{n=1}^{\infty} B(a_n; r_n) \subseteq \bigcap_{n=1}^{\infty} G_n.$$

Since a is also in U , this proves $\left(\bigcap_{n=1}^{\infty} G_n\right) \cap U \neq \emptyset$; hence $\bigcap_{n=1}^{\infty} G_n$ is dense in M . ■

Proof of the Baire Category Theorem. Let $\{A_n\}$ be a sequence of nowhere dense sets in M with $M = \bigcup_{n=1}^{\infty} A_n$. Because A_n is nowhere dense, $\overline{A_n}$ has no interior points, so any open ball in M must intersect $G_n = M \setminus \overline{A_n}$, $n = 1, 2, \dots$. Therefore, $\{G_n\}$ is a family of dense open sets in M . By Lemma 6.3, $\bigcap_{n=1}^{\infty} G_n \neq \emptyset$.

This shows that $\bigcup_{n=1}^{\infty} A_n \neq M$, which is a contradiction. ■

We need one other fact before proving Schur's Lemma. Suppose X is a separable Banach space with unit ball B and let B^* denote the unit ball in X^* . It is easy to see that if S is a dense subset of B and if $\{f_\alpha\}$ is a net in B^* , then

$$\begin{aligned} \text{weak}^* \lim_{\alpha} f_{\alpha} = f \in B^* &\Leftrightarrow \lim_{\alpha} f_{\alpha}(x) = f(x) \text{ for each } x \in B \\ &\Leftrightarrow \lim_{\alpha} f_{\alpha}(x) = f(x) \text{ for each } x \in S. \end{aligned}$$

Now suppose $S = \{x_1, x_2, \dots\}$, and for each $f, g \in B^*$ define

$$\rho(f, g) = \sum_{i=1}^{\infty} \frac{1}{2^i} |f(x_i) - g(x_i)|.$$

Then ρ is a metric on B^* and moreover the metric topology on (B^*, ρ) coincides with the relative weak* topology on B^* (see Exercise 6.14), that is,

$$\text{weak}^* \lim_{\alpha} f_{\alpha} = f \in B^* \Leftrightarrow \lim_{\alpha} \rho(f_{\alpha}, f) = 0.$$

Lemma 6.4 (Schur's Lemma). Let $\{x^n\} = \{(x_1^n, x_2^n, \dots)\}$ be a bounded sequence in ℓ_1 which converges weakly to 0. Then $\lim_{n \rightarrow \infty} \|x^n\|_1 = 0$.

Proof. Let B^* denote the unit ball in ℓ_∞ . Given $\varepsilon > 0$, define

$$F_k = \{f \in B^* : |f(x^n)| \leq \varepsilon/3 \text{ for each } n \geq k\}.$$

The sets F_k are weak* closed and, since $\lim_{n \rightarrow \infty} f(x^n) = 0$ for each $f \in B^*$,

$$B^* \subset \bigcup_{k=1}^{\infty} F_k.$$

Since ℓ_1 is separable, B^* is metrizable in its relative weak* topology. Also, by Alaoglu's Theorem B^* is compact, hence complete, in this topology. This means we may apply the Baire Category Theorem to conclude that some F_k must have nonempty weak* interior. Fix such a k and suppose $f \in \text{weak}^* \text{int}(F_k)$. Now suppose that for each $n > 0$ and for each $N > 0$ there exists $h^n \in B^* \setminus F_k$ such that $|h_i^n - f_i| < 1/n$ for each $1 \leq i \leq N$. By letting $N \rightarrow \infty$ it follows that

$$\lim_{n \rightarrow \infty} |h_i^n - f_i| = 0.$$

Since weak* convergence in B^* coincides with coordinatewise convergence, we conclude that $\text{weak}^* \lim_{n \rightarrow \infty} h^n = f$. This contradicts $f \in \text{weak}^* \text{int}(F_k)$.

So, there must exist $\delta > 0$ and $N \in \mathbb{N}$ such that

$$\{h \in B^* : |h_i - f_i| < \delta, 1 \leq i \leq N\} \subset F_k.$$

Also, since $\text{weak} \lim_{n \rightarrow \infty} x^n = 0$, there exists $p \in \mathbb{N}$ such that for each $n \geq p$,

$$\sum_{i=1}^N |x_i^n| < \varepsilon/3.$$

(Here it is necessary to use the fact that weak convergence implies coordinatewise convergence in ℓ_1 and apply this fact N times.)

Now fix $n \geq \max\{k, p\}$, and define $g \in B^*$ by

$$g_i = \begin{cases} f_i & \text{if } 1 \leq i \leq N, \\ \text{sgn } x_i^n & \text{if } i > N. \end{cases}$$

Then $g \in F_k$. Therefore by the definition of F_k ,

$$|g(x^n)| = \left| \sum_{i=1}^N g_i x_i^n + \sum_{i=N+1}^{\infty} |x_i^n| \right| \leq \varepsilon/3,$$

from which (because $\|g\|_\infty \leq 1$)

$$\sum_{i=N+1}^{\infty} |x_i^n| \leq \varepsilon/3 + \sum_{i=1}^N |x_i^n| < 2\varepsilon/3.$$

It follows that if $n \geq \max\{k, p\}$,

$$\|x^n\|_1 = \sum_{i=1}^{\infty} |x_i^n| = \sum_{i=1}^N |x_i^n| + \sum_{i=N+1}^{\infty} |x_i^n| < \varepsilon.$$

■

Schur's Lemma was instrumental in resolving an early fundamental question in functional analysis, namely, whether every infinite-dimensional Banach space must necessarily contain an *infinite-dimensional* reflexive subspace. The answer is negative because Schur's Lemma implies that every reflexive subspace of ℓ_1 must be finite-dimensional.

We conclude this section with another famous result which is a consequence of the Baire Category Theorem (Theorem 6.6).

Theorem 6.7 (*Closed Graph Theorem*) *Let X and Y be two Banach spaces. Let $T : X \rightarrow Y$ be a linear mapping. Then T is continuous (or bounded) if and only if the graph of T , that is, the set $\{(x, T(x)); x \in X\}$, is closed in $X \times Y$ (endowed with the product topology).*

6.11 More on Schauder bases in Banach spaces

It is certainly true that the concepts of algebraic bases and orthonormal bases are crucial to the study of finite-dimensional vector spaces and Hilbert spaces, respectively. That is why one tries to find a corresponding concept in the study of Banach spaces. Although, using Zorn's Lemma, one can always establish the existence of an algebraic basis of a given Banach space, this is not really sufficient because such a basis does not provide information about the topology and the geometry of the space.

Recall that a sequence $\{x_n\}$ in a Banach space X is called a Schauder basis (or just a basis) for X if for each $x \in X$, there exists a unique sequence of scalars $\{a_n\}$ such that

$$x = \lim_{n \rightarrow \infty} \sum_{k=1}^n a_k x_k = \sum_{n=1}^{\infty} a_n x_n .$$

The *span* of a subset S of X is the collection of all linear combinations $\{\alpha u + \beta v : u, v \in S\}$. If the sequence $\{x_n\}$ does not span the entire space X and at the same time is a Schauder basis for its closed linear span, $\overline{\text{span}}\{x_n\}$, then it is called a *basic sequence*.

We will always assume that basic sequence $\{x_n\}$ is normalized, that is, that $\|x_n\| = 1$ for all $n \geq 1$.

Once it is known that a Banach space has a Schauder basis it is natural to raise the question of its uniqueness. In order to study this question properly, the concept of equivalence of bases is needed.

Definition 6.11 Two bases $\{x_n\}$ and $\{y_n\}$ are called equivalent, provided $\sum_n a_n x_n$ converges if and only if $\sum_n a_n y_n$ converges.

Using the Closed Graph Theorem (Theorem 6.7), one can easily show that $\{x_n\}$ and $\{y_n\}$ are equivalent if and only if there exists an isomorphism T from $\overline{\text{span}}\{x_n\}$ onto $\overline{\text{span}}\{y_n\}$ for which $T(x_n) = y_n$, for all n .

A very useful method of building equivalent bases is based on the notion of block bases.

Definition 6.12 Let $\{x_n\}$ be a basic sequence in a Banach space X . Any non-zero vector $u \in X$ of the form

$$u = \sum_{n=p+1}^q a_n x_n,$$

where $\{a_n\}$ are scalars and $p < q$, is called a block vector. A block basis of $\{x_n\}$ is a sequence of block vectors $\{u_m\}$ such that

$$u_m = \sum_{n=p_m+1}^{p_{m+1}} a_n x_n,$$

where $p_1 < p_2 < \dots$.

It is natural to associate to any vector $x = \sum_{n \geq 1} a_n x_n$, a set of integers, called the support of x , defined by

$$\text{supp}(x) = \{n; a_n \neq 0\}.$$

The basicity of a sequence $\{x_n\}$ can be characterized by a condition on the projections onto finite subsets of the coordinates.

Definition 6.13 Let $\{x_n\}$ be a basic sequence in X . Set $X_0 = \overline{\text{span}}\{x_n\}$. A natural projection on $\{x_n\}$ is a mapping $P_I : X_0 \rightarrow X_0$ defined by

$$P_I \left(\sum_{n=1}^{\infty} a_n x_n \right) = \sum_{n \in I} a_n x_n,$$

where I is a subset of $\mathbb{N}$.

There is a simple and useful criterion for checking whether a given sequence is a basic sequence.

Theorem 6.8 *A sequence $\{x_n\}$ is basic if and only if there exists a number $c > 0$ such that for any positive integers n and p and any sequence of scalars $\{a_n\}$,*

$$\left\| \sum_{k=1}^n a_k x_k \right\| \leq c \left\| \sum_{k=1}^{n+p} a_k x_k \right\|.$$

Proof. Suppose that $\{x_n\}$ is a basic sequence for its closed linear span $X_0 = \overline{\text{span}}\{x_n\}$. Consider the natural projections $\{P_k\}$, defined by

$$P_k \left(\sum_{n=1}^{\infty} a_n x_n \right) = \sum_{n=1}^k a_n x_n.$$

Since the range of each P_k is a finite-dimensional space, they are bounded linear operators. Our assumption implies that for every $x \in X_0$, we have

$$x = \lim_{k \rightarrow \infty} P_k(x).$$

It follows from Banach-Steinhaus theorem (see Proposition 6.12) that $\sup \|P_n\| < \infty$. Thus, we should have

$$\left\| \sum_{k=1}^n a_k x_k \right\| = \left\| P_n \left(\sum_{k=1}^{n+p} a_k x_k \right) \right\| \leq \sup \|P_n\| \left\| \sum_{k=1}^{k=n+p} a_k x_k \right\|$$

for any positive integers n and p and any sequence of scalars $\{a_n\}$. Therefore, the constant c may be taken equal to $\sup \|P_n\|$. Conversely, suppose that $\{x_n\}$ is a sequence such that for any positive integers n and p and any sequence of scalars $\{a_n\}$,

$$\left\| \sum_{k=1}^n a_k x_k \right\| \leq c \left\| \sum_{k=1}^{n+p} a_k x_k \right\|.$$

This condition ensures that linear operators P_n from X_0 into X_0 are uniformly bounded and their norms are less than the constant c . It is easy to check that the vectors $\{x_n\}$ are linearly independent. Let $x \in X_0$ and $\varepsilon > 0$ be given.

Then there exists $y \in \text{span}\{x_n\}$ such that $\|x - y\| < \varepsilon$. We have $y = \sum_{n=1}^{n(\varepsilon)} a_n x_n$, for some $n(\varepsilon)$. But now, if $n \geq n(\varepsilon)$, we have

$$\begin{aligned} \|x - P_n(x)\| &\leq \|x - y\| + \|y - P_n(y)\| + \|P_n(y) - P_n(x)\| \\ &= \|x - y\| + \|y - y\| + \|P_n(y - x)\| \\ &\leq \varepsilon + \|P_n\| \varepsilon \leq (1 + c) \varepsilon. \end{aligned}$$

It follows that $x = \lim_{n \rightarrow \infty} P_n(x)$. Using the properties of the operators P_n , namely, $P_n P_m = P_m P_n = P_{\min(n, m)}$, we deduce that the first n coordinates of $P_m(x)$

are independent of m . This clearly implies the existence of a unique sequence of scalars $\{a_n\}$ such that $x = \sum_{n=1}^{\infty} a_n x_n$. ■

The least number c which satisfies the above inequality is called the *basis constant* of $\{x_n\}$. If the basis constant is 1, then the sequence $\left\{ \left\| \sum_{k=1}^n a_k x_k \right\| \right\}$ is monotone increasing. In this case, the basis $\{x_n\}$ is called *monotone*.

Looking at the proof of Theorem 6.8 closely, we see that the basis constant of $\{x_n\}$ is given by

$$c = \sup_n \|P_n\|.$$

We will say that the basic sequence $\{x_n\}$ is *bimonotone* if

$$\|P_n\| = \|I - P_n\| = 1 \text{ for any } n \in \mathbb{N}.$$

Remark 6.2 One may renorm the space X with an equivalent norm for which the basis constant is 1. Indeed, set

$$|x| = \sup_n \|P_n(x)\|$$

for $x \in X$. Then we have

$$\|x\| \leq |x| \leq c\|x\|$$

for any $x \in X$. Using the basic properties of the natural projections, we can easily show that

$$|P_m(x)| \leq |x|$$

for any $x \in X$ and $m \geq 1$, which means that the basis constant of $\{x_n\}$ with respect to the new norm $|\cdot|$ is 1.

Next we discuss the relationship between basicity and duality.

Definition 6.14 Let $\{x_n\}$ be a Schauder basis for X . For any $n \in \mathbb{N}$, define the linear functional x_n^* on X by

$$x_n^* \left(\sum_{k=1}^{\infty} a_k x_k \right) = a_n.$$

These functionals $\{x_n^*\}$ are called the *biorthogonal functionals* associated with basis $\{x_n\}$. In particular, we have

$$x = \sum_{n=1}^{\infty} x_n^*(x) x_n$$

for any $x \in X$.

It is easy to check that for any n ,

$$\|x_n^*\| \leq 2c,$$

where c is the basis constant of $\{x_n\}$. In fact, let $\{P_n\}$ be the sequence of natural projections associated with $\{x_n\}$. Then for any integers $n < m$, we have

$$P_n^* \left(\sum_{k=1}^m a_k x_k^* \right) = \sum_{k=1}^n a_k x_k^* .$$

Hence, by Theorem 6.8, the sequence $\{x_n^*\}$ is a basic sequence in X^* whose basis constant is also c . But in general $\{x_n^*\}$ is not a Schauder basis of X^* .

Example 6.6 Consider the Banach spaces c_0 , l_1 , and l_∞ . We have seen that $c_0^* = l_1$ and $l_1^* = l_\infty$. The canonical basis $\{e_n\}$ of c_0 is a Schauder basis. The associated biorthogonal basis $\{e_n^*\}$ is the canonical basis of l_1 , which is also a Schauder basis. But the biorthogonal basis $\{e_n^{**}\}$ of the basis $\{e_n^*\}$ is not a Schauder basis of l_∞ , since the Banach space l_∞ is not separable (see Exercise 6.8).

Definition 6.15 A Schauder basis $\{x_n\}$ of X is called *shrinking* if the sequence $\{x_n^*\}$ is a Schauder basis of X^* .

The canonical basis of l_1 has a surprising property which suggests the following.

Definition 6.16 A Schauder basis $\{x_n\}$ of X is called *boundedly complete* if $\sum_{n=1}^{\infty} a_n x_n$ converges whenever the sequence $\left\{ \left\| \sum_{k=1}^n a_k x_k \right\| \right\}$ is bounded.

The natural basis of c_0 is not boundedly complete. A careful study of the spaces c_0 and l_1 suggests a kind of duality between shrinkingness and boundedly completeness.

Theorem 6.9 If the Schauder basis $\{x_n\}$ of X is shrinking, then $\{x_n^*\}$ is a boundedly complete basis of X^* .

Proof. Assume that $\{x_n\}$ is a shrinking basis of X . Let $\{a_n\}$ be a sequence of scalars such that

$$\sup_n \left\| \sum_{k=1}^n a_k x_k^* \right\| < \infty .$$

We need to prove that $\sum_{n=1}^{\infty} a_n x_n^*$ is convergent. Consider the sequence of elements $\{y_n^*\}$ of X^* defined by

$$y_n^* = \sum_{k=1}^n a_k x_k^* .$$

Our assumption implies that $\{y_n^*\}$ is bounded. Since X is separable, the weak*-topology on X^* is sequentially compact. Hence there exists a subsequence $\{y_{n_k}^*\}$ which weak*-converges to an element y^* in X^* . Since $\{x_n\}$ is a shrinking, the sequence $\{x_n^*\}$ is a Schauder basis of X^* . Therefore we have

$$y^* = \sum_{n=1}^{\infty} y^*(x_n) x_n^*.$$

Since $\{y_{n_k}^*\}$ weak*-converges to y^* , we must have

$$\lim_{n_k \rightarrow \infty} y_{n_k}^*(x_i) = y^*(x_i)$$

for any $i \geq 1$. By the definition of the vectors y_n^* , we have

$$y_{n_k}^*(x_i) = a_i$$

whenever $n_k \geq i$. So it must be the case that $y^*(x_i) = a_i$, for any $i \geq 1$. Therefore,

$$y^* = \sum_{n=1}^{\infty} a_n x_n^*,$$

which completes the proof. ■

In the next theorem, we give a nice and useful description of X^{**} when X has a shrinking basis.

Theorem 6.10 *Let $\{x_n\}$ be a shrinking basis of X . Then X^{**} may be identified with the space of all sequences of scalars $\{a_n\}$ such that $\sup_n \left\| \sum_{k=1}^n a_k x_k \right\| < \infty$.*

The correspondence is given by

$$x^{**} \longleftrightarrow (x^{**}(x_1^*), x^{**}(x_2^*), x^{**}(x_3^*), \dots).$$

*The norm of x^{**} is equivalent (and, if the basis constant is 1, even equal) to $\sup_n \left\| \sum_{k=1}^n a_k x_k \right\|$.*

The proof is left as an exercise.

These two dual concepts are crucial in understanding the structure of the dual space. In particular, we have the following characterization of reflexivity.

Theorem 6.11 *Assume that X has a Schauder basis $\{x_n\}$. Then X is reflexive if and only if $\{x_n\}$ is shrinking and boundedly complete.*

In order to prove this theorem, we need the following technical lemma.

Lemma 6.5 *Assume that X has a Schauder basis $\{x_n\}$. Then $\{x_n\}$ is shrinking if and only if for every $x^* \in X^*$, the norm of $x^*|_{[x_i]_{n \leq i}}$ (which is the restriction of x^* to the span of $\{x_i\}_{n \leq i}$) tends to 0 as $n \rightarrow \infty$.*

Proof. If $\{x_n^*\}$ is a basis of X^* then, for every $x^* \in X^*$, we have

$$\lim_{n \rightarrow \infty} \|P_n^*(x^*) - x^*\| = 0.$$

Since $P_{n-1}^*(x^*|_{[x_i]_{n \leq i}}) = 0$, we get

$$\lim_{n \rightarrow \infty} \|x^*|_{[x_i]_{n \leq i}}\| = 0.$$

Conversely, assume that $\|x^*|_{[x_i]_{n \leq i}}\| \rightarrow 0$ as $n \rightarrow \infty$. Let $x \in X$ be such that $\|x\| = 1$. Then, by definition of the adjoint operator, we have

$$(x^* - P_n^*(x^*))(x) = x^*(x - P_n(x)) \leq \|x^*|_{[x_i]_{n+1 \leq i}}\|(K+1),$$

where K is the basis constant of $\{x_n\}$. So we must have

$$\lim_{n \rightarrow \infty} \|x^* - P_n^*(x^*)\| = 0.$$

Back to the proof of Theorem 6.11. Assume that X is reflexive and that $\{x_n\}$ is not shrinking. Then by the technical lemma, there exist $\varepsilon > 0$, y^* and a sequence $\{y_n\}$, with $\|y_n\| = 1$ for any $n \geq 1$, such that

$$|y^*(y_n)| > \varepsilon$$

and $y_n \in \overline{\text{span}}\{x_i, n \leq i\}$. Since X is reflexive, there exists a subsequence $\{y_{n_k}\}$ of $\{y_n\}$ which weakly converges to a point y . Since $\{x_n\}$ is a Schauder basis of X , we can easily see that all the coordinates of y are 0. In other words, $\{y_{n_k}\}$ weakly converges to 0. So we must have

$$\lim_{n_k \rightarrow \infty} |y^*(y_{n_k})| = 0 \geq \varepsilon,$$

which is a contradiction. Therefore, $\{x_n\}$ is shrinking. Moreover, a similar argument used in the proof of Theorem 6.9 will show that $\{x_n\}$ is boundedly complete.

Conversely, assume that $\{x_n\}$ is shrinking and boundedly complete. In order to show that X is reflexive we will show that any bounded sequence has a weakly convergent subsequence. Let $\{y_n\}$ be such that $\|y_n\| \leq 1$, for any $n \geq 1$. Since $\{x_n\}$ is a Schauder basis of X ,

$$y_n = \sum_{i=1}^{\infty} a_i(n)x_i$$

for any $n \geq 1$. We have

$$|a_i(n)| \leq 2K,$$

where K is the basis constant of $\{x_n\}$ (once we assume that $\|x_n\| \equiv 1$). A diagonal argument will help construct a subsequence $\{y_{n_k}\}$ such that $\{a_i(n_k)\}$ converges to a_i , as $n_k \rightarrow \infty$. Since

$$\left\| \sum_{i=1}^{i=N} a_i(n)x_i \right\| \leq K \left\| \sum_{i=1}^{\infty} a_i(n)x_i \right\| \leq K,$$

for any $N \geq 1$, we get

$$\left\| \sum_{i=1}^{i=N} a_i x_i \right\| \leq K.$$

Because $\{x_n\}$ is boundedly complete, there exists $y \in X$ such that

$$y = \sum_{i=1}^{\infty} a_i x_i.$$

In order to complete the proof we will show that y is the weak limit of $\{y_{n_k}\}$. Since $\{x_n\}$ is shrinking, it is enough to show that for any $m \geq 1$ we have

$$x_m^*(y) = \lim_{n_k \rightarrow \infty} x_m^*(y_{n_k}),$$

but this is exactly the assumption on the sequences $\{a_m(n_k)\}$. This completes the proof of Theorem 6.11. $\blacksquare$

Note that the existence of a Schauder basis does not give us very much information about the structure of the space. So if we wish to use bases to study the structure of a Banach space in any detail, we are led to consider bases with various special properties.

Definition 6.17 A Schauder basis $\{x_n\}$ of X is called *unconditional* if whenever $\sum_{n=1}^{\infty} a_n x_n$ converges, it converges unconditionally, that is, $\sum_{n=1}^{\infty} a_{\pi(n)} x_{\pi(n)}$ converges for any permutation π of $\mathbb{N}$.

The following properties more fully describe the unconditional behavior of basic sequences.

Proposition 6.17 Let $\{x_n\}$ be a Schauder basis of X . The following are equivalent:

- (i) $\{x_n\}$ is an unconditional basis.

- (ii) If $\sum_{n=1}^{\infty} a_n x_n$ converges, then for any sequence $\{\varepsilon_n\}$, where $\varepsilon_n = \pm 1$ for all n , the series $\sum_{n=1}^{\infty} \varepsilon_n a_n x_n$ converges.
- (iii) If $\sum_{n=1}^{\infty} a_n x_n$ converges, then for any sequence $\{b_n\}$, where $|b_n| \leq |a_n|$ for all n , the series $\sum_{n=1}^{\infty} b_n x_n$ converges.
- (iv) There exists $c \geq 1$ such that if A and B are finite subsets of $\mathbb{N}$ with $A \subset B$, then for any sequence $\{a_n\}$ we have

$$\left\| \sum_{n \in A} a_n x_n \right\| \leq c \left\| \sum_{n \in B} a_n x_n \right\|.$$

The smallest constant c which satisfies the above inequality is called the unconditional basis constant of $\{x_n\}$. If the unconditional constant is 1, the basis is called unconditionally monotone.

- (v) There exists $\lambda \geq 1$ such that if $\sum_{n=1}^{\infty} a_n x_n$ converges, then for any sequence $\{\varepsilon_n\}$, where $\varepsilon_n = \pm 1$ for all n , we have

$$\left\| \sum_{n=1}^{\infty} \varepsilon_n a_n x_n \right\| \leq \lambda \left\| \sum_{n=1}^{\infty} a_n x_n \right\|.$$

The smallest constant λ which satisfies the above inequality is called the unconditional constant of $\{x_n\}$. Moreover, we have

$$c \leq \lambda \leq 2c,$$

where c is the unconditional basis constant.

The proof is left as an exercise.

Remark 6.3 Let $\{x_n\}$ be an unconditional basis of X . For any $x \in X$, set

$$|x| = \sup \left\{ \left\| \sum_{n=1}^{\infty} x_n^*(x) \varepsilon_n x_n \right\| : \varepsilon_n = \pm 1, \text{ for any } n \geq 1 \right\}.$$

It is easy to check that $|\cdot|$ is a norm equivalent to $\|\cdot\|$ for which the unconditional constant of $\{x_n\}$ is 1.

The conclusion of Corollary 6.11 can be strengthened when unconditional bases are involved.

Theorem 6.12 Let X be a subspace of a space with an unconditional Schauder basis. Then X is reflexive unless it contains a subspace isomorphic to either c_0 or l_1 .

Proof. Taking into account the conclusion of Theorem 6.11, it is enough to establish the following two propositions.

Proposition 6.18 *Assume X has an unconditional Schauder basis. Then the basis is boundedly complete if and only if X does not contain a subspace isomorphic to c_0 .*

Proposition 6.19 *Assume X has an unconditional Schauder basis. Then the basis is shrinking if and only if X does not contain a subspace isomorphic to l_1 .*

The proofs of both propositions are left as an exercise.

6.12 Uniform convexity and reflexivity

In Chapter 9 we will see that the following is actually a special case of a much more general result. However, the proof below illustrates some nice geometric properties of uniformly convex spaces.

Theorem 6.13 *If X is a uniformly convex Banach space, then X is reflexive.*

Proof. We show that Smulian's condition (g) of Proposition 6.16 is satisfied. Suppose $\{K_n\}$ is a descending sequence of nonempty bounded closed convex subsets of X . We rely on exercises at the end of this chapter. For each n there is a unique point $x_n \in K_n$ such that

$$\|x_n\| = \inf\{\|x\| : x \in K_n\}.$$

Since $\{K_n\}$ is descending the sequence $\{\|x_n\|\}$ is nondecreasing. Moreover, for each n , $\|x_n\| \leq \sup\{\|x\| : x \in K_1\}$ and, since K_1 is bounded, the sequence $\{\|x_n\|\}$ is bounded as well. Therefore $\lim_{n \rightarrow \infty} \|x_n\| = r \geq 0$. Now let $m \geq n$. Then

$\frac{x_n + x_m}{2} \in K_n$, so

$$\|x_n\| \leq \left\| \frac{x_n + x_m}{2} \right\| \leq \frac{\|x_n\| + \|x_m\|}{2} \leq r.$$

Thus it must be the case that $\lim_{m, n \rightarrow \infty} \left\| \frac{x_n + x_m}{2} \right\| = r$. On the other hand,

$$\left\| \frac{x_n + x_m}{2} \right\| \leq \left(1 - \delta \left(\frac{\|x_n - x_m\|}{r} \right) \right) r.$$

From this we conclude that

$$\lim_{m, n \rightarrow \infty} \delta \left(\frac{\|x_n - x_m\|}{r} \right) = 0,$$

and since X is uniformly convex it follows that $\lim_{m,n \rightarrow \infty} \|x_n - x_m\| = 0$. This proves that $\{x_n\}$ is a Cauchy sequence. Therefore, $\lim_{n \rightarrow \infty} x_n = x$ with $x \in \bigcap_{n=1}^{\infty} K_n$. ■

Another way to see that $L_p(\mu)$ is reflexive for $1 < p < \infty$ is to show directly that $(L_p(\mu))^{**} = L_p(\mu)$. We will not discuss the details but this follows from the fact that $(L_p(\mu))^* = L_q(\mu)$, where $\frac{1}{p} + \frac{1}{q} = 1$. This instantly gives $(L_q(\mu))^* = (L_p(\mu))^{**} = L_p(\mu)$.

It is also known that $(L_1(\mu))^* = L_\infty(\mu)$ and that $(L_\infty(\mu))^* \neq L_1(\mu)$. Thus $L_1(\mu)$ is not reflexive. (Similarly $L_\infty(\mu)$ is not reflexive.) Since c^* and c_0^* are isometrically isomorphic to ℓ_1 , neither of these spaces is reflexive.

Another thing to keep in mind: It is immediate from the definition that reflexivity of X implies reflexivity of X^* . However, uniform convexity of X does not necessarily imply that X^* is uniformly convex.

A more general result, which we will not prove here, is the following.

Theorem 6.14 *If X is a uniformly nonsquare Banach space, then X is reflexive.*

A useful property of uniformly convex spaces which does not generally hold in reflexive spaces is the following.

Proposition 6.20 *Let X be uniformly convex and let $\{x_\alpha\}$ be a net in X . If $\text{weak-}\lim_\alpha x_\alpha = x$ and if $\lim_\alpha \|x_\alpha\| = \|x\|$, then $\lim_\alpha x_\alpha = x$.*

Thus in a uniformly convex space weak convergence and convergence in norm imply strong convergence. Spaces which have this property are said to have *Kadec-Klee (KK) norm*.

There is another fact which distinguishes reflexive spaces from uniformly convex spaces. If $(X, \|\cdot\|)$ is a Banach space, and if $\|\cdot\|_1$ is another norm on X which is equivalent to the norm $\|\cdot\|$, then it readily follows that $(X, \|\cdot\|)$ is reflexive if and only if $(X, \|\cdot\|_1)$ is reflexive. Thus reflexivity is invariant under equivalent renormings. To see that this is not true of uniform convexity one need look no further than the finite-dimensional spaces $\mathbb{R}_2^n$ and $\mathbb{R}_\infty^n$.

We conclude with one final comment for those interested in specific cases. Let $1 < p < \infty$ and consider the spaces $L_p[0,1]$ where Lebesgue measure is understood. In this case a sequence $\{f_n\}$ in $L_p[0,1]$ converges weakly to $f \in L_p[0,1]$ if and only if $\{f_n\}$ is bounded and for each $x \in [0,1]$,

$$\lim_{n \rightarrow \infty} \int_0^x f_n(t) dt = \int_0^x f(t) dt.$$

6.13 Banach lattices

Let X be a Banach space and suppose $\leq$ is a partial order on X . Recall that this means that:

- (i) $x \leq x$ for all $x \in X$;
- (ii) $x \leq y$ and $y \leq x \Rightarrow x = y$;
- (iii) $x \leq y$ and $y \leq z \Rightarrow x \leq z$.

As usual we adopt the convention $x \geq y \Leftrightarrow y \leq x$. Elements $x \in X$ for which $x \geq 0$ are said to be *positive*.

The linear structure of X is assumed to be compatible with the order structure in the following sense:

- (iv) $x \leq y \Rightarrow x + z \leq y + z$ for all $x, y, z \in X$; and
- (v) $x \leq y \Rightarrow \alpha x \leq \alpha y$ for all $x, y \in X$ and $\alpha \in \mathbb{R}^+$.

Now suppose $S \subseteq X$ and suppose $x \in X$. If x satisfies the following two conditions:

- (a) $y \leq x$ for all $y \in S$, and
- (b) $y \leq z$ for all $y \in S \Rightarrow x \leq z$,

then x is called the *supremum* of S , written

$$x = \sup(S) = \bigvee_{y \in S} y.$$

The *infimum* is defined dually, that is, by replacing $\leq$ with $\geq$, and writing

$$x = \inf(S) = \bigwedge_{y \in S} y.$$

Definition 6.18 If $x \vee y$ and $x \wedge y$ exist for every pair of elements $x, y \in X$ then X is called a *linear lattice* (or *Riesz space*).

Remark 6.4 A good nontrivial example of a linear lattice to keep in mind as we run through the ensuing properties and definitions is the space $B[0, 1]$ of bounded real-valued functions on $[0, 1]$, taking $f \leq g \Leftrightarrow f(x) \leq g(x)$ for all $x \in [0, 1]$. Another good example is the space c_0 of sequences of real numbers which converge

to 0. In this case if $x = \{x_n\}, y = \{y_n\} \in c_0$, take $x \leq y \Leftrightarrow x_n \leq y_n$ for all $n \in \mathbb{N}$. The spaces ℓ_p ($1 \leq p \leq \infty$) with the same order relation provide additional examples. Of course, $\mathbb{R}$ with its usual order relation provides an example as well.

In a linear lattice it is possible to describe the positive and negative parts of elements as follows:

$$x^+ = x \vee 0$$

$$x^- = (-x) \vee 0.$$

Also,

$$|x| = x \vee (-x).$$

This leads to the identities $x = x^+ - x^-$ and $|x| = x^+ + x^- = x^+ \vee x^-$. If $x = u - v$, then $u \wedge v = 0$ if and only if $u = x^+$ and $v = x^-$.

Two elements $x, y \in X$ are said to be *disjoint* if $|x| \wedge |y| = 0$. In particular, x^+ and x^- are disjoint (since $|x^+| = x^+$, $|x^-| = x^-$, and $x^+ \wedge x^- = 0$). In particular, if x and y are disjoint, then

$$\begin{aligned} |x + y| &= |x - y| \\ &= |x| \vee |y| \\ &= |x| + |y|. \end{aligned}$$

Four additional identities are obvious if one thinks about the example spaces. These also link the linear and order structure together.

$$x + y = x \vee y + x \wedge y;$$

$$|x - y| = x \vee y - x \wedge y;$$

$$2(|x| \vee |y|) = |x + y| + |x - y|;$$

$$2(|x| \wedge |y|) = \left| |x + y| - |x - y| \right|.$$

The following familiar identities hold as well.

$$\left| |x| - |y| \right| \leq |x - y| \leq |x| + |y|;$$

$$|x + y| \leq |x| + |y|.$$

Finally, suppose $u, v, w \geq 0$ and suppose u and v are disjoint. Then $u + v = u \vee v$ and, in particular,

$$(u \wedge w) \vee (v \wedge w) = (u \vee v) \wedge w$$

and

$$u \wedge w + v \wedge w = (u + v) \wedge w.$$

Definition 6.19 *If X is a linear lattice which is also a Banach space, then the norm of X is called monotone if*

$$0 \leq x \leq y \Rightarrow \|x\| \leq \|y\|.$$

If in addition, $\| |x| \| = \|x\|$ for all $x \in X$, then X is called a Banach lattice.

A Banach lattice is called *Dedekind complete* if every nonempty order bounded set in X has a supremum and an infimum. If every order bounded sequence has a supremum and an infimum in X then X is said to be σ -Dedekind complete.

If X is σ -Dedekind complete then for each $x \in X$ it is possible to define the *principal band projection*, P_x , by

$$P_x(y) = \bigvee_{n=1}^{\infty} [(n|x|) \wedge y^+] - \bigvee_{n=1}^{\infty} [(n|x|) \wedge y^-].$$

Then $P_x(x) = x$ and both P_x and $I - P_x$ have norm one in the operator norm sense. (The norm of an operator $T : X \rightarrow X$ is given by $\|T\| = \sup\{\|T(x)\|; x \in B_X\}$.)

It might be instructive to look closely at the projection P_x in particular spaces. Suppose $x, y \in B[0, 1]$. Then $P_x(y) \in B[0, 1]$ is the function defined as follows:

$$P_x(y)(t) = \begin{cases} 0 & \text{when } x(t) = 0; \\ y(t) & \text{when } x(t) \neq 0. \end{cases}$$

From this one sees that $y - P_x(y) \in B[0, 1]$ is given by

$$y(t) - P_x(y)(t) = \begin{cases} y(t) & \text{when } x(t) = 0; \\ 0 & \text{when } x(t) \neq 0. \end{cases}$$

Similarly, if $x = \{x_n\}, y = \{y_n\} \in c_0$, then $P_x(y) = \{c_n\}$, where

$$c_n = \begin{cases} 0 & \text{when } x_n = 0; \\ y_n & \text{when } x_n \neq 0. \end{cases}$$

Finally, the norm of a Banach lattice X is said to be *order continuous* if $\inf\{\|x\| : x \in A\} = 0$ for every set $A \subseteq X$ which has the property that $(A, \geq)$ is a directed set with $\inf(A) = 0$.

Exercises

Exercise 6.1 Show that $\ell_\infty(I)$ is complete, for any index set I .

Exercise 6.2 Show that c is complete and that c_0 is a closed subspace of c .

Exercise 6.3 Prove Propositions 6.2 and 6.3.

Exercise 6.4 Show that, in general, $\overline{\text{conv}}(A) \neq \text{conv}(\overline{A})$.

Exercise 6.5 Let K be a compact subset of a Banach space X .

1. Show that $\overline{\text{conv}}(K)$ is also compact. Do we have a similar conclusion when K is weakly compact?
2. Show that there exists a sequence $\{x_n\}$ in X such that $\lim_{n \rightarrow \infty} \|x_n\| = 0$, and

$$K \subset \overline{\text{conv}}(\{x_n\}).$$

Exercise 6.6 (Ascoli-Arzelà theorem) Let (K, d) be a compact metric space, and denote by $C(K)$ the Banach space of continuous real-valued functions defined on K .

1. Show that any totally bounded subset $\mathcal{K}$ of $C(K)$ is equicontinuous, that is, for any $\varepsilon > 0$ there exists $\delta > 0$ such that

$$d(x, y) \leq \delta \Rightarrow |f(x) - f(y)| \leq \varepsilon \text{ for any } f \in \mathcal{K}.$$

2. Show that if $\mathcal{K}$ is a bounded subset of $C(K)$ and D is a countable dense subset of K , then any sequence of elements of $\mathcal{K}$ has a subsequence which converges pointwise on D .
3. Show that any equicontinuous sequence which converges pointwise on the set $S \subset K$, converges uniformly on $\overline{S}$.

Deduce then that a bounded subset $\mathcal{K}$ of $C(K)$ is relatively compact if and only if $\mathcal{K}$ is equicontinuous.

Exercise 6.7 For any p , $1 \leq p < \infty$, show that any bounded subset K of l_p is relatively compact if and only if

$$\lim_{n \rightarrow \infty} \sum_{i=n}^{\infty} |x_i|^p = 0$$

uniformly for $x = \{x_i\} \in K$.

Exercise 6.8 Show that any Banach space with Schauder basis is separable. Deduce from this that l_∞ does not have a Schauder basis.

Hint: Show that l_∞ is not separable.

Exercise 6.9 If $\{x_n\}$ and $\{y_n\}$ are sequences in a uniformly convex space X , show that the condition

$$\lim_{n \rightarrow \infty} \|x_n\| = \lim_{n \rightarrow \infty} \|y_n\| = \lim_{n \rightarrow \infty} \frac{1}{2} \|x_n + y_n\| = r > 0$$

implies $\lim_{n \rightarrow \infty} \|x_n - y_n\| = 0$. In particular, if $x, y, z \in X$, then

$$\|x - z\| = \|y - z\| = \frac{1}{2} \|x - y\| \Rightarrow z = \frac{x + y}{2}.$$

Exercise 6.10 Show that every nonempty closed convex subset of a uniformly convex space contains a unique element of minimal norm. Deduce from this that for any nonempty closed convex subset C of a uniformly convex space X , and for any $x \in X$, there exists a unique point $a \in C$ such that

$$\|x - a\| = \text{dist}(x, C) = \inf\{\|x - y\|; y \in C\}.$$

Exercise 6.11 Show that any finite-dimensional space X is strictly convex if and only if it is uniformly convex.

Exercise 6.12 Show that a Banach space X with modulus of convexity δ is strictly convex if and only if $\delta(2) = 1$.

Exercise 6.13 Show explicitly that ℓ_∞ and ℓ_1 are not strictly convex (thus not uniformly convex).

Exercise 6.14 Let X be a separable Banach space with unit ball B and let B^* denote the unit ball in X^* . Let $S = \{x_1, x_2, \dots\}$ be a dense subset of B . For each $f, g \in B^*$ define

$$\rho(f, g) = \sum_{i=1}^{\infty} \frac{1}{2^i} |f(x_i) - g(x_i)|.$$

Show that ρ is a metric on B^* and that the metric topology on (B^*, ρ) coincides with the relative weak* topology on B^* , that is,

$$\text{weak}^* \lim_{\alpha} f_{\alpha} = f \in B^* \Leftrightarrow \lim_{\alpha} \rho(f_{\alpha}, f) = 0.$$

Exercise 6.15 Let X be a Banach space. Define a type-function by

$$\tau(x) = \limsup_{n \rightarrow \infty} \|x_n - x\|,$$

where $\{x_n\}$ is a bounded sequence. Show that τ is weak lower semi-continuous.

Exercise 6.16 Let X be a Banach space. Let $\{x_n^*\}$ be a bounded sequence in X^* the dual space of X . Show that, in general, the type-function

$$\tau(x^*) = \limsup_{n \rightarrow \infty} \|x_n^* - x^*\|$$

is not weak*-lower semi-continuous.

Exercise 6.17 Let X be a Banach space with a Schauder basis $\{e_n\}$. Assume that $\{e_n\}$ is bimonotone. Show that $\{e_n^*\}$ is also bimonotone. Assume that $\{e_n\}$ is shrinking and bimonotone. Show that any type-function

$$\tau(x^*) = \limsup_{n \rightarrow \infty} \|x_n^* - x^*\|$$

is weak*-lower semi-continuous.

Exercise 6.18 Let X be a Banach space with a Schauder basis $\{e_n\}$ with associated biorthogonal system $\{e_n^*\}$. Let $\{x_k\}$ be a bounded sequence in X such that $\lim_{k \rightarrow \infty} e_n^*(x_k) = 0$, that is, the coefficients of $\{x_k\}$ converge pointwise to 0. Show that there exists a subsequence $\{x_{k_i}\}$ of $\{x_k\}$ and a sequence $\{u_i\}$ of successive blocks of $\{e_n\}$ such that

$$\lim_{i \rightarrow \infty} \|x_{k_i} - u_i\| = 0.$$

Exercise 6.19 Show that $B[0, 1]$ is a Dedekind complete Banach lattice.

Exercise 6.20 Show that c_0 is a Dedekind complete Banach lattice.

Chapter 7

Continuous Mappings in Banach Spaces

7.1 Introduction

In this chapter we take up Brouwer's Fixed Point Theorem (Theorem 1.4 of Chapter 1) and some of the many results that have been inspired by this profound result. We have chosen what seems to us to be one of the most accessible approaches to Brouwer's theorem, taking as our point of departure the following simple fact about the real line.

Proposition 7.1 *Let $\mathcal{P} : (0 = P_0 < P_1 < \cdots < P_k = 1)$ be a partition of the interval $[0, 1]$ with each of the points P_i , $0 < i < k$, 'labeled' either 0 or 1. Then the cardinality of the set $N_{\mathcal{P}} = \{i \in \{1, \dots, k\} : P_{i-1} \text{ and } P_i \text{ are labeled differently}\}$ is an odd integer.*

Proof. This proposition has a straightforward induction proof, which the reader might want to try. However, a 'counting' argument is perhaps more illuminating for what will follow. Begin by labeling $0 = P_0$ and $1 = P_1$, and label points $P_i \in (0, 1)$ arbitrarily with 0's and 1's. Now count the number of 'endpoints' of subintervals defined by the partition which are labeled 0, and do this in two different ways. First, if an endpoint labeled 0 is in $(0, 1)$ then it must be counted twice since it is an endpoint of two different subintervals of the partition. By adding P_0 to this total we see that there are an odd number of such endpoints, say k . Now divide the subintervals of the partition into two different subclasses: (i) Those which have only 0 as endpoints, and (ii) those which have a 0 and a 1 as endpoints. If there are m intervals in class (i) then these m intervals contribute $2m$ endpoints to the total k . If there are n subintervals in class (ii) then, since each such interval contributes only one endpoint to the total, we conclude that $2m + n = k$. This proves that n must be an odd integer. Since n is the cardinality of $N_{\mathcal{P}}$ we are done. ■

Note in particular that the conclusion of Proposition 7.1 implies $N_P \neq \emptyset$.

We first show how to use Proposition 7.1 to give another proof of Brouwer's Theorem in $\mathbb{R}^1$. While the proof given here is no simpler than the one given in Chapter 1, we shall see that this same line of argument leads to a simple proof of Brouwer's Theorem in $\mathbb{R}^2$ and, in turn, it points the way to a proof of Brouwer's theorem in its full generality. The reader might think of quicker ways to proceed in $\mathbb{R}^1$, but please keep in mind that it is the *method* of proof that we are interested in here.

Proposition 7.2 *Let C_0 and C_1 be two closed set in $\mathbb{R}^1$ with $0 \in C_0$ and $1 \in C_1$, and suppose $[0, 1] \subset C_0 \cup C_1$. Then $C_0 \cap C_1 \neq \emptyset$.*

Proof. For each $k \in \mathbb{N}$ let $\mathcal{P}_k$ be a partition of $[0, 1]$ into subintervals of equal length. Assign the label 0 to P_0 and 1 to P_k . For $i \in \{1, \dots, k-1\}$ assign labels as follows: If $P_i \in C_0$ assign the label 0 to P_i ; otherwise assign the label 1 to P_i . Since $[0, 1] \subset C_0 \cup C_1$, this results in a labeling of $\mathcal{P}_k$ as in Proposition 7.1. Consequently, for each k there exists an interval of length $1/k$ whose endpoints $\{P_k, Q_k\}$ are labeled differently, say P_k is labeled 0 and Q_k is labeled 1. Thus for each k , $P_k \in C_0$ and $Q_k \in C_1$. It is possible to choose convergent subsequences $\{P_{k_j}\}$ and $\{Q_{k_j}\}$ with, say, $\lim_{j \rightarrow \infty} P_{k_j} = P$ and $\lim_{j \rightarrow \infty} Q_{k_j} = Q$. Since $\lim_{j \rightarrow \infty} |P_{k_j} - Q_{k_j}| = \lim_{j \rightarrow \infty} 1/k_j = 0$, $P = Q$, and since both C_0 and C_1 are closed this point lies in $C_0 \cap C_1$. ■

It is somewhat remarkable that the above ideas provide the basis for the general proof of Brouwer's theorem. Proposition 7.1 is a special case of a celebrated result known as Sperner's Lemma, whereas Proposition 7.2 is a special case of a result due to Knaster, Kuratowski and Mazurkiewicz, known as the KKM Theorem. We shall discuss more general cases below. First, however, we show how the above results lead another proof of the fact that any continuous mapping $f : [0, 1] \rightarrow [0, 1]$ has a fixed point. Note that every point in $P \in [0, 1]$ can be written in the form

$$P = \lambda_0(P) \cdot 0 + \lambda_1(P) \cdot 1,$$

where $\lambda_0(P) + \lambda_1(P) = 1$, by simply taking $\lambda_1(P) = P$ and $\lambda_0(P) = 1 - P$. Set

$$\begin{aligned} C_0 &= \{P \in [0, 1] : \lambda_0(f(P)) \leq \lambda_0(P)\}; \\ C_1 &= \{P \in [0, 1] : \lambda_1(f(P)) \leq \lambda_1(P)\}. \end{aligned}$$

Thus

$$\begin{aligned} C_0 &= \{P \in [0, 1] : P \leq f(P)\}; \\ C_1 &= \{P \in [0, 1] : f(P) \leq P\}. \end{aligned}$$

Also, $0 \in C_0$ and $1 \in C_1$. The assumption $P \notin C_0 \cup C_1$ for some $P \in [0, 1]$ implies both $f(P) > P$ and $f(P) < P$, which is absurd, so it must be the case that $[0, 1] \subset C_0 \cup C_1$. We now invoke Proposition 7.2 to conclude $C_0 \cap C_1 \neq \emptyset$. But if $P \in C_0 \cap C_1$, then clearly $f(P) = P$.

7.2 Brouwer's Theorem

We now give a detailed proof of Brouwer's Theorem in the case $n = 2$ and we also indicate how to prove the theorem for arbitrary n . The material in this section is inspired by a discussion found in Zeidler [163, p. 798]. Our starting point is Sperner's Lemma in $\mathbb{R}^2$. Let M be a closed triangle in $\mathbb{R}^2$ with vertices P_0, P_1, P_2 . In order to understand the general case it is important to be precise about some definitions. The r -dimensional sides of M are:

The vertices P_0, P_1, P_2 for $r = 1$.

The sides $\overline{P_0P_1}, \overline{P_1P_2}, \overline{P_2P_0}$ for $r = 2$.

The triangle itself for $r = 3$.

We define the *base* of a point P in M to be the side of M of smallest dimension which contains P . An integer i ($i = 0, 1, 2$) is said to be a *Sperner label* for P if P_i belongs to the base of P in M . It is important to understand exactly what this means. The points P_0, P_1, P_2 are labeled 0, 1, 2, respectively. Any *other* point on the side $\overline{P_0P_1}$ may be labeled either 0 or 1, regardless of how any other such points are labeled. The same holds for the remaining sides. Any point *strictly* inside M may be labeled 0, 1 or 2.

The following is Sperner's Lemma [151] for the case $n = 2$.

Lemma 7.1 *Suppose the triangle M of $\mathbb{R}^2$ with vertices P_0, P_1, P_2 is divided into subtriangles, and suppose each vertex of each subtriangle is assigned a Sperner label. Then the cardinality of the set of subtriangles whose vertices have distinct labels is odd.*

Proof. Call a subtriangle of the triangulation of M a *Sperner simplex* if its vertices have distinct labels, and call a side of a subtriangle *distinguished* if its vertices carry the labels $\{0, 1\}$.

Note that if a distinguished side of T has a vertex which lies in the interior of M then this side is also a distinguished side of precisely one other subtriangle in the triangulation of M . Thus there are an even number of distinguished sides having at least one vertex lying strictly inside M . On the other hand, by Proposition 7.1, there are an odd number of distinguished sides lying on $\overline{P_0P_1}$ and there none lying on either $\overline{P_1P_2}$ or $\overline{P_2P_0}$. Therefore, the total number of distinguished sides is an odd integer, say k . At the same time, the vertices of any subtriangle which has distinguished sides and which is *not* a Sperner triangle must be labeled either $\{0, 1, 1\}$ or $\{0, 0, 1\}$. Each such triangle has precisely two distinguished sides. Suppose there are m subtriangles of this type and suppose there are n Sperner triangles. Then $2m + n = k$ and we conclude that n is odd. ■

We now prove the KKM Theorem for the case $n = 2$. Note that here we use M to denote, inclusively, the vertices of the triangle along with its sides and interior. The only new idea to enter at this point in the development is that of a 'triangulation' of a given triangle. It is easy to see how to do this in an orderly way by beginning with an equilateral triangle, partitioning each of

the sides into n subintervals, and joining points of the subintervals with lines which are parallel to the sides of the given triangle. In this way it is possible to achieve a triangulation in which each of the subtriangles is also equilateral and has diameter $1/n$ times the diameter of the original triangle.

Lemma 7.2 *Suppose M is a triangle in $\mathbb{R}^2$ with vertices P_0, P_1, P_2 , and suppose C_0, C_1, C_2 are closed subsets of $\mathbb{R}^2$ which satisfy*

- (i) $P_i \in C_i, i = 0, 1, 2$;
- (ii) $\overline{P_i P_j} \subset C_i \cup C_j, i, j = 0, 1, 2$;
- (iii) $M \subset C_0 \cup C_1 \cup C_2$.

Then $C_0 \cap C_1 \cap C_2 \neq \emptyset$.

Proof. For $k = 1, 2, \dots$ consider a sequence of triangulations $\mathcal{P}_k$ of M where the diameter of the largest triangle in $\mathcal{P}_k$ is ε_k with $\varepsilon_k \rightarrow 0$ as $k \rightarrow \infty$. Assign a Sperner labeling to each triangulation with the added provision that each vertex Q is assigned a label i consistent with $Q \in C_i$. This is possible by (ii) and (iii). Then by Lemma 7.2 each triangulation $\mathcal{P}_k$ contains a Sperner simplex with vertices $\{P_0^{(k)}, P_1^{(k)}, P_2^{(k)}\}$, where $P_i^{(k)} \in C_i, i = 0, 1, 2$. Now choose respective convergent subsequences $\{P_i^{(k_j)}\}_{j \geq 1}$ ($i = 0, 1, 2$) with

$$\lim_{j \rightarrow \infty} P_i^{(k_j)} = Q_i \quad (i = 0, 1, 2).$$

Since the diameters of the subtriangles tend to zero as $j \rightarrow \infty$ it must be the case that $Q_0 = Q_1 = Q_2 = Q$, and since each of the sets C_i is closed, $Q \in C_i$, for $i = 1, 2, 3$. ■

We are now in a position to prove Brouwer's Theorem for the case $n = 2$.

Theorem 7.1 *Suppose M is a triangle in $\mathbb{R}^2$ with vertices P_0, P_1, P_2 , and suppose $f : M \rightarrow M$ is continuous. Then f has a fixed point.*

Proof. Every point P in M can be written in the form

$$P = \lambda_0(P)P_0 + \lambda_1(P)P_1 + \lambda_2(P)P_2,$$

where $0 \leq \lambda_i(P) \leq 1$ ($i = 0, 1, 2$) and $\lambda_0(P) + \lambda_1(P) + \lambda_2(P) = 1$.

Set $C_i = \{P \in M : \lambda_i(f(P)) \leq \lambda_i(P)\}$, $i = 0, 1, 2$. Since f is continuous each of the sets C_i is closed. Also, since $\lambda_i(P_i) = 1$ for each i , $P_i \in C_i$ for $i = 0, 1, 2$. If $\lambda_i(f(P)) > \lambda_i(P)$ for each $i = 0, 1, 2$, then it would follow that

$$\lambda_0(f(P)) + \lambda_1(f(P)) + \lambda_2(f(P)) > 1.$$

Since this contradicts $f : M \rightarrow M$ it must be the case that for each $P \in M$ there exists $i \in \{0, 1, 2\}$ for which $\lambda_i(f(P)) \leq \lambda_i(P)$, that is, $M \subset C_0 \cup C_1 \cup C_2$ and (iii) of Lemma 7.2 holds. Finally, if $P \in \overline{P_0 P_1}$ then $\lambda_2(P) = 0$. However, this fact along with the assumption $P \notin C_0 \cup C_1$ leads to the contradiction

$$\lambda_0(f(P)) + \lambda_1(f(P)) + \lambda_2(f(P)) > \lambda_0(P) + \lambda_1(P) = 1.$$

Similar reasoning applies to the remaining sides, so we conclude that (ii) of Lemma 7.2 also holds. Therefore, we may apply Lemma 7.2 and conclude that there exists $P \in C_0 \cap C_1 \cap C_2$. This implies

$$\lambda_i(f(P)) \leq \lambda_i(P), \quad i = 0, 1, 2,$$

which, in turn, implies $1 = \sum_{i=0}^2 \lambda_i(f(P)) \leq \sum_{i=0}^2 \lambda_i(P) = 1$. Therefore, $\lambda_i(f(P)) = \lambda_i(P)$, $i = 0, 1, 2$, and thus $f(P) = P$. ■

We conclude this section with a few remarks on how the above theorem is extended to higher dimensions. The difficulties largely involve terminology. An n -dimensional closed simplex in $\mathbb{R}^n$ is the convex hull of $n+1$ points $P_0, \dots, P_n$ in $\mathbb{R}^n$ which do not lie in an $n-1$ -dimensional subspace of $\mathbb{R}^n$. As before, the *base* of a point in M is the subsimplex of lowest dimension which contains the point. To see how the induction goes we illustrate how to pass from $n=2$ to $n=3$.

Let M be a closed three-dimensional simplex (i.e., a tetrahedron) in $\mathbb{R}^3$ with vertices labeled $\{0, 1, 2, 3\}$, subdivide M into three-dimensional subsimplexes, and assign to each vertex of the resulting partition a number that belongs to its base in M . Now call a triangular face of a subsimplex *distinguished* if it carries the labels $\{0, 1, 2\}$, and observe that if a vertex of a distinguished triangle lies in interior of M then it is a face of precisely two distinct subsimplexes of the partition. By the case $n=2$ there are an odd number of distinguished triangles on the boundary of M . (Here it is important to observe that *only a face initially labeled* $\{0, 1, 2\}$ can have distinguished triangles.) Therefore, the total number of distinguished triangles is odd. On the other hand, the vertices of a simplex which is *not* a Sperner simplex (i.e., one that does not have distinguished labels) and which *does have a distinguished triangle* must have one of the following labelings: $\{0, 1, 2, 0\}$, $\{0, 1, 2, 1\}$, or $\{0, 1, 2, 2\}$. Each such simplex has precisely two distinguished triangular faces. If there are m such simplexes and if n are the number of Sperner simplexes, then $2m+n$ must equal an odd number; thus n is odd.

In order to state Sperner's Lemma in its full generality we need a little more explanation. Let $M^n = [P_0, \dots, P_n]$ be a closed n -dimensional simplex in $\mathbb{R}^n$. By a subdivision S of M^n we mean a decomposition of M^n into finitely many non-overlapping n -simplexes $s_1, \dots, s_k$ such that (1) the intersection of any two simplexes in S is either empty or a common face of each, and (2) each $n-1$ -simplex in S that is not on the boundary of M^n is a common face of exactly two n -simplexes of S . The *base* of a vertex P in S the face of M^n of lowest dimension which contains P .

Lemma 7.3 [Sperner] *Let S be a subdivision of M^n and label each vertex $P \in S$ with one of the numbers $\{i_0, \dots, i_s\}$ whenever $[P_{i_0}, \dots, P_{i_s}]$ is the base of P in M^n . Then the number of simplexes in S which are labeled $\{0, \dots, n\}$ is odd.*

The proof is by induction with the inductive step applied on the boundary of M^n .

Sperner's Lemma can be used to extend the KKM theorem to higher dimensions, and this, in turn, leads to Brouwer's Theorem. The general version of the KKM theorem needed to carry this out is the following. The proof follows closely that given in the case $n = 2$.

Theorem 7.2 *Suppose $\{P_0, \dots, P_n\}$ are the vertices of an n -simplex in $\mathbb{R}^n$ and suppose $C_0, \dots, C_n$ are closed subsets of $\mathbb{R}^n$ for which*

$$\text{conv}\{P_{i_0}, \dots, P_{i_k}\} \subset \bigcup_{j=0}^k C_{i_j}$$

for every subset $\{P_{i_0}, \dots, P_{i_k}\}$ of $\{P_0, \dots, P_n\}$. Then

$$\bigcap_{i=0}^n C_i \neq \emptyset.$$

7.3 Further comments on Brouwer's Theorem

If X is a topological space and $M \subseteq X$, then a *continuous* mapping $r : X \rightarrow M$ is called a *retraction* if $r(x) = x$ for all $x \in M$. When this occurs, M is said to be a *retract* of X .

We begin with the following simple fact.

Proposition 7.3 *Every closed convex subset K in $\mathbb{R}^n$ is a retract of $\mathbb{R}^n$.*

Proof. This is a special case of our earlier observations about ℓ_2 . Let $p \in \mathbb{R}^n$ and set

$$r = \inf\{\|p - x\| : x \in K\}.$$

Then it is possible to choose a sequence $\{x_n\} \subseteq K$ such that

$$\lim_{n \rightarrow \infty} \|x_n - p\| = r.$$

Since the sequence $\{x_n\}$ is bounded it has a convergent subsequence. (An extension of the Bolzano-Weierstrass Theorem assures that K is compact.) So assume that $\lim_{k \rightarrow \infty} x_{n_k} = x$. Then $x \in K$ because K is closed, and clearly $\|x - p\| = r$.

Now suppose there are two points $x, y \in K$, such that

$$\|x - p\| = \|y - p\| = r.$$

Suppose $x \neq y$ and consider the point $m = \frac{1}{2}(x + y)$. Then by the Pythagorean Theorem,

$$\|p - m\|^2 + \|m - x\|^2 = r^2.$$

If $m \neq x$ this would imply $\|p - m\| < r$. This is a contradiction since convexity of K implies $m \in K$. Therefore $m = x = y$. This assures that for each point $p \in \mathbb{R}^n$ there is a unique point $\rho(p) \in K$ which is nearest to p .

To conclude that ρ is the desired retraction we need only to show that ρ is continuous. However, in view of Theorem 6.1 we know even more, namely, that ρ is nonexpansive, that is, for each $u, v \in \mathbb{R}^n$

$$\|\rho(u) - \rho(v)\| \leq \|u - v\|.$$

■

The above fact permits an extension of Brouwer's Theorem to arbitrary bounded closed convex subsets of $\mathbb{R}^n$. For such a set K select a simplex S sufficiently large that $K \subseteq S$. By Proposition 7.3 there exists a retraction $r : S \rightarrow K$. The composition mapping $f \circ r$ is a continuous mapping of S into itself and therefore must have a fixed point x . Moreover, x must lie in the range of $f \circ r$, which is K . Since $r(x) = x$ for points in K ,

$$x = f \circ r(x) = f(x).$$

Therefore, we have the following.

Theorem 7.3 *Let K be a bounded closed convex subset of $\mathbb{R}^n$ and let $f : K \rightarrow K$ be continuous. Then f has a fixed point.*

One very interesting consequence of Brouwer's Theorem is the following fact, which seems intuitively clear.

Proposition 7.4 *The surface, S , of a nontrivial closed ball B in $\mathbb{R}^n$ is **not** a retract of B .*

Proof. Suppose r is a retraction of B onto S . Then $-r$ would map each point of S to its antipodal point. Thus $-r$ would be a continuous mapping of B into B which is fixed point free, and this contradicts Brouwer's Theorem. ■

In fact, Proposition 7.4 is equivalent to Brouwer's Theorem. To see this, suppose $f : B \rightarrow B$ is continuous. If $f(x) \neq x$ for each $x \in B$ then one could construct a retraction $r : B \rightarrow S$ as follows: For each x follow the directed line segment from $f(x)$ through x and find its intersection with S . Call this point $r(x)$. It is easy to see that the mapping $x \mapsto r(x)$ is continuous, and the existence of such a retraction contradicts Proposition 7.4. Therefore, f must have a fixed point.

The fact that the above proposition fails in an infinite-dimensional Banach space answers a question posed in the famous collection of mathematical problems known as *The Scottish Book* (see [119]). This is a book named after a cafe, in what was at that time Lvov, Poland, where Banach and his collaborators frequently gathered to discuss mathematics. Around 1935 S. Ulam raised the

following question, which appears as Problem 36 in *The Scottish Book*: “Can one transform continuously the solid sphere of a Hilbert space into its boundary such that the transformation should be the identity on the boundary of the ball?” An addendum indicates that Tychonoff had provided an affirmative answer. Subsequently S. Kakutani gave several examples of fixed point free continuous mappings of the unit ball of Hilbert space into itself, any of which yield an affirmative answer to Ulam’s question.

Another nice solution to Ulam’s problem, and one that holds in an arbitrary Banach space, was published by Victor Klee in 1955. In [101] Klee proved that any infinite-dimensional Banach space X is homeomorphic with the ‘punctured’ space $X \setminus \{0\}$. Let $h : X \rightarrow X \setminus \{0\}$ be such a homeomorphism. (Thus h is continuous, one-to-one, and surjective (onto), and h^{-1} is also continuous.) Now assume (as one may) that $h(x) = x$ if $x \in X$ and $\|x\| \geq 1$. The required retraction is now given by the mapping

$$R : x \rightarrow \frac{h(x)}{\|h(x)\|}.$$

Brouwer’s Theorem assures that if K is a closed convex set in $\mathbb{R}^n$ with its usual norm, $\|\cdot\|_2$, then any continuous mapping $f : K \rightarrow K$ has at least one fixed point. However, this fundamental result extends to $\mathbb{R}^n$ with any other norm. In fact, any two norms on $\mathbb{R}^n$ are equivalent. This is a consequence of the Closed Graph Theorem, which assures that the identity mapping $I : (\mathbb{R}^n, \|\cdot\|_2) \rightarrow (\mathbb{R}^n, \|\cdot\|)$ is continuous. (It is also a consequence of a more elementary fact which we prove in Section 9.1 of Chapter 9. *Any linear mapping $T : E \rightarrow F$ where E and F are Banach spaces with E finite-dimensional is necessarily continuous.*)

A consequence of the preceding is the fact that any closed convex set K_1 in $(\mathbb{R}^n, \|\cdot\|)$ is homeomorphic to a closed convex set K_2 in $(\mathbb{R}^n, \|\cdot\|_2)$ via the identity mapping $I : K_2 \rightarrow K_1$. Let $f : K_1 \rightarrow K_1$ be continuous. Now consider the mapping $I^{-1} \circ f \circ I : K_2 \rightarrow K_2$. This mapping is continuous and has a fixed point $x \in B_2$ via Brouwer’s Theorem in $\mathbb{R}^n$. But

$$I^{-1} \circ f \circ I(x) = x \Rightarrow f \circ I(x) = I(x) = x,$$

so x is also a fixed point of f in K_1 .

Since any finite-dimensional Banach space can be viewed as $\mathbb{R}^n$ with an equivalent norm simply by identifying the respective basis vectors, the following is the most useful formulation of Brouwer’s Theorem.

Theorem 7.4 *Suppose K is a nonempty bounded closed convex subset of a finite-dimensional Banach space X , and suppose $f : K \rightarrow K$ is continuous. Then f has at least one fixed point.*

The following facts will be needed in the next section. If $(X, \|\cdot\|)$ is an infinite-dimensional Banach space and if $\{x_1, x_2, \dots, x_n\} \subseteq X$, then the smallest

subspace X_m of X which contains $\{x_1, x_2, \dots, x_n\}$ is, in fact, finite-dimensional (having dimension at most n). (This is because all linear combinations of the form $\sum_{i=1}^n \alpha_i x_i$ form a subspace of X so some subset of $\{x_1, x_2, \dots, x_n\}$ serves as a basis for this subspace.) Therefore $(X_m, \|\cdot\|)$ has the same topological properties as $\mathbb{R}^m$. Consequently, if $K = \text{conv}(\{x_1, x_2, \dots, x_n\})$, then K is a bounded closed convex subset of X_m and therefore K is compact.

7.4 Schauder's Theorem

We begin by observing that Brouwer's Theorem cannot be extended to any Banach space.

Example 7.1 Consider the unit ball B in the Banach space c_0 . For $x = (x_1, x_2, \dots) \in B$ define

$$T(x) = (1 - |x_1|, x_1, x_2, \dots).$$

It is easy to check that $T : B \rightarrow B$. Also, T is continuous; in fact,

$$\|T(x) - T(y)\| = \|x - y\| \text{ for } x, y \in B.$$

However, $T(x) = x$ implies $x_1 = x_2 = \dots = 0$ which, in turn, implies $1 - |x_1| = 0$ —a contradiction.

This shows that Brouwer's Theorem fails in some infinite-dimensional spaces and, in fact, the classical result of Dugundji assures that it fails in *any* infinite-dimensional space. The discussion about retractions in the previous section leads to this conclusion as well. On the other hand, bounded closed convex subsets of $\mathbb{R}^n$ are compact. It turns out that this is precisely the assumption needed to assure the validity of an infinite-dimensional version of Brouwer's Theorem.

Theorem 7.5 (Schauder's Theorem) Let K be a nonempty compact convex subset of a Banach space X , and suppose $f : K \rightarrow K$ is continuous. Then f has at least one fixed point.

Proof. As we have seen before $f(K)$ is precompact, hence totally bounded. Therefore, for each $n \in \mathbb{N}$ there exists $\{y_1, y_2, \dots, y_{N_n}\} \subseteq f(K)$ such that for any $x \in K$

$$\min_{1 \leq i \leq N_n} \|f(x) - y_i\| < \frac{1}{n}.$$

For each such i define

$$a_i(x) = \max \left\{ \frac{1}{n} - \|f(x) - y_i\|, 0 \right\};$$

thus for each $x \in K$ there must exist at least one $i \in \{1, 2, \dots, N_n\}$ for which $a_i(x) \neq 0$. Now define the so-called *Schauder operator* $P_n : K \rightarrow K$ as follows:

$$P_n(x) = \frac{\sum_{i=1}^{N_n} a_i(x)y_i}{\sum_{i=1}^{N_n} a_i(x)}.$$

Note that $P_n(x) \in K$ because $P_n(x)$ is a *convex combination* of the elements $\{y_1, y_2, \dots, y_{N_n}\}$. Also, since f is continuous, each of the functions a_i is as well.

Now let $K_n = \text{conv}(\{y_1, y_2, \dots, y_{N_n}\})$. Then K_n is a bounded closed convex subset of the finite-dimensional Banach space which is spanned by the vectors $\{y_1, y_2, \dots, y_{N_n}\}$, and moreover $P_n : K_n \rightarrow K_n$. Brouwer's Theorem applies. Each of the mappings P_n has a fixed point $x_n \in K_n \subseteq K$. Since K is compact the sequence $\{x_n\}$ has a subsequence $\{x_{n_k}\}$ which converges, say $\lim_{k \rightarrow \infty} x_{n_k} = x \in K$. Now observe that for any n ,

$$\begin{aligned} \|P_n(x) - f(x)\| &= \left\| \left(\sum_{i=1}^{N_n} a_i(x)(y_i - f(x)) \right) / \sum_{i=1}^{N_n} a_i(x) \right\| \\ &\leq \left[\frac{1}{n} \sum_{i=1}^{N_n} a_i(x) \right] / \sum_{i=1}^{N_n} a_i(x) = \frac{1}{n}. \end{aligned}$$

It follows that x is a fixed point of f since

$$\|x_{n_k} - f(x)\| \leq \|P_{n_k}(x_{n_k}) - f(x_{n_k})\| + \|f(x_{n_k}) - f(x)\|$$

and the right-hand side tends to 0 as $k \rightarrow \infty$ while the left-hand side tends to $\|x - f(x)\|$. ■

7.5 Stability of Schauder's Theorem

In some sense continuity is an 'ideal' assumption, often merely added to make otherwise difficult problems manageable. This section deals with situations in which the function under consideration cannot be shown to be continuous, but at the same time can be shown to be 'almost continuous'. For such functions an extension of Schauder's Theorem is useful.

Definition 7.1 Let X be a topological space and M a metric space. A mapping $f : X \rightarrow M$ is said to be ε -continuous if for every $x \in X$ there exists a neighborhood U_x of x such that $\text{diam } f(U_x) \leq \varepsilon$.

We shall apply some basic topological facts to give a detailed proof of the following classical result of V. Klee [102].

Theorem 7.6 *Suppose K is a compact convex subset of a normed linear space, and suppose $f : K \rightarrow K$ is ε -continuous. Then there exists a point $x_0 \in K$ such that $\|f(x_0) - x_0\| \leq \varepsilon$.*

Our proof uses a standard 'partition of unity' argument to show how f may be approximated by a suitable continuous function g to which Schauder's Theorem may be applied. (Klee's original approach involved approximating the domain of f with a polyhedral set, using triangulation to approximate f with a piecewise linear functional, and then applying Brouwer's theorem.)

The proof of Theorem 7.6 is a direct consequence of the following extension theorem.

Theorem 7.7 *Suppose K is a compact convex subset of a normed linear space X , and suppose $f : K \rightarrow K$ is ε -continuous. Then there exists a continuous mapping $\bar{f} : K \rightarrow K$ such that $\|\bar{f}(x) - f(x)\| \leq \varepsilon$ for each $x \in K$.*

Before proving Theorem 7.7 we need some terminology. If X is a topological space and $\phi : X \rightarrow \mathbb{R}$, then the *support* of ϕ is defined to be the closure of the set $\phi^{-1}(\mathbb{R} - \{0\})$. In particular, if $x \notin (\text{support } \phi)$ then there is some neighborhood of x on which ϕ vanishes.

Definition 7.2 *Let $\{U_1, \dots, U_n\}$ be an open covering of the topological space X . A family of continuous functions*

$$\phi_i : X \rightarrow [0, 1], \quad i = 1, \dots, n,$$

is said to be a partition of unity of X dominated by the family $\{U_i\}$ if

- (i) *(support ϕ_i) $\subseteq U_i$ for each i ; and*
- (ii) *$\sum_{i=1}^n \phi_i(x) = 1$ for each $x \in X$.*

The topological fact we need is the following. Its proof can be found in any standard topology text (e.g., [49], [89], [125]).

Theorem 7.8 (Partitions of unity) *Let $\{U_1, \dots, U_n\}$ be a finite open covering of the metric (or, more generally, normal topological) space M . Then there exists a partition of unity of M dominated by the family $\{U_i\}$.*

Proof of Theorem 7.7. Since f is ε -continuous, for each $x \in K$ there exists $\delta_x > 0$ such that $\text{diam}(f(U(x; \delta_x))) \leq \varepsilon$. (Recall that $U(x; \delta_x)$ denotes the open ball centered at x with radius δ_x .) Since K is compact there exists a finite set

$\{x_1, \dots, x_j\} \subseteq K$ such that $K \subseteq \bigcup_{i=1}^j U(x_i; \delta_{x_i}/2)$. Let $\delta = \inf\{\delta_{x_i} : 1 \leq i \leq j\}$.

Note that we now have the following: If $x, y \in K$ and $\|x - y\| < \delta/2$ then there exists an i such that $x, y \in U(x_i; \delta_{x_i})$; hence $\|f(x) - f(y)\| \leq \varepsilon$. Now again use the fact that K is compact to select $A = \{a_1, \dots, a_n\} \subseteq K$ so that $K \subseteq \bigcup_{i=1}^n U_i$, where $U_i = U(a_i; \delta/2)$, $i = 1, \dots, n$. Then the family $\{U_i\}$ is a finite open

covering of K so there exists a partition of unity $\{\phi_i\}_{i=1}^n$ of K dominated by the family $\{U_i\}$. Define the function $\bar{f} : K \rightarrow X$ as follows:

$$\bar{f}(x) = \sum_{i=1}^n \phi_i(x) f(a_i) \text{ for each } x \in K.$$

Since $f(A) \subseteq K$ and $\sum_{i=1}^n \phi_i(x) = 1$ we see that, in fact, $\bar{f}(K) \subseteq \overline{\text{conv}}(f(A)) \subseteq K$.

Furthermore, since each of the functions ϕ_i is continuous, $\bar{f}$ is continuous.

Finally, observe that for $x \in K$,

$$\begin{aligned} \|f(x) - \bar{f}(x)\| &= \left\| f(x) - \sum_{i=1}^n \phi_i(x) f(a_i) \right\| \\ &= \left\| \sum_{i=1}^n \phi_i(x) [f(x) - f(a_i)] \right\| \\ &\leq \sum_{i=1}^n \phi_i(x) \|f(x) - f(a_i)\|. \end{aligned}$$

Since $\phi_i(x) = 0$ if $x \notin U_i$ while $\|f(x) - f(a_i)\| \leq \varepsilon$ if $x \in U_i$, we conclude that

$$\|f(x) - \bar{f}(x)\| \leq \varepsilon.$$

By Schauder's Theorem $\bar{f}$ has a fixed point x_0 , where $\|f(x_0) - x_0\| \leq \varepsilon$. ■

7.6 Banach algebras: Stone Weierstrass Theorem

Certain Banach spaces, especially those whose elements are themselves functions (e.g., the space $C[0, 1]$), possess additional algebraic properties. We pause here to discuss a fundamental fact from real analysis that will be referred to in the next section.

Definition 7.3 A Banach space $(X, \|\cdot\|)$ is said to be a Banach algebra if, in addition to addition and scalar multiplication, there is a product operator xy defined for $x, y \in X$ which satisfies for each $x, y, z \in X$, and $\alpha \in \mathbb{R}$,

- (i) $x(yz) = (xy)z$;
- (ii) $x(y + z) = xy + xz$;
- (iii) $(y + z)x = yx + zx$;
- (iv) $(\alpha x)y = \alpha(xy) = x(\alpha y)$;
- (v) $\|xy\| \leq \|x\| \|y\|$.

If $f, g \in C[0, 1]$ and one defines fg as usual by setting

$$(fg)(t) = f(t)g(t),$$

then all the above properties hold. In particular,

$$\|fg\| = \sup_{t \in [0,1]} |f(t)g(t)| \leq \sup_{t \in [0,1]} |f(t)| \sup_{t \in [0,1]} |g(t)| = \|f\| \|g\|.$$

The following well-known theorem is one of the most useful in all of analysis. A proof based implicitly on Banach's contraction mapping principle can be found in [67].

Theorem 7.9 (Stone-Weierstrass Theorem) *Let S be a compact Hausdorff topological space and let $C(S)$ be the Banach algebra of all bounded, continuous, real-valued functions on S with the usual norm: $\|f\| = \sup_{s \in S} |f(s)|$. Suppose X is a subalgebra of $C(S)$ which contains the constant functions and which separates points of S . Then X is dense in $C(S)$.*

We should be clear about what the theorem says. To say that X is a subalgebra of $C(S)$ means that X is a subspace of $C(S)$ which has the property that $f, g \in X$ and $\alpha \in \mathbb{R}$ implies $fg \in X$ and $\alpha f \in X$. A constant function is a function with the property $f(s) \equiv \alpha$ for $s \in S$. The condition that X separates points of S means that if $u, v \in S$ and $u \neq v$ then there exists $f \in X$ such that $f(u) \neq f(v)$. The conclusion is that the topological closure $\overline{X}$ of X in $C(S)$ coincides with $C(S)$.

The following is the familiar special case of the Stone-Weierstrass Theorem we allude to in the next section.

Corollary 7.1 *The space $P[0, 1]$ of polynomial functions is dense in the space $C[0, 1]$.*

To see that the corollary is a special case of the general theorem one need only to observe that $P[0, 1]$ is a subalgebra of $C[0, 1]$ and that the identity function separates points in $[0, 1]$.

In view of the corollary we know that if $f \in C[0, 1]$ and $\varepsilon > 0$, then there is a polynomial $\bar{f} \in P[0, 1]$ such that $\|f - \bar{f}\| < \varepsilon$.

7.7 Leray-Schauder degree

One of the principal tools in applications of nonlinear functional analysis is the concept of degree of a mapping. We give an *outline* of the development of this theory here.

Classical degree theory, in its most general sense, is the study of mapping degree for various classes of continuous mappings with domains contained in one Banach space X and which take values in a (possibly different) Banach space Y . Such a theory consists of the algebraic count of the number of solutions of the equation $f(x) = y_0$, where f is defined on the closure $\overline{G}$ of an open set G in X and y_0 is a given point in Y which is assumed to lie in the complement of the image of the boundary ∂G of G in Y (i.e., $y_0 \notin f(\partial G)$). The value of the degree function for any such count is an ordinary integer, either positive or negative.

Degree theory in $\mathbb{R}^n$ was introduced by L. E. J. Brouwer in 1912. In this section we seek to make the above observations more intuitively clear in $\mathbb{R}^n$ by beginning with the case $n = 1$ and $G = (0, 1)$. (There is nothing special about the interval $(0, 1)$ —we could just as well consider $G = (a, b)$.)

Suppose $f \in C[0, 1]$ and $f(0) \neq 0 \neq f(1)$. Then by Weierstrass's approximation theorem if $\varepsilon > 0$ there is a polynomial function $\bar{f}$ such that $\|f - \bar{f}\| < \varepsilon$. In particular, any such $\bar{f}$ has at most a finite number of zeros, $x_1, \dots, x_m$ in $[0, 1]$. Also, since $\bar{f}$ has at most a finite number of critical points ($n - 1$ if its degree is n), by adding an arbitrarily small constant to $\bar{f}$ if necessary, we can retain the condition $\|f - \bar{f}\| < \varepsilon$ and at the same time insure that $\bar{f}'(x_i) \neq 0$ for all i . Having done all this, let $G = (0, 1)$ and define $\deg(\bar{f}, G) = 0$ if f has no zeros in $(0, 1)$; otherwise set

$$\deg(\bar{f}, G) = \sum_{i=1}^m \operatorname{sgn}(\bar{f}'(x_i)),$$

where sgn is the function

$$\operatorname{sgn}(x) = \begin{cases} 1 & \text{if } x > 0, \\ -1 & \text{if } x < 0. \end{cases}$$

Now define

$$\deg(f, G) = \deg(\bar{f}, G).$$

In order to see that the function $\deg$ is well defined, it is necessary to show that it is independent of the choice of $\bar{f}$. In other words, it needs to be shown that if $\varepsilon > 0$ is sufficiently small and if $\bar{f}$ and $\bar{g}$ are polynomials for which $\|f - \bar{f}\| < \varepsilon$ and $\|f - \bar{g}\| < \varepsilon$ then $\deg(\bar{f}, G) = \deg(\bar{g}, G)$. However, this is easy to see if one thinks about how polynomials behave. Note that since we are assuming $f(0) \neq 0 \neq f(1)$, we can take $\varepsilon = \min\{|f(0)|, |f(1)|\}$. Then $\bar{f}$ and $\bar{g}$ will have the same respective signs at 0 and 1 if $\|f - \bar{f}\| < \varepsilon$. There are three cases.

Case 1: $f(0) \cdot f(1) > 0$. In this case either $\bar{f}$ will have no zeros, or $\bar{f}$ will have an *even* number of zeros, say $x_1 < x_2 < \dots < x_m$ with $\bar{f}'(x_i)$ and $\bar{f}'(x_{i+1})$ having opposite signs. Thus either $\deg(\bar{f}, G) = 0$ because $\bar{f}$ has no zeros, or

$$\deg(\bar{f}, G) = \sum_{i=1}^m \operatorname{sgn}(\bar{f}'(x_i)) = [f'(x_1) + f'(x_2)] + \dots + [f'(x_{m-1}) + f'(x_m)] = 0.$$

Case 2: $f(0) < 0$ and $f(1) > 0$. In this case $\bar{f}$ will have an *odd* number of zeros, say $x_1 < x_2 < \dots < x_m$, again with $\bar{f}'(x_i)$ and $\bar{f}'(x_{i+1})$ having opposite signs, but with $\bar{f}'(x_1) > 0$ and $\bar{f}'(x_m) > 0$. Thus

$$\deg(\bar{f}, G) = \sum_{i=1}^m \operatorname{sgn}(\bar{f}'(x_i)) = 1.$$

Case 3: $f(0) > 0$ and $f(1) < 0$. This is the same as case (2), but with $\bar{f}'(x_1) < 0$ and $\bar{f}'(x_m) < 0$. Thus

$$\deg(\bar{f}, G) = \sum_{i=1}^m \operatorname{sgn}(\bar{f}'(x_i)) = -1.$$

Since the above analysis holds for *any* polynomial function $\bar{f}$ for which $\|f - \bar{f}\| < \varepsilon$, $\deg(f, G)$ is well defined and takes on the values 0, 1, or -1 according to the cases prescribed.

Mapping Degree in $\mathbb{R}^n$.

In passing to $\mathbb{R}^n$ the situation is less obvious, but the principle is the same. We only present an outline of the procedure here.

Let G be a bounded open subset of $\mathbb{R}^n$ with $f : \bar{G} \rightarrow \mathbb{R}^n$ continuous, let $y \in \mathbb{R}^n$, and assume $f(x) \neq y$ for $x \in \partial G$. As in the case $n = 1$ it is possible to approximate f with a function $\bar{f} : \bar{G} \rightarrow \mathbb{R}^n$, which is continuously differentiable on G and which satisfies

$$\sup_{x \in \partial G} \|f(x) - \bar{f}(x)\| < \inf_{x \in \partial G} \|f(x) - y\|,$$

and for which the equation $\bar{f}(x) = y$ either has no solutions in G or has at most finitely many solutions, $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m$, in G . For each such solution $\mathbf{x}_k$ the derivative $\bar{f}'(\mathbf{x}_k)$ is given by the linear transformation represented by the $n \times n$ matrix

$$\bar{f}'(\mathbf{x}_k) = \left[\frac{\partial \bar{f}_i}{\partial x_j}(\mathbf{x}_k) \right].$$

It may be further assumed (this is a much deeper fact) that each of the solutions is *regular*, that is,

$$\det(\bar{f}'(\mathbf{x}_k)) \neq 0, \quad k = 1, \dots, m.$$

Note that the boundary condition assures that $\bar{f}(x) \neq y$ for each $x \in \partial G$.

If $f(x) = y$ has no solutions, set $\deg(f, G, y) = 0$. Otherwise set

$$\deg(f, G, y) = \sum_{k=1}^m \operatorname{sgn}(\det(\bar{f}'(\mathbf{x}_k))).$$

It can be shown that the number $\deg(f, G, y)$ is independent of the approximating function $\bar{f}$ and that the function $\deg$ is the *unique* function defined for all continuous $f : \bar{G} \rightarrow \mathbb{R}^n$ which satisfies the following properties.

Property 1: If $y \notin \partial G$, then $\deg(f, G, y)$ is an integer, and if $\deg(f, G, y) \neq 0$, then the equation $f(x) = y$ has a solution.

Property 2: If $f = I$ (the identity), then $\deg(I, G, y) = 1$ if $y \in G$ and $\deg(I, G, y) = 0$ if $y \notin \overline{G}$.

Property 3 (Additivity): $\deg(f, \cup_i G_i, y) = \sum_i \deg(f, G_i, y)$ if $G_i, i = 1, \dots, m$ are disjoint open sets with $y \notin \partial G_i$ for each i .

Property 4 (Homotopy): If $h : [0, 1] \times G \rightarrow \mathbb{R}^n$ is continuous, and if $h_t(x) \equiv h(t, x) \neq y$ for all $x \in \partial G$, then

$$\deg(h_0, G, y) = \deg(h_1, G, y).$$

Property 5 (Excision): If K is a closed subset of G and if $y \notin f(K)$, then

$$\deg(f, G, y) = \deg(\tilde{f}, G \setminus K, y),$$

where $\tilde{f}$ denotes the restriction of f to $G \setminus K = \{x \in G : x \notin K\}$.

To illustrate the usefulness of the above theory, we use it to prove a simple extension of the Brouwer fixed point theorem. Properties (1), (2), and (4) play the key roles.

Theorem 7.10 *Suppose $B = B(0; r)$ is a closed ball in $\mathbb{R}^n$ and suppose $T : B \rightarrow \mathbb{R}^n$ is a continuous mapping which satisfies*

$$(LS) \quad T(x) \neq \lambda x \text{ for all } x \in \partial B \text{ and } \lambda > 1.$$

Then T has a fixed point in B .

Proof. For $t \in [0, 1]$ let h_t be the mapping defined by taking $h_t(x) = x - tT(x)$, $x \in B$, that is, $h_t = I - tT$. First note that $h_t(x) \neq 0$ for $x \in \partial B$ and $t \in [0, 1]$ since $h_t(x) = 0$ implies $tT(x) = x$, from which $T(x) = t^{-1}x$, contradicting (LS). If $h_1(x) = 0$ for some $x \in \partial B$, then $f(x) = x$ and we are finished. Otherwise $0 \notin h_t(\partial B)$ for all $t \in [0, 1]$, so by (1), $\deg(h_t, B, 0)$ is defined. By (4) and (2)

$$1 = \deg(h_0, B, 0) = \deg(h_1, B, 0),$$

so by (1) again, $0 \in h_1(B)$. Clearly this implies T has a fixed point in B . ■

Note that in the above theorem one could replace B with the ball $B(x_0; r)$ and replace (LS) with the condition

$$T(x) - x_0 \neq \lambda(x - x_0) \text{ for all } x \in \partial B \text{ and } \lambda > 1.$$

A final comment about Brouwer's Theorem. In 1904, Bohl proved a theorem in [14] which can be formulated as follows: Suppose $B = B(0; r)$ is a closed ball in $\mathbb{R}^n$ and suppose $f : B \rightarrow \mathbb{R}^n$ is a continuous mapping for which $f(x) \neq 0$ for each $x \in B$. Then $f(x) = \mu x$ for some $x \in \partial B$ and $\mu < 0$. Taking $T = I - f$ and $\lambda = 1 - \mu$ this becomes the contrapositive formulation of Theorem 7.10.

Mapping Degree in Infinite Dimensions

We now turn briefly to the theory developed by Leray and Schauder (and for whom the boundary condition (LS) is named). Let X be an arbitrary Banach space. A mapping T defined on a subset D of X and taking values in X is said to be *compact* if T is continuous and if T maps bounded sets into sets whose closures are compact. Using approximations similar to the one used in the proof of the Schauder Fixed Point Theorem it is possible to extend degree theory to mappings of the form $I - T$ where T is a compact mapping. This theory of degree is characterized by the same properties that characterize the degree function in $\mathbb{R}^n$ and leads to the following theorem.

Theorem 7.11 (*Leray-Schauder Fixed Point Theorem*) *Let G be a nonempty bounded open subset of a Banach space X and let $T : \overline{G} \rightarrow X$ be a compact mapping which satisfies for some $x_0 \in G$:*

$$(LS) \quad T(x) - x_0 \neq \lambda(x - x_0) \text{ for all } x \in \partial G \text{ and } \lambda > 1.$$

Then T has a fixed point in G .

Subsequently degree theories have been established for many classes of mappings, including those we take up in the next section.

7.8 Condensing mappings

A well-known generalization of Schauder's theorem which bridges the gap between the metric and topological theories involves the notion of a 'measure of noncompactness'.

Definition 7.4 *Let D be a bounded subset of a metric space (M, d) . If $D = \bigcup_{i=1}^n P_i$, then we say that $P = \{P_1, \dots, P_n\}$ is a finite partition of D , and we set*

$$\|P\| = \sup\{\text{diam}(P_i) : i = 1, \dots, n\}.$$

The Kuratowski measure of noncompactness $\chi(D)$ of D is defined as follows:

$$\chi(D) = \inf\{\|P\| : P \text{ is a finite partition of } D\}.$$

Certain fundamental properties of χ are easy to see. For example:

- (i) $\chi(D) = \chi(\overline{D})$, and
- (ii) $\chi(D) = 0$ if and only if $\overline{D}$ is compact.
- (iii) $A \subseteq B \Rightarrow \chi(A) \leq \chi(B)$.
- (iv) $\chi(A \cup B) = \max\{\chi(A), \chi(B)\}$.

The above properties all follow immediately from the definition, although for (ii) one should remember that a set is compact if and only if it is closed and totally bounded.

There are other important properties of the noncompactness measure χ that hold for bounded subsets A, B of a Banach space.

- (v) $\chi(A + B) \leq \chi(A) + \chi(B)$.
- (vi) $\chi(\lambda A) = |\lambda| \chi(A), \quad \lambda \in \mathbb{R}$.
- (vii) $\chi(\text{conv}(A)) = \chi(A)$.

Properties (v) and (vi) are easy and left to the reader. Property (vii) is less obvious, and since its validity is a critical step in the proof of the main result of this section we give a detailed proof.

Step 1. For any bounded set A , $\chi(N_\varepsilon(A)) \leq \chi(A) + 2\varepsilon$, where

$$N_\varepsilon(A) = \bigcup_{x \in A} U(x; \varepsilon).$$

Proof. This follows from the trivial fact that for any bounded set S in a metric space, $\text{diam}(N_\varepsilon(S)) \leq \text{diam}(S) + 2\varepsilon$.

Step 2. Suppose $C = A \cup B$ where A and B are bounded and convex. Then

$$\chi(\text{conv}(C)) \leq \max\{\chi(A), \chi(B)\}.$$

Proof. Let $x \in \text{conv}(C)$ and suppose $x \notin A \cup B$. Then there exist $x_i \in C$ and $\lambda_i \geq 0, i = 1, \dots, n, \sum_{i=1}^n \lambda_i = 1$, such that $x = \sum_{i=1}^n \lambda_i x_i$. By regrouping we may

suppose that $x_i \in A$ for $1 \leq i \leq m$ and $x_i \in B$ for $m+1 \leq i \leq n$. Therefore,

$$x = \left(\sum_{i=1}^m \lambda_i \right) \frac{\sum_{i=1}^m \lambda_i x_i}{\sum_{i=1}^m \lambda_i} + \left(\sum_{i=m+1}^n \lambda_i \right) \frac{\sum_{i=m+1}^n \lambda_i x_i}{\sum_{i=m+1}^n \lambda_i},$$

so it follows that every such x can be written in the form $x = \alpha u + (1 - \alpha)v$ for $u \in A$, $v \in B$, where $0 < \alpha < 1$. Let $\varepsilon > 0$ and choose $N \in \mathbb{N}$ so that $\frac{1}{N} < \varepsilon$. For each $i = 0, 1, \dots, N$, let

$$C_i = \left\{ x \in \text{conv}(C) : x = \frac{i}{N}u + \left(1 - \frac{i}{N}\right)v \text{ for some } u \in A, v \in B \right\}.$$

Next observe that if $\alpha \in (0, 1)$, then $\alpha \in \left[\frac{i}{N}, \frac{i+1}{N}\right]$ for some $0 \leq i \leq N-1$, so it follows that if $x \in \text{conv}(C)$ and $x \notin A \cup B$, then $x \in N_\varepsilon(C_i)$ for some $i = 0, \dots, N$. Therefore,

$$\text{conv}(C) \subseteq \left(\bigcup_{i=1}^N N_\varepsilon(C_i) \right) \cup A \cup B,$$

from which

$$\begin{aligned} \chi(\text{conv}(C)) &\leq \max \left\{ \chi(A), \chi(B), \max_{1 \leq i \leq N} \chi(N_\varepsilon(C_i)) \right\} \\ &\leq \max \left\{ \chi(A), \chi(B), \max_{1 \leq i \leq N} [\chi(C_i) + 2\varepsilon] \right\}. \end{aligned}$$

But by (v) and (vi), for each i ,

$$\chi(C_i) = \chi \left(\frac{i}{N}A + \left(1 - \frac{i}{N}\right)B \right) \leq \frac{i}{N}\chi(A) + \left(1 - \frac{i}{N}\right)\chi(B),$$

which implies

$$\chi(C_i) \leq \max\{\chi(A), \chi(B)\}.$$

Therefore,

$$\chi(\text{conv}(C)) \leq \max\{\chi(A), \chi(B)\} + 2\varepsilon.$$

Step 2 now follows from the fact that $\varepsilon > 0$ is arbitrary.

Step 3. Suppose $C = \bigcup_{i=1}^{i=n} A_i$ where each A_i is bounded and convex. Then

$$\chi(\text{conv}(C)) \leq \max_{1 \leq i \leq n} \chi(A_i).$$

Proof. This follows from Step 2 and an obvious induction argument.

Proof of (vii). Let $a > \chi(A)$. Then A can be written as a finite union $A \subseteq \bigcup_{i=1}^n A_i$, where $\text{diam}(A_i) \leq a$ and since $\text{diam}(\text{conv}(A_i)) = \text{diam}(A_i)$ we may assume that each of the sets A_i is convex. By Step 3,

$$\chi(\text{conv}(A)) \leq \max_{1 \leq i \leq n} \chi(A_i) \leq \max_{1 \leq i \leq n} \text{diam}(A_i) \leq a.$$

Since this is true for all $a > \chi(A)$ the proof of (vii) is complete. $\blacksquare$

Definition 7.5 Let K be a subset of a metric space M . A mapping $T : K \rightarrow M$ is said to be a k -set contraction, $k \in (0, 1)$, if T is bounded and continuous, and if for all bounded subsets D of M

$$\chi(T(D)) \leq k\chi(D).$$

T is said to be condensing if T is bounded and continuous, and if

$$\chi(T(D)) < \chi(D)$$

for all bounded subsets D of M for which $\chi(D) > 0$.

Example 7.2 Let K be subset of a Banach space X . Suppose $T_1 : K \rightarrow X$ is a contraction mapping with Lipschitz constant $k < 1$, and suppose $T_2 : K \rightarrow X$ is a compact mapping (T is continuous and $\overline{T(D)}$ is compact for every bounded $D \subseteq X$). Then it is easy to see that the mapping $T = T_1 + T_2$ is a k -set contraction.

The following theorem was proved by Sadovskii in 1976. In 1965 Darbo proved the same theorem for k -set contractions, $k < 1$. Such mappings are obviously condensing.

Theorem 7.12 Suppose K is a nonempty bounded closed and convex subset of a Banach space and suppose $T : K \rightarrow K$ is condensing. Then T has a fixed point.

Proof. Fix $x \in K$ and let Σ denote the family of all closed convex subsets D of K for which $x \in D$ and $T : D \rightarrow D$. Now set

$$B = \bigcap_{D \in \Sigma} D,$$

and let

$$C = \overline{\text{conv}}\{T(B) \cup \{x\}\}.$$

Since $x \in B$ and $T : B \rightarrow B$ it must be the case that $C \subseteq B$. This, in turn, implies that $T(C) \subseteq T(B) \subseteq C$. Since $x \in C$ it follows that $C \in \Sigma$. Therefore, $B \subseteq C$, from which we conclude $B = C$.

We now have $T(C) = T(B) \subseteq C$ and

$$\chi(C) = \chi(\overline{\text{conv}}\{T(B) \cup \{x\}\}) = \chi(\{T(B) \cup \{x\}\}) = \chi(T(B)) = \chi(T(C)).$$

Since T is condensing this can only happen if $\chi(C) = 0$, that is, if C is compact. Therefore T is a continuous mapping of the compact convex set C into itself, so by Schauder's theorem T must have a fixed point. ■

The above theorem has applications in abstract hyperconvex spaces as well.

Theorem 7.13 *Let (M, d) be a bounded hyperconvex metric space, and suppose $T : M \rightarrow M$ is condensing. Then T has a fixed point.*

Proof. Since M is a complete metric space, M can be isometrically embedded in a Banach space X . If we use M to denote its image in X as well, then M as a hyperconvex subspace of X , and as such it is a nonexpansive retract of X . In particular, there exists a nonexpansive retraction R which maps $\overline{\text{conv}}(M)$ onto M . Note that now we have

$$T \circ R : \overline{\text{conv}}(M) \rightarrow M.$$

Now let $D \subseteq \overline{\text{conv}}(M)$ and suppose $\chi(D) > 0$. If $\chi(R(D)) = 0$ then since T is continuous, T maps compact sets into compact sets and so $\chi(T \circ R(D)) = 0 < \chi(D)$. On the other hand, if $\chi(R(D)) > 0$, then since R is nonexpansive, $\chi(R(D)) \leq \chi(D)$, and since T is condensing,

$$\chi(T \circ R(D)) < \chi(R(D)) \leq \chi(D).$$

We therefore conclude that $T \circ R$ is a *condensing* mapping of the closed convex set $\overline{\text{conv}}(M)$ into itself. Hence by Sadovskii's theorem, there exists $x \in \overline{\text{conv}}(M)$ such that $T \circ R(x) = x$. This implies that $x \in M$ from which $R(x) = x = T(x)$. ■

7.9 Continuous mappings in hyperconvex spaces

In this section we look more closely at compact hyperconvex spaces. For this we need to recall some notation of Chapter 4. Let (M, d) be a metric space and let $D \subseteq M$ be nonempty and bounded. Set:

$$\text{cov}(D) = \bigcap \{B : B \text{ is a closed ball and } D \subseteq B\}.$$

As before let $\mathcal{A}(M) = \{A \subseteq M : A = \text{cov}(A)\}$ denote the class of admissible sets in M .

We begin with the following consequence of Theorem 7.13.

Theorem 7.14 *Let M be a compact hyperconvex metric space and let $f : M \rightarrow M$ be continuous. Then f has a fixed point.*

Our next result specifically concerns the class $\mathfrak{S}$ of nonempty admissible sets of M endowed with the Hausdorff metric.

Theorem 7.15 *Let M be a compact hyperconvex metric space and let $f : M \rightarrow M$ be continuous. Define the mapping $\bar{f} : \mathfrak{S} \rightarrow \mathfrak{S}$ by setting*

$$\bar{f}(D) = \text{cov}(f(D)).$$

Set $D_0 = M$, let $D_n = \bar{f}(D_{n-1}) = \bar{f}^n(M)$, and suppose $D = \bigcap_{n=1}^{\infty} D_n$. Then $\bar{f}(D) = D$, and $D = \lim_{n \rightarrow \infty} D_n$, where the limit is taken relative to the Hausdorff metric H on $\mathfrak{S}$. In particular, if $f(x) = x$ then $x \in D$.

Remark 7.1 *The existence of a largest (relative to set inclusion) $D \in \mathfrak{S}$ such that $\bar{f}(D) = D$ (even without assuming continuity of f) follows easily from the fact that $\bar{f}$ is monotone ($D_1 \subseteq D_2 \Rightarrow \bar{f}(D_1) \subseteq \bar{f}(D_2)$) and every descending chain in $\mathfrak{S}$ has an infimum, namely, the intersection of its members. From this the existence of such a D is immediate from a fixed point theorem of Amman (see Theorem 11.D of [162]). It is the fact that this largest fixed point can be obtained by iterating $\bar{f}$ that is of interest here.*

Example 7.3 *Theorem 7.15 can be thought of as an abstract formulation of a well-known fact in interval analysis, the simplest case of which is the following. Let $[a, b]$ be an interval on the real line and let $f : [a, b] \rightarrow [a, b]$ be a continuous function. If $M(a, b)$ denotes the set of all nonempty closed subintervals of $[a, b]$, then $M(a, b)$ corresponds to the set $\mathfrak{S}$ of all nonempty admissible sets described above, and the Hausdorff metric H is given by*

$$H([x, y], [u, v]) = \max\{|x - u|, |y - v|\}.$$

In this case one can define $\bar{f} : M(a, b) \rightarrow M(a, b)$ by taking $\bar{f}(J) = f(J)$ for $J \in M(a, b)$, since by continuity of f the latter set is always in $M(a, b)$. Then the mapping $\bar{f}$ has a greatest fixed point $J \in M(a, b)$ (relative to set inclusion) and for this fixed point J one has $J = \bigcap_{n=1}^{\infty} J_n$ and $J = \lim_{n \rightarrow \infty} J_n$, where $J_0 = [a, b]$ and $J_n = \bar{f}^n(J_0)$. (This example is taken from Chapter 11, Section 10 of [162].)

We now turn to the proof of Theorem 7.15. Suppose M is hyperconvex and $\rho \geq 0$, and for $D \in \mathcal{A}(M)$ set

$$N_\rho(D) = \{z \in D : \text{dist}(z, D) \leq \rho\}.$$

Since the hyperconvexity of M implies that each point $z \in M$ has at least one 'nearest' point in D it follows that

$$N_\rho(D) = \cup\{B(x; \rho) : x \in D\}.$$

The Hausdorff metric H on $\mathfrak{S}$ may be described as follows: For $U, V \in \mathfrak{S}$,

$$H(U, V) = \inf\{\rho \geq 0 : U \subseteq N_\rho(V) \text{ and } V \subseteq N_\rho(U)\}.$$

The proof is based on the fact that a hyperconvex space is complete and on several lemmas, the first of which is a restatement of Lemma 4.2 of Chapter 4 (Section 4.6).

Lemma 7.4 *If M is hyperconvex and if $D = \bigcap_{i \in I} B(z_i; r_i) \in \mathcal{A}(M)$, then for $\rho \geq 0$,*

$$N_\rho(D) = \bigcap_{i \in I} B(z_i; r_i + \rho).$$

The following lemma may be of independent interest since it does not require compactness of the space.

Lemma 7.5 *Let M be a bounded hyperconvex metric space and $\mathfrak{S}$ the collection of all nonempty admissible sets of M endowed with the Hausdorff metric H . Then $\mathfrak{S}$ is hyperconvex.*

Proof. Let $\{D_\alpha\}_{\alpha \in A} \subseteq \mathfrak{S}$, and suppose for $\rho_\alpha \geq 0$ the family $\{B(D_\alpha; \rho_\alpha)\}$ satisfies

$$H(D_\alpha, D_\beta) \leq \rho_\alpha + \rho_\beta.$$

We need to show that there exists $D \in \mathfrak{S}$ such that

$$D \in \bigcap_{\alpha \in A} B(D_\alpha; \rho_\alpha).$$

Since each of the sets D_α is admissible, for each $\alpha \in A$ there exists an index set I_α and a family of closed balls $\{B(x_{\alpha_i}; r_{\alpha_i})\}_{i \in I_\alpha}$ in M such that

$$D_\alpha = \bigcap_{i \in I_\alpha} B(x_{\alpha_i}; r_{\alpha_i}).$$

Moreover, if $i \in I_\alpha$ and $j \in I_\beta$, since $H(D_\alpha, D_\beta) \leq \rho_\alpha + \rho_\beta$ it follows that

$$d(x_{\alpha_i}, x_{\beta_j}) \leq r_{\alpha_i} + \rho_\alpha + r_{\beta_j} + \rho_\beta.$$

Since M is hyperconvex the family

$$\{B(x_{\alpha_i}; r_{\alpha_i} + \rho_\alpha) : i \in I_\alpha; \alpha \in A\}$$

has a nonempty intersection D , and clearly $D \in \mathfrak{S}$. In view of Lemma 7.4,

$$D = \bigcap_{\alpha \in A} N_{\rho_\alpha}(D_\alpha). \quad (*)$$

In order to show that $D \in \bigcap_{\alpha \in A} B(D_\alpha; \rho_\alpha)$ it suffices to show that for fixed $\beta \in A$, $D \subseteq N_{\rho_\beta}(D_\beta)$ and $D_\beta \subseteq N_{\rho_\beta}(D)$ (thus $H(D, D_\beta) \leq \rho_\beta$). The first inclusion is obvious from (*). For the second, since $H(D_\alpha, D_\beta) \leq \rho_\alpha + \rho_\beta$, Lemma 7.4 implies that for each $\alpha \in A$, $\alpha \neq \beta$,

$$D_\beta \subseteq N_{\rho_\alpha + \rho_\beta}(D_\alpha) = \bigcap_{i \in I_\alpha} B(x_{\alpha_i}; r_{\alpha_i} + \rho_\alpha + \rho_\beta).$$

Therefore, since $D_\beta = \bigcap_{i \in I_\beta} B(x_{\beta_i}; r_{\beta_i})$, again using Lemma 7.4

$$D_\beta \subseteq N_{\rho_\beta}(D) = \bigcap \{B(x_{\alpha_i}; r_{\alpha_i} + \rho_\alpha + \rho_\beta) : i \in I_\alpha; \alpha \in A\}.$$

■

Lemma 7.6 *Let M be a bounded hyperconvex metric space, let $\mathcal{B}$ be the collection of all nonempty closed subsets of M endowed with the Hausdorff metric H , and let $A, B \in \mathcal{B}$. Then $H(\text{cov}(A), \text{cov}(B)) \leq H(A, B)$.*

Proof. Suppose $H(A, B) = \rho > 0$. Then if $B(x; r) \supseteq A$ it follows that $B(x; r + \rho) \supseteq N_\rho(A) \supseteq B$. Therefore, if $\text{cov}(A) = \bigcap_{i \in I} B(x_i; r_i)$, by Lemma 7.4

$$N_\rho(\text{cov}(A)) = \bigcap_{i \in I} B(x_i; r_i + \rho) \supseteq N_\rho(A) \supseteq B.$$

From this it follows that $\text{cov}(B) \subseteq N_\rho(\text{cov}(A))$. Similarly, $\text{cov}(A) \subseteq N_\rho(\text{cov}(B))$, from which $H(\text{cov}(A), \text{cov}(B)) \leq \rho$. ■

We also need the following elementary fact.

Lemma 7.7 *Suppose $\{D_n\}$ is a sequence of nonempty closed subsets of a compact metric space M and suppose $D_1 \supseteq D_2 \supseteq \cdots \supseteq D_n \supseteq \cdots$. Then, relative to the Hausdorff metric H on the space $\mathcal{B}$ of nonempty closed subsets of M ,*

$$\lim_{n \rightarrow \infty} D_n = \bigcap_{n=1}^{\infty} D_n.$$

Proof. Let $D = \bigcap_{n=1}^{\infty} D_n$ and let $\varepsilon > 0$. Clearly $D \subseteq N_\varepsilon(D_n)$ for each n . Suppose for each n there exists $x_n \in D_n$ such that $x_n \notin N_\varepsilon(D)$. Since M is compact $\{x_n\}$ has a subsequence which converges to a point x which is necessarily in D . However, $\text{dist}(x_n, D) \geq \varepsilon$ implies $\text{dist}(x, D) \geq \varepsilon$. Since this is a contradiction, it follows that $D_{n_0} \subseteq N_\varepsilon(D)$ for some integer n_0 ; hence $H(D_n, D) \leq \varepsilon$ if $n \geq n_0$. ■

Proof of Theorem 7.15. Let $\mathcal{B}$ denote the family of all nonempty closed subsets of M . Since $f : M \rightarrow M$ is (uniformly) continuous, it is easy to see that the induced mapping $D \mapsto f(D)$ is a continuous mapping from $\mathfrak{F}$ to $\mathcal{B}$. Since Lemma 7.6 shows that the mapping $A \mapsto \text{cov}(A)$ from $\mathcal{B}$ to $\mathfrak{F}$ is also continuous, it follows that $\bar{f}$ is a continuous mapping from $\mathfrak{F} \rightarrow \mathfrak{F}$. Also, since M is compact, the space $\mathcal{B}$ with the Hausdorff metric is also compact [107, p. 38]. Since $\mathfrak{F}$ is hyperconvex (Lemma 7.5), $\mathfrak{F}$ is a complete (hence closed) subspace of $\mathcal{B}$. Thus $\mathfrak{F}$ itself is compact. Let $D_0 = M$ and let $D_n = \bar{f}(D_{n-1})$, $n = 1, 2, \dots$. Then $D_1 = \text{cov}(\bar{f}(D_0)) \subseteq D_0$, so it follows that $\bar{f}(D_1) \subseteq \bar{f}(D_0)$; hence $D_2 = \text{cov}(\bar{f}(D_1)) \subseteq \text{cov}(\bar{f}(D_0)) = D_1$. Continuing, $D_0 \supseteq D_1 \supseteq \dots \supseteq D_n \supseteq \dots$. By Lemma 7.7, $\lim_{n \rightarrow \infty} D_n = D = \bigcap_{n=1}^{\infty} D_n$, and since $\bar{f}$ is continuous $\bar{f}(D) = D$. ■

Remark 7.2 If f is assumed to be a compact mapping, then Theorem 7.14 remains true without the compactness assumption on M , provided M has the property that for any compact subset S of M it is the case that $\text{cov}(S)$ is compact as well. We mention also that this problem has recently been studied by R. Espínola and G. López (Houston J. Math. 25 (1999), 371–378) using the Isbell hull discussed in Chapter 4, Section 4.7.

Exercises

Exercise 7.1 Prove that if M is a compact metric space, then the metric space $\mathcal{M}$ consisting of all nonempty closed subsets of M endowed with the Hausdorff metric is also compact.

Exercise 7.2 Prove the opening statement in the proof of Theorem 7.1; namely, that each point of a triangular region has a unique expression as a convex combination of the vertices of the triangle.

Exercise 7.3 Let $B = \{x \in \mathbb{R}^2 : \|x\| \leq 1\}$ denote the closed unit ball in $\mathbb{R}^2$, and let $S = \{x \in B : \|x\| = 1\}$. Suppose $f : B \rightarrow B$ is a continuous mapping which has the property $f(x) = x$ for all $x \in S$. Show that f is surjective.

Exercise 7.4 With B and S as in the previous exercise, suppose $g : B \rightarrow B$ is a continuous mapping satisfying $g(x) = -x$ for each $x \in S$. Show that there exists $x \in B$ such that $g(x) = 0$. [Hint: Use the previous exercise.]

Exercise 7.5 Show that if a subset X of $\mathbb{R}^n$ has the fixed point property (for continuous mappings) and if Y is a retract of X , then Y also has the fixed point property.

Exercise 7.6 Consider whether the following subsets of $\mathbb{R}^2$ have the fixed point property:

- (a) A figure eight (two circles with exactly one point in common);
- (b) A solid figure eight (two discs with exactly one point in common);
- (c) A cross ($\{(x, y) \in \mathbb{R}^2 : |x| \leq 1, |y| \leq 1, \text{ and } x \cdot y = 0\}$).

Exercise 7.7 Can you think of a subset of $\mathbb{R}^2$ which is not a retract of $\mathbb{R}^2$ but which does have the fixed point property?

Exercise 7.8 Let B be the closed unit ball in any infinite-dimensional Banach space. Show that given any $\alpha \in [0, 1)$ there exists a continuous mapping $f : B \rightarrow B$ which has the property: $\|x - f(x)\| \equiv \alpha$. [You may use Dugundji's result that Brouwer's Theorem fails in any infinite-dimensional space.]

Exercise 7.9 Prove properties (v) and (vi) of the measure of noncompactness χ (Section 7.8).

Chapter 8

Metric Fixed Point Theory

8.1 Contraction mappings

In this chapter we examine metric fixed point theory in a Banach space setting, taking an application of Banach's contraction mapping principle as a point of departure.

Since Banach's Theorem is couched in a complete metric space setting, one might wonder what more can be said if the underlying space is richer. The following is a fundamental domain invariance result which is an easy consequence of Banach's Theorem, yet it requires an underlying linear structure for its very formulation.

Theorem 8.1 *Let G be an open subset of a Banach space X and let $T : G \rightarrow X$ be a contraction mapping. Then $(I - T)(G)$ is an open subset of X .*

Proof. Let $f = I - T$ and let $x_0 \in G$. What we must show is that for $\rho > 0$ sufficiently small, the ball $B(f(x_0); \rho) \subseteq f(G)$. The approach is classical. Choose $r > 0$ so that $B(x_0; r) \subseteq G$, and choose $\rho = (1 - k)r$, where k is the Lipschitz constant of T . Now fix $z \in B(f(x_0); \rho)$. We wish to show that there exists $w \in B(x_0; r)$ such that $f(w) = w - T(w) = z$, from which

$$B(f(x_0); \rho) \subseteq f(G).$$

Define the mapping $T_z : G \rightarrow X$ by setting

$$T_z(x) = T(x) + z, \quad x \in G.$$

Note that the problem has now been reduced to showing that the (contraction) mapping T_z has a fixed point which lies in $B(x_0; r)$.

We assert that $T_z : B(x_0; r) \rightarrow B(x_0; r)$, from which the conclusion follows

via Banach's Theorem. To see this, let $x \in B(x_0; r)$. Then

$$\begin{aligned} \|T_z(x) - x_0\| &= \|T(x) + z - x_0\| \\ &= \|T(x) + T(x_0) - T(x_0) + z - x_0\| \\ &\leq \|T(x) - T(x_0)\| + \|z - (x_0 - T(x_0))\| \\ &\leq k\|x - x_0\| + \|z - f(x_0)\| \\ &\leq kr + \rho = r. \end{aligned}$$

We now conclude that for some $w \in B(x_0; r)$, $T_z(w) = w$ and the proof is complete. $\blacksquare$

The above argument provides an elementary and elegant example of how it is possible to reduce a mapping theorem to a fixed point theorem.

We look at another result in which an underlying linear structure is required. Recall that we use the symbol ∂K to denote the boundary of K . (A point $x \in K$ is in the boundary of $K \subset X$ if each ball centered at x contains both points of K and points of $X \setminus K$.)

Theorem 8.2 *Let K be a closed convex subset of a Banach space X with $\text{int}(K) \neq \emptyset$, and let $T : K \rightarrow X$ be a contraction mapping which satisfies $T : \partial K \rightarrow K$. Then T has a unique fixed point in K .*

Results more general than the above are known. In fact, the theorem holds for more general mappings (e.g., the condensing mappings) under even weaker boundary assumptions. However, we state Theorem 8.2 to motivate the following observation due to Felix Browder, from which it follows immediately. Browder made this observation several years ago yet it appears to be not so widely known.

Theorem 8.3 *Let K be a bounded closed convex subset of a Banach space X with $\text{int}(K) \neq \emptyset$, and let $T : K \rightarrow X$ be a Lipschitzian mapping which satisfies $T : \partial K \rightarrow K$. Then for $\lambda > 0$ sufficiently small, the mapping*

$$T_\lambda(x) = (1 - \lambda)x + \lambda T(x), \quad x \in K,$$

maps K into K (and has the same fixed points as T).

Proof. Without loss of generality we may suppose $0 \in \text{int}(K)$. Let p denote the Minkowski functional of K associated with 0 , that is,

$$p(x) = \inf\{\mu > 0 : x \in \mu K\}, \quad x \in X.$$

Suppose $B(0; r) \subseteq K$. Then since K is bounded there exists $t > 0$ such that $K \subseteq B(0; tr)$. Consequently, since $p(x)^{-1}x \in \partial K$ for each $x \neq 0$,

$$r \leq p(x)^{-1} \|x\| \leq tr$$

and thus there exist constants c_1 and c_2 such that

$$c_1 \|x\| \leq p(x) \leq c_2 \|x\|$$

for each $x \in X$.

Now let k be the Lipschitz constant of T and let x be any point of K . If $y = p(x)^{-1}x$ then $y \in \partial K$ and y is also on the ray joining 0 to x . Therefore,

$$p(y - x) = p((p(x)^{-1} - 1)x) = (p(x)^{-1} - 1)p(x) = 1 - p(x).$$

Now let $\lambda \in (0, 1)$. Then

$$p(T_\lambda(x)) = p((1 - \lambda)x + \lambda T(x)) \leq (1 - \lambda)p(x) + \lambda p(T(x)).$$

On the other hand, since $T(y) \in K$, it follows that $p(T(y)) \leq 1$ and

$$\begin{aligned} p(T(x)) &= p(T(y) + T(x) - T(y)) \\ &\leq p(T(y)) + p(T(x) - T(y)) \\ &\leq 1 + c_2 \|T(y) - T(x)\| \\ &\leq 1 + c_2 k \|y - x\| \\ &\leq 1 + c_2 k c_1^{-1} p(y - x) \\ &\leq 1 + k_1 (1 - p(x)), \end{aligned}$$

where $k_1 = c_2 k c_1^{-1}$. From this we have

$$\begin{aligned} p(T_\lambda(x)) &\leq (1 - \lambda)p(x) + \lambda(1 + k_1(1 - p(x))) \\ &= (1 - \lambda(1 + k_1))p(x) + \lambda(1 + k_1). \end{aligned}$$

Now observe that if $p(x) \leq 1$ (i.e., if $x \in K$) and if λ is chosen so small that $\lambda(1 + k_1) \leq 1$, then $p(T_\lambda(x)) \leq 1$ and we conclude $T_\lambda : K \rightarrow K$. ■

Now suppose G is a subset of a Banach space X with $0 \in \text{int}(G)$, suppose $T : \overline{G} \rightarrow X$, and consider the condition

$$T(x) \neq \mu x \text{ for } x \in \partial G \text{ and } \mu > 1. \quad (8.1)$$

Theorem 8.4 *Let G be a subset of a Banach space X with $0 \in \text{int}(G)$, and let $T : \overline{G} \rightarrow X$ be a contraction mapping which satisfies (8.1) on ∂G . Then T has a fixed point in $\overline{G}$.*

See Exercise 8.2.

8.2 Basic theorems for nonexpansive mappings

In this section we discuss applications of the results and techniques of Chapters 4 and 5 to Banach spaces. The following definition goes back to a 1948 paper of Brodskii and Milman [20]. We specialize the definition here to allow for the fact that only weakly compact sets or, if the underlying space is a conjugate space, the weak* compact sets may be of interest.

Definition 8.1 A convex subset K of a (possibly conjugate) Banach space is said to have normal structure [resp., weak, weak* normal structure] if any bounded closed [resp., weak, weak* closed] convex subset H of K for which $\text{diam}(H) > 0$ contains a point x_0 for which

$$\sup\{\|x_0 - x\| : x \in H\} < \text{diam}(H).$$

Such a point x_0 is called a nondiametral point of H .

We adopt notation consistent with that used previously in metric spaces. For a bounded closed convex subset K of a Banach space X , let

$$r_x(K) = \sup\{\|x - y\| : y \in K\},$$

let

$$r(K) = \inf\{r_x(K) : x \in K\},$$

and let

$$C(K) = \{x \in K : r_x(K) = r(K)\}.$$

Thus the assertion that X has normal structure means that for any bounded closed convex subset K of X it is the case that $r(K) < \text{diam}(K)$ whenever $\text{diam}(K) > 0$. If K is weakly [resp., weak*] compact, and if X has weak [resp., weak*] normal structure then $C(K)$ is a nonempty *proper* weakly [resp., weak*] compact convex subset of K whenever $\text{diam}(K) > 0$. This latter assertion follows from the fact that $C(K)$ is admissible in the sense that it can be expressed as an intersection of closed balls centered at points of K , and closed balls are both weak and weak* closed. Specifically:

$$C(K) = \bigcap_{\epsilon > 0} C_\epsilon(K),$$

where

$$C_\epsilon(K) = (\bigcap_{x \in K} (B(x; r(K) + \epsilon))) \cap K.$$

Many spaces which have normal structure satisfy an even stronger condition.

Definition 8.2 A convex subset K of a Banach space is said to have uniform normal structure if there exists a constant $c < 1$ such that any bounded closed convex subset H of K for which $\text{diam}(H) > 0$ contains a point x_0 for which

$$\sup\{\|x_0 - x\| : x \in H\} \leq c \text{diam}(H).$$

It should be noted that the above definition is more restrictive than its metric counterpart of Chapter 5 since here it is required that the condition apply to *all* convex subsets of K —not just the admissible subsets of K .

We begin with one of the first observations to evolve from the introduction of the concept of normal structure. A bounded sequence $\{x_n\}$ in a Banach space is said to be a *diametral sequence* if

$$\lim_{n \rightarrow \infty} \text{dist}(x_{n+1}, \overline{\text{conv}}\{x_1, x_2, \dots, x_n\}) = \text{diam}\{x_1, x_2, \dots\}.$$

In the paper [20] in which the notion of normal structure was introduced, Brodskii and Milman proved the following useful fact.

Proposition 8.1 *A bounded convex subset K of a Banach space has normal structure if and only if it does not contain a diametral sequence.*

Proof. If K contains a diametral sequence $\{x_n\}$, then

$$H = \overline{\text{conv}}\{x_1, x_2, \dots\}$$

is a convex subset of K each point of which is a diametral point; hence K fails to have normal structure.

Now suppose K contains a convex subset H which contains more than one point and each point of which is diametral, and let $\varepsilon \in (0, d)$ where $d = \text{diam}(H)$. Select $x_1 \in H$, let $y_0 = x_1$, and assume $x_n \in H$ has been defined. Set

$$y_{n-1} = \sum_{i=1}^n \frac{x_i}{n}.$$

By assumption, y_{n-1} is a diametral point, so there exists a point $x_{n+1} \in H$ such that

$$\|x_{n+1} - y_{n-1}\| > d - \frac{\varepsilon}{n^2}.$$

Let $x \in \text{conv}\{x_1, \dots, x_n\}$, say $x = \sum_{j=1}^n \alpha_j x_j$, where $\alpha_j \geq 0$ and $\sum_{j=1}^n \alpha_j = 1$. Now take

$$\alpha = \max\{\alpha_1, \dots, \alpha_n\}.$$

Since $\alpha > 0$, it is possible to rewrite y_{n-1} as follows:

$$y_{n-1} = \frac{1}{n\alpha}x + \sum_{j=1}^n \left(\frac{1}{n} - \frac{\alpha_j}{n\alpha}\right)x_j.$$

It is easy to check that $\left(\frac{1}{n} - \frac{\alpha_j}{n\alpha}\right) \geq 0$ and that $\frac{1}{n\alpha} + \sum_{j=1}^n \left(\frac{1}{n} - \frac{\alpha_j}{n\alpha}\right) = 1$. Consequently,

$$\begin{aligned}
 d - \frac{\varepsilon}{n^2} &< \|x_{n+1} - y_{n-1}\| \\
 &\leq \frac{1}{n\alpha} \|x_{n+1} - x\| + \sum_{j=1}^n \left(\frac{1}{n} - \frac{\alpha_j}{n\alpha}\right) \|x_{n+1} - x_j\| \\
 &\leq \frac{1}{n\alpha} \|x_{n+1} - x\| + \sum_{j=1}^n \left(\frac{1}{n} - \frac{\alpha_j}{n\alpha}\right) d \\
 &= \frac{1}{n\alpha} \|x_{n+1} - x\| + \left(1 - \frac{1}{n\alpha}\right) d.
 \end{aligned}$$

Therefore,

$$\begin{aligned}
 \|x_{n+1} - x\| &\geq n\alpha \left(d - \frac{\varepsilon}{n^2} - \left(1 - \frac{1}{n\alpha}\right) d \right) \\
 &= n\alpha \left(\frac{d}{n\alpha} - \frac{\varepsilon}{n^2} \right) \\
 &= d - \frac{\varepsilon\alpha}{n} \geq d - \frac{\varepsilon}{n}.
 \end{aligned}$$

Thus $\text{dist}(x_{n+1}, \text{conv}\{x_1, \dots, x_n\}) \geq d - \frac{\varepsilon}{n}$ with ε arbitrary, so $\{x_n\}$ is a diametral sequence. ■

Corollary 8.1 *Every compact convex subset K of a Banach space X has normal structure. In particular, every finite-dimensional Banach space has normal structure.*

Proof. Let K be a compact and convex subset of X . If K does not have normal structure then by Proposition 8.1 K contains a diametral sequence $\{x_n\}$, and clearly $\{x_n\}$ can have no convergent subsequence. This contradicts compactness of K . ■

Proposition 8.2 *Every uniformly convex Banach space X has uniform normal structure.*

Proof. Let H be a bounded closed convex subset of X for which $d = \text{diam}(H) > 0$. Then there must exist $u, v \in H$ such that $\|u - v\| \geq d/2$. Let $m = \frac{1}{2}(u + v)$, and let $x \in H$. Then (see Chapter 6, Section 6.4)

$$\left. \begin{aligned} \|u - x\| &\leq d \\ \|v - x\| &\leq d \\ \|u - v\| &\geq d/2 \end{aligned} \right\} \Rightarrow \|x - m\| \leq \left(1 - \delta\left(\frac{d/2}{d}\right)\right) d = \left(1 - \delta\left(\frac{1}{2}\right)\right) d.$$

Since X is uniformly convex, $\delta\left(\frac{1}{2}\right) > 0$, so we conclude that

$$\sup_{x \in H} \|m - x\| \leq \left(1 - \delta\left(\frac{1}{2}\right)\right) d < d.$$

Thus m is a nondiametral point of H which satisfies the uniform normal structure condition for $c = \left(1 - \delta\left(\frac{1}{2}\right)\right)$. $\blacksquare$

An examination of the above proof reveals even more. Let $\varepsilon > 0$ and choose $u, v \in H$ such that $\|u - v\| \geq d(1 - \varepsilon)$. Then if $m = \frac{1}{2}(u + v)$, one obtains

$$\left. \begin{array}{l} \|u - x\| \leq d \\ \|v - x\| \leq d \\ \|u - v\| \geq d(1 - \varepsilon) \end{array} \right\} \Rightarrow \|x - m\| \leq \left(1 - \delta\left(\frac{d(1 - \varepsilon)}{d}\right)\right) d = (1 - \delta(1 - \varepsilon))d.$$

Note that m will be a nondiametral point of H if it is the case that $\delta(1 - \varepsilon) > 0$, where ε can be chosen arbitrarily small. Recall that the characteristic of convexity $\varepsilon_0(X)$ of a Banach space is the number

$$\varepsilon_0(X) = \sup\{t : \delta(t) = 0\}.$$

Therefore, if $\varepsilon_0(X) < 1$, then it is always possible to choose $\varepsilon > 0$ small enough that $\delta(1 - \varepsilon) > 0$. This gives the following extension of the previous proposition.

Proposition 8.3 *Suppose X is a Banach space for which $\varepsilon_0(X) < 1$. Then X has uniform normal structure.*

Now suppose K is either a weakly compact convex subset of a Banach space X or a weak* compact convex subset of a dual space X^* . In either case K is bounded. Let $\mathcal{A}(K)$ denote the set of all admissible subsets of K . Thus $D \in \mathcal{A}(K)$ if $D = K \cap (\cap B(x_i; r_i))$, where $\{B(x_i; r_i)\}$ is a family of closed balls centered at points of K . Each such set D is itself closed and convex, hence weakly compact if K is weakly compact. If K is weak* compact then, since closed balls in X^* are themselves weak* closed (indeed by the Alaoglu theorem they are weak* compact) each $D \in \mathcal{A}(K)$ is weak* compact. In either case $\mathcal{A}(K)$ is compact in the sense of Section 5.1, Chapter 5. If X has normal structure (or more particularly, if K has normal structure), then it follows that $\mathcal{A}(K)$ is normal as well. Therefore the following are immediate consequences of Theorem 5.1 of Chapter 5.

Theorem 8.5 *Let K be a weakly compact convex subset of a Banach space, and suppose K has normal structure. Then every nonexpansive mapping $T : K \rightarrow K$ has at least one fixed point.*

Theorem 8.6 *Let K be a weak* compact convex subset of a dual Banach space, and suppose K has weak* normal structure. Then every nonexpansive mapping $T : K \rightarrow K$ has at least one fixed point.*

A Banach space in which every bounded closed convex subset has the fixed point property for nonexpansive self-mappings is said to have the FPP. (In metric spaces the FPP designation was reserved for spaces in which the admissible sets have the fixed point property.) If we restrict the convex subsets to be compact for the weak or weak* topologies, we refer to the weak- or weak*-FPP.

As a consequence of Theorem 8.5 (which was proved in 1965) the search for spaces having the FPP centered on identifying reflexive spaces which have normal structure. However, it was soon discovered (in 1974) that normal structure is not an essential condition for the FPP. (Whether or not reflexivity is essential remains unknown.) Later (in 1981) it was shown that the classical space $L_1[0, 1]$ fails to have the weak-FPP (Alspach), while reflexive subspaces of $L_1[0, 1]$ do have the FPP (Maurey). Also it was shown that c_0 has the weak-FPP (Maurey) but, of course, fails the FPP. As we shall see momentarily, ℓ_1 provides an example of a space which fails the FPP yet has the weak*-FPP.

Some of the history of this development is discussed, for example, in [1] and [67].

We proceed with another consequence of the metric development. It is known that a Banach space X is hyperconvex if and only if it is isometrically isomorphic to a space $C(K)$ of continuous scalar valued functions where K is a compact extremally disconnected (the closure of each open set is itself open) Hausdorff space. Such spaces are of interest in a classical sense because they include the L_∞ (ℓ_∞) spaces. From Theorem 4.8 (Chapter 4) we have the following.

Theorem 8.7 *Let $B = B(0; 1)$ be the closed unit ball (more generally, an admissible set) in a hyperconvex Banach space (e.g., L_∞) and let $T : B \rightarrow B$ be nonexpansive. Then the fixed point set of T is a nonempty nonexpansive retract of B .*

Remark 8.1 *It is worth noting that the argument of Lemma 5.2 (Chapter 5) can be modified to prove the following.*

Theorem 8.8 *If X is a Banach space which has uniform normal structure, then X is reflexive.*

The proof involves replacing “cov” with “ $\overline{\text{conv}}$ ” in the proof of Lemma 5.2 and then applying property (g) of Proposition 6.16 (Chapter 6).

Corollary 8.2 *Suppose X is a Banach space for which $\varepsilon_0(X) < 1$. Then X is reflexive.*

Corollary 8.3 *Every uniformly convex space X is reflexive.*

8.3 A closer look at ℓ_1

Using the fact that a bounded sequence in ℓ_1 is weak* convergent relative to $\ell_1 = c_0^*$ if and only if it converges coordinate-wise (Section 6.8, Chapter 6) we immediately have the following.

Lemma 8.1 *If $\{x^k\}_{k=1}^\infty$ is a bounded sequence in ℓ_1 which is weak* convergent to x relative to $\ell_1 = c_0^*$, then for any $y \in \ell_1$,*

$$r(y) = r(x) + \|x - y\|,$$

where $r(y) = \limsup_{k \rightarrow \infty} \|x^k - y\|$.

Proof. As usual, let $\{e^1, e^2, \dots\}$ be the standard unit basis for ℓ_1 and for each i let P_i denote the natural projection of ℓ_1 onto its subspace spanned by $\{e^1, \dots, e^i\}$; specifically,

$$P_i(x_1, x_2, \dots) = (x_1, x_2, \dots, x_i, 0, 0, \dots).$$

Also let $Q_i = I - P_i$. Then

$$Q_i(x_1, x_2, \dots) = (0, 0, \dots, 0, x_{i+1}, x_{i+2}, \dots).$$

For any $i = 1, 2, \dots$,

$$r(x) = \limsup_{k \rightarrow \infty} \|x^k - x\| = \lim_{i \rightarrow \infty} \sup_{k \rightarrow \infty} \|Q_i(x^k - x)\|.$$

Since

$$\|P_i(x^k - y)\| = \sum_{n=1}^i |x_n^k - y_n|,$$

then we have

$$\|x - y\| = \lim_{i \rightarrow \infty} \left(\lim_{k \rightarrow \infty} \|P_i(x^k - y)\| \right).$$

Also,

$$\lim_{i \rightarrow \infty} \|Q_i(x - y)\| = \lim_{i \rightarrow \infty} \sum_{n=i}^{\infty} |x_n - y_n| = 0.$$

Now

$$\begin{aligned} \|P_i(x^k - y)\| + \|Q_i(x^k - x)\| - \|Q_i(x - y)\| &\leq \|P_i(x^k - y)\| + \|Q_i(x^k - y)\| \\ &= \|x^k - y\| \end{aligned}$$

and letting $k \rightarrow \infty$ and $i \rightarrow \infty$, we get

$$\|x - y\| + r(x) \leq r(y).$$

On the other hand,

$$\|x^k - y\| \leq \|x^k - x\| + \|x - y\|,$$

so $r(y) \leq r(x) + \|x - y\|$. ■

Theorem 8.9 *Weak* compact convex subsets of ℓ_1 viewed as the dual of c_0 have weak* normal structure. Hence ℓ_1 has the weak*-FPP relative to this weak* topology.*

Proof. Let $K \subseteq \ell_1$ be weak* compact and convex and let H be a weak* closed and convex subset of K . We need to show that if $\text{diam}(H) > 0$ then H contains a nondiametral point. We have already seen that this is the case if H is compact. If H is not compact, then H contains a sequence $\{x^k\}$ such that

$$\text{weak}^* \lim_{k \rightarrow \infty} x^k = x_0 \in H$$

but for which $\{x^k\}$ does not converge in norm to x_0 . For each $y \in H$, let

$$r(y) = \limsup_{k \rightarrow \infty} \|x^k - y\|.$$

By the previous lemma

$$r(y) = r(x_0) + \|x_0 - y\|,$$

where $r(x_0) > 0$. Thus

$$\|x_0 - y\| = r(y) - r(x_0) \leq \text{diam}(H) - r(x_0).$$

This proves that x_0 is a nondiametral point of H . ■

Since the admissible subsets of ℓ_1 are weak* compact relative to *any* weak* topology on ℓ_1 , we have the following.

Corollary 8.4 *Let D be an admissible subset of ℓ_1 and let $T : D \rightarrow D$ be nonexpansive. Then T has a fixed point in D .*

At this point it might be interesting to note that there exist weak*-compact convex subsets of ℓ_1 viewed as the dual of c which *fail* to have the fixed point property. An example can be found in [67].

8.4 Stability results in arbitrary spaces

This topic is a little specialized but it highlights a ‘metric’ approach to the stability result discussed in connection with Brouwer’s Theorem.

Let K be a bounded closed convex subset of a Banach space X and let $T : K \rightarrow K$ be a mapping which satisfies, for some $h > 0$ and $p > 0$, the condition

$$\|T(x) - T(y)\| \leq h \|x - y\|^p, \quad (x, y \in K). \quad (8.2)$$

These mappings are well understood for various choices of h and p . If $p = 1$ and $h < 1$, then such mappings always have a fixed point because of the Banach contraction mapping principle, and if $p = 1$ and $h = 1$ these are the nonexpansive mappings. In the case $p = 1$ and $h > 1$ (the usual Lipschitz condition), a fair amount is known about minimal displacement properties of such mappings (e.g., see [67], [69]) although a number of questions remain open. Since mappings satisfying (8.2) are degenerate if $p > 1$ regardless of the value of h , what remains is the case $p < 1$. This is the case we take up here. Central to our observations is the fact that mappings which satisfy the Hölder condition (8.2) for $0 < h, p \leq 1$, also satisfy the following more general condition:

$$\|T(x) - T(y)\| \leq \max\{h, \|x - y\|\}, \quad (x, y \in K). \quad (8.3)$$

Thus such mappings are h -nonexpansive in the terminology of Section 5.6, Chapter 5. As was the case there, our objective is not to assure that such mappings have fixed points (indeed, mappings satisfying (8.3) may not even be continuous), but rather to find conditions under which there always exists at least one $x \in K$ for which $\|x - T(x)\| < h$.

This section is motivated also by the results of Chapter 7. It was proved in Section 7.5 of Chapter 7 that if K is a compact convex subset of a normed linear space and if $\varphi : K \rightarrow K$ is ε -continuous in the sense that each point $x \in K$ has a neighborhood U_x for which $\text{diam}(\varphi(U_x)) \leq \varepsilon$, then for every $\varepsilon' > \varepsilon$ there exists $z \in K$ such that $\|z - \varphi(z)\| < \varepsilon'$. It is our purpose here to show that by adding the assumption $\|\varphi(u) - \varphi(v)\| \leq \|u - v\|$ for $u, v \notin U_x$ the same conclusion holds in noncompact settings, and in many cases (including the compact case) a slightly stronger conclusion holds.

For convenience we take as our point of departure the following variant of Klee’s definition introduced in Bula [34]. Let (M, d) be a metric space, $D \subseteq M$, and $f : D \rightarrow M$. Then f is said to be w -continuous at $x_0 \in D$ (for $w > 0$) if for each $\varepsilon > 0$ there exists $\delta = \delta(\varepsilon) > 0$ such that whenever $x \in D$ and $d(x_0, x) < \delta$ then $d(f(x_0), f(x)) < \varepsilon + w$. f is said to be w -continuous (on D) if f is w -continuous at each $x_0 \in D$. (The analog of Klee’s result in Bula’s terminology is that if $\varphi : K \rightarrow K$ is uniformly w -continuous, then for every $w' > w$ there exists $x_0 \in K$ such that $\|x_0 - \varphi(x_0)\| < w'$.)

We begin by considering an analog of w -continuity for contractions.

Definition 8.3 Let (M, d) be a metric space. A mapping $T : M \rightarrow M$ is said to be a w -contraction, $w > 0$, if there exists $k \in (0, 1)$ such that for each $x, y \in M$, $d(T(x), T(y)) \leq kd(x, y) + w$.

Note that if $k \in (0, 1)$ and if $\ell \in (k, 1)$ then for $x, y \in M$,

$$kd(x, y) + w \leq \ell d(x, y) \Leftrightarrow d(x, y) \geq \frac{w}{\ell - k}.$$

This fact leads to the following related definition.

Definition 8.4 A mapping $T : M \rightarrow M$ is said to be a h_ℓ -contractive if for $h > 0$ and $\ell \in (0, 1)$,

$$d(T(x), T(y)) \leq \ell \max\{d(x, y), h\} \quad x, y \in M. \quad (8.4)$$

Two facts are immediate.

Proposition 8.4 If $T : M \rightarrow M$ is a w -contraction with contractive constant $k \in (0, 1)$, then T is a h_ℓ -contractive for $h = \frac{w}{\ell - k}$ and $\ell \in (k, 1)$.

Proof. If $d(x, y) \geq h$ then $d(T(x), T(y)) \leq kd(x, y) + w \leq \ell d(x, y)$. Otherwise $d(T(x), T(y)) \leq kd(x, y) + w < kh + w = \ell h$. ■

Proposition 8.5 If $T : M \rightarrow M$ is a w -contraction with contractive constant k , then

$$\inf\{d(x, T(x)) : x \in M\} \leq \frac{w}{1 - k}.$$

Proof. Let $\ell \in (k, 1)$, let $h(\ell) = \frac{w}{\ell - k}$, and let $x \in M$. Suppose

$$d(T^n(x), T^{n+1}(x)) \geq h(\ell), n = 1, 2, \dots$$

Then Definition 8.4 in conjunction with the geometric series argument for proving Banach's contraction principle implies $\{T^n(x)\}$ is a Cauchy sequence, and clearly this leads to a contradiction. Therefore, for some n it must be the case that $d(T^n(x), T^{n+1}(x)) \leq h(\ell)$. Since $\ell \in (k, 1)$ is arbitrary, the conclusion follows. ■

In the limiting case $k = 1$, the above result becomes meaningless. However, in the limiting case in Definition 8.4 it seems that a great deal can be said. For this case we return to the definition introduced in Chapter 5.

Definition 8.5 If $T : M \rightarrow M$ is h_1 -contractive then we say that T is h -nonexpansive.

Now let K be a nonempty bounded closed convex subset of a Banach space X and suppose $T : K \rightarrow K$ is h -nonexpansive. For $t \in (0, 1)$ and fixed $z \in K$, define $T_t : K \rightarrow K$ by setting

$$T_t(x) = (1 - t)z + tT(x), \quad x \in K.$$

Then it follows that

$$\inf \{ \|x - T(x)\| : x \in K \} \leq h.$$

To see this, note that for each $x, y \in K$,

$$\|T_t(x) - T_t(y)\| = t \|T(x) - T(y)\| \leq t \max \{ \|x - y\|, h \}.$$

Therefore, for any $x \in K$ and $n \geq 1$,

$$\begin{aligned} \|T_t^n(x) - T_t^{n+1}(x)\| &\leq t \max \{ \|T^{n-1}(x) - T^n(x)\|, h \} \\ &\leq t \max \{ t \max \{ \|T^{n-2}(x) - T^{n-1}(x)\|, h \}, h \} \\ &= \max \{ t^2 \|T^{n-2}(x) - T^{n-1}(x)\|, th \} \\ &\leq \dots \\ &\leq \max \{ t^n \|x - T(x)\|, th \}. \end{aligned}$$

Since K is bounded and $\lim_{n \rightarrow \infty} t^n = 0$, this proves

$$\inf \{ \|x - T_t(x)\| : x \in K \} \leq th.$$

But

$$\|x - T(x)\| \leq t^{-1} \|x - T_t(x)\| + t^{-1}(1 - t) \|x - z\|.$$

Therefore, $\inf \{ \|x - T(x)\| : x \in K \} \leq h + t^{-1}(1 - t) \|x - z\|$, so upon letting $t \rightarrow 1^-$, we have the following.

Theorem 8.10 *If K is a nonempty bounded closed convex subset of a Banach space X and if $T : K \rightarrow K$ is h -nonexpansive, then*

$$\inf \{ \|x - T(x)\| : x \in K \} \leq h.$$

If T satisfies a Hölder condition, the conclusion is sharper.

Theorem 8.11 *Suppose K is a nonempty bounded closed convex subset of a Banach space X , and suppose $T : K \rightarrow K$ satisfies*

$$\|T(x) - T(y)\| \leq h \|x - y\|^p, \quad (x, y \in K),$$

for some $h, p \in (0, 1)$. Then

$$\inf \{ \|x - T(x)\| : x \in K \} \leq h^{1/(1-p)}.$$

Proof. Since $h < 1$, Theorem 8.10 implies there exists $x \in K$ such that $\|x - T(x)\| < 1$. Thus

$$\|T(x) - T^2(x)\| \leq h \|x - T(x)\|^p < h,$$

from which

$$\|T^2(x) - T^3(x)\| \leq h \|T(x) - T^2(x)\|^p < h^{1+p},$$

and by induction

$$\|T^{n+1}(x) - T^{n+2}(x)\| < h^{(1+p+\cdots+p^n)}, \quad n = 1, 2, \dots$$

Therefore,

$$\lim_{n \rightarrow \infty} \|T^{n+1}(x) - T^{n+2}(x)\| \leq \lim_{n \rightarrow \infty} h^{(1+p+\cdots+p^n)} = h^{1/(1-p)},$$

and the conclusion follows. ■

Since $h^{1/(1-p)} < h$ for $h, p \in (0, 1)$, the minimal displacement estimate of Theorem 8.11 is better than that of Theorem 8.10.

As we have seen, if K is a closed convex subset of a Banach space X and if K is weakly compact and has normal structure, then the family of all admissible subsets of K is compact (via weak compactness of K) and normal. Therefore the following is a direct consequence of the principal result of Section 5.6, Chapter 5.

Theorem 8.12 *Suppose K is a weakly compact convex subset of a Banach space, suppose K also has normal structure, and suppose $T : K \rightarrow K$ is h -nonexpansive. Then there exists a point $z \in K$ such that $\|z - T(z)\| < h$.*

Corollary 8.5 *Suppose K is a compact convex subset of a Banach space and suppose $T : K \rightarrow K$ is h -nonexpansive. Then there exists a point $z \in K$ such that $\|z - T(z)\| < h$.*

Proof. As we have seen, compact convex sets have normal structure. ■

Theorem 8.13 *Let X be a Banach space which has uniform normal structure with coefficient c , let K be a bounded closed convex subset of X , and suppose $T : K \rightarrow K$ is h -nonexpansive. Then there exists $z \in K$ such that $\|z - T(z)\| \leq ch$.*

Again, if T satisfies a Hölder condition a sharper conclusion holds.

Theorem 8.14 *Let X be a Banach space which has uniform normal structure with coefficient c , let K be a bounded closed convex subset of X , and suppose $T : K \rightarrow K$ satisfies*

$$\|T(x) - T(y)\| \leq h \|x - y\|^p, \quad (x, y \in K),$$

where $h, p \in (0, 1)$. Then if p is sufficiently near 1,

$$\inf \{\|x - T(x)\| : x \in K\} < ch.$$

Proof. By Theorem 8.13 there exists $z \in K$ such that $\|z - T(z)\| \leq ch$. Thus

$$\|T(z) - T^2(z)\| \leq h \|z - T(z)\|^p \leq c^p h^{1+p}.$$

Since $c < 1$, $c^p h^{1+p} < ch$ for all p sufficiently near 1. ■

The above estimate invites a comparison with the displacement estimate of Theorem 8.11. Clearly for certain values of p, h and c it will be the case that $ch < h^{1/(1-p)}$; in particular, this is always true if $c < h$ and $p \leq 1/2$, and also for all $h < 1$ if p is sufficiently small.

Since it is known that ℓ_∞ is hyperconvex, the following is an immediate consequence of Corollary 5.1 (Section 5.6, Chapter 5).

Theorem 8.15 *Let K be an admissible subset of ℓ_∞ , and let $T : K \rightarrow K$ be h -nonexpansive. Then there exists $z \in K$ such that $\|z - T(z)\| \leq h/2$.*

As a very special case of the above we have:

Corollary 8.6 *Suppose $K = [a, b] \subset \mathbb{R}^1$ and suppose $T : K \rightarrow K$ is h -nonexpansive. Then there exists $z \in K$ such that $|z - T(z)| \leq h/2$.*

The results of this section are qualitative in that no effort has been made to determine optimal minimal displacement for each of the classes of mappings and spaces considered. A number of questions in the spirit of those raised in [69] remain open.

Also there is another question which the above analysis leaves open:

Question. Let K be a bounded closed convex subset of a Banach space, and suppose $T : K \rightarrow K$ is h -nonexpansive for $h > 0$. Under what conditions can one conclude that there exists $z \in K$ such that $\|z - T(z)\| \leq h$?

As we have seen, if K is weakly compact and has normal structure, then one obtains even more, namely, the existence of $z \in K$ for which $\|z - T(z)\| < h$. This leaves the tempting suggestion that weak compactness alone suffices for an affirmative answer to the above question. The limiting case of the above (when $h = 0$) involves a question about fixed points of nonexpansive mappings which remains unresolved, and it is plausible that the two problems may be related.

8.5 The Goebel-Karlovitz Lemma

Any nonempty set K of a Banach space which is minimal with respect to being closed, convex and invariant under any mapping $T : K \rightarrow K$ must satisfy $\overline{\text{conv}}(T(K)) = K$. This has nothing to do with properties of the mapping T . However, if T is nonexpansive and if K is also bounded, then much more can be said.

Lemma 8.2 *Each point of K is a diametral point.*

Proof. Suppose there exists a point $z \in K$ such that $r = r_z(K) < \text{diam}(K)$, and let

$$C_r(K) = \left\{ u \in K : u \in \bigcap_{x \in K} B(x; r) \right\} = K \cap \left(\bigcap_{x \in K} B(x, r) \right).$$

Then obviously $C_r(K)$ is nonempty ($z \in C_r(K)$), closed and convex. At the same time note that $u \in C_r(K)$ if and only if $u \in K$ and $K \subseteq B(u; r)$. In particular, if $u \in C_r(K)$ then by nonexpansiveness of T , $T(K) \subseteq B(T(u); r)$. But this, in turn, implies that

$$K = \overline{\text{conv}}(T(K)) \subseteq B(T(u); r).$$

Therefore, $T(u) \in C_r(K)$, that is, $T : C_r(K) \rightarrow C_r(K)$. However, this contradicts the minimality of K . $\blacksquare$

Since a nonexpansive mapping is h -nonexpansive for any $h > 0$, the following is a consequence of Theorem 8.14.

Lemma 8.3 *If K is a bounded closed and convex subset of a Banach space, and if $T : K \rightarrow K$ is nonexpansive, then there exists a sequence $\{x_n\} \subseteq K$ such that $\lim_{n \rightarrow \infty} \|x_n - T(x_n)\| = 0$.*

There is a fundamental fact which, in conjunction with the previous lemma, has proved extremely useful in extending fixed point results for nonexpansive mappings beyond those spaces which possess normal structure. It appears to have been discovered independently and at about the same time by Goebel [66] and Karlovitz [88].

Lemma 8.4 (Goebel-Karlovitz) *Let K be a subset of a Banach space X which is minimal with respect to being nonempty, weakly compact, convex, and T -invariant for some nonexpansive mapping T , and suppose $\{x_n\} \subseteq K$ satisfies*

$$\lim_{n \rightarrow \infty} \|x_n - T(x_n)\| = 0.$$

Then for each $x \in K$, $\lim_{n \rightarrow \infty} \|x - x_n\| = \text{diam}(K)$.

Proof. As we have just seen, if $\text{diam}(K) > 0$ then each point of K is a diametral point. Now suppose there exists $\{x_n\} \subseteq K$ for which $\lim_{n \rightarrow \infty} \|x_n - T(x_n)\| = 0$, but for which $\lim_{n \rightarrow \infty} \|x - x_n\| = r < \text{diam}(K)$ for some $x \in K$. Let

$$C = \{z \in K : \limsup_{n \rightarrow \infty} \|z - x_n\| \leq r\}.$$

It is easy to see that C is convex and not difficult to see that C is closed. Indeed, if $\{u_n\} \subseteq C$ and if $\lim_{i \rightarrow \infty} u_i = u$, then for each $i = 1, 2, \dots$,

$$\limsup_{n \rightarrow \infty} \|u - x_n\| \leq \|u - u_i\| + \limsup_{n \rightarrow \infty} \|u_i - x_n\| \leq \|u - u_i\| + r.$$

Letting $i \rightarrow \infty$ gives $\limsup_{n \rightarrow \infty} \|u - x_n\| \leq r$. Thus $u \in C$ and C is closed. Also if $u \in K$,

$$\begin{aligned} \limsup_{n \rightarrow \infty} \|T(u) - x_n\| &\leq \limsup_{n \rightarrow \infty} (\|T(u) - T(x_n)\| + \|T(x_n) - x_n\|) \\ &= \limsup_{n \rightarrow \infty} \|T(u) - T(x_n)\| \\ &\leq \limsup_{n \rightarrow \infty} \|u - x_n\| \leq r. \end{aligned}$$

This proves that $T : C \rightarrow C$. Since any closed convex subset of a weakly compact set is itself weakly compact, we conclude that $C = K$ by minimality of K . However, if $\varepsilon > 0$ is chosen so that $r + \varepsilon < \text{diam}(K)$, then the family $\{B(x; r + \varepsilon) : x \in K\}$ has the finite intersection property since each such ball contains all but a finite number of terms of the sequence $\{x_n\}$. Since these balls are weakly compact,

$$K \cap \left(\bigcap_{x \in K} B(x; r + \varepsilon) \right) \neq \emptyset.$$

But this implies $r(K) \leq r + \varepsilon < \text{diam}(K)$, contradicting the fact that K is a diametral set (all its points are diametral). ■

8.6 Orthogonal convexity

The Goebel-Karlovitz Lemma has been applied extensively in the study of fixed point theory for nonexpansive mappings, both as formulated in Lemma 8.4 and as stated in the language of ultraproducts (see [1] and references cited therein). This section is devoted to a typical application of the Goebel-Karlovitz Lemma. The approach is a recent one due to A. Jiménez-Melado and E. Lloréns-Fuster in [85]. It arises in connection with their introduction and study of a generalization of uniform convexity called *orthogonal convexity*.

We begin by describing the concept of orthogonal convexity. For points x, y of a Banach space X and $\lambda > 0$, let

$$M_\lambda(x, y) = \left\{ z \in X : \max \{ \|z - x\|, \|z - y\| \} \leq \frac{1}{2}(1 + \lambda) \|x - y\| \right\}.$$

If A is a bounded subset of X , let $|A| = \sup\{\|x\| : x \in A\}$, and for a bounded sequence $\{x_n\}$ in X and $\lambda > 0$, let

$$D(\{x_n\}) = \limsup_{i \rightarrow \infty} (\limsup_{j \rightarrow \infty} \|x_i - x_j\|);$$

$$A_\lambda(\{x_n\}) = \limsup_{i \rightarrow \infty} (\limsup_{j \rightarrow \infty} |M_\lambda(x_i, x_j)|).$$

Definition 8.6 A Banach space is said to be *orthogonally convex (OC)* if for each sequence $\{x_n\}$ in X which converges weakly to 0 and for which $D(\{x_n\}) > 0$, there exists $\lambda > 0$ such that $A_\lambda(\{x_n\}) < D(\{x_n\})$.

It is shown in [86] that every uniformly convex Banach space is OC. Other examples given in [86] include Banach spaces with the Schur property (hence ℓ_1 , c_0 , c , and James's space J , described in Section 6.1 of Chapter 6).

Theorem 8.16 Let K be a nonempty weakly compact convex subset of a Banach space X , and suppose K is orthogonally convex. Then every nonexpansive mapping $T : K \rightarrow K$ has a fixed point.

Proof. Invoking Zorn's Lemma we may assume K satisfies the assumptions of Lemma 8.4. We assume that $\text{diam}(K) > 0$ and show that this leads to a contradiction. Let $\{x_n\} \subset K$ satisfy $\lim_{n \rightarrow \infty} \|x_n - T(x_n)\| = 0$. By a suitable translation and passing to a subsequence, we may assume $0 \in K$, with $\{x_n\}$ converging weakly to 0. Also, for convenience, we assume that $\text{diam}(K) = 1$. The argument in the general case merely requires multiplying everything by d^{-1} , where d is the diameter of K .

Now observe that the Goebel-Karlovitz Lemma implies that $D(\{x_n\}) = 1$. Hence by orthogonal convexity there exist positive real numbers λ and b such that $A_\lambda(\{x_n\}) < b < 1$. Thus there exists a subsequence of $\{x_n\}$, which we again label $\{x_n\}$, such that $|M_\lambda(x_n, x_m)| < b$ for all $m \neq n$, and again by passing to a subsequence we may assume

$$\lim_{n \rightarrow \infty} \|x_n - x_{n+1}\| = 1; \quad \lim_{n \rightarrow \infty} n \|x_n - T(x_n)\| = 0. \quad (*)$$

Apply Banach's contraction mapping principle to obtain the (unique) solution z_n to the equation:

$$z_n = \left(1 - \frac{1}{n}\right) T(z_n) + \frac{1}{n} m_n,$$

where $m_n = \frac{x_n + x_{n+1}}{2}$. For $i = 0, 1$, we have

$$z_n - x_{n+i} = z_n - \left(1 - \frac{1}{n}\right) T(x_{n+i}) + \left(1 - \frac{1}{n}\right) T(x_{n+i}) - x_{n+i}.$$

Using the definition of z_n , we get

$$\begin{aligned} z_n - x_{n+i} &= \left(1 - \frac{1}{n}\right) (T(z_n) - T(x_{n+i})) + \left(1 - \frac{1}{n}\right) (T(x_{n+i}) - x_{n+i}) \\ &\quad + \frac{1}{n} (m_n - x_{n+i}), \end{aligned}$$

from which (using the fact that T is nonexpansive)

$$\begin{aligned} \|z_n - x_{n+i}\| &\leq \left(1 - \frac{1}{n}\right) \|z_n - x_{n+i}\| + \left(1 - \frac{1}{n}\right) \|T(x_{n+i}) - x_{n+i}\| \\ &\quad + \frac{1}{n} \|m_n - x_{n+i}\|. \end{aligned}$$

Multiplying by n and using the definition of m_n , we have (again for $i = 0, 1$)

$$\|z_n - x_{n+i}\| \leq (n-1) \|T(x_{n+i}) - x_{n+i}\| + \frac{1}{2} \|x_{n+1} - x_n\|.$$

In view of $(*)$, we have for n sufficiently large:

$$\max \left\{ \|z_n - x_n\|, \|z_n - x_{n+1}\| \right\} \leq \frac{(1+\lambda)}{2} \|x_{n+1} - x_n\|.$$

This proves that $\{z_n\}$ is eventually in $M_\lambda(x_n, x_{n+1})$; hence

$$\|z_n\| \leq A_\lambda(\{x_n\}) \leq b < D(\{x_n\}) = 1.$$

On the other hand, $\lim_{n \rightarrow \infty} \|z_n - T(z_n)\| = 0$, so by the Goebel-Karlovitz Lemma,

$$\lim_{n \rightarrow \infty} \|z_n\| = 1.$$

The resulting contradiction establishes the theorem. ■

8.7 Structure of the fixed point set

It is useful to know that the fixed point set of a nonexpansive mapping is a nonexpansive retract of its domain. The following simple facts serve to illustrate this. Throughout this section we use $F(T)$ to denote the fixed point set of a mapping T .

Theorem 8.17 *Suppose M is a metric space which has the property that the fixed point set of every nonexpansive mapping of M into M is a nonempty nonexpansive retract of M , and suppose T and G are commuting nonexpansive mappings of M into M . Then $F(T) \cap F(G) \neq \emptyset$.*

Proof. First observe that if $x \in F(T)$ then $T \circ G(x) = G \circ T(x) = G(x)$. This proves $G : F(T) \rightarrow F(T)$. By assumption there is a nonexpansive retraction r of M onto the nonempty set $F(T)$. Thus $G \circ r : M \rightarrow F(T)$ is nonexpansive and by assumption has a nonempty fixed point set. However, since r is a retraction onto $F(T)$, it must be the case that $F(G \circ r) = F(G) \cap F(T)$. This proves that $F(T) \cap F(G) \neq \emptyset$ (and at the same time that $F(T) \cap F(G)$ is itself a nonexpansive retract of M). ■

A mapping $T : M \rightarrow M$ is said to be *eventually nonexpansive* if there exists an integer N such that for each $n \geq N$, T^n is nonexpansive.

Theorem 8.18 *Suppose M is a metric space which has the property that the fixed point set of every nonexpansive mapping of M into M is a nonexpansive retract of M , and suppose $T : M \rightarrow M$ is eventually nonexpansive. Then $F(T) \neq \emptyset$.*

Proof. For N sufficiently large the mappings T^N and T^{N+1} are commuting nonexpansive mappings of M into M . By the previous theorem $F(T^N) \cap F(T^{N+1}) \neq \emptyset$. But it is trivial to check that

$$F(T) = F(T^N) \cap F(T^{N+1}).$$

■

Of course, the above results are not really about nonexpansive mappings. All one needs to know is that there is a retraction from the domain of the mapping onto its fixed point set which can be drawn from a class of mappings for which the original domain has the fixed point property. It turns out, however, that the class of nonexpansive mappings often works—a fact we have already noted in hyperconvex spaces. We now take up this question in a Banach space setting, beginning with the separable case since most of the difficulties in this case have been disposed of in Section 5.8 of Chapter 5. It is perhaps surprising that in any separable space the following simple assumption is enough to assure the desired result.

Definition 8.7 *A bounded closed convex subset K of a Banach space is said to have the generic fixed point property for nonexpansive mappings (GFPP) if every nonexpansive mapping $f : K \rightarrow K$ has a fixed point set in every nonempty bounded closed convex f -invariant subset of K .*

By what we have already seen, any bounded closed convex subset of a reflexive Banach space which has normal structure satisfies (GFPP), as does any weakly compact convex subset of an orthogonally convex Banach space. Other examples will be discussed later.

Our first result is the following.

Theorem 8.19 *Suppose K is a bounded closed convex subset of a separable Banach space, suppose K has (GFPP), and let $T : K \rightarrow K$ be nonexpansive. Then the fixed point set $F(T)$ of T is a (nonempty) nonexpansive retract of K .*

The above follows quickly from Theorem 5.12 of Chapter 5 which, for convenience, we restate here.

Theorem 8.20 (5.12) *Let M be a separable complete metric space and let S be a semigroup of nonexpansive self-mappings of M . Then there exists in $\overline{S}$ a nonexpansive retraction r of M onto $F(S)$ if and only if one of the following two (equivalent) conditions holds:*

- (i) *Each nonempty closed S -invariant subset of M contains a fixed point of S .*
- (ii) *Whenever $x \in M$ then $\text{cl } S(x) \cap F(S) \neq \emptyset$.*

Proof of Theorem 8.19. Let K^K denote the family of all mappings of K into K and put

$$S = \{s \in K^K : s \text{ is nonexpansive and } F(s) \supseteq F(T)\}.$$

Obviously, S is a semigroup on K and $F(S) \supseteq F(T)$. Since $T \in S$ we conclude that $F(S) = F(T)$. If $s_1, s_2 \in S$ and $\lambda \in (0, 1)$ then $\lambda s_1 + (1 - \lambda)s_2 \in S$. Thus for each $x \in K$ the set $S(x) = \{s(x) : s \in S\}$ is clearly nonempty, convex, and T -invariant. Since T is continuous, $\text{cl } S(x)$ is a nonempty bounded closed convex T -invariant subset of K , so by (GFPP) $\text{cl } S(x) \cap F(T) \neq \emptyset$. This shows that S satisfies (ii). The conclusion now follows from Theorem 5.12. ■

The following is now a direct consequence of the above and the theorem stated at the beginning of the section.

Corollary 8.7 *Suppose K is a bounded closed convex subset of a separable Banach space, suppose that K has (GFPP), and let $\mathcal{F}$ be a finite family of commuting nonexpansive mappings of K into K . Then the common fixed point set $F(\mathcal{F})$ of $\mathcal{F}$ is a (nonempty) nonexpansive retract of K .*

The above corollary can be extended to arbitrary families $\mathcal{F}$. Rather than pursue this further, however, we take up the question of what can be said in the nonseparable case. The central observation is that the separability assumption on the space can be replaced with reflexivity or, more generally, with the assumption that the domain is weakly compact. The ideas are due to Bruck [29].

Theorem 8.21 *Let K be a nonempty weakly compact convex subset of a Banach space X and suppose $\mathcal{T} : K \rightarrow K$ is a family of nonexpansive mappings with a nonempty common fixed point set A . Suppose further that*

$$A \text{ intersects every nonempty } \mathcal{T}\text{-invariant closed convex subset of } K. \quad (8.5)$$

Then A is a nonexpansive retract of K .

Proof. As before, let K^K denote the family of all mappings of K into K and let

$$N = \{f \in K^K : f \text{ is nonexpansive and } F(f) \supseteq A\},$$

where $F(f)$ denotes the fixed point set of f . Obviously, $\mathcal{T} \subseteq N$ so $N \neq \emptyset$. Notice also that by Tychonoff's Theorem K^K is compact in the topology of weak pointwise convergence, since this is the product topology on the space $\prod_{x \in K} K$ induced by the weak topology on K .

Now for $f, f' \in N$, we say $f \sim f'$ if $\|p - f(x)\| = \|p - f'(x)\|$ for each $p \in A$ and $x \in K$, and for $f \in N$ let

$$[f] = \{f' \in N : f' \sim f\}.$$

Let $[N] = \{[f] : f \in N\}$ and introduce the partial order $\preceq$ on $[N]$ by defining $[f] \preceq [g]$ if and only if $\|p - f(x)\| \leq \|p - g(x)\|$ for each $p \in A$ and $x \in K$. To see that $([N], \preceq)$ has a minimal element, let $\{[f_\alpha], \alpha \in I\}$ be a descending chain in $([N], \preceq)$. Since K^K is compact in the topology of weak pointwise convergence the net $\{f_\alpha\}$ has a subnet $\{f_{\alpha_\xi}\}$ which converges in this topology to some $f \in K^K$. Thus for each $x \in K$,

$$\text{weak-}\lim_{\xi} f_{\alpha_\xi}(x) = f(x).$$

For $x, y \in K$, $\{f_{\alpha_\xi}(x) - f_{\alpha_\xi}(y)\}$ converges weakly to $f(x) - f(y)$ (algebraic operations are always weakly continuous) so

$$\|f(x) - f(y)\| \leq \lim_{\xi} \|f_{\alpha_\xi}(x) - f_{\alpha_\xi}(y)\| \leq \|x - y\|.$$

Also, if $x \in A$ then $f_\alpha(x) = x$ for each $\alpha \in I$ so it follows that $f(x) = x$. Thus $F(f) \supseteq A$ and this proves that $f \in N$.

Now let $x \in K$ and $p \in A$. Since $\{[f_\alpha], \alpha \in I\}$ is descending in $\preceq$ it is the case that the limit $\lim_{\xi} \|p - f_{\alpha_\xi}(x)\|$ exists, and since norm-closed balls in X are weakly closed it must be the case that

$$\|p - f(x)\| \leq \lim_{\xi} \|p - f_{\alpha_\xi}(x)\| = \lim_{\alpha} \|p - f_\alpha(x)\|.$$

Since $\lim_{\alpha} \|p - f_\alpha(x)\| = \lim_{\alpha'} \|p - f_{\alpha'}(x)\|$ if $f_\alpha \sim f_{\alpha'}$, $\{[f_\alpha], \alpha \in I\}$ is bounded below by $[f]$ in $([N], \preceq)$. We may now invoke Zorn's lemma to conclude that $([N], \preceq)$ contains a minimal element $[r]$. We show that $r \in [r]$ is the desired retraction.

Now suppose there exists a point $z \in K$ such that $r(z) \notin A$. Since

$$\|p - r \circ r(x)\| \leq \|p - r(x)\|$$

for each $p \in A$ and $x \in K$, $[r \circ r] \preceq [r]$. But since $[r]$ is minimal, $[r \circ r] = [r]$. In particular, if $z_0 = r(z)$ then for all $p \in A$,

$$\|p - r(z_0)\| = \|p - z_0\| > 0.$$

Let

$$C = \{f \circ r(z_0) : f \in N\}.$$

Since $f, g \in N$ implies $\lambda f + (1 - \lambda)g \in N$ for any $\lambda \in [0, 1]$, C is convex. To see that C is weakly compact we only need to show that C is closed. Let $\{x_\alpha\}$ be a net in C for which $\lim_{\alpha} x_\alpha = x$. Then for each α there exists $f_\alpha \in N$ such that $x_\alpha = f_\alpha \circ r(z_0)$. But since K^K is compact in the topology of weak pointwise convergence $\{f_\alpha\}$ has a subnet $\{f_{\alpha_\xi}\}$ which converges to some $f \in K^K$. Arguing as before, f is nonexpansive and $F(f) \supseteq A$. Hence $f \in N$, so $f \circ r(z_0) \in C$. But

$$x = \text{weak-}\lim_{\xi} f_{\alpha_\xi} \circ r(z_0) = f \circ r(z_0).$$

This proves that $x \in C$; hence C is closed.

Let $f \in N$. Then $F(f) \supseteq A$ so for each $p \in A$,

$$\|p - f \circ r(z_0)\| = \|f(p) - f \circ r(z_0)\| \leq \|p - r(z_0)\|.$$

Thus $[f \circ r] \preceq [r]$. But since $[r]$ is minimal this implies $\|p - f \circ r(z_0)\| = \|p - r(z_0)\| > 0$ for each $p \in A$. Therefore, $A \cap C = \emptyset$. On the other hand, we have already observed that $T \subseteq N$, so $s : C \rightarrow C$ for each $s \in T$. Thus $A \cap C \neq \emptyset$ by our basic hypothesis. This is a contradiction, so we conclude that $r(z) \in A$ for each $z \in K$. ■

The following is an obvious special case of the preceding theorem.

Corollary 8.8 *Suppose K is a weakly compact convex subset of a Banach space, suppose K has (GFPP), and let $T : K \rightarrow K$ be nonexpansive. Then the fixed point set $F(T)$ of T is a (nonempty) nonexpansive retract of K .*

8.8 Asymptotically regular mappings

In this section we show that in terms of fixed point properties of nonexpansive mappings defined on a convex subset of a Banach space, it always suffices to study only a special class of mappings, the so-called asymptotically regular mappings. Let D be a subset of a Banach space X .

Definition 8.8 *A mapping $f : D \rightarrow D$ is said to be asymptotically regular if for each $x \in D$ it is the case that $\lim_{n \rightarrow \infty} \|f^n(x) - f^{n+1}(x)\| = 0$.*

The key observation of this section is the following: If D is bounded and convex, and if $T : D \rightarrow D$ is nonexpansive, then the mapping $f = (I + T)/2$ has the same fixed point set as T and moreover, f is asymptotically regular. This very useful fact was discovered by Ishikawa in 1976 [76]. It is an easy consequence of the following lemma.

Lemma 8.5 *Suppose $\{x_n\}$ and $\{y_n\}$ are sequences in a Banach space X which satisfy for all $n \in \mathbb{N}$:*

$$(i) \quad x_{n+1} = \frac{1}{2}(x_n + y_n)$$

$$(ii) \quad \|y_{n+1} - y_n\| \leq \|x_{n+1} - x_n\|.$$

Then for all $i, n \in \mathbb{N}$,

$$\left(1 + \frac{n}{2}\right) \|y_i - x_i\| \leq 2^n \left[\|y_i - x_i\| - \|y_{i+n} - x_{i+n}\| \right] + \|y_{i+n} - x_i\|.$$

Proof. The proof is by induction on n . Observe that the lemma is trivially true for all i if $n = 0$, and make the inductive assumption that the theorem holds for a given $n \in \mathbb{N}$ and all i . This gives (upon replacing i with $i + 1$ in the inequality)

$$\left(1 + \frac{n}{2}\right) \|y_{i+1} - x_{i+1}\| \leq 2^n \left[\|y_{i+1} - x_{i+1}\| - \|y_{i+n+1} - x_{i+n+1}\| \right] \\ + \|y_{i+n+1} - x_{i+1}\|.$$

Next observe that by (i), the triangle inequality, and (ii), respectively:

$$\|y_{i+n+1} - x_{i+1}\| \leq (1/2) \left[\|y_{i+n+1} - x_i\| + \|y_{i+n+1} - y_i\| \right] \\ \leq (1/2) \left[\|y_{i+n+1} - x_i\| + \sum_{k=0}^n \|y_{i+k+1} - y_{i+k}\| \right] \\ \leq (1/2) \left[\|y_{i+n+1} - x_i\| + \sum_{k=0}^n \|x_{i+k+1} - x_{i+k}\| \right].$$

Combined with the previous inequality, this gives

$$\left(1 + \frac{n}{2}\right) \|y_{i+1} - x_{i+1}\| \leq 2^n \left[\|y_{i+1} - x_{i+1}\| - \|y_{i+n+1} - x_{i+n+1}\| \right] \\ + (1/2) \left[\|y_{i+n+1} - x_i\| + \sum_{k=0}^n \|x_{i+k+1} - x_{i+k}\| \right].$$

Thus

$$2 \left(1 + \frac{n}{2}\right) \|y_{i+1} - x_{i+1}\| \leq 2^{n+1} \left[\|y_{i+1} - x_{i+1}\| - \|y_{i+n+1} - x_{i+n+1}\| \right] \\ + \left[\|y_{i+n+1} - x_i\| + \sum_{k=0}^n \|x_{i+k+1} - x_{i+k}\| \right].$$

■

Next we observe that (i) and (ii) imply

$$\|y_{n+1} - x_{n+1}\| \leq \|y_{n+1} - y_n\| + \|y_n - x_{n+1}\| \\ \leq \|x_{n+1} - x_n\| + \|y_n - x_{n+1}\| \\ = \|y_n - x_n\|.$$

This means that the sequence $\{\|y_n - x_n\|\}$ is monotone nonincreasing. Thus

$$\sum_{k=0}^n \|x_{i+k+1} - x_{i+k}\| = \frac{1}{2} \sum_{k=0}^n \|y_{i+k} - x_{i+k}\| \leq \frac{(n+1)}{2} \|y_i - x_i\|,$$

and we have

$$\begin{aligned}
 2\left(1 + \frac{n}{2}\right) \|y_{i+1} - x_{i+1}\| &\leq 2^{n+1} \left[\|y_{i+1} - x_{i+1}\| - \|y_{i+n+1} - x_{i+n+1}\| \right] \\
 &\quad + \left[\|y_{i+n+1} - x_i\| + ((n+1)/2) \|y_i - x_i\| \right] \\
 &= 2^{n+1} \left[\|y_i - x_i\| - \|y_{i+n+1} - x_{i+n+1}\| \right] \\
 &\quad + 2^{n+1} \|y_{i+1} - x_{i+1}\| \\
 &\quad + [((n+1)/2) - 2^{n+1}] \|y_i - x_i\| + \|y_{i+n+1} - x_i\|
 \end{aligned}$$

Since $2\left(1 + \frac{n}{2}\right) - 2^{n+1} \leq 0$,

$$\begin{aligned}
 &\left[2\left(1 + \frac{n}{2}\right) - 2^{n+1} \right] \|y_{i+1} - x_{i+1}\| + \left[2^{n+1} - ((n+1)/2) \right] \|y_i - x_i\| \\
 &\geq \left[2\left(1 + \frac{n}{2}\right) - 2^{n+1} \right] \|y_i - x_i\| + \left[2^{n+1} - ((n+1)/2) \right] \|y_i - x_i\| \\
 &= \left[1 + (n+1)/2 \right] \|y_i - x_i\|,
 \end{aligned}$$

and we have

$$\begin{aligned}
 \left[1 + (n+1)/2 \right] \|y_i - x_i\| &\leq 2^{n+1} \left[\|y_{i+1} - x_{i+1}\| - \|y_{i+n+1} - x_{i+n+1}\| \right] \\
 &\quad + \|y_{i+n+1} - x_i\|.
 \end{aligned}$$

This completes the induction. ■

We now prove the main result of this section.

Theorem 8.22 *Let K be a bounded convex subset of a Banach space and suppose $T : K \rightarrow K$ is nonexpansive. Then the mapping $f : K \rightarrow K$ defined by setting $f(x) = (x + T(x))/2$ is asymptotically regular.*

Proof. Select $x_0 \in K$, let $y_0 = f(x_0)$, and having defined x_n take $y_n = f(x_n)$ and set $x_{n+1} = (x_n + y_n)/2, n = 1, 2, \dots$. It is easy to see that the resulting sequences $\{x_n\}, \{y_n\}$ satisfy the assumptions of the lemma. Thus for each $i, n \in \mathbb{N}$,

$$\begin{aligned}
 \left(1 + \frac{n}{2}\right) \|y_i - x_i\| &\leq 2^n \left[\|y_i - x_i\| - \|y_{i+n} - x_{i+n}\| \right] \\
 &\quad + \|y_{i+n} - x_i\|.
 \end{aligned}$$

Since $\{\|x_n - y_n\|\} = \{\|x_n - f(x_n)\|\}$ is monotone nonincreasing, there exists a number $r \geq 0$ such that $\lim_{n \rightarrow \infty} \|x_{n+1} - f(x_{n+1})\| = r$. Now let $i \rightarrow \infty$ in the above inequality. Then

$$\left(1 + \frac{n}{2}\right) r \leq \text{diam}(K).$$

Clearly this implies $r = 0$. ■

Two remarks are in order. First, a slight modification of the preceding argument yields the fact that each of the mappings $f_\alpha = (1 - \alpha)I + \alpha T$ are asymptotically regular for each $\alpha \in (0, 1)$. Moreover, it can be shown that the convergence in Theorem 8.22 is uniform in the sense that the rate of convergence does not depend on the initial point x_0 , nor indeed on the mapping T . Details may be found in [67].

8.9 Set-valued mappings

We conclude this chapter with some observations about set-valued nonexpansive mappings. For this we need some preliminary facts. For a bounded sequence $\{x_n\}$ in a Banach space X and K a bounded subset of X we associate the number

$$r = r_K(\{x_n\}) = \inf \left\{ \limsup_{n \rightarrow \infty} \|x_n - y\| : y \in K \right\}$$

and the set

$$A = A_K(\{x_n\}) = \{y \in K : \limsup_{n \rightarrow \infty} \|x_n - y\| = r_K(\{x_n\})\}.$$

r and A are called, respectively, the *asymptotic radius* of $\{x_n\}$ and *asymptotic center* of $\{x_n\}$ relative to K .

If X is reflexive and K is closed and convex, then $A_K(\{x_n\})$ is always a nonempty closed convex subset of K for any bounded sequence $\{x_n\}$ in X . To see this observe that for each $\varepsilon > 0$ the set

$$C_\varepsilon = \{y \in K : \limsup_{n \rightarrow \infty} \|x_n - y\| \leq r + \varepsilon\}$$

is nonempty by definition of r , and straightforward arguments show that each of the sets C_ε is closed and convex. Hence

$$A_K(\{x_n\}) = \bigcap_{\varepsilon > 0} C_\varepsilon,$$

and the latter set is nonempty by weak compactness of K .

Definition 8.9 A bounded sequence $\{x_n\}$ is said to be *regular with respect to a bounded subset K of X* if $r_K(\{x_n\}) = r_K(\{x_{n_k}\})$ for every subsequence $\{x_{n_k}\}$ of $\{x_n\}$, and $\{x_n\}$ is said to be *asymptotically uniform* if $A_K(\{x_n\}) = A_K(\{x_{n_k}\})$ for every subsequence $\{x_{n_k}\}$ of $\{x_n\}$.

Lemma 8.6 Let K be a bounded subset of X and $\{x_n\}$ a bounded sequence in X . Then $\{x_n\}$ has a subsequence which is regular with respect to K .

Proof. To avoid double subscripts we use the notation $\{v_n\} \prec \{x_n\}$ to indicate that $\{v_n\}$ is a subsequence of $\{x_n\}$. We begin by setting

$$r_0 = \inf\{r_k(\{v_n\}) : \{v_n\} \prec \{x_n\}\}.$$

Obviously, there exists $\{v_n^1\} \prec \{x_n\}$ such that

$$r_K(\{v_n^1\}) < r_0 + 1.$$

Set

$$r_1 = \inf\{r_k(\{v_n\}) : \{v_n\} \prec \{v_n^1\}\}$$

and proceed inductively. Having defined $\{v_n^i\} \prec \{v_n^{i-1}\}$, set

$$r_i = \inf\{r_k(\{v_n\}) : \{v_n\} \prec \{v_n^i\}\}$$

and choose $\{v_n^{i+1}\} \prec \{v_n^i\}$ so that

$$r_K(\{v_n^{i+1}\}) < r_i + \frac{1}{i+1}.$$

Note that the sequence $\{r_i\}$ is nondecreasing and bounded above, so $r = \lim_{i \rightarrow \infty} r_i$ exists. Since $r_i \leq r_K(\{v_n^{i+1}\}) < r_i + \frac{1}{i+1}$ it follows that

$$\lim_{i \rightarrow \infty} r_K(\{v_n^i\}) = r.$$

Now consider the diagonal sequence $\{\bar{v}_n\} = \{v_1^1, v_2^2, \dots, v_n^n, \dots\}$ and let $\rho = r_K(\{\bar{v}_n\})$. Since $\{\bar{v}_n\} \prec \{v_n^i\}$ for each i , $\rho \geq r_i$. On the other hand, since $\{\bar{v}_n\} \prec \{v_n^{i+1}\}$,

$$\rho \leq r_K(\{v_n^{i+1}\}) < r_i + \frac{1}{i+1}.$$

Therefore, $r_i \leq \rho \leq r_i + 1/(i+1)$ for each i , and letting $i \rightarrow \infty$, we conclude that $\rho = r$. Also, if $\{u_n\} \prec \{\bar{v}_n\}$ then $\{u_n\} \prec \{v_n^i\}$ and $\{u_n\} \prec \{v_n^{i+1}\}$. This implies $r_K(\{u_n\}) = r$, completing the proof. ■

It is worth noting that if in addition X is uniformly convex and K is nonempty closed and convex, then $A_K(\{x_n\})$ consists of exactly one point. Thus every regular sequence in X is asymptotically uniform. However, much more can be said.

Lemma 8.7 *In addition to the assumptions of the previous lemma, assume K is separable. Then any bounded sequence $\{x_n\}$ in X has an asymptotically uniform subsequence.*

Proof. In view of the preceding lemma we may assume at the outset that $\{x_n\}$ is regular. Let $S = \{y_1, y_2, \dots\}$ be a countable dense subset of K . There exists a subsequence $\{x_n^{(1)}\}$ of $\{x_n\}$ such that $\lim_{n \rightarrow \infty} \|y_1 - x_n^{(1)}\|$ exists. Similarly there exists a subsequence $\{x_n^{(2)}\}$ of $\{x_n^{(1)}\}$ such that $\lim_{n \rightarrow \infty} \|y_2 - x_n^{(2)}\|$ exists. Proceed in this way to obtain a subsequence $\{x_n^{(k+1)}\}$ of $\{x_n^{(k)}\}$ such that $\lim_{n \rightarrow \infty} \|y_{k+1} - x_n^{(k+1)}\|$ exists. Then the diagonal sequence $\{v_n\} = \{x_n^{(n)}\}$ is a subsequence of $\{x_n\}$ for which $\lim_{n \rightarrow \infty} \|y_{k+1} - v_n\|$ exists for each $k = 1, 2, \dots$, and since S is dense in K , it follows that $\lim_{n \rightarrow \infty} \|y - v_n\|$ exists for each $y \in K$. Now let $\{u_n\}$ be any subsequence of $\{v_n\}$.

Since $\{x_n\}$ is regular, $r_K(\{u_n\}) = r_K(\{x_n\})$ and since $\{u_n\}$ is a subsequence of $\{x_n\}$ it must be the case that

$$A_K(\{x_n\}) \subseteq A_K(\{u_n\}).$$

On the other hand, for each $y \in A_K(\{u_n\})$

$$\lim_{n \rightarrow \infty} \|y - v_n\| = r_K(\{u_n\}) = r_K(\{x_n\}).$$

This proves that $A_K(\{u_n\}) \subseteq A_K(\{x_n\})$. ■

Our principal result in this section is the following.

Theorem 8.23 *Let X be a uniformly convex Banach space, let K be a bounded closed convex subset of X , and let T be a nonexpansive mapping of K into the nonempty compact subsets of K endowed with the Hausdorff metric H . Then there exists $x \in K$ such that $x \in T(x)$.*

Proof. If $t \in (0, 1)$ and if $z \in K$ is fixed, the mapping T_t defined by

$$T_t(x) = (1 - t)z + tT(x)$$

is a set-valued contraction mapping defined on K (with closed values in K), so by Theorem 3.20 of Chapter 3, for each such t there exists $x_t \in K$ such that

$$x_t \in T(x_t).$$

Therefore, there exists $x'_t \in T(x_t)$ such that

$$x_t = (1 - t)z + tx'_t.$$

Thus $\|x_t - x'_t\| = (1 - t)\|z - x'_t\| \rightarrow 0$ as $t \rightarrow 1$. It follows that there exists a sequence $\{x_n\}$ in K such that

$$\lim_{n \rightarrow \infty} \text{dist}(x_n, T(x_n)) = 0.$$

Moreover, by passing to a subsequence, we may assume $\{x_n\}$ is regular. Since X is uniformly convex, $A_K(\{x_n\})$ is a singleton, say $\{v\}$. Let $r = r_K(\{x_n\})$. For each n select $y_n \in T(x_n)$ so that $\lim_{n \rightarrow \infty} \|x_n - y_n\| = 0$, and for each n select $z_n \in T(v)$ so that $\|y_n - z_n\| = \text{dist}(y_n, T(v))$. Then

$$\|y_n - z_n\| \leq H(T(x_n), T(v)) \leq \|x_n - v\|.$$

Since $T(v)$ is compact, $\{z_n\}$ has a convergent subsequence, say $\{z_{n_k}\}$, with $\lim_{k \rightarrow \infty} z_{n_k} = w \in T(v)$. On the other hand,

$$\|x_{n_k} - w\| \leq \|x_{n_k} - y_{n_k}\| + \|y_{n_k} - z_{n_k}\| + \|z_{n_k} - w\|$$

and

$$\|y_{n_k} - z_{n_k}\| \leq \|x_{n_k} - v\|.$$

From this we have

$$\limsup_{k \rightarrow \infty} \|w - x_{n_k}\| \leq \limsup_{k \rightarrow \infty} \|v - x_{n_k}\| = r.$$

Since $\{x_n\}$ is regular this proves $v = w \in T(v)$. ■

In the above theorem we only used Lemma 8.6. Lemma 8.7 can be used in conjunction with a set-valued extension of Schauder's Theorem to extend Theorem 8.23 to spaces which have the property that $A_K(\{x_n\})$ is always compact for bounded sequences $\{x_n\}$. Details may be found in [67].

8.10 Fixed point theory in Banach lattices

In this section we illustrate briefly how the lattice structure of a Banach lattice can be exploited to prove fixed point theorems for nonexpansive mappings defined on closed convex subsets of such spaces. This section has no bearing on material covered in Chapter 9 and may be omitted. However, the techniques are interesting in that they require intertwining the geometric and order theoretic properties of such spaces in order to attain the goal. Since such classical spaces as c_0 , c , ℓ_p , and L_p ($1 \leq p \leq \infty$) are all Banach lattices, it is not surprising that such an approach would yield positive results, if only to show how similar results can be achieved by diverse methods.

Many of the results discussed in this section were first recorded in the 1997 University of Newcastle (Australia) thesis of T. Dalby [39].

8.10.1 Weak orthogonality

The concept of weak orthogonality in Banach lattices dates back to Borwein and Sims [16] who, in turn, derived it from the known fact mentioned earlier that weakly compact convex subsets of c_0 always have the fixed point property for nonexpansive mappings.

Definition 8.10 A Banach lattice X (dual Banach lattice X^*) is said to be weakly orthogonal (weak* orthogonal) if whenever $\{x_n\}$ converges weakly (weak*) to 0, it follows that

$$\lim_{n \rightarrow \infty} \left\| |x_n| \wedge |x| \right\| = 0$$

for all $x \in X$.

Note that in the weak* case it may be necessary to view $\{x_n\}$ as a net. In this discussion we confine our remarks only to the weakly orthogonal case.

Borwein and Sims [16] used a slightly weaker definition of weak orthogonality, namely, that whenever $\{x_n\}$ converges weakly to 0, then

$$\liminf_{n \rightarrow \infty} \liminf_{m \rightarrow \infty} \left\| |x_n| \wedge |x_m| \right\| = 0.$$

By embedding the problem in the quotient space $\tilde{X} = \ell^\infty(X)/c_0(X)$ they showed that a weakly orthogonal Banach lattice with Riesz angle $\alpha(X) < 2$ has the weak fixed point property. The Riesz angle of X is the number

$$\alpha(X) = \sup \left\{ \left\| |x| \vee |y| \right\| : x, y \in B(0; 1) \right\}.$$

It was later observed by Sims [51] that the requirement $\alpha(X) < 2$ is superfluous if the stronger definition of weak orthogonality is used, and he also obtained stability results in the sense that Banach spaces sufficiently near X in the sense of Banach-Mazur distance (Chapter 9, Definition 9.2) also have the weak fixed point property. Similar results were obtained by Khamsi and Turpin in [94] by different methods. It is these methods we employ below.

The two main ideas in [94] are the definition of a mapping that is not linear but mimics to some extent the principal band projections in Banach lattices, and the use of an auxiliary nonexpansive mapping F_λ (discovered by K. Goebel; see [67]) which is created from T .

Suppose X is a Banach lattice, and for $u, v \in X$, let

$$S_u(x) = x^+ \wedge |u| - x^- \wedge |u|.$$

We pause to look at S_u in c_0 . If $u = \{u_n\}, x = \{x_n\} \in c_0$ then $S_u(x) = c_n$, where

$$c_n = \begin{cases} 0 & \text{if } u_n = 0; \\ \min\{|u_n|, x_n\} & \text{if } x_n \geq 0; \\ \max\{-|u_n|, x_n\} & \text{if } x_n < 0. \end{cases}$$

Now some general properties of S_u : Since $(x^+ \wedge |u|) \wedge (x^- \wedge |u|) = 0$,

$$S_u(x)^+ = x^+ \wedge |u| \text{ and } S_u(x)^- = x^- \wedge |u|.$$

Thus

$$\begin{aligned} |S_u(x)| &= (x^+ \wedge |u|) + (x^- \wedge |u|) \\ &= (x^+ \wedge |u|) \vee (x^- \wedge |u|) \\ &= |x| \wedge |u|, \end{aligned}$$

and in particular this implies

$$\|S_u(x)\| \leq \inf\{\|x\|, \|u\|\}. \quad (8.6)$$

Now we turn to the mapping $I - S_u$. If $x \in X$ then

$$x - S_u(x) = (x^+ - x^+ \wedge |u|) - (x^- - x^- \wedge |u|),$$

where

$$(x^+ - x^+ \wedge |u|) \wedge (x^- - x^- \wedge |u|) = 0.$$

Therefore,

$$(x - S_u(x))^+ = x^+ - x^+ \wedge |u|$$

and

$$(x - S_u(x))^- = x^- - x^- \wedge |u|.$$

This leads to the following calculation:

$$\begin{aligned} |x - S_u(x)| &= x^+ + x^- - (x^+ \wedge |u| + x^- \wedge |u|) \\ &= |x| - |x| \wedge |u| \\ &= |x| + |x| \vee |u| - |x| - |u| \\ &= |x| \vee |u| - |u| \\ &\leq |x| \vee |u| - |u| \wedge |x| \\ &= \left| |x| - |u| \right| \\ &\leq |x - u|. \end{aligned}$$

By monotonicity of the norm

$$\|x - S_u(x)\| \leq \|x - u\|. \quad (8.7)$$

Also,

$$\begin{aligned} |S_u(x) - S_u(y)| &= \left| |x^+ \wedge |u| - x^- \wedge |u| - y^+ \wedge |u| + y^- \wedge |u| \right| \\ &\leq |x^+ - y^+| \wedge |u| + |y^- - x^-| \wedge |u| \\ &\leq 2(|x - y| \wedge |u|). \end{aligned}$$

Hence

$$\|S_u(x) - S_u(y)\| \leq 2 \left\| |x - y| \wedge |u| \right\|. \quad (8.8)$$

Now

$$S_u(x) + S_v(x) = S_{|u| \wedge |v|}(x) + S_{|u| \vee |v|}(x),$$

so

$$\|S_u(x) + S_v(x)\| \leq \|S_{|u| \wedge |v|}(x)\| + \|S_{|u| \vee |v|}(x)\|.$$

Using (8.6),

$$\begin{aligned} \|S_u(x) + S_v(x)\| &\leq \inf \left\{ \|x\|, \left\| |u| \wedge |v| \right\| \right\} + \inf \left\{ \|x\|, \left\| |u| \vee |v| \right\| \right\} \\ &\leq \|x\| + \inf \left\{ \left\| |u| \wedge |v| \right\|, \left\| |u| \vee |v| \right\| \right\} \\ &= \|x\| + \left\| |u| \wedge |v| \right\|. \end{aligned} \tag{8.9}$$

We are now ready to prove a key lemma.

Lemma 8.8 *Let X be a weakly orthogonal Banach lattice and let $\{u_n\}, \{v_n\}$ be sequences in X which converge weakly to some $c \in X$. Suppose*

$$\lim_{n \rightarrow \infty} \left\| |u_n - c| \wedge |v_n - c| \right\| = 0.$$

Then for every sequence $\{w_n\}$ in X and for every $x \in X$,

$$\begin{aligned} 2 \limsup_{n \rightarrow \infty} \|w_n - c\| &\leq \limsup_{n \rightarrow \infty} \|w_n - x\| + \limsup_{n \rightarrow \infty} \|w_n - u_n\| \\ &\quad + \limsup_{n \rightarrow \infty} \|w_n - v_n\|. \end{aligned}$$

Proof. Without loss of generality assume $c = 0$. Since

$$2w_n = w_n - S_{u_n}(w_n) + S_{u_n}(w_n) + w_n - S_{v_n}(w_n) + S_{v_n}(w_n),$$

by the triangle inequality and (8.7)

$$\begin{aligned} 2\|w_n\| &\leq \|S_{u_n}(w_n) + S_{v_n}(w_n)\| + \|w_n - S_{u_n}(w_n)\| + \|w_n - S_{v_n}(w_n)\| \\ &\leq \|S_{u_n}(w_n) + S_{v_n}(w_n)\| + \|w_n - u_n\| + \|w_n - v_n\|. \end{aligned}$$

Using (8.8) and weak orthogonality,

$$\limsup_{n \rightarrow \infty} \|S_{u_n}(w_n) - S_{u_n}(w_n - x)\| \leq 2 \limsup_{n \rightarrow \infty} \left\| |x| \wedge |u_n| \right\| = 0.$$

Similarly,

$$\limsup_{n \rightarrow \infty} \|S_{v_n}(w_n) - S_{v_n}(w_n - x)\| = 0.$$

Therefore, using (8.9),

$$\begin{aligned}
 \limsup_{n \rightarrow \infty} \|S_{u_n}(w_n) + S_{v_n}(w_n)\| &\leq \limsup_{n \rightarrow \infty} \|S_{u_n}(w_n) - S_{v_n}(w_n - x)\| \\
 &\quad + \limsup_{n \rightarrow \infty} \|S_{u_n}(w_n - x) + S_{v_n}(w_n - x)\| \\
 &\quad + \limsup_{n \rightarrow \infty} \|S_{v_n}(w_n - x) - S_{v_n}(w_n)\| \\
 &= \limsup_{n \rightarrow \infty} \|S_{u_n}(w_n - x) + S_{v_n}(w_n - x)\| \\
 &\leq \limsup_{n \rightarrow \infty} \|w_n - x\| + \limsup_{n \rightarrow \infty} \left\| |u_n| \wedge |v_n| \right\| \\
 &= \limsup_{n \rightarrow \infty} \|w_n - x\|,
 \end{aligned}$$

which completes the proof. ■

We now prove a technical lemma which invokes the Goebel-Karlovitz Lemma.

Lemma 8.9 *Let X be a weakly orthogonal Banach lattice with C a nonempty weakly compact convex subset of X and suppose $T : C \rightarrow C$ nonexpansive. Assume C is a minimal invariant set for T with $\text{diam}(C) = 1$, and assume that $\{w_n\}$ is an approximate fixed point sequence for T which converges weakly to 0. Then there exists a subsequence $\{u_n\}$ of $\{w_n\}$ for which*

$$\lim_{k \rightarrow \infty} \left\| |u_k| \wedge |u_{k+1}| \right\| = 0 \text{ and } \lim_{k \rightarrow \infty} \|u_k - u_{k+1}\| = 1.$$

Proof. For any $x \in C$, by weak orthogonality and minimal invariance,

$$\lim_{n \rightarrow \infty} \left\| |w_n| \wedge |x| \right\| = 0 \tag{8.10}$$

and

$$\lim_{n \rightarrow \infty} \|x - w_n\| = 1. \tag{8.11}$$

In particular, for fixed $m \in \mathbb{N}$

$$\lim_{n \rightarrow \infty} \left\| |w_n| \wedge |w_m| \right\| = 0 \tag{8.12}$$

and

$$\lim_{n \rightarrow \infty} \|w_m - w_n\| = 1. \tag{8.13}$$

Let $\{\varepsilon_k\} \subseteq \mathbb{R}^+$ satisfy $\lim_{k \rightarrow \infty} \varepsilon_k = 0$. By (8.12) and (8.13) there exists $n_1 \in \mathbb{N}$ such that for all $n \geq n_1$,

$$\left\| |w_n| \wedge |w_1| \right\| < \varepsilon_1 \text{ and } \left| \|w_1 - w_n\| - 1 \right| < \varepsilon_1.$$

Also there exists $n_2 > n_1$ such that for all $n \geq n_2$,

$$\left\| |w_n| \wedge |w_{n_1}| \right\| < \varepsilon_2 \quad \text{and} \quad \left| \|w_{n_1} - w_n\| - 1 \right| < \varepsilon_2.$$

Having defined n_k choose $n_{k+1} > n_k$ so that for all $n \geq n_{k+1}$

$$\left\| |w_n| \wedge |w_{n_k}| \right\| < \varepsilon_{k+1} \quad \text{and} \quad \left| \|w_{n_k} - w_n\| - 1 \right| < \varepsilon_{k+1}.$$

Since

$$\left\| |w_{n_{k+1}}| \wedge |w_{n_k}| \right\| < \varepsilon_{k+1} \quad \text{and} \quad \left| \|w_{n_k} - w_{n_{k+1}}\| - 1 \right| < \varepsilon_{k+1},$$

the sequence $\{w_{n_k}\}$ clearly has the desired properties. ■

We now digress to make some general observations. Let C be a bounded closed and convex subset of a Banach space X , and let $T : C \rightarrow C$ be non-expansive. Fix $x \in C$ and $\lambda \in (0, 1)$, and define the mapping $T_\lambda : C \rightarrow C$ by

$$T_\lambda(y) = (1 - \lambda)x + \lambda T(y) \quad y \in C.$$

As we have seen, T_λ is a contraction mapping and hence there exists a unique point $F_\lambda(x)$ for which $T_\lambda(F_\lambda(x)) = F_\lambda(x)$. This, in turn, induces a mapping $F_\lambda : C \rightarrow C$, which has a number of interesting properties. In particular,

- (a) F_λ is nonexpansive.
- (b) $T(F_\lambda(x)) - x = \lambda^{-1}(F_\lambda(x) - x)$.
- (c) $\|F_\lambda(x) - x\| \leq \lambda(1 - \lambda)^{-1} \|T(x) - x\|$.

Among other things, an approximate fixed point sequence for T is also an approximate fixed point sequence for F_λ and the fixed point set of T coincides with the fixed point set of F_λ . However,

$$(d) \|F_\lambda(x) - T(F_\lambda(x))\| = (1 - \lambda)\lambda^{-1} \|x - F_\lambda(x)\|.$$

So it is also the case that if $\{x_n\}$ is an approximate fixed point sequence for F_λ , then $\{F_\lambda(x_n)\}$ is an approximate fixed point sequence for T .

Finally, if C is a minimal invariant weakly compact convex subset for T then, by the Goebel-Karlovitz Lemma, for all $x, y \in C$,

$$(e) \lim_{\lambda \rightarrow 1} \|F_\lambda(x) - y\| = \text{diam}(C).$$

This brings us to the main result of this section.

Theorem 8.24 *Let X be a weakly orthogonal Banach lattice. Then X has the weak fixed point property.*

Proof. Assume the theorem is false. Then there exists a nonempty weakly compact convex subset C of X and a nonexpansive mapping $T : C \rightarrow C$ which has no fixed point. We may further assume that C is a minimal such set with $0 \in C$ and $\text{diam}(C) = 1$.

Let $\{a_n\}$ be an approximate fixed point sequence for T and assume weak $\lim_{n \rightarrow \infty} a_n = 0$. Fix $\lambda \in (0, 1)$ and set $u_n = F_\lambda(a_n)$. Then if $x^* \in X^*$,

$$\begin{aligned} |x^*(u_n)| &= |x^*(F_\lambda(a_n))| \\ &= |(1 - \lambda)x^*(a_n) + \lambda x^*(T(F_\lambda(a_n)))| \\ &\leq |x^*(a_n)| + \lambda |x^*(T(F_\lambda(a_n))) - x^*(a_n)| \\ &\leq |x^*(a_n)| + \lambda \|x^*\| \|T(F_\lambda(a_n)) - a_n\| \\ &= |x^*(a_n)| + \|x^*\| \|F_\lambda(a_n) - a_n\|. \end{aligned}$$

Also, by (c) and (d) above, $\{u_n\}$ is an approximate fixed point sequence for T . Since the last term tends to 0 as $n \rightarrow \infty$ we conclude that weak- $\lim_{n \rightarrow \infty} u_n = 0$.

By Lemma 8.9 $\{u_n\}$ has a subsequence, which we again label $\{u_n\}$, for which

$$\lim_{n \rightarrow \infty} \|u_n - u_{n+1}\| = 1 \quad \text{and} \quad \lim_{n \rightarrow \infty} \left\| |u_n| \wedge |u_{n+1}| \right\| = 0.$$

Now set $w_n = \frac{1}{2}(u_n + u_{n+1})$. Then

$$\lim_{n \rightarrow \infty} \|w_n - u_n\| = \lim_{n \rightarrow \infty} \frac{1}{2} \|u_n - u_{n+1}\| = \frac{1}{2}.$$

Similarly, $\lim_{n \rightarrow \infty} \|w_n - u_{n+1}\| = \frac{1}{2}$. Also,

$$\begin{aligned} \|F_\lambda(w_n) - u_n\| &\leq \|F_\lambda(w_n) - F_\lambda(u_n)\| + \|F_\lambda(u_n) - u_n\| \\ &\leq \|w_n - u_n\| + \lambda(1 - \lambda)^{-1} \|T(u_n) - u_n\|, \end{aligned}$$

from which $\limsup_{n \rightarrow \infty} \|F_\lambda(w_n) - u_n\| \leq \frac{1}{2}$. Similarly,

$$\limsup_{n \rightarrow \infty} \|F_\lambda(w_n) - u_{n+1}\| \leq \frac{1}{2}.$$

Since $2(|u_n| \wedge |u_{n+1}|) = \left| |u_n + u_{n+1}| - |u_n - u_{n+1}| \right|$ (lattice identity) and

$$\lim_{n \rightarrow \infty} (|u_n| \wedge |u_{n+1}|) = 0$$

we conclude

$$\lim_{n \rightarrow \infty} \left| \|u_n + u_{n+1}\| - \|u_n - u_{n+1}\| \right| \leq \lim_{n \rightarrow \infty} \left\| |u_n + u_{n+1}| - |u_n - u_{n+1}| \right\| = 0.$$

Therefore $\lim_{n \rightarrow \infty} \|u_n + u_{n+1}\| = \lim_{n \rightarrow \infty} \|u_n - u_{n+1}\| = 1$, from which

$$\lim_{n \rightarrow \infty} \|w_n\| = \frac{1}{2}.$$

By Lemma 8.8

$$\begin{aligned} 2 \limsup_{n \rightarrow \infty} \|F_\lambda(w_n)\| &\leq \limsup_{n \rightarrow \infty} \|F_\lambda(w_n) - F_\lambda(0)\| + \limsup_{n \rightarrow \infty} \|F_\lambda(w_n) - u_n\| \\ &\quad + \limsup_{n \rightarrow \infty} \|F_\lambda(w_n) - u_{n+1}\| \\ &\leq \limsup_{n \rightarrow \infty} \|w_n\| + \frac{1}{2} + \frac{1}{2} \\ &= \frac{3}{2}. \end{aligned}$$

This, in turn, implies that

$$\limsup_{n \rightarrow \infty} \|F_\lambda(w_n)\| \leq \frac{3}{4}.$$

Now, let $\lambda_n \in (0, 1)$ with $\lim_{n \rightarrow \infty} \lambda_n = 1$. Since the above inequality holds for all $\lambda \in (0, 1)$, for each n there exists $z_n \in C$ such that

$$\|F_{\lambda_n}(z_n)\| \leq \frac{3}{4} + \frac{1}{n}.$$

On the other hand, by (d),

$$\|F_{\lambda_n}(z_n) - T(F_{\lambda_n}(z_n))\| = (1 - \lambda_n)\lambda_n^{-1} \|z_n - F_\lambda(z_n)\| \rightarrow 0$$

as $n \rightarrow \infty$. Therefore, $\{F_{\lambda_n}(z_n)\}$ is an approximate fixed point sequence for T so by the Goebel-Karlovitz Lemma $\lim_{n \rightarrow \infty} \|F_{\lambda_n}(z_n)\| = 1$. This obviously contradicts

$$\limsup_{n \rightarrow \infty} \|F_{\lambda_n}(z_n)\| \leq \frac{3}{4}. \quad \blacksquare$$

Having proved the above theorem, where does it apply? It is known ([39]) that the class of Banach lattices whose lattice operations ($\wedge$ and $\vee$) are weakly sequentially continuous and which do not contain subspaces isomorphic to c_0 are weakly orthogonal. This fact follows rather directly from results which can be found, for example, in [122]. The most familiar spaces with this property are the ℓ_p spaces, $1 \leq p < \infty$.

8.10.2 Uniformly monotone Banach lattices

The first recognition that uniform monotonicity plays a role in fixed point theory seems to occur in [54], where it is shown that if a Banach lattice X has a uniformly monotone norm and if ℓ_1 is not finitely representable in X (see Chapter 9) then X has the weak fixed point property. At the same time it is known that L_1 has uniformly monotone norm, while Alspach's example [2] shows that L_1 does not have the weak fixed point property. Thus the following condition alone is not sufficient for the weak fixed point property.

Definition 8.11 *The norm of a Banach lattice X is said to be uniformly monotone if there exists a strictly increasing continuous modulus $\delta : [0, 1] \rightarrow \mathbb{R}$ with $\delta(0) = 0$ such that if $x, y \geq 0$ satisfy $1 = \|y\| \geq \|x\|$, then $\|x + y\| \geq 1 + \delta(\|x\|)$.*

In this section we show that with an additional assumption there is a connection between uniform monotonicity of a Banach lattice and normal structure; hence the weak fixed point property. In proving this result we need to recall (Proposition 8.1 of Chapter 8) that a bounded convex subset K of a Banach space has normal structure if and only if it does not contain a diametral sequence, that is, a bounded sequence $\{x_n\}$ for which

$$\lim_{n \rightarrow \infty} \text{dist}(x_{n+1}, \overline{\text{conv}}\{x_1, x_2, \dots, x_n\}) = \text{diam}\{x_1, x_2, \dots\}.$$

We also need another direct consequence of weak orthogonality. If $\{x_n\}$ converges weakly to 0 in a weakly orthogonal Banach lattice then by definition for each $x \in X$,

$$\lim_{n \rightarrow \infty} \left\| |x_n| \wedge |x| \right\| = 0.$$

As noted in the previous argument, in conjunction with the lattice identity

$$2(|x| \wedge |y|) = \left| |x + y| - |x - y| \right|$$

this gives

$$\lim_{n \rightarrow \infty} \left\| |x + x_n| - |x - x_n| \right\| = 0.$$

This is precisely the observation that leads to the definition of the weak orthogonality condition labeled WORTH, which we take up in the next section.

Theorem 8.25 *Suppose X is a Banach lattice with uniformly monotone norm and suppose X is weakly orthogonal. Then weakly compact convex subsets of X have normal structure.*

Proof. Let K be a weakly compact convex subset of X with $\text{diam}(K) > 0$ and assume K does not have normal structure. Then by Proposition 8.1 K contains a diametral sequence $\{x_n\}$, and by weak compactness we may assume that $\{x_n\}$ converges weakly to $p \in H = \overline{\text{conv}}\{x_1, x_2, \dots\}$. By translating H we may assume $p = 0$ and by normalization we may assume $\text{diam}(H) = 1$. Thus $\|x_n\| \leq 1$ for each n . Since

$$\lim_{n \rightarrow \infty} \text{dist}(x_{n+1}, \overline{\text{conv}}\{x_1, x_2, \dots, x_n\}) = 1$$

and $0 \in \overline{\text{conv}}\{x_1, x_2, \dots\}$, we also have

$$\lim_{n \rightarrow \infty} \|x_n\| = 1.$$

Now let $x \in X$ with $x \neq 0$ and assume $\liminf_{n \rightarrow \infty} \|x_n - x\| \leq 1$. Uniform monotonicity of the norm of X implies that for all n ,

$$\delta(1) + 1 \leq \left\| \frac{|x_n|}{\|x_n\|} + \frac{|x|}{\|x\|} \right\|.$$

Thus (using weak orthogonality in the second step and using $|a| - |b| \leq |a - b|$ in the third step)

$$\begin{aligned} \|x\| (\delta(1) + 1) &\leq \liminf_{n \rightarrow \infty} \left\| \frac{\|x\|}{\|x_n\|} |x_n| + |x| \right\| \\ &= \liminf_{n \rightarrow \infty} \left\| \frac{\|x\|}{\|x_n\|} |x_n| - |x| \right\| \\ &\leq \liminf_{n \rightarrow \infty} \left\| \frac{\|x\|}{\|x_n\|} x_n - x \right\| \\ &\leq \liminf_{n \rightarrow \infty} \left\| \frac{\|x\|}{\|x_n\|} x_n - x_n \right\| + \liminf_n \|x_n - x\| \\ &\leq |1 - \|x\|| + 1. \end{aligned}$$

Since $\{x_n - x\}$ converges weakly to x , $\|x\| \leq \liminf_{n \rightarrow \infty} \|x_n - x\| \leq 1$. If $\|x\| = 1$ then $\delta(1) \leq 0$ which is a contradiction. So $\|x\| < 1$ and

$$\|x\| (\delta(1) + 1) \leq 2 - \|x\|.$$

Solving for $\|x\|$ we obtain

$$\|x\| \leq \frac{2}{\delta(1) + 2} < 1.$$

Taking $x = x_n$ for n sufficiently large, we see that this leads to a contradiction. ■

8.10.3 WORTH

In this section we study a weak orthogonality condition labeled WORTH, introduced independently by H. Rosenthal in 1983 [?] and B. Sims in 1988 [?]. As we just observed, this condition arises as an outgrowth of the concept of weak orthogonality in a Banach lattice.

Definition 8.12 *A Banach space X is said to have WORTH if whenever $\{x_n\}$ is a sequence in X which converges weakly to 0, then for any $x \in X$*

$$\limsup_{n \rightarrow \infty} \|x_n - x\| = \limsup_{n \rightarrow \infty} \|x_n + x\|.$$

If weak convergence is replaced with weak convergence in a dual space X^* then X^* is said to have WORTH*.*

We begin with the following.

Proposition 8.6 *If X is a separable Banach space and if X^* has $WORTH^*$, then X has $WORTH$.*

Proof. Assume X^* has $WORTH^*$ while X fails $WORTH$. Then there exists a sequence $\{x_n\}$ in X and an element $u \in X$, for which $\text{weak-}\lim_{n \rightarrow \infty} x_n = 0$ while

$$\limsup_{n \rightarrow \infty} \|x_n + u\| > \limsup_{n \rightarrow \infty} \|x_n - u\|.$$

Let $\{x_{n_k}\}$ be a subsequence of $\{x_n\}$ for which

$$\limsup_{n \rightarrow \infty} \|x_n + u\| = \lim_{k \rightarrow \infty} \|x_{n_k} + u\|.$$

Then

$$\begin{aligned} \limsup_{n \rightarrow \infty} \|x_n + u\| &= \lim_{k \rightarrow \infty} \|x_{n_k} + u\| \\ &> \limsup_{n \rightarrow \infty} \|x_n - u\| \\ &\geq \limsup_{k \rightarrow \infty} \|x_{n_k} - u\|, \end{aligned}$$

so by again passing to a subsequence we may assume

$$\lim_{n \rightarrow \infty} \|x_n + u\| > \lim_{n \rightarrow \infty} \|x_n - u\|.$$

For each $n \in \mathbb{N}$ choose $x_n^* \in S_{X^*}$ such that $x_n^*(x_n + u) = \|x_n + u\|$. Since (by separability of X) X^* is metrizable in the weak* topology, B_{X^*} is weak* sequentially compact so by yet again passing to a subsequence we may suppose that

$$\text{weak}^* - \lim_{n \rightarrow \infty} x_n^* = x^* \in B_{X^*}.$$

Therefore, recalling that $\text{weak} \lim_{n \rightarrow \infty} x_n = 0$, and then using $WORTH^*$,

$$\begin{aligned} \lim_{n \rightarrow \infty} \|x_n + u\| &= \lim_{n \rightarrow \infty} x_n^*(x_n + u) \\ &= \lim_{n \rightarrow \infty} x_n^*(x_n) + \lim_{n \rightarrow \infty} x_n^*(u) - 2 \lim_{n \rightarrow \infty} x^*(x_n) \\ &= \lim_{n \rightarrow \infty} (x_n^* - x^*)(x_n) + \lim_{n \rightarrow \infty} x^*(u - x_n) \\ &= \lim_{n \rightarrow \infty} ((x_n^* - x^*) - x^*)(x_n - u) \\ &\leq \limsup_{n \rightarrow \infty} \|(x_n^* - x^*) - x^*\| \|x_n - u\| \\ &= \limsup_{n \rightarrow \infty} \|(x_n^* - x^*) + x^*\| \|x_n - u\| \\ &= \limsup_{n \rightarrow \infty} \|x_n^*\| \|x_n - u\| \\ &= \limsup_{n \rightarrow \infty} \|x_n - u\|. \end{aligned}$$

This contradicts our original assumption. ■

We now use the preceding result to establish a connection between $WORTH^*$ and normal structure.

Theorem 8.26 *Suppose X is a separable Banach space for which X^* is uniformly nonsquare and has WORTH*. Then X has normal structure.*

Proof. By assumption, $\varepsilon_0(X^*) < 2$ and X^* (hence X) is super-reflexive. Moreover, by Proposition 8.6 X has WORTH. It can be shown that there exists a sequence $\{x_n\}$ in B_X such that $x_1 = 0$, $\text{weak-}\lim_{n \rightarrow \infty} x_n = 0$, $\lim_{n \rightarrow \infty} \|x_n\| = 1$, $\text{diam}(\{x_1, x_2, \dots\}) = 1$, and for each n

$$\text{dist}(x_n, \text{conv}\{x_1, \dots, x_{n-1}\}) > 1 - 1/n.$$

(This is a refinement of Proposition 8.1 due to B. Turett, which is left as an exercise; alternatively, see Step 1 in the proof of Proposition 3 of [159].)

Since $\text{conv}\{x_1, \dots, x_{n-1}\}$ is compact there exists $m_n \in \text{conv}\{x_1, \dots, x_{n-1}\}$ such that

$$\text{dist}(x_n, \text{conv}\{x_1, \dots, x_{n-1}\}) = \|x_n - m_n\|.$$

By the Hahn-Banach Theorem, there exists $x_n^* \in S_{X^*}$ such that

$$x_n^*(x_n - m_n) = \|x_n - m_n\|.$$

Thus for $j = 1, \dots, n$,

$$\begin{aligned} 1 - \frac{1}{n} &< \inf\{\|x_n - x\|; x \in \text{conv}\{x_1, \dots, x_n\}\} \\ &= x_n^*(x_n - m_n) = x_n^*(x_n) - x_n^*(m_n) \\ &\leq x_n^*(x_n) - x_n^*(x_j) \leq 1 - x_n^*(x_j). \end{aligned}$$

Since $x_1 = 0$, we conclude that for each $n \geq 2$,

$$1 \geq x_n^*(x_n) > 1 - 1/n.$$

Also, since B_{X^*} is weak* sequentially compact we may assume

$$\text{weak}^* - \lim_{n \rightarrow \infty} x_n^* = x^* \in B_{X^*}.$$

Now fix $\varepsilon > 0$ and select $j_0 \in \mathbb{N}$ so that $j_0 > 1/\varepsilon$, $1/j_0 < \varepsilon/2$ and so that $|x^*(x_{j_0})| < \varepsilon/4$. Then for $i > j_0$ sufficiently large

$$|(x_i^* - x^*)(x_{j_0})| < \varepsilon/4, \quad |x_{j_0}^*(x_i)| < \varepsilon/2,$$

and, since X has WORTH, $\|x_{j_0} - x_i\| + \varepsilon \geq \|x_{j_0} + x_i\|$. In particular,

$$|x_i^*(x_{j_0})| - |x^*(x_{j_0})| \leq |(x_i^* - x^*)(x_{j_0})|,$$

so $|x_i^*(x_{j_0})| < \varepsilon/2$. Hence (for $\varepsilon < 1$)

$$\begin{aligned} -x_{j_0}^* \left(\frac{x_i - x_{j_0}}{\|x_i - x_{j_0}\|} \right) &\geq -x_{j_0}^*(x_i - x_{j_0}) \\ &= -x_{j_0}^*(x_i) + x_{j_0}^*(x_{j_0}) \\ &> -\varepsilon/2 + 1 - 1/j_0 \\ &> 1 - \varepsilon. \end{aligned}$$

Also,

$$\begin{aligned}
 x_i^* \left(\frac{x_i - x_{j_0}}{\|x_i - x_{j_0}\|} \right) &\geq x_i^*(x_i - x_{j_0}) \\
 &= x_i^*(x_i) - x_i^*(x_{j_0}) \\
 &> 1 - 1/i - \varepsilon/2 \\
 &> 1 - \varepsilon.
 \end{aligned}$$

Now for $\mu \in (0, 1)$ and $x \in S_X$, let

$$S(B_{X^*}, x, \mu) = \{x^* \in B_{X^*} : x(x^*) > 1 - \mu\}$$

denote the x, μ -slice of B_{X^*} . Then in view of the above inequalities,

$$-x_{j_0}^*, x_i^* \in S \left(B_{X^*}, \frac{x_i - x_{j_0}}{\|x_i - x_{j_0}\|}, \varepsilon \right).$$

Moreover, since

$$\|x_i + x_{j_0}\| \leq \|x_i - x_{j_0}\| + \varepsilon \leq 1 + \varepsilon,$$

and

$$\|x_i^* + x_{j_0}^*\| \geq (x_i^* + x_{j_0}^*) \left(\frac{x_i + x_{j_0}}{\|x_i + x_{j_0}\|} \right)$$

we have (for $\varepsilon < 1$)

$$\begin{aligned}
 \|x_i^* - (-x_{j_0}^*)\| &= \|x_i^* + x_{j_0}^*\| \\
 &\geq (1 + \varepsilon)^{-1} (x_i^* + x_{j_0}^*)(x_i + x_{j_0}) \\
 &= (1 + \varepsilon)^{-1} (x_i^*(x_i) + x_i^*(x_{j_0}) + x_{j_0}^*(x_i) + x_{j_0}^*(x_{j_0})) \\
 &> (1 + \varepsilon)^{-1} (1 - 1/i - \varepsilon/2 - \varepsilon/2 + 1 - 1/j_0) \\
 &> (1 + \varepsilon)^{-1} (2 - 2\varepsilon) = 2(1 - \varepsilon)/(1 + \varepsilon).
 \end{aligned}$$

If $\mu > \varepsilon$ then $\text{diam } S \left(B_{X^*}, \frac{x_i - x_{j_0}}{\|x_i - x_{j_0}\|}, \mu \right) \geq \text{diam } S \left(B_{X^*}, \frac{x_i - x_{j_0}}{\|x_i - x_{j_0}\|}, \varepsilon \right)$.

With the above, this implies that for all $\mu \in (0, 2)$,

$$\text{diam } S \left(B_{X^*}, \frac{x_i - x_{j_0}}{\|x_i - x_{j_0}\|}, \mu \right) \geq 2.$$

Now let $\varepsilon_0(X^*) < \varepsilon' < 2$. Then $\delta(\varepsilon') > 0$. On the other hand, for any $\mu > 0$, there exist $u^*, v^* \in S \left(B_{X^*}, \frac{x_i - x_{j_0}}{\|x_i - x_{j_0}\|}, \mu \right)$ such that $\|u^* - v^*\| > \varepsilon'$. At the same time,

$$\begin{aligned}
 \left\| \frac{u^* + v^*}{2} \right\| &\geq \left(\frac{u^* + v^*}{2} \right) \left(\frac{x_i - x_{j_0}}{\|x_i - x_{j_0}\|} \right) \\
 &= \frac{1}{2} \left[u^* \left(\frac{x_i - x_{j_0}}{\|x_i - x_{j_0}\|} \right) + v^* \left(\frac{x_i - x_{j_0}}{\|x_i - x_{j_0}\|} \right) \right] \\
 &\geq \frac{1}{2} [1 - \mu + 1 - \mu] \\
 &= 1 - \mu.
 \end{aligned}$$

This proves that $\delta(\varepsilon') \leq \mu$, which is obviously a contradiction if $\mu < \delta(\varepsilon')$. Therefore, it must be the case that $\varepsilon_0(X^*) = 2$, contradicting the original assumption that X^* is uniformly nonsquare. ■

Exercises

Exercise 8.1 Show that if G is convex and $T : \partial G \rightarrow \overline{G}$, then T satisfies condition (8.1).

Exercise 8.2 Prove Theorem 8.4. [Hint: Let $f = (I - T)$, let $G_t = (I - tT)(G)$ for $t \in [0, 1]$. Let

$$H = \{t \in [0, 1] : 0 \in G_t\}.$$

Observe that $H \neq \emptyset$. Use Theorem 8.1 to show that H is open in $[0, 1]$. Then show that H is closed in $[0, 1]$. From this it follows that $H = [0, 1]$ and therefore $0 \in G_1$.]

Exercise 8.3 Let K be a convex subset of a Banach space and $p > 1$. Show that any mapping $T : K \rightarrow K$ which satisfies $\|T(x) - T(y)\| \leq \|x - y\|^p$, $x, y \in K$, is a constant mapping.

Exercise 8.4 Let X be a Banach space. Define the modulus of uniform convexity in the direction $z \in X$ ($\|z\| = 1$) by

$$\delta(z, \varepsilon) = \inf \left\{ 1 - \left\| \frac{x + y}{2} \right\| : x, y \in B_X, x - y = \alpha z, \text{ and } \|x - y\| \geq \varepsilon \right\}$$

for any $\varepsilon \in [0, 2)$. X is said to be uniformly convex in every direction (U.C.E.D.) if $\delta(z, \varepsilon) > 0$, for any z and $\varepsilon > 0$. Show that if X is U.C.E.D., then X has normal structure.

Exercise 8.5 Let K be a weakly compact convex subset of a Banach space which is U.C.E.D. Show that the asymptotic center $A_k(\{x_n\})$ of any sequence $\{x_n\}$ in K is a singleton.

Exercise 8.6 Let X be a Banach space. Define the modulus of smoothness ρ_X of X by

$$\rho(\eta) = \sup \left\{ \left\| \frac{x + \eta y}{2} \right\| + \left\| \frac{x - \eta y}{2} \right\| - 1 : \|x\| \leq 1, \|y\| \leq 1 \right\}$$

for any $\eta > 0$. X is said to be uniformly smooth (U.S.) if

$$\lim_{\eta \rightarrow 0} \frac{\rho_X(\eta)}{\eta} = 0.$$

Show that

$$\lim_{\eta \rightarrow 0} \frac{\rho_X(\eta)}{\eta} = \frac{\varepsilon_0(X^*)}{2},$$

where $\varepsilon_0(X^*)$ is the characteristic of convexity of X^* .

Exercise 8.7 Use the previous exercises to show that if

$$\lim_{\eta \rightarrow 0} \frac{\rho_X(\eta)}{\eta} < \frac{1}{2},$$

then X and X^* have uniform normal structure property. In particular, uniformly smooth Banach spaces have uniform normal structure property.

Exercise 8.8 Let X be a Banach space. X is said to be nearly uniformly convex if for every $\varepsilon > 0$, there exists $\delta(\varepsilon) > 0$ such that if $\|x_n\| \leq 1$ and

$$\inf \left\{ \|x_n - x_m\| : n \neq m \right\} \geq \varepsilon$$

then

$$\overline{\text{conv}}\{x_n\} \cap B(0; 1 - \delta(\varepsilon)) \neq \emptyset.$$

Show that if X is nearly uniformly convex, then X has normal structure.

Exercise 8.9 Let X be a Banach space. X is said to satisfy Opial's condition if for every sequence $\{x_n\}$ which converges weakly to ω ,

$$\liminf_{n \rightarrow \infty} \|x_n - \omega\| < \liminf_{n \rightarrow \infty} \|x_n - x\|$$

holds for any $x \neq \omega$. Show that if X satisfies Opial's condition, then X has weak normal structure.

Exercise 8.10 Let X be a Banach space. For any nonempty bounded subset A of X , define the Kuratowski's measure of noncompactness α of A by

$$\alpha(A) = \inf \left\{ r > 0, A \subset \bigcup_{1 \leq i \leq n} A_i \text{ with } \text{diam}(A_i) \leq r \right\}.$$

The modulus of noncompact convexity of X is defined by

$$\Delta_X(\varepsilon) = \inf \left\{ 1 - \inf_{x \in A} \|x\| : A \text{ is convex subset of the unit ball with } \alpha(A) \geq \varepsilon \right\}$$

for any $\varepsilon \in (0, 2)$. The characteristic of noncompact convexity ε_1 of X is given by

$$\varepsilon_1(X) = \sup \{ \varepsilon : \Delta_X(\varepsilon) = 0 \}.$$

Show that if $\varepsilon_1(X) < 1$, then X is reflexive and has normal structure.

Exercise 8.11 Let X be a Banach space with a Schauder basis $\{x_n\}$. Define $\beta_p(X)$, for $p \geq 1$, to be the infimum of the set of numbers λ such that

$$\left(\|x\|^p + \|y\|^p\right)^{1/p} \leq \lambda\|x + y\|$$

for every $x, y \in X$ which satisfy $\text{supp}(x) + 1 < \text{supp}(y)$, that is, for every $i \in \text{supp}(x)$ and any $j \in \text{supp}(y)$, we have $i + 1 < j$. Show that X has weak normal structure provided $\beta_p(X) < 2^{1/p}$.

Exercise 8.12 (James space) Let c_0 denote the space of all real sequences that converge to 0. The James space J (resp. J_1) consists of all sequences $(\alpha_n) \in c_0$ for which $\|(\alpha_n)\|_J < \infty$ (resp. $\|(\alpha_n)\|_{J_1} < \infty$), where

$$\|(\alpha_n)\|_J = \sup \left\{ \sum_{1 \leq i \leq n} (\alpha_{p_i} - \alpha_{p_{i+1}})^2 + (\alpha_{p_{n+1}} - \alpha_{p_1})^2 \right\},$$

resp.

$$\|(\alpha_n)\|_{J_1} = \sup \left\{ \sum_{1 \leq i \leq n} (\alpha_{p_i} - \alpha_{p_{i+1}})^2 + \alpha_{p_{n+1}}^2 \right\},$$

the supremum being taken over all finite increasing sequences of positive numbers $p_1, p_2, \dots, p_{n+1}$.

1. Show that $\beta_2(J_1) = 1$. Deduce from it that J_1 has weak normal structure.
2. Show that for any sequence $\{x_n\}$ in J which converges weakly to 0, there exists a subsequence $\{x_{n'}\}$ of $\{x_n\}$ such that

$$\sup_q \left\{ \limsup_{n' \rightarrow \infty} \|x_{n'} - x_q\| \right\} > \liminf_{n' \rightarrow \infty} \|x_{n'}\|.$$

Deduce that J has weak normal structure.

Exercise 8.13 Show that any dual Banach space, with a shrinking unconditional basis with the unconditional constant equal to 1, has the weak*-fixed point property.

Hint: Use the conclusion of Exercise 6.17.

Exercise 8.14 Let X be a Banach space. Let C be a nonempty closed convex subset of X and $T : C \rightarrow C$ be nonexpansive.

1. Show that T has a bounded orbit if and only if any orbit of T is bounded.

2. Assume that T has bounded orbits. Show that there exists a bounded closed convex subset K of C which is T -invariant, that is, $T(K) \subset K$.
3. Show that if X has the fixed point property and T is a nonexpansive mapping defined on a closed convex set, then T has a fixed point if and only if it has bounded orbits.

Exercise 8.15 Show that the mapping F_λ defined in Section 8.10.1 satisfies properties (a)–(d).

Chapter 9

Banach Space Ultrapowers

9.1 Finite representability

The central concept in this chapter derives from the fundamental notion of finite representability introduced by R. C. James [78] in 1972.

Definition 9.1 *A Banach space Y is said to be finitely representable in a Banach space X if for every $\xi > 0$ and every finite-dimensional subspace Y_0 of Y there is a finite-dimensional subspace X_0 of X (with $\dim X_0 = \dim Y_0$) and an isomorphism (one-to-one linear mapping) $T : Y_0 \rightarrow X_0$ for which*

$$(1 - \xi) \|y\| \leq \|T(y)\| \leq (1 + \xi) \|y\|$$

for all $y \in Y_0$.

It is possible to formulate the above definition in a slightly different way. Recall that, in general, if X and Y are Banach (or normed) spaces and if $T : Y \rightarrow X$ is a linear mapping, then T is continuous if and only if $\|T\| < \infty$, where

$$\|T\| = \sup\{\|T(y)\| : y \in Y \text{ and } \|y\| = 1\}.$$

Now suppose Y is finitely representable in X and let $\eta > 0$. Choose $\xi > 0$ so that $\frac{1+\xi}{1-\xi} = 1 + \eta$. Then with $T : Y_0 \rightarrow X_0$ as in Definition 9.1, $\|T\| \leq 1 + \xi$ and $\|T^{-1}\| \leq (1 - \xi)^{-1}$, so $\|T\| \|T^{-1}\| \leq \frac{1+\xi}{1-\xi} = 1 + \eta$. Consequently, Y is finitely representable in X if and only if for every $\eta > 0$ and every finite-dimensional subspace Y_0 of Y there is a finite-dimensional subspace X_0 of X and an isomorphism T of Y_0 onto X_0 with

$$\|T\| \|T^{-1}\| \leq 1 + \eta.$$

This version of the definition is sometimes more useful, and even it can be refined. Having such an isomorphism T let $F = \|T^{-1}\|T$. Then $F^{-1} = T^{-1}/\|T^{-1}\|$ and thus

$$\|F\| \leq 1 + \eta \text{ and } \|F^{-1}\| = 1.$$

This rather minor observation will be useful later in this section.

These facts can be expressed in terms of the Banach-Mazur distance.

Definition 9.2 *Let X and Y be Banach spaces. The Banach-Mazur distance between X and Y is denoted by $d(X, Y)$ and is defined as*

$$d(X, Y) = \inf\{\|T\| \|T^{-1}\| : T \text{ is an isomorphism from } X \text{ onto } Y\}.$$

When X and Y are not isomorphic we simply set $d(X, Y) = \infty$.

Therefore, Y is finitely representable in X if for any $0 < \varepsilon < 1$ and for any finite-dimensional subspace Y_0 of Y , there exists a subspace X_0 of X such that $\dim X_0 = \dim Y_0$ and $d(X_0, Y_0) \leq 1 + \varepsilon$.

It is immediate from the definition that finite representability is transitive: If Z is finitely representable in Y and if Y is finite representable in X , then Z is finitely representable in X . Also, if Y is finitely representable in X then Y possesses any Banach space properties which are 'locally' determined, that is, determined by uniform behavior on finite-dimensional subspaces of X .

Definition 9.3 *Let X be a Banach space and let P be a Banach space property. Then X is said to be super- P if every Banach space which is finitely representable in X is P .*

For example, we have the following.

Definition 9.4 *A Banach space X is super-reflexive if it is the case that any Banach space Y which is finitely representable in X is necessarily reflexive.*

Since a Banach space is always finitely representable in itself, if X is super- P for any P then X is P . Thus any super-reflexive space is necessarily reflexive.

Properties which are preserved under finite representability are called 'super' properties. Finite dimensionality is a trivial super property. Also a celebrated theorem of Dvoretzky assures that ℓ_2 is finitely representable in *every* infinite-dimensional Banach space. On the other hand, reflexivity is not a super property. In particular, there exist reflexive Banach spaces in which ℓ_1 is finitely representable; hence there exist reflexive Banach spaces which are not super-reflexive. An example of such a space is the space

$$X = \left(\prod_{n \in \mathbb{N}} \ell_1^{(n)} \right)_2.$$

The elements of X are sequences $x = (x_1, x_2, \dots)$, where $x_n \in \ell_1^{(n)}$ (the n -dimensional ℓ_1 , i.e., the space $\mathbb{R}_1^n$) and

$$\|x\| = \left(\sum_{n=1}^{\infty} \|x_n\|_1^2 \right)^{1/2} < \infty.$$

It is not difficult to see that X is reflexive. In fact, X is isomorphic to X^* , where

$$X^* = \left(\prod_{n \in \mathbb{N}} \ell_{\infty}^{(n)} \right)_2.$$

To see that ℓ_1 is finitely representable in X we begin with an elementary fact about finite-dimensional spaces mentioned in Chapter 7.

Lemma 9.1 *Suppose E and F are Banach spaces with E finite-dimensional, and suppose $T : E \rightarrow F$ is linear. Then T is continuous.*

Proof. Suppose E is m -dimensional and let $\{e_1, \dots, e_m\}$ be an algebraic basis for E . The unit sphere $S(\ell_1^{(m)})$ of $\ell_1^{(m)}$ is compact and the mapping $\Phi : S(\ell_1^{(m)}) \rightarrow \mathbb{R}$ that sends $(u_1, \dots, u_m) \in S(\ell_1^{(m)})$ into the number $\left\| \sum_{j=1}^m u_j e_j \right\|$ is continuous and nonzero (since the vectors $\{e_1, \dots, e_m\}$ are linearly independent). Thus there exists a number $\delta > 0$ such that for all such $(u_1, \dots, u_m) \in S(\ell_1^{(m)})$,

$$\left\| \sum_{j=1}^m u_j e_j \right\| \geq \delta.$$

From this it follows that for every sequence $(a_1, \dots, a_m) \in \ell_1^{(m)}$,

$$\delta \sum_{j=1}^m |a_j| \leq \left\| \sum_{j=1}^m a_j e_j \right\|.$$

Therefore,

$$\begin{aligned} \|T\| &= \sup \left\{ \left\| T \left(\sum_{j=1}^m a_j e_j \right) \right\| / \left\| \sum_{j=1}^m a_j e_j \right\| : (a_1, \dots, a_m) \in \ell_1^{(m)} \right\} \\ &= \sup \left\{ \left\| \sum_{j=1}^m a_j T(e_j) \right\| / \left\| \sum_{j=1}^m a_j e_j \right\| : (a_1, \dots, a_m) \in \ell_1^{(m)} \right\} \\ &\leq \sup \{ \|T(e_j)\| / \delta : j = 1, \dots, m \} < \infty. \end{aligned}$$

■

Now let $\varepsilon > 0$ and let Y_0 be a subspace of ℓ_1 , say of dimension m , with basis $\{y_1, \dots, y_m\}$. Further assume $\|y_i\| = 1$, $i = 1, \dots, m$. As in the above proof there exists $\delta > 0$ such that for every sequence $(a_1, \dots, a_m) \in \ell_1^{(m)}$,

$$\delta \sum_{j=1}^m |a_j| \leq \left\| \sum_{j=1}^m a_j y_j \right\|.$$

Choose $\mu > 0$ so that

$$\frac{1 + \frac{\mu}{\delta}}{1 - \frac{\mu}{\delta}} < 1 + \varepsilon,$$

and for $i = 1, \dots, m$, choose $y'_i \in \ell_1$ so that y'_i has only finitely many nonzero coordinates, and so that $\|y_i - y'_i\| < \mu$. Then

$$\left\| \sum_{i=1}^m a_i (y_i - y'_i) \right\| \leq \left(\sum_{i=1}^m |a_i| \right) \mu,$$

from which

$$\left\| \sum_{i=1}^m a_i y'_i \right\| \leq \left\| \sum_{i=1}^m a_i y_i \right\| + \mu \sum_{i=1}^m |a_i| \leq \left(1 + \frac{\mu}{\delta} \right) \left\| \sum_{i=1}^m a_i y_i \right\|.$$

Also,

$$\left\| \sum_{i=1}^m a_i y'_i \right\| \geq \left(1 - \frac{\mu}{\delta} \right) \left\| \sum_{i=1}^m a_i y_i \right\|,$$

hence

$$\left(1 - \frac{\mu}{\delta} \right) \left\| \sum_{i=1}^m a_i y_i \right\| \leq \left\| \sum_{i=1}^m a_i y'_i \right\| \leq \left(1 + \frac{\mu}{\delta} \right) \left\| \sum_{i=1}^m a_i y_i \right\|.$$

This means that the operator T defined from Y_0 onto $\text{span}\{y'_1, \dots, y'_m\}$, defined by

$$T(y_1) = y'_1, \dots, T(y_m) = y'_m$$

(and extended by taking linear combinations), satisfies

$$\|T\| \leq 1 + \frac{\mu}{\delta} \text{ and } \|T^{-1}\| \leq \left(1 - \frac{\mu}{\delta} \right)^{-1}.$$

Let n_k be the index of the last nonzero term in y'_k , $k = 1, \dots, m$, and let $n = \max\{n_k : 1 \leq k \leq m\}$. Then $\text{span}\{y'_1, \dots, y'_m\}$ is isometric to an m -dimensional subspace of $\ell_1^{(n)}$ and $\ell_1^{(n)}$ is a subspace of X . Therefore, T composed with this embedding is the desired isomorphism.

Uniform convexity provides a nontrivial example of a super-property. This is a consequence of the following result, which illustrates the 'two-dimensional' character of the modulus of convexity.

Theorem 9.1 *Let X and Y be Banach spaces with respective moduli of convexity δ_X and δ_Y , and suppose Y is finitely representable in X . Then for each $\varepsilon \in [0, 2)$, $\delta_Y(\varepsilon) \geq \delta_X(\varepsilon)$.*

Proof. Let $\varepsilon > 0$ and let $x, y \in Y$ with $\|x\|_Y = \|y\|_Y = 1$ and $\|x - y\|_Y \geq \varepsilon$. Suppose $\eta > 0$ and let F be an isomorphism of $\text{span}\{x, y\}$ onto some two-dimensional subspace X_0 of X which satisfies $\|F\| \leq 1 + \eta$ and $\|F^{-1}\| = 1$. If

$$x' = F(x) \text{ and } y' = F(y),$$

then $\|x'\|_X \leq 1 + \eta$ and $\|y'\|_X \leq 1 + \eta$, while

$$\|x' - y'\|_X = \|F(x - y)\|_X \geq \|x - y\|_Y \geq \varepsilon.$$

According to the definition of δ_X ,

$$\left\| \frac{1}{2} \left(\frac{x'}{1 + \eta} + \frac{y'}{1 + \eta} \right) \right\|_X \leq 1 - \delta_X \left(\frac{\varepsilon}{1 + \eta} \right).$$

Using the fact that δ_X is continuous¹ and letting $\eta \rightarrow 0$, we see that

$$\left\| \frac{1}{2} (x' + y') \right\|_X \leq 1 - \delta_X(\varepsilon).$$

But since $\|F^{-1}\| = 1$,

$$\left\| \frac{1}{2} (x + y) \right\|_Y = \left\| F^{-1} \left(\frac{1}{2} (x' + y') \right) \right\|_Y \leq \left\| \frac{1}{2} (x' + y') \right\|_X.$$

The definition of δ_Y now gives $\delta_Y(\varepsilon) \geq \delta_X(\varepsilon)$. ■

Corollary 9.1 *If X is a uniformly convex Banach space and if Y is finitely representable in X , then Y is uniformly convex.*

Corollary 9.2 *If X is a Banach space and if Y is finitely representable in X , then $\varepsilon_0(Y) \leq \varepsilon_0(X)$.*

A Banach space X is said to be *uniformly nonsquare* if $\varepsilon_0(X) < 2$. Thus we conclude that this is also a super-property:

Corollary 9.3 *If X is a uniformly nonsquare Banach space and if Y is finitely representable in X , then Y is uniformly nonsquare.*

We conclude this section with a collection of important facts which follow from results proved elsewhere.

First, since uniformly convex spaces are reflexive it follows from Corollary 9.1 that uniformly convex spaces are super-reflexive. In fact, since it is known that uniformly nonsquare spaces are reflexive (R. C. James [81]), the following is true.

¹A proof of this fact can be found in [67].

Theorem 9.2 *Every uniformly nonsquare Banach space is super-reflexive.*

Dvoretzky's result gives another corollary to Theorem 9.1.

Corollary 9.4 *If X is any infinite-dimensional Banach space, then $\delta_X \leq \delta_{\ell_2}$.*

The following deep result of P. Enflo [55] reveals a surprising connection between super-reflexivity and uniform convexity.

Theorem 9.3 *A Banach space is super-reflexive if and only if it can be given an equivalent uniformly convex norm.*

There have been a number of more technical characterizations of super-reflexivity, many of which are due to R. C. James. One such characterization is the following, which we shall use later.

Theorem 9.4 (James) *A Banach space X is not super-reflexive if and only if the following condition holds:*

If $0 < \theta < 1$ and $0 < \varepsilon < 1$, then for every positive integer n there exist subsets $\{z_1, \dots, z_n\} \subseteq B_X$ and $\{g_1, \dots, g_n\} \subseteq B_{X^}$ such that*

$$g_i(z_j) = \theta \text{ if } i \leq j \text{ and } g_i(z_j) = 0 \text{ if } i > j,$$

and, for every sequence of numbers $\{a_i\}$ and every $k \leq n$,

$$\left\| \sum_{i=1}^n a_i z_i \right\| \geq \frac{1}{2} \varepsilon \left\| \sum_{i=1}^k a_i z_i \right\|.$$

9.2 Convergence of ultranets

The purpose of this section is to lay the groundwork for how to construct for any Banach space X a *much larger* Banach space $\tilde{X}$ which has the property that $\tilde{X}$ both contains an isometric copy of X and yet is always finitely representable in X . The fact that $\tilde{X}$ is 'much larger' than X is important; otherwise X itself would do. The usefulness of this particular type of construction will be made apparent in subsequent sections.

Recall (see Appendix) that a *net* $\{x_\alpha\}_{\alpha \in A}$ in a set X is a mapping $\varphi : A \rightarrow X$, where A is a directed set and $x_\alpha = \varphi(\alpha)$, $\alpha \in A$. A net is an *ultranet* if given any subset $G \subseteq X$ either $\{x_\alpha\}$ is eventually in G or $\{x_\alpha\}$ is eventually in $X \setminus G$ (the complement of G). Thus given any subset G of X one of the two following mutually exclusive alternatives must hold.

(a) There exists α_0 such that $\alpha \geq \alpha_0$ implies $x_\alpha \in G$.

(b) There exists α_0 such that $\alpha \geq \alpha_0$ implies $x_\alpha \notin G$.

One nice property of ultranets is that they automatically pass from one set to another.

Proposition 9.1 *Let X and Y be sets and let $f : X \rightarrow Y$. Suppose $\{x_\alpha\}$ is an ultranet in X . Then $\{f(x_\alpha)\}$ is an ultranet in Y .*

Proof. Let H be any subset of Y and let $G = \{x \in X : f(x) \in H\}$. Since $\{x_\alpha\}$ is an ultranet in X , $\{x_\alpha\}$ is eventually in G or in $X \setminus G$. If $\{x_\alpha\}$ is eventually in G , then $\{f(x_\alpha)\}$ is eventually in H ; otherwise $\{f(x_\alpha)\}$ is eventually in $Y \setminus H$. ■

Proposition 9.2 *If $\{x_\alpha\}_{\alpha \in A}$ is an ultranet in a topological space X and if some subnet $\{x_{\alpha_\xi}\}_{\xi \in E}$ of $\{x_\alpha\}$ converges to $x \in X$, then $\lim_\alpha x_\alpha = x$.*

Proof. Let U be any open set in X which contains x . Since $\{x_{\alpha_\xi}\}$ converges to x there exists $\xi_0 \in E$ such that $\xi \geq \xi_0$ implies $x_{\alpha_\xi} \in U$. Also, since $\{x_{\alpha_\xi}\}$ is a subnet of $\{x_\alpha\}$, given any $\alpha \in A$ there exists $\xi_1 \in E$ such that if $\xi \geq \xi_1$, then $\alpha_\xi \geq \alpha$. Since E is directed there exists $\xi \in E$ such that $\xi \geq \xi_0$ and $\xi \geq \xi_1$. For such ξ , we have $\alpha_\xi \geq \alpha$ and $x_{\alpha_\xi} \in U$. This proves that $\{x_\alpha\}$ is not eventually in $X \setminus U$; hence $\{x_\alpha\}$ is eventually in U , proving $\lim_\alpha x_\alpha = x$. ■

Our next fact relies on the following characterization: A subset K of a topological space is compact if and only if every net $\{x_\alpha\}$ in K has a subnet which converges to a point of K .

Proposition 9.3 *Let K be a compact subset of a topological space and let $\{x_\alpha\}$ be an ultranet in K . Then $\{x_\alpha\}$ converges (to a point of K).*

Proof. Since K is compact $\{x_\alpha\}$ has a subnet which converges to a point $x \in K$. In view of the previous proposition, this implies that $\{x_\alpha\}$ itself converges to x . ■

We now bring to bear a fundamental fact discussed in the Appendix. We may take this as an axiom since, indeed, it is known to be equivalent to the Axiom of Choice.

Lemma 9.2 *Every net $\{x_\alpha\}$ in a set S has a subnet $\{x_{\alpha_\xi}\}$ which is an ultranet.*

9.3 The Banach space ultrapower $\tilde{X}$

It is not our intention here to study Banach space ultrapowers in general. This theory has a rich history and elegant treatments may be found elsewhere, for example, in [73], [147]. Also, an excellent exposition which deals with fixed point theory of nonexpansive mappings is found in [1]. However, a few facts have emerged more recently.

For our immediate purpose we shall only need to have at hand one Banach space ultrapower associated with each Banach space, and the one described

below will be adequate. It is common practice to construct Banach space ultrapowers using ultrafilters but we use ultranets instead. The end result is the same.

It suffices to begin with the smallest infinite ordinal ω . As a well-ordered set this set is linearly ordered, therefore a net. By Lemma 9.2 it has a subnet $\{\varphi, E\}$ which is an ultranet. Thus E is a directed set and $\varphi : E \rightarrow \omega$ satisfies the following condition.

(i) Given any $n \in \omega$ there exists $\xi_0 \in E$ such that if $\xi \geq \xi_0$ then $\varphi(\xi) \geq \varphi(\xi_0)$.

We follow the usual convention and denote $\varphi(\xi) = n_\xi$, $\xi \in E$. In particular, we use $\{n_\xi\}$ to denote $\{\varphi, E\}$. Throughout this section the ultranet subnet $\{n_\xi\}$ of ω will remain fixed.

Now let X be a Banach space and let

$$\ell_\infty(X) = \{x = (x_n) \in X : \sup_{1 \leq i < \infty} \|x_i\| < \infty\}.$$

It is known (and easily checked) that $\ell_\infty(X)$ is a Banach space with the norm defined by

$$\|(x_n)\|_\infty = \sup_{1 \leq i < \infty} \|x_i\|$$

for $\{x_n\} \in \ell_\infty(X)$. Set

$$\mathcal{N} = \{x = (x_n) \in \ell_\infty(X) : \lim_\xi \|x_{n_\xi}\| = 0\}.$$

The Banach space ultrapower $\tilde{X}$ of X (relative to the ultranet $\{n_\xi\}$) is the quotient space $\ell_\infty(X)/\mathcal{N}$. Thus the elements of $\tilde{X}$ are equivalence classes $[(x_n)]$ of bounded sequences $(x_n) \subset X$, where one agrees that two such sequences (x_n) and (y_n) are equivalent if and only if

$$\lim_\xi \|x_{n_\xi} - y_{n_\xi}\| = 0.$$

The norm $\|\cdot\|_\xi$ in $\tilde{X}$ is the usual quotient norm. Thus for $\tilde{x} = [(x_n)] \in \tilde{X}$,

$$\|\tilde{x}\|_\xi = \inf\{\|(x_n + y_n)\|_\infty : (y_n) \in \mathcal{N}\}.$$

There is another approach which leads to the same thing. Note that since $\{\|x_n\|\}$ is bounded it lies in a compact subset of $\mathbb{R}$. Therefore $\lim_\xi \|x_{n_\xi}\|$ always exists. Thus one could just as well define for $\tilde{x} = [(x_n)] \in \tilde{X}$,

$$\|\tilde{x}\|_\xi = \lim_\xi \|x_{n_\xi}\|.$$

We check that the second definition coincides with the first.

Proposition 9.4 *Let $\|\tilde{x}\|_\xi = \inf\{\|(x_n + y_n)\|_\infty : (y_n) \in \mathcal{N}\}$. Then*

$$\|\tilde{x}\|_\xi = \lim_\xi \|x_{n_\xi}\|.$$

Proof. Let $\tilde{x} = [(x_n)] \in \tilde{X}$. Then $\tilde{x} = \{(x_n + y_n) : (y_n) \in \mathcal{N}\}$. Therefore, if $(y_n) \in \mathcal{N}$,

$$\lim_{\xi} \|x_{n_{\xi}}\| = \lim_{\xi} \left| \|x_{n_{\xi}}\| - \|y_{n_{\xi}}\| \right| \leq \lim_{\xi} \|x_{n_{\xi}} + y_{n_{\xi}}\| \leq \|(x_n + y_n)\|_{\infty}.$$

This proves that $\lim_{\xi} \|x_{n_{\xi}}\| \leq \|\tilde{x}\|_{\xi}$.

For the reverse implication, let $\varepsilon > 0$ and let

$$I = \{n \in \mathbb{N} : \|x_n\| \leq \lim_{\xi} \|x_{n_{\xi}}\| + \varepsilon\}.$$

Clearly $\{n_{\xi}\}$ is eventually in I . Define $(y_n) \in \ell_{\infty}(X)$ by setting

$$y_n = \begin{cases} -x_n & \text{if } n \notin I, \\ 0 & \text{otherwise.} \end{cases}$$

Then $\lim_{\xi} y_{n_{\xi}} = 0$ and it follows that

$$\|\tilde{x}\|_{\xi} \leq \|(x_n + y_n)\|_{\infty} = \sup_n \|(x_n + y_n)\| = \sup_{n \in I} \|x_n\| \leq \lim_{\xi} \|x_{n_{\xi}}\| + \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary, this completes the proof. ■

Proposition 9.5 *If X is a Banach space, then $\tilde{X}$ is a Banach space.*

Proof. There are two ways to see this. One way is to show that $\mathcal{N}$ is a closed subspace of $\ell_{\infty}(X)$ and invoke the fact that the norm on $\tilde{X}$ is the quotient norm on the quotient space $\ell_{\infty}(X)/\mathcal{N}$. (It is well known that the quotient space of a Banach space over one of its closed subspaces is itself a Banach space.) However, we proceed with a direct proof.

Let $\{\tilde{x}_k\}_{k=1}^{\infty}$ be a Cauchy sequence in $\tilde{X}$. Since a Cauchy sequence converges if it has a convergent subsequence, it suffices to show that such is the case for $\{\tilde{x}_k\}$.

We pass to a subsequence and assume at the outset that

$$\|\tilde{x}_k - \tilde{x}_{k-1}\|_{\xi} < \frac{1}{2^k}, \quad k = 1, 2, \dots$$

Now set

$$\tilde{u}_1 = \tilde{x}_1, \quad \tilde{u}_2 = \tilde{x}_2 - \tilde{x}_1, \dots, \tilde{u}_k = \tilde{x}_k - \tilde{x}_{k-1}.$$

Then $\tilde{x}_k = \sum_{j=1}^k \tilde{u}_j$ and $\|\tilde{u}_k\|_{\xi} < \frac{1}{2^k}$ for all $k = 1, 2, \dots$.

For each k we can think of $\tilde{u}_k$ as an equivalence class of the form $[(u_n^{(k)})]$. Moreover, since $\|\tilde{u}_k\|_{\xi} = \lim_{\xi} \left\| (u_{n_{\xi}}^{(k)}) \right\| < \frac{1}{2^k}$ we can select $(v_n^{(k)}) \in [(u_n^{(k)})]$ by setting, for each n , $v_n^{(k)} = u_n^{(k)}$ if $\left\| (u_n^{(k)}) \right\| < \frac{1}{2^k}$, and $v_n^{(k)} = 0$ otherwise. (Note

that one then has $\lim_{\xi} \|u_{n\xi}^{(k)} - v_{n\xi}^{(k)}\| = 0$.) The sequence thus obtained satisfies $\|v_n^{(k)}\| < \frac{1}{2^k}$ for all $n \geq 1$. It now follows that for each n the sequence $\left\{ \sum_{j=1}^k v_n^{(j)} \right\}_{k=1}^{\infty}$ is a Cauchy sequence in X and hence it has a limit $u_n \in X$. (Let $w_k = \sum_{j=1}^k v_n^{(j)}$, $k = 1, 2, \dots$. Then $\|w_{k+1} - w_k\| = \|v_n^{(k+1)}\| < \frac{1}{2^{k+1}}$; hence $\sum_{k=1}^{\infty} \|w_{k+1} - w_k\| < \infty$.) Let $\tilde{u} = [(u_n)]$. We complete the proof by showing that

$$\lim_{k \rightarrow \infty} \|\tilde{x}_k - \tilde{u}\|_{\xi} = 0.$$

Let $k < m$. Then

$$\left\| \sum_{j=1}^k v_n^{(j)} - \sum_{j=1}^m v_n^{(j)} \right\| = \left\| \sum_{j=k+1}^m v_n^{(j)} \right\| \leq \sum_{j=k+1}^m \|v_n^{(j)}\| < \frac{1}{2^k}.$$

Letting $m \rightarrow \infty$ gives

$$\left\| \sum_{j=1}^k v_n^{(j)} - u_n \right\| \leq \frac{1}{2^k}.$$

But

$$\sum_{j=1}^k (v_n^{(j)}) \in \sum_{j=1}^k [(u_n^{(j)})] = \sum_{j=1}^k \tilde{u}_j = \tilde{x}_k.$$

Thus

$$\lim_{\xi} \left\| \sum_{j=1}^k v_{n\xi}^{(j)} - u_{n\xi} \right\| = \|\tilde{x}_k - \tilde{u}\|_{\xi} \leq \frac{1}{2^k}.$$

Therefore, $\{\tilde{x}_k\}$ converges to $\tilde{u}$. ■

9.4 Some properties of $\tilde{X}$

We continue with the notation of the previous section, with X a given Banach space.

Proposition 9.6 *$\tilde{X}$ contains a subspace isometrically isomorphic to X .*

Proof. For each $x \in X$ let (x_n) denote the sequence for which $x_n \equiv x$, and let $\dot{x} = [(x_n)] \in \tilde{X}$. Then the subspace

$$\dot{X} = \{\dot{x} : x \in X\}$$

is isometric to X via the mapping $x \rightarrow \dot{x}$ since $\|\dot{x}\|_\xi = \lim_\xi \|x_{n_\xi}\| = \lim_\xi \|x\| = \|x\|$. ■

The following is less obvious. However, it ties the observations of this section to those of the first section.

Theorem 9.5 $\tilde{X}$ is finitely representable in X .

Proof. Let $\tilde{X}_0$ be any finite-dimensional subspace of $\tilde{X}$, say of dimension m , and let $\{\tilde{x}^1, \dots, \tilde{x}^m\} \subseteq \tilde{X}_0$ be an algebraic basis of $\tilde{X}_0$ for which $\|\tilde{x}^k\|_\xi = 1$, $k = 1, \dots, m$. For each $k = 1, \dots, m$, let $\tilde{x}^k = [(x_n^k)]$ and for each $n \in \mathbb{N}$ let X_n be the subspace of X spanned by $\{x_n^1, \dots, x_n^m\}$. Now define $T_n : \tilde{X}_0 \rightarrow X_n$ by setting $T_n(\tilde{x}^j) = x_n^j$, $j = 1, \dots, m$ and extending T_n linearly to all of $\tilde{X}_0$. Thus for

$$\tilde{x} = \sum_{j=1}^m a_j \tilde{x}^j \in \tilde{X}_0$$

set

$$T_n(\tilde{x}) = \sum_{j=1}^m (T_n a_j \tilde{x}^j) = \sum_{j=1}^m a_j T_n(\tilde{x}^j) = \sum_{j=1}^m a_j x_n^j \in X_n.$$

What we propose to show is that given $\varepsilon > 0$ there exists an $n \in \mathbb{N}$ such that T_n is an isomorphism of $\tilde{X}_0$ onto X_n for which

$$(1 - \varepsilon) \|\tilde{x}\|_\xi \leq \|T_n(\tilde{x})\| \leq (1 + \varepsilon) \|\tilde{x}\|_\xi \quad (\tilde{x} \in \tilde{X}_0).$$

To this end, let $\varepsilon > 0$ and let $\tilde{x} = \sum_{j=1}^m a_j \tilde{x}^j \in \tilde{X}_0$ be arbitrary. Since

$$\|T_n(\tilde{x})\| = \left\| \sum_{j=1}^m a_j x_n^j \right\|,$$

we have

$$\lim_\xi \|T_{n_\xi}(\tilde{x})\| = \lim_\xi \left\| \sum_{j=1}^m a_j x_{n_\xi}^j \right\| = \|\tilde{x}\|_\xi.$$

In particular, for each $\tilde{x} \in \tilde{X}_0$ there exists an index $\xi_{\tilde{x}}$ (in some directed set) such that if $\xi \geq \xi_{\tilde{x}}$,

$$\left(1 - \frac{\varepsilon}{2}\right) \|\tilde{x}\|_\xi \leq \|T_{n_\xi}(\tilde{x})\| \leq \left(1 + \frac{\varepsilon}{2}\right) \|\tilde{x}\|_\xi.$$

Moreover, since T_{n_ξ} is (uniformly) continuous (Lemma 9.1), each $\tilde{x}$ in the unit sphere $\tilde{S}_0$ of $\tilde{X}_0$ has a neighborhood $V_{\tilde{x}}$ with the property that $\tilde{y} \in V_{\tilde{x}}$ and $\xi \geq \xi_{\tilde{x}}$ imply

$$(1 - \varepsilon) \|\tilde{y}\|_\xi \leq \|T_{n_\xi}(\tilde{y})\| \leq (1 + \varepsilon) \|\tilde{y}\|_\xi.$$

Since $\tilde{S}_0$ is compact there exists a finite set $\{\tilde{z}_1, \dots, \tilde{z}_p\} \subseteq \tilde{S}_0$ such that

$$\tilde{S}_0 \subseteq \bigcup_{i=1}^p V_{\tilde{z}_i}.$$

Now, if $n = n_\xi$ where ξ is chosen so that $\xi \geq \xi_{\tilde{z}_i}$ for all $i = 1, \dots, p$, then for each $\tilde{y} \in \tilde{S}_0$,

$$(1 - \varepsilon) \|\tilde{y}\|_\xi \leq \|T_n(\tilde{y})\| \leq (1 + \varepsilon) \|\tilde{y}\|_\xi.$$

■

The preceding fact is important because it assures that $\tilde{X}$ possesses all the super-properties that X possesses. The following is a special case.

Corollary 9.5 *If X is uniformly convex then $\tilde{X}$ is uniformly convex.*

In fact, one can say more.

Theorem 9.6 *For any Banach space X , $\delta_{\tilde{X}} = \delta_X$; hence $\varepsilon_0(X) = \varepsilon_0(\tilde{X})$.*

Proof. $\delta_{\tilde{X}}(\varepsilon) \geq \delta_X(\varepsilon)$ by Theorem 9.1. The reverse inequality follows from the fact that the spaces X and $\tilde{X} \subseteq \tilde{X}$ are isometric. ■

We also have the following.

Proposition 9.7 *If X is super-reflexive, then $\tilde{X}$ is also super-reflexive.*

Proof. $\tilde{X}$ is reflexive because $\tilde{X}$ is finitely representable in X . If Y is finitely representable in $\tilde{X}$, then since finite representability is transitive and $\tilde{X}$ is finitely representable in X it follows that Y is finitely representable in X . Since X is super-reflexive it must be the case that Y is reflexive. Hence $\tilde{X}$ is also super-reflexive. ■

In fact, the above proof suggests a more general result. Indeed, if X is super-P, then any Banach space Y which is finitely representable in X is also super-P. In particular, $\tilde{X}$ is also super-P.

Another important fact about super-reflexive spaces is the following result of J. Stern [155]. We will use this fact later.

Theorem 9.7 *If X is super-reflexive, then $(\tilde{X})^* = \widetilde{X^*}$.*

What this means precisely is the following: Each functional $f \in (\tilde{X})^*$ is the form $\tilde{f} = [(f_n)] \in \tilde{X}^*$, and $\tilde{f}(\tilde{x}) = \lim_{\xi} f_{n_{\xi}}(x_{n_{\xi}})$ for each $\tilde{x} \in \tilde{X}$.

So far we have been talking about a very special type of ultrapower $\tilde{X}$ because this is all that is needed for the specific results we have in mind. However, in order to consider the class of all Banach spaces which are finitely representable in a given Banach space, a more general definition is needed; namely, one in which the ultrapower is built by using arbitrary ultranets rather than just those which are subnets of ω . If $(\tilde{X})^* = \tilde{X}^*$ holds for *any* such Banach space ultrapower of X , then it can be shown that X is necessarily super-reflexive. So with this extended definition the converse of Theorem 9.7 is also true. In fact, it can be shown that any space Y which is finitely representable in X is a subspace of some ultrapower of X . Another fact is the following (cf. [9, pp. 228–232]).

Theorem 9.8 *If a Banach space ultrapower $\tilde{X}$ of X is reflexive, then X is super-reflexive.*

9.5 Extending mappings to $\tilde{X}$

Now let $K \subseteq X$ be bounded closed and convex, and suppose $T : K \rightarrow K$. Letting

$$\tilde{K} = \{\tilde{x} = [(x_n)] \in \tilde{X} : x_n \in K \text{ for each } n\},$$

there is a canonical way to extend T to a mapping $\tilde{T} : \tilde{K} \rightarrow \tilde{K}$ by setting for $\tilde{x} = [(x_n)] \in \tilde{K}$,

$$\tilde{T}(\tilde{x}) = [(T(x_n))].$$

It is immediate that $\tilde{K}$ is a bounded closed and convex subset of $\tilde{X}$. Now assume that T is nonexpansive. Then for $\tilde{x}, \tilde{y} \in \tilde{K}$,

$$\|\tilde{T}(\tilde{x}) - \tilde{T}(\tilde{y})\|_{\xi} = \lim_{\xi} \|T(x_{n_{\xi}}) - T(y_{n_{\xi}})\| \leq \lim_{\xi} \|x_{n_{\xi}} - y_{n_{\xi}}\| = \|\tilde{x} - \tilde{y}\|_{\xi},$$

so $\tilde{T}$ is nonexpansive as well. Moreover, since K is bounded and T is nonexpansive there always exist sequences $(x_n) \subset K$ such that

$$\lim_{n \rightarrow \infty} \|x_n - T(x_n)\| = 0.$$

From this it follows that $\tilde{T}(\tilde{x}) = \tilde{x}$, where $\tilde{x} = [(x_n)]$, that is, $\text{Fix}(\tilde{T}) \neq \emptyset$.

Not only does the mapping $\tilde{T}$ always have fixed points, but in fact $\text{Fix}(\tilde{T})$ also has a nice structure.

Theorem 9.9 $\text{Fix}(\tilde{T})$ is metrically convex.

Proof. All we need to do is show that for any $\tilde{x}, \tilde{y} \in \text{Fix}(\tilde{T})$ there exists $\tilde{z} \in \text{Fix}(\tilde{T})$ such that

$$\|\tilde{x} - \tilde{z}\|_{\xi} = \|\tilde{y} - \tilde{z}\|_{\xi} = \frac{1}{2} \|\tilde{x} - \tilde{y}\|_{\xi}.$$

Suppose $\tilde{x} = [(x_n)]$, $\tilde{y} = [(y_n)]$ and fix $n \in \mathbb{N}$. Set

$$\begin{cases} \delta_n &= \max\{\|x_n - T(x_n)\|, \|y_n - T(y_n)\|\}; \\ d_n &= \frac{1}{2} \|x_n - y_n\|. \end{cases}$$

Now choose $\{\varepsilon_n\}$ with $\varepsilon_n \in (0, 1)$ so that

$$\lim_{n \rightarrow \infty} \varepsilon_n = \lim_{n \rightarrow \infty} \frac{\delta_n}{\varepsilon_n} = 0,$$

and set

$$\eta_n = \frac{1 - \varepsilon_n}{\varepsilon_n} \delta_n.$$

For each n let

$$K_n = B(x_n; d_n + \eta_n) \cap B(y_n; d_n + \eta_n) \cap K.$$

Then obviously K_n is closed and convex, and since $\eta_n > 0$, K_n is also nonempty (since, in particular, $\frac{1}{2}(x_n + y_n) \in K_n$).

Next define for each n a mapping $T_n : K_n \rightarrow K$ by setting for each $z \in K_n$,

$$T_n(z) = (1 - \varepsilon_n)T(z) + \varepsilon_n \left(\frac{1}{2}(x_n + y_n) \right).$$

It is easy to see that T_n is a strict contraction with Lipschitz constant $1 - \varepsilon_n$. Less obvious is the fact that T_n maps K_n into K_n . To see this write

$$x_n - T_n(z) = (1 - \varepsilon_n)(x_n - T(z)) + \varepsilon_n \left(\frac{1}{2}(x_n - y_n) \right).$$

Thus (using in the last step the fact that $(1 - \varepsilon_n)\delta_n = \varepsilon_n\eta_n$)

$$\begin{aligned} \|x_n - T_n(z)\| &\leq (1 - \varepsilon_n) \left(\|x_n - T(x_n)\| + \|T(x_n) - T(z)\| \right) + \frac{\varepsilon_n}{2} \|x_n - y_n\| \\ &\leq (1 - \varepsilon_n)(\delta_n + d_n + \eta_n) + \varepsilon_n d_n \\ &= d_n + (1 - \varepsilon_n)\delta_n + (1 - \varepsilon_n)\eta_n \\ &= d_n + \eta_n. \end{aligned}$$

Replacing x_n with y_n in the above gives $\|y_n - T_n(z)\| \leq d_n + \eta_n$ as well. Therefore, $T_n : K_n \rightarrow K_n$ and so there exists a unique $z_n \in K_n$ such that $T_n(z_n) = z_n$. In particular,

$$\|x_n - z_n\| \leq d_n + \eta_n \text{ and } \|y_n - z_n\| \leq d_n + \eta_n.$$

Let $\tilde{z} = [(z_n)]$. Since $\lim_{\xi} d_{n_{\xi}} = \frac{1}{2} \|\tilde{x} - \tilde{y}\|_{\xi}$ and $\lim_{\xi} \eta_{n_{\xi}} = 0$, the above gives

$$\begin{aligned} \|\tilde{x} - \tilde{z}\|_{\xi} &= \lim_{\xi} \|x_{n_{\xi}} - z_{n_{\xi}}\| = \frac{1}{2} \|\tilde{x} - \tilde{y}\|_{\xi}; \\ \|\tilde{y} - \tilde{z}\|_{\xi} &= \lim_{\xi} \|y_{n_{\xi}} - z_{n_{\xi}}\| = \frac{1}{2} \|\tilde{x} - \tilde{y}\|_{\xi}. \end{aligned}$$

We now only need to show that $\tilde{z} \in \text{Fix}(\tilde{T})$. However, this is a direct consequence of the fact that $\lim_{n \rightarrow \infty} \varepsilon_n = 0$ and the fact that

$$z_n = (1 - \varepsilon_n)T(z_n) + \varepsilon_n \left(\frac{1}{2}(x_n + y_n) \right),$$

from which

$$\|z_n - T(z_n)\| = \varepsilon_n \left\| \left(\frac{1}{2}(x_n + y_n) \right) - T(z_n) \right\| \leq \varepsilon_n \text{diam}(K).$$

■

9.6 Some fixed point theorems

It was Maurey's surprising discovery [121] that all bounded closed convex subsets of reflexive subspaces of L^1 have the fixed point property for nonexpansive mappings that set the stage to some amazing fixed point theorems. These results are deep and nonconstructive in the sense that they involve ultrapowers of Banach spaces. One of the ingredients behind most of these new results is a property satisfied by the ultraproduct of the minimal convex set.

Indeed, let X be a Banach space and C be a nonempty weakly compact convex subset of X . Assume that there exists a nonexpansive mapping $T : C \rightarrow C$ whose fixed point set $\text{Fix}(T)$ is empty. Since C is weakly compact, there exists K a closed convex subset of C , which is T -invariant and minimal. In Chapter 8, we have seen some of the properties of this minimal set. Define $\tilde{K}$ as usual to be

$$\tilde{K} = \left\{ \tilde{x} = [(x_n)] \in \tilde{X} : x_n \in K \text{ for each } n \right\}.$$

We have the following result:

Lemma 9.3 *Let $\{\tilde{w}_n\}$ be an approximate fixed point sequence (a.f.p.s.) for $\tilde{T}$ in $\tilde{K}$, that is, $\|\tilde{T}(\tilde{w}_n) - \tilde{w}_n\|_\xi \rightarrow 0$ as n goes to ∞ . Then for any $x \in K$, we have*

$$\lim_{n \rightarrow \infty} \|\tilde{w}_n - x\|_\xi = \text{diam}(K).$$

Proof. Let $\{\tilde{w}_n\}$ be an a.f.p.s. for $\tilde{T}$ in $\tilde{K}$ and $x \in K$. We will show that $\text{diam}(K)$ is the only possible limit for $\{\|\tilde{w}_n - x\|_\xi\}$. Without loss of generality, we can assume that $\lim_{n \rightarrow \infty} \|\tilde{w}_n - x\|_\xi = d$. Write $\delta_n = \|\tilde{T}(\tilde{w}_n) - \tilde{w}_n\|_\xi$ and $\dot{w}_n = [(w_n(m))]$ with $w_n(m) \in K$. Let $\{\varepsilon_n\}$, with $\varepsilon > 0$, such that $\lim_{n \rightarrow \infty} \varepsilon_n = 0$. Set

$$A_k = \left\{ m \in \mathbb{N}; \|w_n(m) - x\| \leq d + 2\varepsilon_k \text{ and } \|w_n(m) - T(w_n(m))\| \leq \varepsilon_k + \delta_n \right\}.$$

From the properties of the ultranet ξ , we can construct increasing sequences of integers $\{n_k\}$ and $\{m_k\}$ such that

$$\|w_{n_k}(m_k) - x\| \leq d + 2\varepsilon_k \text{ and } \|w_{n_k}(m_k) - T(w_{n_k}(m_k))\| \leq \varepsilon_k + \delta_{n_k}$$

hold for any $k \in \mathbb{N}$. The sequence $\{w_{n_k}(m_k)\}$ is an a.f.p.s. for T . Using the Goebel-Karlovitz Lemma (Lemma 8.4) we get

$$\text{diam}(K) = \lim_{k \rightarrow \infty} \|w_{n_k}(m_k) - x\| \leq d.$$

By definition of $\tilde{K}$, we obviously have $d \leq \text{diam}(K)$ which completes the proof of Lemma 9.3. $\blacksquare$

A direct consequence of this lemma is the following result:

Corollary 9.6 *Let $\tilde{W}$ be any nonempty closed convex subset of $\tilde{K}$ which is invariant under $\tilde{T}$, that is, $\tilde{T}(\tilde{W}) \subset \tilde{W}$. Then*

$$\sup \left\{ \|\tilde{w} - x\|_\xi; \tilde{w} \in \tilde{W} \right\} = \text{diam}(K)$$

holds for any $x \in K$.

The proof is left as an exercise.

The first of these new results was discovered by Lin [113] in Banach spaces with unconditional bases.

Theorem 9.10 *Let X be a Banach space with an unconditional basis $\{e_n\}$. Assume that the constant of unconditionality is 1. Then for any weakly compact convex subset C of X and any nonexpansive mapping $T : C \rightarrow C$, the fixed point set $\text{Fix}(T)$ is not empty.*

Proof. Assume that there exists a nonexpansive mapping $T : C \rightarrow C$ such that $\text{Fix}(T) = \emptyset$. Since C is weakly compact, there exists a closed nonempty convex subset K of C which is minimal invariant under T . Without loss of generality, we may assume that $\text{diam}(K) = 1$. By approximation, there exists an approximate fixed point sequence $\{x_n\}$. We may assume that $\{x_n\}$ weakly converges to $0 \in K$. In view of Goebel-Karlovitz Lemma we have

$$\lim_{n \rightarrow \infty} \|x_n\| = 1$$

and by passing to a subsequence we may further assume

$$\lim_{n \rightarrow \infty} \|x_n - x_{n+1}\| = 1$$

or, even stronger,

$$\lim_{n \rightarrow \infty} \text{dist}(x_n, \{x_{n+1}, x_{n+2}, \dots\}) = 1.$$

Let P_n denote the natural projection of X onto the subspace spanned by $\{e_1, \dots, e_n\}$. Since $\{x_n\}$ weakly converges to 0, then we must have

$$\lim_{n \rightarrow \infty} \|P_k(x_n)\| = 0$$

for any $k \geq 1$. Also, for each $n \geq 1$,

$$\lim_{k \rightarrow \infty} \|(I - P_k)(x_n)\| = 0.$$

By passing to a subsequence, we may assume that there exists a sequence $\{F_n\}$ of finite subsets of $\mathbb{N}$ such that $\max F_i < \min F_{i+1}$ and satisfy

$$(i) \quad \lim_{n \rightarrow \infty} \|P_{F_n}(x_n)\| = \lim_{n \rightarrow \infty} \|x_n\| = 1;$$

$$(ii) \quad \lim_{n \rightarrow \infty} \|(I - P_{F_n})(x_n)\| = 0;$$

and obviously we have $P_{F_n} \circ P_{F_m} = 0$, for $n \neq m$.

Now we turn our attention to the ultrapower of X . First we translate all of the above into $\tilde{X}$. Indeed, consider the set $\tilde{K}$, the mapping $\tilde{T} : \tilde{K} \rightarrow \tilde{K}$. Set $\tilde{x} = [(x_n)]$ and $\tilde{y} = [(x_{n+1})]$. From the above properties, we have

$$\tilde{T}(\tilde{x}) = \tilde{x}, \quad \tilde{T}(\tilde{y}) = \tilde{y}, \quad \text{and} \quad \|\tilde{x} - \tilde{y}\|_{\xi} = 1.$$

Define the two projections (in $\tilde{X}$) by

$$\tilde{P} = [(P_{F_n})] \quad \text{and} \quad \tilde{Q} = [(P_{F_{n+1}})].$$

We have $\tilde{P} \circ \tilde{Q} = 0$, and

$$\tilde{P}(\tilde{x}) = \tilde{x}, \quad \tilde{Q}(\tilde{y}) = \tilde{y}, \quad \text{and} \quad \tilde{Q}(\tilde{x}) = \tilde{P}(\tilde{y}) = 0.$$

Finally, for any $x \in K \subset \tilde{K}$, we have

$$\tilde{P}(x) = \tilde{Q}(x) = 0.$$

At this stage, we will make use of the constant of unconditionality of $\{e_n\}$ being equal to 1. Indeed, we have

$$\|P_{F_n}(x_n) - P_{F_{n+1}}(x_{n+1})\| = \|P_{F_n}(x_n) + P_{F_{n+1}}(x_{n+1})\|$$

for any $n \geq 1$. This clearly implies that

$$\|\tilde{P}(\tilde{x}) - \tilde{Q}(\tilde{y})\|_\xi = \|\tilde{P}(\tilde{x}) + \tilde{Q}(\tilde{y})\|_\xi.$$

So we have

$$\|\tilde{x} - \tilde{y}\|_\xi = \|\tilde{P}(\tilde{x}) - \tilde{Q}(\tilde{y})\|_\xi = 1$$

and

$$\|\tilde{x} + \tilde{y}\|_\xi = \|\tilde{P}(\tilde{x}) + \tilde{Q}(\tilde{y})\|_\xi = \|\tilde{P}(\tilde{x}) - \tilde{Q}(\tilde{y})\|_\xi = 1.$$

Thus, finally,

$$\|\tilde{x} + \tilde{y}\|_\xi = \|\tilde{x} - \tilde{y}\|_\xi = 1.$$

Next we will show that the above fact leads to a contradiction. Indeed, consider the set

$$\tilde{W} = \left\{ \tilde{w} \in \tilde{K}; \max\{\|\tilde{w} - \tilde{x}\|_\xi, \|\tilde{w} - \tilde{x}\|_\xi, \|\tilde{w} - \tilde{y}\|_\xi\} \leq \frac{1}{2} \text{ for some } x \in K \right\}.$$

Clearly, $\frac{\tilde{x} + \tilde{y}}{2} \in \tilde{W}$ (where $x = 0$). Also since $\tilde{x}$ and $\tilde{y}$ are fixed point for $\tilde{T}$, we have

$$\begin{cases} \|\tilde{T}(\tilde{w}) - \tilde{T}(\tilde{x})\|_\xi \leq \|\tilde{w} - \tilde{x}\|_\xi \\ \|\tilde{T}(\tilde{w}) - \tilde{x}\|_\xi = \|\tilde{T}(\tilde{w}) - \tilde{T}(\tilde{x})\|_\xi \leq \|\tilde{w} - \tilde{x}\|_\xi \\ \|\tilde{T}(\tilde{w}) - \tilde{y}\|_\xi = \|\tilde{T}(\tilde{w}) - \tilde{T}(\tilde{y})\|_\xi \leq \|\tilde{w} - \tilde{y}\|_\xi \end{cases}$$

showing that $\tilde{W}$ is invariant under $\tilde{T}$. Let $\tilde{w} \in \tilde{W}$. Then there exists $x \in K$ such that

$$\max\{\|\tilde{w} - \tilde{x}\|_\xi, \|\tilde{w} - \tilde{x}\|_\xi, \|\tilde{w} - \tilde{y}\|_\xi\} \leq \frac{1}{2}.$$

On the other hand, we have

$$2\tilde{w} = (\tilde{P} + \tilde{Q})(\tilde{w}) + (I - \tilde{P})(\tilde{w}) + (I - \tilde{Q})(\tilde{w}),$$

which implies

$$\|\tilde{w}\|_\xi \leq \frac{1}{2} \left(\|(\tilde{P} + \tilde{Q})(\tilde{w})\|_\xi + \|(I - \tilde{P})(\tilde{w})\|_\xi + \|(I - \tilde{Q})(\tilde{w})\|_\xi \right).$$

But

$$(\tilde{P} + \tilde{Q})(\tilde{x}) = (I - \tilde{P})(\tilde{x}) = (I - \tilde{Q})(\tilde{y}) = 0,$$

so

$$\|\tilde{w}\|_\xi \leq \frac{1}{2} \left(\|(\tilde{P} + \tilde{Q})(\tilde{w} - \tilde{x})\|_\xi + \|(I - \tilde{P})(\tilde{w} - \tilde{x})\|_\xi + \|(I - \tilde{Q})(\tilde{w} - \tilde{y})\|_\xi \right).$$

Since the constant of unconditionality of $\{e_n\}$ is 1, then we must have

$$\|\tilde{P} + \tilde{Q}\| \leq 1, \|I - \tilde{P}\| \leq 1, \|I - \tilde{Q}\| \leq 1.$$

Thus all of the above yield

$$\|\tilde{w}\|_\xi \leq \frac{1}{2} \left(\|\tilde{w} - \tilde{x}\|_\xi + \|\tilde{w} - \tilde{x}\|_\xi + \|\tilde{w} - \tilde{y}\|_\xi \right) \leq \frac{3}{4}.$$

This contradicts the conclusion of Corollary 9.6, which gives

$$\sup \left\{ \|\tilde{w}\|_\xi, \tilde{w} \in \tilde{W} \right\} = \text{diam}(K) = 1.$$

The proof of Theorem 9.10 is therefore complete. ■

Example 9.1 Consider the Hilbert space l_2 with its usual norm. For any $\beta \geq 1$ and $x \in l_2$, set

$$\|x\|_\beta = \max \left(\|x\|_{l_2}; \beta \|x\|_{l_\infty} \right).$$

The functional $\|\cdot\|_\beta$ is a norm which is equivalent to $\|\cdot\|_{l_2}$. Indeed, we have

$$\|\cdot\|_{l_2} \leq \|\cdot\|_\beta \leq \beta \|\cdot\|_{l_2}.$$

Write $X_\beta = (l_2, \|\cdot\|_\beta)$. These spaces have been studied extensively. First it is easy to see that $X_{\sqrt{2}}$ fails to have normal structure. This answers the question whether any reflexive Banach space has normal structure. In fact, we can show that X_β has normal structure property if and only if $\beta < \sqrt{2}$. It was then natural to ask whether X_β has the fixed point property. This question was open until Lin proved Theorem 9.10. Indeed, the constant of unconditionality of the canonical basis of l_2 for $\|\cdot\|_\beta$ is 1.

Remark 9.1 In fact, Lin has improved the conclusion of Theorem 9.10. Indeed, if X is a Banach space with an unconditional basis $\{e_n\}$ with λ as its constant of unconditionality, then the conclusion of Theorem 9.10 holds, provided

$$\lambda < \frac{\sqrt{33} - 3}{2}.$$

It is still unknown whether the conclusion of Theorem 9.10 holds for any value of λ .

9.7 Asymptotically nonexpansive mappings

Asymptotic fixed point theorems are those theorems from which fixed point properties are derived from the behavior of large iterates of the mapping. For the nonexpansive mappings this assumption, in its most direct case, takes the following form. Let X be a Banach space, $D \subseteq X$, and $T : D \rightarrow D$. If there exists a sequence $\{k_i\} \subset \mathbb{R}^+$ with $k_i \rightarrow 1$ as $i \rightarrow \infty$ for which

$$\|T^i(x) - T^i(y)\| \leq k_i \|x - y\| \text{ for all } x, y \in D,$$

then T is said to be *asymptotically nonexpansive*.

If T is asymptotically nonexpansive then $\tilde{T}$ will be asymptotically nonexpansive as well, but $\tilde{T}$ may not be nonexpansive. However, the mapping $\hat{T}$ defined as follows is *nonexpansive*. For $\tilde{x} = [(x_n)] \in \tilde{K}$ set

$$\hat{T}(\tilde{x}) = [\{T(x_1), T^2(x_2), T^3(x_3), \dots\}].$$

Then

$$\|\hat{T}(\tilde{x}) - \hat{T}(\tilde{y})\|_\xi = \lim_\xi \|T^{n_\xi}(x_{n_\xi}) - T^{n_\xi}(y_{n_\xi})\| \leq \lim_\xi k_{n_\xi} \|x_{n_\xi} - y_{n_\xi}\| = \|\tilde{x} - \tilde{y}\|_\xi.$$

We show how this observation can be used to give a very quick proof of the original fixed point theorem of Goebel-Kirk [68] about the existence of fixed points for asymptotically nonexpansive mappings.

Theorem 9.11 *Suppose K is a bounded closed convex subset of a uniformly convex Banach space X and suppose $T : K \rightarrow K$ is asymptotically nonexpansive. Then T has a fixed point.*

Proof. We adopt the notation $(\tilde{T}$ and $\tilde{K})$ as above. Since the modulus of convexity is a super-property, $\tilde{X}$ is uniformly convex. We first show that $\tilde{T}$ has a fixed point in $\tilde{K}$. Consider the mapping $\hat{T} : \tilde{K} \rightarrow \tilde{K}$ defined by

$$\hat{T}([(x_n)]) = [(T^n(x_n))], \quad ([x_n]) \in \tilde{K}.$$

Since $\hat{T}$ is nonexpansive and $\tilde{X}$ is uniformly convex, $\hat{T}$ has a nonempty fixed point set F in $\tilde{K}$, and further, F itself is closed and convex. Now consider the mapping $\check{H} : \tilde{K} \rightarrow \tilde{K}$, where $\check{H} = \tilde{T} \circ \hat{T}$. Then if $\tilde{x} = [(x_n)]$, $\tilde{y} = [(y_n)] \in \tilde{K}$,

$$\begin{aligned} \|\check{H}(\tilde{x}) - \check{H}(\tilde{y})\| &= \lim_\xi \|T^{n_\xi+1}(x_{n_\xi}) - T^{n_\xi+1}(y_{n_\xi})\| \\ &\leq \lim_\xi (k_{n_\xi+1}) \|x_{n_\xi} - y_{n_\xi}\| \\ &= \|\tilde{x} - \tilde{y}\|. \end{aligned}$$

Thus $\check{H}$ is also nonexpansive, and further $\check{H} \circ \hat{T} = \hat{T} \circ \check{H}$. It follows that $\check{H} : F \rightarrow F$ and therefore $\check{H}$ has a fixed point, say $\tilde{x} \in F$. We now have $\hat{T} \circ \hat{T}(\tilde{x}) = \tilde{x} = \hat{T}(\tilde{x})$, from which $\tilde{T}(\tilde{x}) = \tilde{x}$.

Now let $\dot{K}$ be the closed convex subset of $\tilde{K}$ defined as follows:

$$\dot{K} = \{\dot{x} = [(x_n)] : x_n \equiv x \in K\}.$$

As we have seen, $\tilde{T}$ has a fixed point $\tilde{x} \in \tilde{K}$. Note that if $\tilde{x} = \dot{x} \in \dot{K}$ then $\|\dot{x} - \tilde{T}(\dot{x})\|_\xi = \lim_\xi \|T(x) - x\| = 0$, and it follows that $T(x) = x$. On the other hand, if $\tilde{x} \notin \dot{K}$, then there is a unique point $\dot{z} \in \dot{K}$ such that $\|\tilde{x} - \dot{z}\|_\xi = \text{dist}(\tilde{x}, \dot{K})$. Since $\tilde{T}$ is asymptotically nonexpansive

$$\lim_{n \rightarrow \infty} \|\tilde{T}^n \dot{z} - \tilde{x}\|_\xi = \lim_{n \rightarrow \infty} \|\tilde{T}^n \dot{z} - \tilde{T}^n \tilde{x}\|_\xi \leq \lim_{n \rightarrow \infty} k_n \|\tilde{T}^n \dot{z} - \tilde{T}^n \tilde{x}\|_\xi = \|\dot{z} - \tilde{x}\|_\xi,$$

and since $\tilde{X}$ is uniformly convex this implies $\lim_{n \rightarrow \infty} \|(\tilde{T})^n(\dot{z}) - \dot{z}\|_\xi = 0$. Therefore, $\tilde{T}(\dot{z}) = \dot{z}$. Hence $T(z) = z$ and in either case we are finished. ■

9.8 The demiclosedness principle

A fundamental result in the theory of nonexpansive mappings is Browder's demiclosedness principle [24].

Definition 9.5 *Let X be a Banach space and $K \subseteq X$. A mapping $f : K \rightarrow X$ is demiclosed (at y) if the conditions $\{x_j\}$ converges weakly to x and $\{f(x_j)\}$ converges strongly to y then $x \in K$ and $f(x) = y$.*

Browder's demiclosedness principle asserts that if K is a closed convex subset of a uniformly convex Banach space X and if $T : K \rightarrow X$ is nonexpansive, then $I - T$ is demiclosed.

The following fact is proved in [24] using sharp properties of the modulus of convexity and a clever subsequential thinning process. The ultrapower approach given here is direct and simple (although it does require the deep fact mentioned earlier that if X is a super-reflexive space then $(\tilde{X})^* = \widetilde{X^*}$).

Theorem 9.12 *Suppose K is a bounded closed convex subset of a uniformly convex Banach space X and suppose $T : K \rightarrow K$ is asymptotically nonexpansive. Then $I - T$ is demiclosed at 0.*

Proof. We continue with the notation used previously and we use the fact that $\tilde{X}$ is uniformly convex.

Let $\dot{K}$ be the closed convex subset of $\tilde{K}$ defined as follows:

$$\dot{K} = \{\dot{x} = [(x_n)] : x_n \equiv x \in K\}.$$

Let $\{x_n\}$ be a sequence in K satisfying

$$\text{weak-} \lim_{n \rightarrow \infty} x_n = x \text{ and } \lim_{n \rightarrow \infty} \|x_n - T(x_n)\| = 0.$$

By assumption $[(x_n)] \in \text{Fix}(\tilde{T})$. Since $\tilde{T}$ is asymptotically nonexpansive, it is easy to see that $\text{Fix}(\tilde{T})$ is closed and convex. Let $\varepsilon > 0$ and let $\tilde{f} \in \tilde{X}^*$. Since X is super-reflexive there exists $(f_n) \subset X^*$ such that for each $\tilde{u} = [(u_n)] \in \tilde{X}$, $\tilde{f}(\tilde{u}) = \lim_{\xi} f_{n_{\xi}}(u_{n_{\xi}})$. Also, since $\{x_n\}$ converges weakly to x , for each $n \in \mathbb{N}$ there exists an integer k_n such that

$$|f_n(x_{k_n}) - f_n(x)| \leq \varepsilon$$

and clearly we may assume $\{k_n\}$ is increasing. Thus

$$\tilde{x} = [(x_{k_n})] \in \text{Fix}(\tilde{T}).$$

We now have

$$|\tilde{f}(\tilde{x}) - \tilde{f}(\tilde{x})| = \lim_{\xi} |f_{n_{\xi}}(x_{k_{n_{\xi}}}) - f_{n_{\xi}}(x)| \leq \varepsilon.$$

Thus $\inf \left\{ |\tilde{f}(\tilde{u}) - \tilde{f}(\tilde{x})| : \tilde{u} \in \text{Fix}(\tilde{T}) \right\} = 0$. Since $\text{Fix}(\tilde{T})$ is weakly compact, this implies $\tilde{f}(\tilde{u}) = \tilde{f}(\tilde{x})$ for some $\tilde{u} \in \text{Fix}(\tilde{T})$. By Theorem 6.5 (Separation Theorem) this, in turn, implies $\dot{x} \in \text{Fix}(\tilde{T})$; hence $T(x) = x$. ■

9.9 Uniformly non-creasy spaces

A Banach space X is said to have the FPP if every bounded closed convex subset of X has the fixed point property for nonexpansive self-mappings. The results of the previous two sections suggest that there are implications for asymptotically nonexpansive mappings in the class of spaces X for which FPP is a super-property, in particular, for which $\tilde{X}$ has the FPP. Of course, the uniformly convex spaces belong to this class, as do Banach spaces X for which $\varepsilon_0(X) < 1$, but in general, the class of spaces X for which $\tilde{X}$ has the FPP is not yet well understood. In particular, it is not even known whether uniform normal structure is a super-property. Recently S. Prus [135] has introduced a class of spaces, which he calls uniformly non-creasy, which also have the super-fixed point property. We take up this fact here.

There are two stages to the approach. First it is shown that uniformly non-creasy is a super-property. Then, it is shown (via an ultrapower approach) that uniformly non-creasy spaces have the FPP.

We need some more notation. We shall use B_X and S_X to denote, respectively, the unit ball and unit sphere of a Banach space X . For any $f \in S_{X^*}$ and $\delta \in [0, 1]$ we define the *slice* $S(f, \delta)$ to be the set

$$S(f, \delta) = \{x \in B_X : f(x) \geq 1 - \delta\}.$$

Certain geometric properties of Banach spaces are easily thought of in terms of slices; indeed, uniform convexity can be characterized in such a way.

Proposition 9.8 *A Banach space X is uniformly convex if and only if*

(a): *for every $\varepsilon > 0$ there exists $\delta > 0$ such that $\text{diam}(S(f, \delta)) \leq \varepsilon$ for each $f \in S_{X^*}$.*

Proof. Assume X is uniformly convex, let $\varepsilon > 0$ and let $\delta = \delta(\varepsilon)$. Suppose $x, y \in S(f, \delta)$ for some $f \in S_{X^*}$. Then

$$1 - \delta \leq f\left(\frac{1}{2}(x+y)\right) \leq \frac{1}{2}\|x+y\|\|f\| = \frac{1}{2}\|x+y\|.$$

Since X is uniformly convex it must be the case that $\|x-y\| \leq \varepsilon$, that is,

$$\text{diam}(S(f, \delta)) \leq \varepsilon.$$

Now assume (a) holds and assume X is not uniformly convex. Then $\delta(4\varepsilon) = 0$ for some $\varepsilon > 0$. Let $\delta > 0$ be the number in condition (a) which corresponds to ε . Since $\delta(4\varepsilon) = 0$ there exist $x, y \in B_X$ such that $\|x-y\| \geq 4\varepsilon$ but for which $\frac{1}{2}\|x+y\| \geq 1 - \delta$. Choose $f \in S_{X^*}$ so that

$$f\left(\frac{1}{2}(x+y)\right) = \frac{1}{2}\|x+y\|.$$

Then $\frac{1}{2}(x+y) \in S(f, \delta)$, and it also must be the case that either $x \in S(f, \delta)$ or $y \in S(f, \delta)$, from which $\text{diam}(S(f, \delta)) \geq 2\varepsilon > \varepsilon$. ■

Banach spaces X for which X^* is uniformly convex can be characterized by the intersection of slices. For this we need another definition.

Definition 9.6 *A Banach space X is said to be uniformly smooth if*

$$f_x(h) = \lim_{t \rightarrow 0} \frac{\|x+th\| - \|x\|}{t}$$

exists uniformly for each $(x, h) \in S_X \times S_X$. (It is easy to check that this limit defines a functional $f_x \in X^$.)*

A Banach space property (P) is said to be *self-dual* if X has property (P) if and only if X^* has property (P) . It is well known that a Banach space X is reflexive if and only if X^* is reflexive; that is, reflexivity is self dual. On the other hand, there exist uniformly convex spaces X for which X^* is not uniformly convex. However, uniform convexity does entail another type of duality. It is known that a Banach space X is uniformly convex if and only if X^* is uniformly smooth (and conversely, X is uniformly smooth if and only if X^* is uniformly convex).

Now consider two functionals $f, g \in S_{X^*}$ and a scalar $\delta \in [0, 1]$, and set

$$S(f, g, \delta) = S(f, \delta) \cap S(g, \delta).$$

Proposition 9.9 *A Banach space X is uniformly smooth if and only if*

- (b): *for every $\varepsilon > 0$ there exists $\delta > 0$ such that if $f, g \in S_{X^*}$ and $\|f - g\| \geq \varepsilon$, then $S(f, g, \delta) = \emptyset$.*

In view of the comments immediately preceding its statement, the proposition will follow from a conjunction of the following two facts:

(i) If X satisfies (a), then X^* satisfies (b).

(ii) If X^* satisfies (b), then X satisfies (a).

Proof. (i) Suppose (a) holds for X and (b) does not hold for X^* . Then (since (a) implies $X^{**} = X$) there exists $\varepsilon > 0$ such that for any $\delta > 0$ there exist $x, y \in B_X$ for which $\|x - y\| \geq 2\varepsilon$ and $S(x, y, \delta) \neq \emptyset$. Since $f \in S(x, \delta) \cap S(y, \delta)$ implies $x, y \in S(f, \delta)$ and hence $\text{diam } S(f, \delta) \geq 2\varepsilon > \varepsilon$, this clearly contradicts (a) for δ sufficiently small.

(ii) Now suppose (b) holds for X^* and (a) does not hold for X . Then there exists $\varepsilon > 0$ such that for any $\delta > 0$, $\text{diam } S(f, \delta) \geq 2\varepsilon$ for some $f \in S_{X^*}$. Choose $x, y \in S(f, \delta)$ such that $\|x - y\| > \varepsilon$. Then $f \in S(x, y, \delta)$. This contradicts (b) for $\delta > 0$ sufficiently small. ■

A Banach space is said to have a *crease* if there exist two distinct functionals $f, g \in S_{X^*}$ such that $\text{diam } S(f, g, 0) > 0$. Since $f(x) \leq 1$ for each $f \in S_{X^*}$ and $x \in B_X$, given $f \in S_{X^*}$ the *hyperplane* $\{x \in X : f(x) = 1\}$ supports the unit ball B_X . Thus to say that X has a crease means that the sphere S_X of X contains a segment of positive length that lies on two different hyperplanes which support the unit ball B_X .

A space X is called *noncreasy* if its unit sphere does not have a crease. Any Banach space X with $\dim(X) \leq 2$ is trivially noncreasy, as is any strictly convex space.

Definition 9.7 *A Banach space is said to be uniformly noncreasy (UNC) if given any $\varepsilon > 0$ there is a $\delta > 0$ such that if $f, g \in S_{X^*}$ and $\|f - g\| \geq \varepsilon$, then*

$$\text{diam} \left(S(f, g, \delta) \right) \leq \varepsilon.$$

Also, Propositions 9.8 and 9.9 show that UNC is implied by both uniform convexity and uniform smoothness. As we shall soon see, UNC is a self-dual property. However, we first prove the following.

Theorem 9.13 *If a Banach space X is UNC, then X is super-reflexive.*

Proof. Assume X is not super-reflexive. Then by James's characterization of Theorem 9.4 for any number $\delta \in (0, 1)$ there exist points $x_i \in S_X$ and functionals $f_i \in S_{X^*}$ for $i = 1, 2, 3$ for which $f_i(x_j) = 1 - \delta$ if $j \geq i$ and $f_i(x_j) = 0$ if $i > j$.

This implies $\|f_1 - f_2\| \geq (f_1 - f_2)(x_1) = 1 - \delta$. Moreover, $x_2, x_3 \in S(f_1, f_2, \delta)$ so $\text{diam } S(f_1, f_2, \delta) \geq \|x_3 - x_2\| \geq f_3(x_3 - x_2) = 1 - \delta$. This proves that X is not UNC. ■

In contrast to uniform convexity and uniform smoothness, UNC is a self-dual property.

Theorem 9.14 *A Banach space X is UNC if and only if X^* is UNC.*

Proof. It suffices to show that if X is UNC then X^* is as well. The converse will then follow from the fact that UNC implies reflexivity. So, suppose X is UNC and X^* is not. Then (making the identification $X^{**} = X$) there exists a constant $\varepsilon > 0$ such that for each $\delta \in (0, \frac{\varepsilon}{4})$ there exist elements $x_1, x_2 \in S_X$ with $\|x_1 - x_2\| \geq \varepsilon$ and functionals $f_1, f_2 \in S(x_1, x_2, \delta)$ for which $\|f_1 - f_2\| > \varepsilon$. In particular, $\|f_1\| \geq f_1(x_1) \geq 1 - \delta$ and $\|f_2\| \geq f_2(x_2) \geq 1 - \delta$.

Now set $f = f_1 / \|f_1\|$ and $g = f_2 / \|f_2\|$. Then

$$\begin{aligned} \|f - g\| &= \left\| f \|f_1\| + f(1 - \|f_1\|) - g \|f_2\| - g(1 - \|f_2\|) \right\| \\ &\geq \left\| f \|f_1\| - g \|f_2\| \right\| - \|f(1 - \|f_1\|)\| - \|g(1 - \|f_2\|)\|. \end{aligned}$$

Thus

$$\|f - g\| \geq \|f_1 - f_2\| - (1 - \|f_1\|) - (1 - \|f_2\|) > \varepsilon - 2\delta > \frac{\varepsilon}{2}.$$

Moreover, $x_1, x_2 \in S(f, g, \delta)$. This contradicts the assumption that the space X is UNC. ■

Finally, we observe that since the ultrapower $\tilde{X}$ of a Banach space inherits all finite-dimensional geometric properties of X we have the following.

Theorem 9.15 *A Banach space X is UNC if and only if $\tilde{X}$ is UNC.*

Proof. In view of Theorem 9.13 we may assume X is reflexive. Now suppose every five-dimensional subspace of X has UNC (for the same δ corresponding to a given ε), and suppose X fails the UNC assumption. Then there exists $\varepsilon > 0$ such that for any δ (take $\delta = \delta(\varepsilon)$) there exist $f, g \in S_{X^*}$ with $\|f - g\| \geq \varepsilon$ and for which

$$\text{diam} \left(S(f, g, \delta) \right) > \varepsilon.$$

Now pick $x_1, x_2, x_3 \in X$ such that

$$\|f\| = f(x_1), \quad \|g\| = g(x_2), \quad \|f - g\| = (f - g)(x_3).$$

Also pick $x_4, x_5 \in S(f, g, \delta)$ such that $\|x_4 - x_5\| > \varepsilon$. Then the five-dimensional space spanned by $\{x_1, \dots, x_5\}$ fails the UNC assumption. This shows that the property of UNC has five-dimensional character, so any space which is finitely representable in X must also be UNC. ■

This fact can now be used to show that X (hence $\tilde{X}$) has the fixed point property for nonexpansive mappings.

Theorem 9.16 *Let K be a bounded closed convex subset of a Banach space X which is UNC, and let $T: K \rightarrow K$ be a nonexpansive mapping. Then T has a fixed point.*

Proof. Assume T does not have a fixed point. Since X is reflexive, we may use Zorn's Lemma and assume that K is minimal with respect to being nonempty bounded closed convex and T -invariant. Also, for convenience, we may assume that $0 \in K$ and that $\text{diam}(K) = 1$. Now let $\{x_n\} \subseteq K$ satisfy $\lim_{n \rightarrow \infty} \|x_n - T(x_n)\| = 0$. By passing to a subsequence and translating we may also assume that $\{x_n\}$ converges weakly to 0. Now for each $n \in \mathbb{N}$ select a functional $f_n \in S_{X^*}$ such that $f_n(x_n) = \|x_n\|$. Since X is reflexive, then X^* is also reflexive. Hence $\{f_n\}$ has a weak limit point, say f , in B_{X^*} . We now need some detailed calculations.

Since $\lim_{n \rightarrow \infty} f(x_n) = 0$ it is possible to choose n_1 so that $|f(x_n)| \leq \frac{1}{4}$ for all $n \geq n_1$. Again use the fact that $\{x_n\}$ converges weakly to 0 to choose $n_2 \in \mathbb{N}$, $n_2 > n_1$, so that $|f_{n_1}(x_n)| \leq \frac{1}{2}$ and $|f(x_n)| \leq \frac{1}{8}$ for all $n \geq n_2$, and also so that $|(f_{n_2} - f)(x_{n_i})| \leq \frac{1}{4}$ for $i = 1, 2$. Furthermore, in view of the Goebel-Karlovitz Lemma, n_2 can also be chosen so that $\|x_{n_1} - x_{n_2}\| \geq 1 - \frac{1}{2}$. Note that we now have

$$|f_{n_1}(x_{n_2})| \leq \frac{1}{2}$$

and

$$|f_{n_2}(x_{n_1})| \leq |f(x_{n_1})| + |(f_{n_2} - f)(x_{n_1})| \leq \frac{1}{4} + \frac{1}{4} = \frac{1}{2}.$$

We proceed one step further. Choose $n_3 \in \mathbb{N}$, $n_3 > n_2$, so that $|f_{n_2}(x_n)| \leq \frac{1}{4}$ and $|f(x_n)| \leq \frac{1}{16}$ for all $n \geq n_3$, and also so that $|(f_{n_3} - f)(x_{n_i})| \leq \frac{1}{8}$ for $i = 1, 2, 3$. Furthermore, in view of the Goebel-Karlovitz Lemma, n_2 can also be chosen so that $\|x_{n_2} - x_{n_3}\| \geq 1 - \frac{1}{4}$. Note that we now have

$$|f_{n_2}(x_{n_3})| \leq \frac{1}{4}$$

and

$$|f_{n_3}(x_{n_2})| \leq |f(x_{n_2})| + |(f_{n_3} - f)(x_{n_2})| \leq \frac{1}{8} + \frac{1}{8} = \frac{1}{4}.$$

Proceeding by induction it is possible to choose $n_1 < n_2 < n_3, \dots$ so that for $i = 1, \dots, k-1$,

$$|f_{n_{k+1}}(x_{n_k})| \leq \frac{1}{2^k} \text{ and } |f_{n_k}(x_{n_{k+1}})| \leq \frac{1}{2^k}$$

and also so that $\|x_{n_k} - x_{n_{k+1}}\| \geq 1 - \frac{1}{2k}$.

Now set $\tilde{x} = [(x_{n_{2k-1}})]$, $\tilde{y} = [(x_{n_{2k}})] \in \tilde{X}$, and $\tilde{f} = [(f_{n_{2k-1}})]$, $\tilde{g} = [(f_{n_{2k}})] \in \tilde{X}^*$. Then $\tilde{x}, \tilde{y} \in S_{\tilde{X}}$, and

$$\|\tilde{f}\|_{\xi} = \tilde{f}(\tilde{x}) = \lim_{\xi} f_{n_{2k_{\xi}-1}}(x_{n_{2k_{\xi}-1}}) = \lim_{\xi} \|x_{n_{2k_{\xi}-1}}\| = 1.$$

Similar calculations show that $\|\tilde{g}\|_{\xi} = \tilde{g}(\tilde{y}) = 1$, $\tilde{f}(\tilde{y}) = \tilde{g}(\tilde{x}) = 0$, and $\|\tilde{x} - \tilde{y}\|_{\xi} = 1$.

Now, since $\text{Fix}(\tilde{T})$ is metrically convex (Theorem 9.9) there exists $\tilde{w} \in \tilde{K}$ such that $\|\tilde{w}\|_{\xi} = 1$ (another consequence of the Goebel-Karlovitz Lemma) and such that

$$\|\tilde{w} - \tilde{x}\|_{\xi} = \|\tilde{w} - \tilde{y}\|_{\xi} = \frac{1}{2}.$$

Also, $\tilde{f}(\tilde{w}) = \tilde{f}(\tilde{w} - \tilde{y}) \leq \|\tilde{w} - \tilde{y}\|_{\xi} = 1/2$. On the other hand, $1/2 = \|\tilde{x} - \tilde{w}\|_{\xi} \geq \tilde{f}(\tilde{x} - \tilde{w}) = 1 - \tilde{f}(\tilde{w})$ so $\tilde{f}(\tilde{w}) \geq 1/2$. Therefore, $\tilde{f}(\tilde{w}) = 1/2$. Similarly, $\tilde{g}(\tilde{w}) = 1/2$.

Thus

$$\tilde{f}(2(\tilde{x} - \tilde{w})) = 1 \text{ and } \tilde{g}(2(\tilde{x} - \tilde{w})) = -1.$$

This shows that

$$2(\tilde{x} - \tilde{w}) \in S(\tilde{f}, -\tilde{g}, 0)$$

and similarly

$$2(\tilde{w} - \tilde{y}) \in S(\tilde{f}, -\tilde{g}, 0).$$

Moreover, $\tilde{f} \neq -\tilde{g}$ since $\|\tilde{f} - (-\tilde{g})\|_{\xi} = \|\tilde{f} + \tilde{g}\|_{\xi} \geq (\tilde{f} + \tilde{g})(\tilde{x}) = 1$. So, since X is UNC it must be the case that $\tilde{x} - \tilde{w} = \tilde{w} - \tilde{y}$, that is,

$$\tilde{w} = \frac{1}{2}(\tilde{x} + \tilde{y}).$$

Now choose a functional $h_k \in S_{X^*}$ for each k such that

$$h_k(x_{n_{2k-1}} + x_{n_{2k}}) = \|x_{n_{2k-1}} + x_{n_{2k}}\|.$$

There is a subsequence $\{x_{m_k}\}$ of $\{x_n\}$ for which

$$\lim_{k \rightarrow \infty} h_k(x_{m_k}) = \lim_{k \rightarrow \infty} f_{n_{2k-1}}(x_{m_k}) = 0$$

and

$$\lim_{k \rightarrow \infty} \|x_{n_{2k-1}} - x_{m_k}\| = 1.$$

Put $\tilde{z} = (x_{m_k}) \in \tilde{X}$ and $\tilde{h} = (h_k) \in \tilde{X}^*$. Then clearly $\|\tilde{h}\|_\xi = 1$; moreover,

$$\tilde{h}(\tilde{x} + \tilde{y}) = \lim_{\xi} h_{k_\xi}(x_{n_{2k_\xi-1}} + x_{n_{2k_\xi}}) = \lim_{\xi} \|x_{n_{2k_\xi-1}} + x_{n_{2k_\xi}}\| = \|\tilde{x} + \tilde{y}\|_\xi.$$

Also, $\tilde{h}(\tilde{z}) = \tilde{f}(\tilde{z}) = 0$ and $\|\tilde{x} - \tilde{z}\|_\xi = 1$.

Therefore,

$$1 \geq \tilde{h}(\tilde{x}) = 2\tilde{h}(\tilde{w}) - \tilde{h}(\tilde{y}) \geq 2 - \|\tilde{h}\|_\xi \|\tilde{y}\|_\xi = 1.$$

Similarly, $\tilde{h}(\tilde{y}) = 1$. Thus

$$\tilde{x}, \tilde{x} - \tilde{z} \in S(\tilde{f}, \tilde{g}, 0),$$

where $\|\tilde{h} - \tilde{f}\|_\xi \geq |(\tilde{h} - \tilde{f})(\tilde{y})| = \tilde{h}(\tilde{y}) = 1$. Therefore,

$$\|\tilde{z}\|_\xi = \|\tilde{x} - (\tilde{x} - \tilde{z})\|_\xi \leq \text{diam}(S(\tilde{f}, \tilde{g}, 0)) = 0,$$

from which $\tilde{z} = 0$, which, of course, is a contradiction since $\|\tilde{z}\|_\xi = 1$ by the Goebel-Karlovitz Lemma. $\blacksquare$

Exercises

Exercise 9.1 Show that X is a finite-dimensional Banach space if and only if $\tilde{X}$ is a finite-dimensional Banach space. In this case, show that $\dim(X) = \dim(\tilde{X})$.

Exercise 9.2 Show that if X is not a finite-dimensional Banach space, then $\tilde{X}$ is not separable.

Exercise 9.3 Show that $\tilde{X}$ is strictly convex if and only if $\tilde{X}$ is uniformly convex.

Exercise 9.4 Let $\{x_n(t)\}_{n \geq 1}$ be a Schauder basis of $C([0, 1])$, the space of continuous functions on $[0, 1]$. Set X_n to be the space spanned by $x_1, \dots, x_n$. Let

$$X = \left(\prod_{n=1}^{\infty} X_n \right)_2.$$

(i) Show that $C([0, 1])$ is finitely representable in X .

(ii) Show that X is separable and reflexive.

(iii) Show that any separable Banach space is finitely representable in X .

Exercise 9.5 Let X, Y, X_1, Y_1 be Banach spaces, and $T : X \rightarrow Y, T_1 : X_1 \rightarrow Y_1$ operators. We say that T_1 is finitely representable in T if, for every $\varepsilon > 0$, every finite-dimensional subspace X_1^o of X_1 , one can find a finite-dimensional subspace X^o of X , and (setting $Y_1^o = T(X_1^o), Y^o = T(X^o)$), two isomorphism $U : X_1^o \rightarrow X^o$ and $V : Y_1^o \rightarrow Y^o$ such that $T \circ U = V \circ T_1$ and

$$\begin{cases} \|U\| \cdot \|U^{-1}\| & \leq 1 + \varepsilon; \\ \|V\| \cdot \|V^{-1}\| & \leq 1 + \varepsilon. \end{cases}$$

Show that this definition is transitive. Show that $\tilde{T} : \tilde{X} \rightarrow \tilde{Y}$ is finitely representable in T .

Exercise 9.6 (Ultrapower of Banach lattices) Let X be a Banach lattice. Consider the ultrapower $\tilde{X}$ of X . For $\tilde{x} \in \tilde{X}$, we write $\tilde{x} \geq 0$, if there exists a sequence $\{x_n\}$ of positive elements in X such that $\tilde{x} = [(x_n)]$. Show that $\tilde{X}$ is a Banach lattice with

$$[(x_n)] \vee [(y_n)] = [(x_n \vee y_n)] \quad \text{and} \quad [(x_n)] \wedge [(y_n)] = [(x_n \wedge y_n)].$$

Show also that $\left| [(x_n)] \right| = [|x_n|]$.

Exercise 9.7 Let X be a Banach space with a Schauder basis $\{e_n\}$. Set

- $\mu = \sup\{\|u - v\|; u \text{ and } v \text{ are disjoint blocks with } \|u + v\| \leq 1\};$
- $c_1 = \sup\{\|I - P_n\|; P_n \text{ is the natural projection on the interval } [1, n]\};$
- $c_2 = \sup\{\|I - P_F\|; F \text{ is any interval in } \mathbb{N}\}.$

The constant c is the basis constant of $\{e_n\}$. Show that if

$$c_1\mu + c + c_2 < 4,$$

then X has the fixed point property.

Hint: Use the same ideas developed in the proof of Theorem 9.10.

For example, for the James space J , show that $\mu = \sqrt{2}$, $c_2 = \sqrt{2}$, and $c = c_1 = 1$. In particular, we have $c_1\mu + c + c_2 = \sqrt{2} + \sqrt{2} + 1 < 4$.

Exercise 9.8 Show that if $\tilde{X}$ has normal structure, then $\tilde{X}$ has uniform normal structure. Deduce that if X has super-normal structure, then X is super-reflexive.

Appendix

Set Theory

A.1 Mappings

Familiarity with the basic rudiments of elementary set notation, and concepts such as set inclusion, intersection, union, and so on, is assumed throughout this discussion. We pause here to look a little more closely at some of the underlying set-theoretic concepts that arise in the text. Obviously it is important in mathematics to be precise about what is meant by a mapping (or function) from one set to another.

Definition A.1 *If A and B are sets, then the Cartesian product of A and B is the set $A \times B$ of all ordered pairs (a, b) with $a \in A$ and $b \in B$.*

Of course, the above definition raises the question of what is meant by an 'ordered pair'. Commonly this means that one should distinguish between the 'left' element a from the 'right' element b . To make this idea mathematically rigorous, logicians define $(a, b) = \{a, \{a, b\}\}$, where the bracket notation $\{u, v\}$ denotes an unordered pair. Thus one can identify a with the element a , and b with the element $\{a, b\}$ of the unordered pair $\{a, \{a, b\}\}$. This validates the accuracy of the intuitive approach.

Definition A.2 *A mapping (or function) f from A to B , denoted $f : A \rightarrow B$, is a subset of $A \times B$ which satisfies:*

- (i) *If $a \in A$ then there exists $b \in B$ such that $(a, b) \in f$; and*
- (ii) *$(a, b) \in f$ and $(a, c) \in f \Rightarrow b = c$.*

If $(a, b) \in f$ one denotes $b = f(a)$.

If only (i) is assumed then f is called a relation (with domain A).

Thus in common language, corresponding to each $a \in A$ there is exactly one element $b \in B$ such that $b = f(a)$.

A.2 Order relations and Zermelo's Theorem

Let $\mathcal{R}$ be a *relation* on a set S . By this we mean that $\mathcal{R}$ is a subset of $S \times S$. Next, say $a \leq b$ for $a, b \in S$ if the pair $(a, b) \in \mathcal{R}$. The mere introduction of this notation prompts one to refer to $\mathcal{R} = (S, \leq)$ as an order relation. There are several types of order relations that arise in analysis; we list the most common.

Definition A.3 $(S, \leq)$ is said to be a *partially ordered set (POS)* if for each $a, b, c \in S$:

- (i) $a \leq a$;
- (ii) $a \leq b$ and $b \leq a \Rightarrow a = b$;
- (iii) $a \leq b$ and $b \leq c \Rightarrow a \leq c$.

Definition A.4 $(S, \leq)$ is said to be a *directed set (DS)* if $(S, \leq)$ is a (POS) and if, in addition:

- (iv) $a, b \in S \Rightarrow$ there exists $c \in S$ such that $a \leq c$ and $b \leq c$.

Definition A.5 $(S, \leq)$ is said to be a *linearly ordered set (LOS)* if $(S, \leq)$ is a (DS) and if, in addition:

- (v) $a, b \in S \Rightarrow a \leq b$ or $b \leq a$.

Definition A.6 $(S, \leq)$ is said to be a *well-ordered set (WOS)* if $(S, \leq)$ is a (LOS) and if, in addition:

- (vi) every nonempty subset of S has a smallest element.

Thus it is immediate from the definitions that for a given ordered set $(S, \leq)$ the following implications hold:

$$(WOS) \Rightarrow (LOS) \Rightarrow (DS) \Rightarrow (POS).$$

The set $\mathbb{N}$ of natural numbers (with usual ordering) is a well-ordered set. The set $\mathbb{R}$ of real numbers (again with usual ordering) is a linearly ordered set which is not well ordered.

The set $\mathcal{P}(S)$ (or 2^S) of subsets of a given set is a directed set when ordered by set inclusion ($A \leq B \Leftrightarrow A \subseteq B$) but it is not a linearly ordered set (if S contains at least two elements). Finally, the nonempty subsets of $\mathcal{P}(S)$ is a partially ordered set when ordered by reverse set inclusion ($A \leq B \Leftrightarrow B \subseteq A$) which, in general, is not a linearly ordered set.

Each of these concepts play fundamental roles in analysis.

Linearly ordered sets are often called *chains*, especially when referring to families of sets ordered by set inclusion. It is possible for a partially ordered set to contain a linearly ordered subset. When this is the case, such a subset is also called a chain. The concepts of 'bounded above', 'least upper bound', and

so on, which were defined initially for the real numbers, carry over verbatim to the more abstract setting of either a linearly or partially ordered set.

We are now in a position to state yet another fixed point theorem. The roots of this theorem go back to a 1908 result of Zermelo [168], and proofs are readily at hand to those who have access to, for example, [52], p. 5; [166], p. 504. (In the latter citation the result is attributed to both Kneser and Bourbaki. The idea is only implicit in [168].)

A quick proof of this theorem based on Zorn's Lemma is given in the next section. At the end of this Appendix we add a 'constructive' proof.

Theorem A.1 (Zermelo) *Suppose $(S, \leq)$ is a partially ordered set in which each chain has a least upper bound, and suppose $f : S \rightarrow S$ is a mapping which satisfies $x \leq f(x)$ for each $x \in S$. Then f has a fixed point.*

A.3 Zorn's Lemma and the Axiom Of Choice

The abstract discussion of the previous section would have been impossible without language involving "sets". The concept is so basic that it requires only a brief explanation. A *set* is a collection of objects, usually called elements or points, along with the primitive concept of *belonging*; that is, a set is determined by the collection of objects "belonging" to it. If an object x belongs to a set A we write $x \in A$. Given any object imaginable and any set A , either the object belongs to A or it does not belong to A —the framework of mathematical logic allows for no ambiguity; indeed the core of mathematics as an intellectual discipline rests upon this premise.

In Section A.1 the cartesian product $A \times B$ of two sets A and B was defined as the collection of all ordered pairs (a, b) with $a \in A$ and $b \in B$. It is clear that $A \times B$ is itself a set since we have described the objects belonging to it (especially since we have defined 'ordered pair' as well). Suppose one has a finite number of sets, say $A_1, \dots, A_n$. Surely one would want to define the cartesian product $A_1 \times A_2 \times \dots \times A_n$ to be the collection $(a_1, a_2, \dots, a_n)$ of all ordered n -tuples where $a_i \in A_i$, $i = 1, 2, \dots, n$. Mathematical induction can make this definition precise. Having defined $A_1 \times A_2$, suppose $A_1 \times \dots \times A_{n-1}$ has been defined, and define

$$A_1 \times A_2 \times \dots \times A_n = (A_1 \times \dots \times A_{n-1}) \times A_n.$$

Beginning with an infinite number of sets, say $A_1, A_2, \dots$, then surely it would make sense to define the cartesian product $A_1 \times A_2 \times \dots$ to be the collection of all sequences $(a_1, a_2, \dots, a_n, \dots)$ where $a_i \in A_i$, $i = 1, 2, \dots$. Something subtle arises here, for in order to make such a definition it is necessary to *simultaneously* select an infinite number of elements $a_i \in A_i$, $i = 1, 2, \dots$. While the basic axioms of classical Zermelo-Fraenkel set theory do not allow for such a selection, most mathematicians accept the validity of such a procedure

by admitting the so-called *Countable Axiom of Choice*. Since the concept of 'sequence' seems so fundamental to mathematics this is usually not viewed as a serious problem. Very little mathematics as we know it could be accomplished without an assumption of this type.

The general Axiom of Choice is a different matter entirely. Suppose $\mathcal{F} = \{A_\alpha\}_{\alpha \in I}$ is a collection of sets with no restriction placed on the index set I . Thus I may be very large, especially uncountable (see Section A.6 below). The *Axiom of Choice* states that there is a function $f : \mathcal{F} \rightarrow \bigcup_{\alpha \in I} A_\alpha$ such that $f(A_\alpha) \in A_\alpha$ for each $\alpha \in I$. Such a function f is called a *choice function*. Upon knowing that such a function exists, it is now possible to define the cartesian product $\prod_{\alpha \in I} A_\alpha$ to be the collection of all such choice functions. The Countable Axiom of Choice now becomes simply the assertion that there exists a choice function for each countable family of sets.

Mathematicians study both the assumptions upon which the subject is based and the objects about which those assumptions are made. As Moore observes in [125], rarely in history has there been more controversy about one of the central premises of mathematics than the one which arose when Zermelo proposed the Axiom of Choice in 1908 [169]. The Axiom of Choice has had profound implications in the recent development of mathematics and without it mathematics would be much different than we know it today. Indeed, if the most severe 'constructivists' were to prevail, mathematics would be reduced to the study of a collection of algorithms. Consequently, most mathematicians accept the Axiom of Choice without question. Indeed, it is such a natural assumption that its use often goes unnoticed. But it is deceptive in its simplicity. Try, for example, to *describe* a choice function for the family of all nonempty subsets of the real numbers. (It is easy for the family of all closed subsets—take an element of each set which is nearest 0.)

On the one hand, the Axiom of Choice is different from other generally accepted assumptions in mathematics because it can be used to prove theorems which run deeply counter to intuition. For example, it can be used to show that in a euclidean space of dimension $n \geq 3$ any two arbitrary sets with nonempty interior can be decomposed into the same (finite) number of mutually disjoint subsets that are, respectively, isometric to one another. This is the astonishing Banach-Tarski paradox. (For an excellent discussion, see [84]; also see [125] for a discussion of the impact of the Axiom of Choice in mathematics and the controversy surrounding it.) At the same time, the Axiom of Choice validates many intuitively attractive propositions. Because of the efficiency it brings to many proofs we have used the Axiom of Choice in this text without reservation (although not necessarily without comment), despite the fact that many of the results we discuss can be shown to be true using weaker axioms.

One of the most useful consequences of the Axiom of Choice (in fact, it is equivalent to Axiom of Choice) is the celebrated Zorn's Lemma [170].

Lemma A.1 (Zorn) *Let $(S, \leq)$ be a partially ordered set in which every chain has an upper bound. Then S has a maximal element relative to $\leq$.*

It is important to be precise. A *maximal element* of S is an element $m \in S$ which has the property that if $x \in S$ and if $m \leq x$, then $x = m$. In particular, nothing in S is *larger* than m ; it is not asserted that every element of S is $\leq m$.

We return to Zermelo's Theorem of the previous section to illustrate the power of the Axiom of Choice. Despite the fact that it can be proved without recourse to the Axiom of Choice, we show here that it is a trivial consequence of Zorn's Lemma.

Theorem A.2 (Zermelo) *Suppose $(S, \leq)$ is partially ordered set in which each chain has a least upper bound, and suppose $f : S \rightarrow S$ is a mapping which satisfies $x \leq f(x)$ for each $x \in S$. Then f has a fixed point.*

Proof. By Zorn's Lemma S has a maximal element $x_0 \in S$. Since $x_0 \leq f(x_0)$ it must be the case that $f(x_0) = x_0$. ■

Given that it is such an immediate consequence of Zorn's Lemma, it might seem remarkable that the above theorem can be proved *without* recourse to the Axiom of Choice (see the proof at the end of this Appendix).

A.4 Nets and subnets

There usually comes a time in the study of analysis when it becomes essential to confront the concept of convergence of nets (or filters). The concept of sequential convergence is all that is needed to describe continuity and various topological notions in metric spaces. It is convenient, however, to use nets to describe topological concepts in those non-metric settings in which sequential convergence is not adequate. We do this in the discussion of Tychonoff's Theorem below, and we did so in Chapter 6 in discussing weak topologies.

We begin with a special kind of partial order.

Definition A.7 *A directed set is a partially ordered set $(I, \leq)$ which has the property*

$$\alpha, \beta \in I \Rightarrow \alpha \leq \gamma \text{ and } \beta \leq \gamma \text{ for some } \gamma \in I.$$

Obviously, any linearly ordered set is a directed set. The collection of all subsets of a given set X ordered by set inclusion ($A \leq B \Leftrightarrow A \subseteq B$) is directed. Also, if $x \in X$ then the family of all subsets $A \subseteq X$ with the property $x \in A$ is directed when ordered either by set inclusion or reverse set inclusion ($A \leq B \Leftrightarrow A \supseteq B$). The family of all finite subset of a given set is also directed when ordered by set inclusion. There are many such examples.

A subset J of a directed set I is *cofinal* in I if for each $\alpha \in I$ there is a $\beta \in J$ such that $\alpha \leq \beta$. It is easy to see that under these circumstances J itself is directed.

We now come to the principal definition.

Definition A.8 *Let S be a set. A net in S is a mapping $\phi : I \rightarrow S$ where I is a directed set.*

Observe that every sequence is a net, a fact that gives rise to some standard notation. If $\alpha \in I$ we usually write $\phi(\alpha) = x_\alpha$, and we use $\{x_\alpha\}_{\alpha \in I}$, or simply $\{x_\alpha\}$ when there is no confusion, to denote the net ϕ .

Recall that a sequence in S is a mapping $\phi : \mathbb{N} \rightarrow S$, where $\mathbb{N}$ denotes the natural numbers. Therefore, any sequence is a net, and while sequences and nets share many common properties, there is a fundamental difference between subsequences and subnets. If $\phi : \mathbb{N} \rightarrow S$ is a sequence and if $f : \mathbb{N} \rightarrow \mathbb{N}$ satisfies $f(n_1) > f(n_2)$ whenever $n_1 > n_2$, then $\psi = \phi \circ f : \mathbb{N} \rightarrow S$ is called a subsequence of ϕ . In the analogous definition for nets the mapping f is far less restricted.

Definition A.9 *If $\phi : I \rightarrow S$ is a net in S and if $f : J \rightarrow I$ is a mapping of a (possibly different) directed set J into I which satisfies*

for each $\alpha \in I$ there exists $j_0 \in J$ such that $f(j) \geq \alpha$ for all $j \geq j_0$

then the net $\phi \circ f : J \rightarrow S$ is called a subnet of ϕ .

The essential difference, of course, is the fact that the domain of the subnet may differ from that of the original net. This is not the case for subsequences. Consequently, one *cannot* say that a subnet of a sequence is a subsequence (although a subsequence is a subnet).

There is one very surprising consequence of Zorn's Lemma which involves nets. Let S be a set and $H \subseteq S$. A net $\{x_\alpha\}_{\alpha \in I}$ in S is said to be *eventually in* H if there exists $\alpha_0 \in I$ such that $x_\alpha \in H$ for all $\alpha \geq \alpha_0$.

Definition A.10 *A net $\{x_\alpha\}$ in a set S is called an *ultranet* (or *universal net*) if, given any subset $G \subseteq S$, either $\{x_\alpha\}$ is eventually in G or $\{x_\alpha\}$ is eventually in $S \setminus G$.*

It can be shown that the following is a consequence of the Axiom of Choice via Zorn's Lemma. (Indeed it can be shown to be equivalent to the Axiom of Choice.) This observation is crucial to the development of Chapter 9.

Proposition A.1 *Any net $\{x_\alpha\}$ in a set S has a subnet which is an ultranet.*

A.5 Tychonoff's Theorem

Tychonoff's Theorem states quite simply that the cartesian product of an arbitrary family of compact topological spaces is itself compact in the so-called 'product topology', and it is well known that this result is equivalent to the Axiom of Choice. The concept of net convergence offers a convenient way of understanding Tychonoff's Theorem and in fact the proposition about ultranets of the previous section can be used to give a quick proof of Tychonoff's Theorem.

Since Tychonoff's Theorem is a topological fact we briefly review some concepts. Here τ denotes the collection of open sets which defines the topology on X .

Definition A.11 A net $\{x_\alpha\}$ in a topological space (X, τ) is said to converge to $x \in X$, written $\lim_\alpha x_\alpha = x$, if $x \in U \in \tau$ implies there exists α_0 such that $x_\alpha \in U$ whenever $\alpha \geq \alpha_0$. Thus the net $\{x_\alpha\}$ is eventually in U for any open set U containing x .

A topological space (X, τ) is said to be *Hausdorff* if given any two points $x, y \in X$ with $x \neq y$ there exist (open) sets $U, V \in \tau$ such that $x \in U$, $y \in V$ and $U \cap V = \emptyset$. All the topological spaces encountered in this text will be Hausdorff. The significance of this concept lies in the fact that in a Hausdorff space limits are unique: If $\lim_\alpha x_\alpha = x$ and $\lim_\alpha x_\alpha = y$ then $x = y$.

Proposition A.2 A subset K of a topological space X is closed if and only if for any net $\{x_\alpha\} \subseteq K$, $\lim_\alpha x_\alpha = x$ implies $x \in K$.

Definition A.12 A subset K of a topological space (X, τ) is compact if whenever K is contained in the union of any family $\mathcal{F}$ of sets in τ then the union of some finite subfamily of $\mathcal{F}$ also contains K .

This leads to the following result. This fact is true in metric spaces for sequences, but its validity in general topological space requires the use of nets.

Proposition A.3 A subset K of a topological space is compact if and only if every net $\{x_\alpha\}$ in K has a subnet which converges to a point in K .

Tychonoff's Theorem involves the product of a family of topological spaces. A convenient way to describe the product topology in such a setting is to first define its closed sets. Then one can say a set is *open* in the product topology if and only if its complement is closed.

Let $(X_i, \tau_i)_{i \in I}$ be a family of topological spaces (I is some index set) and consider the cartesian product $\prod_{i \in I} X_i$ where, as before,

$$\prod_{i \in I} X_i = \left\{ f : I \rightarrow \bigcup_{i \in I} X_i \text{ with } f(i) \in X_i \text{ for each } i \in I \right\}.$$

One now says that a net $\{g_\alpha\}$ in $\prod_{i \in I} X_i$ converges to $g \in \prod_{i \in I} X_i$ (in the product topology) if and only if the net $\{g_\alpha(i)\}$ converges to $g(i)$ for each $i \in I$. By letting $f(i) = x_i$ one can think of points $f \in \prod_{i \in I} X_i$ as I -tuples of the form $(x_i)_{i \in I}$. Thought of in this way, convergence in the product topology is the same a 'coordinatewise' convergence.

In order to define what is meant by a closed set in the product topology, we take our cue from Proposition A.2.

Definition A.13 A subset K of the product space $X = \prod_{i \in I} X_i$ is said to be closed (in the product topology) if and only if for any net $\{f_\alpha\} \subseteq K$, $\lim_\alpha f_\alpha = f \in X$ implies $f \in K$. A set is open in X if and only if its complement is closed.

We now come to the point of this section.

Theorem A.3 (Tychonoff) *If $(X_i)_{i \in I}$ is any family of compact topological spaces, then $X = \prod_{i \in I} X_i$ is compact in the product topology.*

In his original paper Tychonoff proved only that the product of any number of copies of the closed interval $[0, 1]$ is compact, but his proof lent itself to the above generalization. Hence the theorem carries his name.

It can be shown rather easily that Tychonoff's Theorem is consequence of the following three facts.

(1) A topological space X is compact if and only if every net in X has a convergent subnet.

(2) Every net in any set S has a subnet which is an ultranet.

(3) If X is compact and if $\{x_\alpha\}$ is an ultranet in X , then $\{x_\alpha\}$ converges to a point of X .

A.6 Cardinal numbers

At times it is necessary to make distinctions between certain classes of infinite sets. Given two infinite sets, A and B , there are precisely three mutually exclusive possibilities. Either there is a one-to-one mapping from A onto B , in which case we say that A and B have the same *cardinality* and write $|A| = |B|$, or no such mapping exists. If no such mapping exists, then either there is a one-to-one mapping from A onto a proper subset of B or there is a one-to-one mapping from B onto a proper subset of A . In the former instance we write $|A| < |B|$ and in the latter $|B| < |A|$. (**Caution:** Note that $|A| < |B|$ means that there is a one-to-one mapping from A onto a proper subset of B and at the same time there does not exist a one-to-one mapping from A onto B .) For any two sets A and B the following trichotomy holds: $|A| < |B|$ or $|B| < |A|$ or $|A| = |B|$. This is another of the many fundamental results in mathematics which is known to be equivalent to the Axiom of Choice. It is this fact that enables us in some sense to order infinite sets according to their 'size'. (The fact that for infinite sets A and B , $|A| < |B|$ and $|B| < |A| \Rightarrow |A| = |B|$ is known as the Schröder-Bernstein Theorem. Its proof does not require the Axiom of Choice.)

At this point the notion of cardinality may still seem vague, so we begin with something familiar. The cardinality of any finite set is simply the number of elements in the set. This suggests the use of the term "cardinal number" to refer to the cardinality of a set. We say that a set A is *countable* if it is either finite or if $|A| = |\mathbb{N}|$. By the trichotomy of cardinality, any infinite set S contains an infinite countable (countably infinite) subset; thus $|\mathbb{N}| \leq |S|$ for any infinite set S . It is traditional to denote $|\mathbb{N}| = \aleph_0$ where the symbol $\aleph$ (aleph) is the first letter of the Hebrew alphabet. Thus $\aleph_0$ is the smallest infinite cardinal number. As we observed in the previous section there does not exist a one-to-one mapping from the set $\mathbb{N}$ onto $\mathbb{R}$. Thus $|\mathbb{N}| < |\mathbb{R}|$. Often the letter c (which

stands for “continuum”) is used to denote the cardinality of $\mathbb{R}$; thus there are at least two infinite cardinal numbers. In fact, there are many. It can be easily shown (using induction) that there does not exist a one-to-one mapping from any finite set S onto the set $\mathcal{P}(S)$ of all subsets of S . Thus for any finite set S , $|S| < |\mathcal{P}(S)|$.

The above fact extends to infinite sets as well. There is a one-to-one correspondence between $\mathcal{P}(S)$ and the set 2^S of all functions from S into the two-element set $\{0, 1\}$, and it can be shown that there does not exist a function mapping S onto $\mathcal{P}(S)$. (This might be an interesting exercise for those who have not studied any set theory.) Therefore, in general, one has $|S| < |\mathcal{P}(S)|$ (or, equivalently, $|S| < |2^S|$). In terms of cardinality, there is no ‘largest’ set.

There is a lot more that can be said about cardinal numbers but this is about all that is needed for an understanding of anything in the present text. This is fortunate because at this point things get a bit tricky. For example, it is known that relative to the order relation defined above there is a smallest cardinal number, denoted $\aleph_1$, which is larger than $\aleph_0$, and Cantor conjectured that $c = \aleph_1$. To this day this conjecture, known as the *continuum hypothesis*, remains unproved. However Kurt Gödel has shown that it cannot be proved to be *false* within the standard axioms of set theory. There is something subtle here. The fact that it cannot be proved false does not mean it is true. Another option remains, namely, that it is *unprovable* within the axioms of set theory.

We have been admittedly vague about the use of the term “cardinal number” except to hint that “numbers” of the form $\aleph_1, \aleph_2, \aleph_3, \dots$ can be assigned to classes of sets of different cardinalities. In the next section we discuss ordinal numbers and the well-ordering theorem—facts which will shed further light on the concept of cardinal numbers.

A.7 Ordinal numbers and transfinite induction

It is usually possible to avoid recourse to transfinite methods by using some other approach (e.g., Zorn’s Lemma). But for some arguments, especially ones involving iterative processes, transfinite induction seems the natural avenue. The idea is based on an extension of ordinary mathematical induction and the fact that any nonempty set of positive integers has a smallest element. Recall that a linearly ordered set $(S, \leq)$ is said to be *well ordered* if every nonempty subset of S has a smallest element.

It is important to distinguish in the above definition the notion of ‘smallest element’ from that of an infimum. If B is a nonempty subset of the nonnegative real numbers, then B is bounded below and hence $\inf B$ exists. However $\inf B$ is a smallest element of B only if $\inf B \in B$, a fact which is not generally true.

There is a notion analogous to cardinality for well-ordered sets. Two well-ordered sets A and B are said to have the same *order type* if there is a one-to-one order-preserving mapping from one onto the other. This leads to a classification of well-ordered sets in terms of special well-ordered sets called ‘ordinal numbers’ which are obtained by extending the natural numbers.

An *ordinal number* is a well-ordered set α (more precisely, an equivalence class of such sets all having the same order type) which has the property that for each $\xi \in \alpha$, $\xi = \{\eta \in \alpha : \eta < \xi\}$. To see how the natural numbers can be viewed as ordinal numbers, begin with 0 (which satisfies the condition vacuously) and let $1 = \{0\}$, $2 = \{0, 1\}, \dots, n+1 = \{0, 1, \dots, n\}$. Viewed this way each $n \in \mathbb{N}$ is an ordinal number since n has the property that for all $k \in n$

$$k = \{i \in n : i < k\}.$$

Also for each n , $n+1$ is the smallest element of the set $\{x \in \mathbb{N} : n < x\}$. Any finite well-ordered set has the same 'order type' as the set $\{1, 2, \dots, n\}$ for some natural number n . Now observe that $\omega = \{0, 1, 2, \dots\}$ is also an ordinal number. One can continue this procedure by defining

$$\begin{aligned}\omega + 1 &= \{0, 1, 2, \dots, \omega\}, \\ \omega + 2 &= \{0, 1, 2, \dots, \omega, \omega + 1\}, \\ \omega + n + 1 &= \{0, 1, 2, \dots, \omega, \omega + 1, \dots, \omega + n\}.\end{aligned}$$

Now let

$$\omega 2 = \{0, 1, 2, \dots, \omega, \omega + 1, \omega + 2, \dots\}.$$

At this point it is worth observing that both ω and $\omega 2$ are countable sets and thus they have the same 'cardinality'. However, $\omega \neq \omega 2$ because they have different order types. There is an easy way to see this. Suppose Γ is an ordinal number and suppose $\alpha \in \Gamma$. Then by our scheme of definition if $\alpha \in \Gamma$ and if the set $\{x \in \Gamma : \alpha < x\}$ is not empty, then it has a smallest element called the *successor* of α , and this element is denoted $\alpha + 1$. This shows that the nonzero elements of Γ fall into two categories—those which are successors of other elements of Γ and those which are not. If a nonzero element of Γ is not a successor it is called a *limit ordinal*. To see that $\omega \neq \omega 2$ note that ω does not have any limit ordinals while $\omega 2$ has exactly one, namely ω .

It is possible to actually list the first few ordinals as follows (but this is only the beginning!).

$$\begin{array}{ccccccc}0, & 1, & 2, & \dots, & \omega, & \omega + 1, & \dots, \\ \omega 2, & \omega 2 + 1, & \omega 2 + 2, & \dots, & \omega 3, & \omega 3 + 1, & \dots, \\ \dots & & & & & & \\ \omega n, & \omega n + 1, & \omega n + 2, & \dots, & \omega^2, & \omega^2 + 1, & \dots, \\ \dots & & & & & & \\ \omega^n, & \omega^n + 1, & \omega^n + 2, & \dots, & \omega^{n2}, & \omega^{n2} + 1, & \dots, \\ \dots & & & & & & \\ \omega^\omega, & \omega^\omega + 1, & \omega^\omega + 2, & \dots, & \omega^{(\omega^\omega)}, & \omega^{(\omega^\omega)} + 1, & \dots, \\ \dots & & & & & & \\ \varepsilon_0, & \varepsilon_0 + 1, & \varepsilon_0 + 2, & \dots, & \varepsilon_0 \omega, & \varepsilon_0 \omega + 1, & \dots,\end{array}$$

Examples of limit ordinals in the above diagram include $\omega, \omega n, \omega^2, \omega^\omega, \varepsilon_0$, and so forth.

To summarize, each well-ordered set has an order type which coincides with the order type of some generic well-ordered set called an ordinal number. Each ordinal number can be thought of as a set consisting of the union of all strictly smaller ordinal numbers. And any ordinal number is either a successor or a limit ordinal.

Here is the principle of transfinite induction. Its usefulness rests on the premise that there exist well-ordered sets of arbitrarily large cardinality.

Proposition A.4 (*Principle of transfinite induction*) Suppose Γ is an ordinal number and suppose $S \subseteq \Gamma$. Suppose for each $\beta \in \Gamma$ the following is true.

(i) If $\alpha \in S$ for each $\alpha < \beta$ then $\beta \in S$.

Then $S = \Gamma$.

The proof of this fact is surprisingly simple. Suppose $S \neq \Gamma$. Then the set $U = \{\gamma \in \Gamma : \gamma \notin S\}$ is nonempty and thus has a smallest element γ_0 . Since $\beta \in U$, $\gamma_0 \leq \beta$. But $\gamma_0 < \beta$ implies $\gamma_0 \in S$ by (i)—a contradiction. On the other hand, $\gamma_0 = \beta$ implies $\beta \in S$ by (i)—again a contradiction.

In practice it is the verification of (i) that differs from ordinary mathematical induction. It is typical to have to consider two separate cases. (a) $\beta = \alpha + 1$ for some $\alpha \in \Gamma$, and (b) β is a limit ordinal. The verification (a) usually mimics the corresponding step in ordinary induction, and indeed, if $\Gamma = \omega$ this is all there is to do and transfinite induction coincides with ordinary induction. (Note that $0 \in S$ by assumption since the statement “ $\alpha \in S$ for each $\alpha < 0$ ” is vacuously true.) However, it may be the case, in general, that β is a limit ordinal, and the verification in this case usually involves taking intersections or passing to a limit.

The usefulness of transfinite induction stems largely from the following famous theorem, which is due to Zermelo.

Theorem A.4 (*Well-ordering theorem*) If S is a set, then there exists an order relation on S that is a well-ordering.

It is well known that the above theorem is equivalent to the Axiom of Choice. One of its consequences is that ordinal numbers exist which are larger (in the sense of cardinality) than any given set. In particular (and often this is all that is needed) there exists an uncountable ordinal, and therefore a *smallest* uncountable ordinal. (Curiously, this weaker fact does not require the full Axiom of Choice.)

We conclude this section with the following useful fact. Let Ω denote the smallest *uncountable* ordinal. In terms of cardinality discussed in the previous section, $|\Omega| = \aleph_1$.

Proposition A.5 If A is a countable subset of Ω then there exists $\beta \in \Omega$ such that $\beta \geq \alpha$ for all $\alpha \in A$.

Proof. Since A is countable one can write $A = \{\alpha_1, \alpha_2, \dots\}$. Let $\beta = \bigcup_{i=1}^{\infty} \alpha_i$. Then β is a countable ordinal since β is a countable union of countable sets. Hence $\beta \in \Omega$, and $\beta \geq \alpha_n$, $n = 1, 2, \dots$. ■

The above fact is often used in conjunction with the real numbers as follows.

Proposition A.6 *Suppose $\{x_\alpha\}_{\alpha \in \Omega} \subseteq \mathbb{R}$ is bounded and either nonincreasing or nondecreasing. Then there exists β such that $x_\alpha \equiv x_\beta$ for all $\alpha \geq \beta$.*

Proof. Suppose $\{x_\alpha\}_{\alpha \in \Omega}$ is nonincreasing. Then there exists $x \in \mathbb{R}$ such that

$$\lim_{\alpha} x_\alpha = x = \inf\{x_\alpha : \alpha \in \Omega\}.$$

Assume $x_\alpha \neq x$ for all $\alpha \in \Omega$ (otherwise there is nothing to prove). Then for each $n \in \mathbb{N}$ there exist $x_{\alpha_n} \in \left(x, x + \frac{1}{n}\right)$. By the previous proposition there exists $\beta \in \Omega$ such that $\beta \geq \alpha_n$, $n = 1, 2, \dots$. Then if $\alpha \geq \beta$ it must be the case that $x_\alpha = x_\beta \equiv x$. ■

A.8 Zermelo's Fixed Point Theorem

Most of the purely constructive aspects of metric fixed point theory (i.e., those which can be proved wholly within the basic axioms of Zermelo and Fraenkel) essentially rely on at least some version of the 1908 result of Zermelo. These results include many of those for nonexpansive mappings discussed in this text. This is an observation that seems to have originated with Fuchssteiner [58].

The following is a revised formulation of Zermelo's Theorem. We use the notation E^E to denote the collection of all mappings $f : E \rightarrow E$, and we use $\text{Fix}(f)$ to denote the fixed point set of f . (Thus $\text{Fix}(f) = \{x \in E : f(x) = x\}$.) Note that implicit in the conclusion of the following theorem is the fact that $\text{Fix}(f) \neq \emptyset$.

Theorem A.5 *Let $(E, \leq)$ be a partially ordered set in which each chain has a least upper bound, and let*

$$\Phi_E = \{f \in E^E : f(x) \geq x \text{ for each } x \in E\}.$$

Then there exists a mapping $\varphi : \Phi_E \rightarrow \Phi_E$ such that for each $f \in \Phi_E$, $\varphi(f)$ is a retraction of E onto $\text{Fix}(f)$.

Proof. Let $f \in \Phi_E$. First we claim that for each $x \in E$ there exists $w_x \in E$ such that $x \leq w_x$ and $f(w_x) = w_x$. To this end, with $x \in E$ fixed, we call a set $B \subseteq E$ (x, f) -admissible if (i) $x \in B$; (ii) $f(B) \subseteq B$; (iii) every least upper bound of any totally ordered subset of B lies in B . Note that E itself is (x, f) -admissible and also that if B is (x, f) -admissible, then $\{u \in B : x \leq u\}$ is also (x, f) -admissible. Now let

$$A = \bigcap \{B : B \text{ is } (x, f)\text{-admissible}\}.$$

Then A is also (x, f) -admissible, and $x \leq u$ for each $u \in A$. Now set

$$P = \{u \in A : y \in A \text{ and } y < u \Rightarrow f(y) \leq u\}.$$

$P \neq \emptyset$ since $x \in P$ by vacuous hypothesis. Let

$$A_1 = \{z \in A : u \in P \Rightarrow z \leq u \text{ or } z \geq f(u)\}.$$

The next step is to show that A_1 is (x, f) -admissible; hence $A_1 = A$. Obviously $x \in A_1$ since $x \leq u$ for all $u \in A$. Also, if $z \in A_1$ and $u \in P$ then either $z \leq u$ or $z \geq f(u)$. If $z \geq f(u)$ then $f(z) \geq z \geq f(u)$. If $z \leq u$, either $z < u$ in which case $f(z) \leq u$ since $u \in P$, or $z = u$ and $f(z) = f(u)$. In either case either $f(z) \leq u$ or $f(z) \geq f(u)$, and from this we conclude $f(A_1) \subseteq A_1$. Now let $u \in P$ and let $\{z_\alpha\}$ be a totally ordered subset of A_1 with $z = \sup z_\alpha$. If $z_\alpha \geq f(u)$ for some α then $z \geq z_\alpha \geq f(u)$. Otherwise $z_\alpha \leq u$ for all α from which $z \leq u$. Thus $z \in A_1$ and $A_1 = A$.

We now know that if $u \in P$ and $z \in A$, either $z \leq u$ or $f(u) \leq z$. We use this fact to show that P is (x, f) -admissible. We have already observed that $x \in P$. Now let $u \in P$ and $y \in A$ and suppose $y < f(u)$. Then $y \not\geq f(u)$ so it must be the case that $y \leq u$. If $y < u$ then $f(y) \leq u \leq f(u)$ since $u \in P$. On the other hand, if $y = u$ then $f(y) = f(u)$, so in either case $y < f(u) \Rightarrow f(y) \leq f(u)$. This proves that $f(P) \subseteq P$. Finally, let $\{u_\alpha\}$ be a totally ordered subset of P with $u = \sup u_\alpha$. Then if $y < u$ it must be the case that $y < u_\alpha$ for some α from which $f(y) \leq u_\alpha \leq u$. Hence we conclude that P is (x, f) -admissible, so $P = A$.

Therefore, for $u, z \in A$, either $z \leq u$ or $z \geq f(u) \geq u$ and this, in turn, implies that A is totally ordered. By assumption A has a least upper bound w_x . Then since $w_x \in A$, $w_x \leq f(w_x) \leq w_x$, yielding $f(w_x) = w_x$ and $x \leq w_x$. At this point we already have the conclusion of Zermelo's Theorem as originally stated.

Now define $\varphi : \Phi_E \rightarrow \Phi_E$ by setting, for each $f \in \Phi_E$,

$$\varphi(f)(x) = w_x.$$

Then since $w_x \geq x$, $\varphi(f)(x) \geq x$ so $\varphi(f) \in \Phi_E$. Furthermore, if $x \in \text{Fix}(f)$, then $\varphi(f)(x) = x$. To see this observe that A in the intersection of all (x, f) -admissible sets and for $x \in \text{Fix}(f)$, $\{x\}$ itself is an (x, f) -admissible set. It follows that $A = \{x\}$ and $w_x = x$. Hence $\varphi(f)$ leaves each point of $\text{Fix}(f)$ fixed, and for arbitrary $x \in E$ we have $\varphi(f)(x) = w_x \in \text{Fix}(f)$. ■

We have chosen to include the above argument in detail for two reasons. First, it illustrates the power of the Axiom of Choice when compared with the efficient Zorn's Lemma proof given in Section A.3. On the other hand, it illustrates that deep results can be obtained without recourse to the Axiom of Choice.

A.9 A remark about constructive mathematics

Although Theorem 1.4 is generally attributed to Brouwer (1912) (the citation is [21]), his correspondence suggests that it was known to him even earlier, and it was almost surely known earlier to others as well, most notably by P. Bohl [14]. It is curious to note, however, that Brouwer later adopted an intuitionist view of logic that led him to reject the Bolzano-Weierstrass Theorem, the premise upon which the theorem bearing his name rests. Brouwer expressed his position as follows:

The belief in the universal validity of the principle of the excluded third in mathematics is considered by the intuitionists as a phenomenon of the history of civilization of the same kind as the former belief in the rationality of π , or in the rotation of the firmament about the earth. [22]

The constructive approach grows out of the fundamental idea of interpreting 'existence' in terms of an intuitive notion of 'computability'. We shall not be concerned with this approach here, except to note in passing that there are essentially three modern constructive schools of mathematics which enjoy at least a limited following. In addition to Brouwer's intuitionism (I), there is the constructive approach developed by Erret Bishop (B), and the algorithmic approach of the Russian school founded by Markov (R). The constructive approach which seems closest to the classical approach we shall take is (B). This is because every proposition in (B) has an immediate classical interpretation, and a proof in (B) is also a classical proof, a fact which is true in neither (I) nor (R). In fact, it is possible to pass from (B) to classical mathematics by simply adopting the law of the excluded middle. (This law asserts that given any proposition P , either ' P ' holds or ' $\text{not } P$ ' holds.)

Brouwer's Theorem in (B) is the following:

Theorem A.6 *If $f : B \rightarrow B$ (B the unit ball in $\mathbb{R}^n$) is uniformly continuous, then for each $\varepsilon > 0$ there exists $x \in B$ such that*

$$d(x, f(x)) < \varepsilon.$$

Obviously this is no longer a 'fixed point theorem'. However fixed point theory does exist within (B). If in addition to the above assumptions it is assumed that f is contractive ($d(f(x), f(y)) < d(x, y)$ for each $x \neq y$) then f has a unique fixed point (see [19]).

To illustrate how these three constructive approaches differ, consider the following assertion.

Proposition A.7 *Every pointwise continuous function $f : [0, 1] \rightarrow \mathbb{R}$ is uniformly continuous.*

This proposition is true in (I), false in (R), and unprovable in (B).

For a detailed account of the constructive approaches, see Bridges and Richman [18].

Exercises

Exercise A.1 *If X is compact and if $\{x_\alpha\}$ is an ultranet in X , then $\{x_\alpha\}$ converges to a point of X .*

Exercise A.2 *Now use (1), (2), and (3) to prove Tychonoff's Theorem (Theorem A.3).*

Exercise A.3 *Prove that if a set S has n elements, then the set of all subsets of S has 2^n elements. [Try induction, and remember that $\emptyset$ is a subset of S .]*

Exercise A.4 *Prove there is a one-to-one correspondence between $\mathcal{P}(S)$ and the set 2^S of all functions from S into the two-element set $\{0, 1\}$.*

Hint: For each $A \in \mathcal{P}(S)$ consider the characteristic function $\chi_A : S \rightarrow \{0, 1\}$ defined by

$$\chi_A(x) = \begin{cases} 1 & \text{if } x \in A; \\ 0 & \text{otherwise.} \end{cases}$$

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